PROBABILITY & STATISTICS · COMMON CORE HIGH SCHOOL MATH

Answer Key

Main • Backward Branches • Forward 1 • Forward 2 (later courses)

54 question sets • 1435 questions

Page numbers match the Question Collection.

Every page shows its question ID (for example S40-F2-Q3).

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Represent data with dot plots

MAIN	Probability & Statistics main question	**HSS-ID.A.1**
BACKWARD 1	Count how often each value occurs	**6.SP.B.5.a**
BACKWARD 2	Locate values on a number-line scale	**6.NS.C.6.c**
BACKWARD 3	Read counts from a dot plot	**6.SP.B.4**
FORWARD 1	Find the median and mean from a dot plot	**HSS-ID.A.2**
FORWARD 2	Choose the display that keeps the information you need	**HSS-ID.A.1**

SET 1 • HSS-ID.A.1

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

The numbers of books 15 students read over the summer are 2, 4, 3, 2, 5, 3, 2, 6, 3, 4, 2, 3, 1, 3, 4. A dot plot shows one dot for each student above the number of books. How many dots are above 2, above 3, and above 4?

Dots above 2: _____

Dots above 3: _____

Dots above 4: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • MAIN • Q1/5

8

QUESTION ID

S1-M-Q1

TYPE

Short answer

CORRECT ANSWER

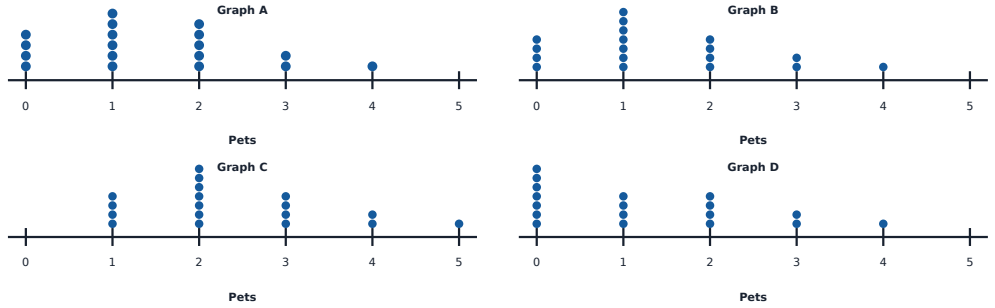
4; 5; 3

GRADING NOTE Form required: all three. Counts: 1 → 1, 2 → 4, 3 → 5, 4 → 3, 5 → 1, 6 → 1 (15 dots in all).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • MAIN • Q1/5

The numbers of pets in 18 households are 0, 1, 1, 2, 0, 3, 1, 2, 1, 0, 4, 1, 2, 1, 0, 2, 1, 3. Which dot plot shows the data?



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • MAIN • Q2/5

9

QUESTION ID

S1-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE Counts: 0 → 4, 1 → 7, 2 → 4, 3 → 2, 4 → 1. The others move one dot from 1 to 2, shift every value up 1, or swap the counts of 0 and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • MAIN • Q2/5

Fourteen quiz scores (out of 10) are 7, 9, 8, 10, 7, 8, 8, 6, 9, 8, 7, 10, 8, 9. In a dot plot of the scores, which score has the tallest stack of dots, and how many dots are in that stack?

Score: _____

Dots in that stack: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S1-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • MAIN • Q3/5

10

QUESTION ID

S1-M-Q3

TYPE

Short answer

CORRECT ANSWER

8; 5

GRADING NOTE Form required: both. Counts: 6 → 1, 7 → 3, 8 → 5, 9 → 3, 10 → 2.

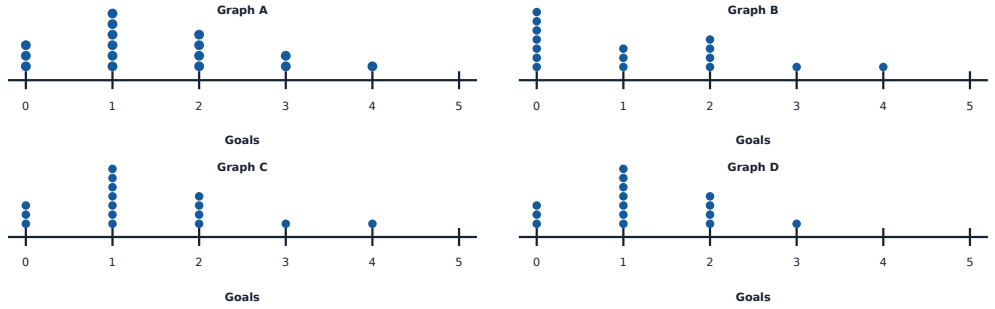
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S1-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • MAIN • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

A team scored these numbers of goals in 16 games: 1, 0, 2, 1, 3, 1, 0, 2, 2, 1, 4, 1, 0, 1, 2, 1. Which dot plot shows the data?



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • MAIN • Q4/5

11

QUESTION ID

S1-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE

Counts: 0 → 3, 1 → 7, 2 → 4, 3 → 1, 4 → 1 (16 dots). The others move a dot from 1 to 3, swap the counts of 0 and 1, or leave out the 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • MAIN • Q4/5

Twenty students reported their numbers of siblings: 1, 2, 0, 1, 3, 2, 1, 1, 0, 2, 5, 1, 2, 0, 1, 2, 3, 1, 1, 2. In a dot plot of the data, how many dots are at values of 2 or more, and how many dots are there in all?

Dots at 2 or more: _____

Dots in all: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • MAIN • Q5/5

12

QUESTION ID

S1-M-Q5

TYPE

Short answer

CORRECT ANSWER

9; 20

GRADING NOTE Form required: both. Counts: 0 → 3, 1 → 8, 2 → 6, 3 → 2, 5 → 1; 6 + 2 + 1 = 9 at 2 or more.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • MAIN • Q5/5

SET 1 • HSS-ID.A.1

BACKWARD 1

Count how often each value occurs • Grade 6 • Statistics & Probability • 6.SP.B.5.a

ANSWER KEY Reduced copy of the question page, with its answer

In the list 3, 5, 3, 4, 3, 5, how many times does 3 appear?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q1/5

14

QUESTION ID

S1-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE 3 appears in positions 1, 3 and 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q1/5

In the list 1, 2, 2, 4, 2, 1, 2, how many times does 2 appear?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q2/5

15

QUESTION ID

S1-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Four of the seven values are 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q2/5

True or false?

The list 6, 6, 7, 8, 6 has three values equal to 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q3/5

16

QUESTION ID

S1-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 6 appears three times.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q3/5

How many values are in the list 2, 4, 4, 5, 2, 4, 4?

A. 3

B. 5

C. 4

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q4/5

17

QUESTION ID

S1-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 7

GRADING NOTE Count every value, repeats included. 4 is how often the value 4 appears; 3 and 5 are other miscounts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In the list 0, 1, 0, 2, 1, 0, how many values are 0 or 1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q5/5

18

QUESTION ID

S1-B1-Q5

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Three 0s and two 1s.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q5/5

SET 1 • HSS-ID.A.1

BACKWARD 2

Locate values on a number-line scale • Grade 6 • The Number System • 6.NS.C.6.c

A number line has marks labeled 0, 1, 2, 3, 4, 5, 6.
Where does a dot for the value 4 go?

A. at the right end of the number line

B. above the fourth mark from the left, labeled 3

C. above the mark labeled 4

D. above the mark labeled 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q1/5

20

QUESTION ID

S1-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. above the mark labeled 4

GRADING NOTE Place the dot at the label, not by counting marks; the fourth mark from the left is 3 because the first is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q1/5

True or false?

On a number line with marks at 0, 2, 4, 6, 8, the value 5 is halfway between the marks 4 and 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q2/5

21

QUESTION ID

S1-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 5 is 1 more than 4 and 1 less than 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q2/5

A number line has marks at 10, 15, 20, 25, 30. What value is halfway between the marks 15 and 20?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q3/5

22

QUESTION ID

S1-B2-Q3

TYPE

Short answer

CORRECT ANSWER

17.5

GRADING NOTE (15 + 20)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q3/5

A number line runs from 0 to 10 with a mark at every whole number. How many marks are there?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q4/5

23

QUESTION ID

S1-B2-Q4

TYPE

Short answer

CORRECT ANSWER

11

GRADING NOTE 0, 1, ..., 10 is eleven marks.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q4/5

True or false?

On a horizontal number line, a larger value is placed to the left of a smaller value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q5/5

24

QUESTION ID

S1-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Values increase to the right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

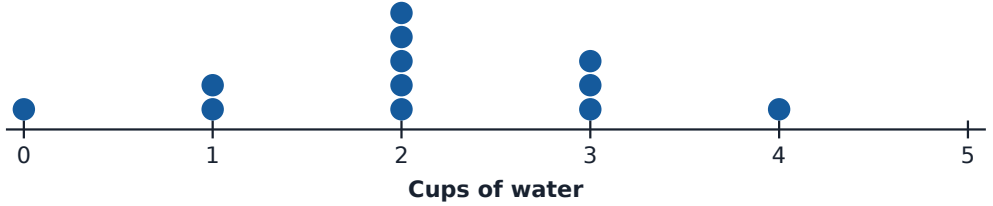
Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q5/5

SET 1 • HSS-ID.A.1

BACKWARD 3

Read counts from a dot plot • Grade 6 • Statistics & Probability • 6.SP.B.4

The dot plot shows the cups of water 12 workers drank before noon. How many workers drank 3 cups?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q1/5

26

QUESTION ID

S1-B3-Q1

TYPE

Short answer

CORRECT ANSWER

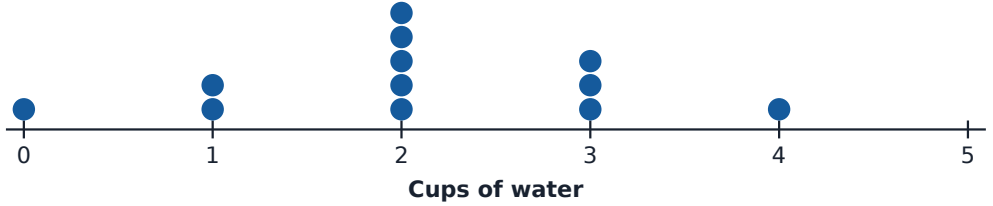
3

GRADING NOTE The stack above 3 has 3 dots.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q1/5

The dot plot shows the cups of water some workers drank before noon. How many workers are shown in all?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q2/5

27

QUESTION ID

S1-B3-Q2

TYPE

Short answer

CORRECT ANSWER

12

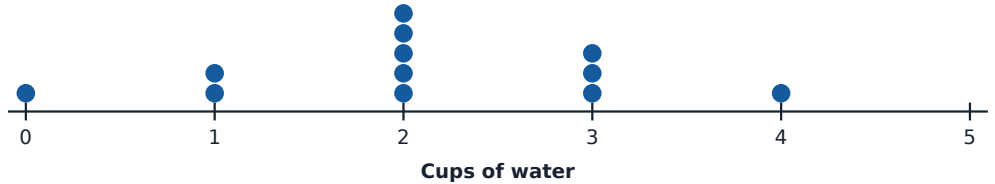
GRADING NOTE 1 + 2 + 5 + 3 + 1 = 12 dots.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q2/5

True or false?

The dot plot shows the cups of water 12 workers drank before noon. No worker drank 5 cups.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q3/5

28

QUESTION ID

S1-B3-Q3

TYPE

True or false

CORRECT ANSWER

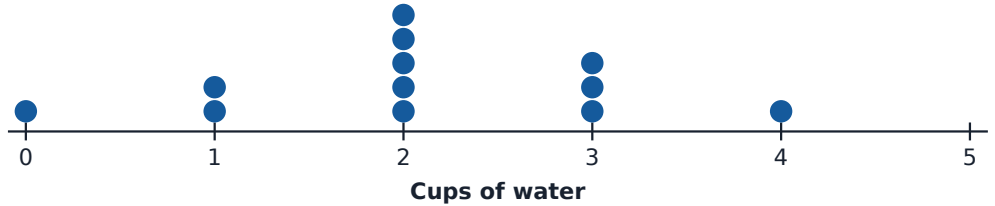
A. True

GRADING NOTE There are no dots above 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q3/5

The dot plot shows the cups of water 12 workers drank before noon. Which number of cups occurs most often?



A. 3 **B.** 12 **C.** 2 **D.** 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q4/5

29

QUESTION ID

S1-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

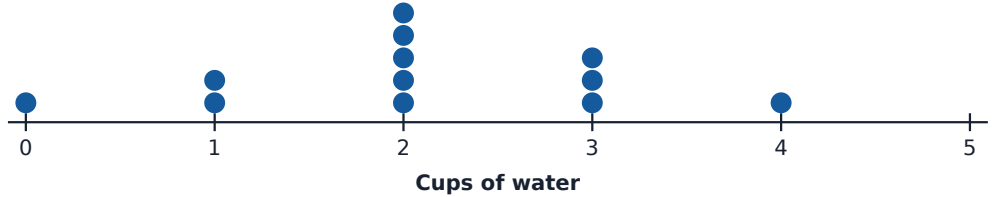
C. 2

GRADING NOTE The tallest stack (5 dots) is above 2. 12 is the number of workers; 5 is the height of the stack.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q4/5

The dot plot shows the cups of water 12 workers drank before noon. How many workers drank fewer than 2 cups?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q5/5

30

QUESTION ID

S1-B3-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE 1 at 0 and 2 at 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q5/5

SET 1 • HSS-ID.A.1

FORWARD 1

Find the median and mean from a dot plot • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

The dot plot shows rainy days in 10 two-week periods.
Find the median number of rainy days.



Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 1 • FORWARD 1 • Q1/5

32

QUESTION ID
S1-F1-Q1

TYPE
Short answer

CORRECT ANSWER
4

GRADING NOTE Ordered: 3, 3, 4, 4, 4, 4, 5, 5, 6, 7; the middle two values are 4 and 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q1**
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 1 • FORWARD 1 • Q1/5

The dot plot shows rainy days in 10 two-week periods. Find the mean number of rainy days. Give a decimal.



Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 1 • FORWARD 1 • Q2/5

33

QUESTION ID

S1-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4.5

GRADING NOTE Sum 45 over 10 values = 4.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 1 • FORWARD 1 • Q2/5

True or false?

A dot plot shows the values 3, 3, 4, 4, 4, 4, 5, 5, 6, 7. Adding one more dot at 20 changes the median more than it changes the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 1 • FORWARD 1 • Q3/5

34

QUESTION ID

S1-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The median stays 4; the mean moves from 4.5 to 65/11 ≈ 5.9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 1 • FORWARD 1 • Q3/5

The dot plot shows the cups of water 12 workers drank before noon. Which statement is correct?



A. The mean is 12 cups, because 12 workers are shown in the plot.

B. The median is 2.5 cups, halfway between the smallest and largest values.

C. The median is 5 cups, the height of the tallest stack in the plot.

D. The median is 2 cups and the mean is 2.08 cups (to the nearest hundredth).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 1 • FORWARD 1 • Q4/5

35

QUESTION ID

S1-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

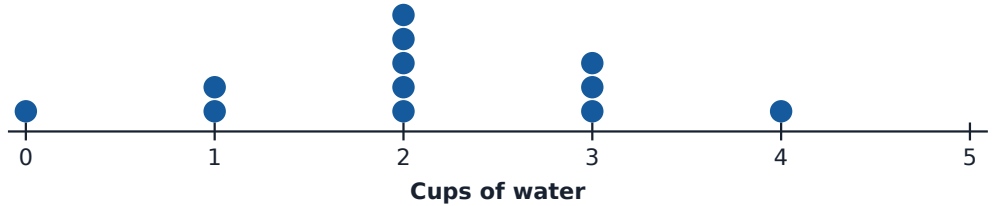
D. The median is 2 cups and the mean is 2.08 cups (to the nearest hundredth).

GRADING NOTE Ordered values: 0, 1, 1, 2, 2, 2, 2, 2, 3, 3, 3, 4; median (2 + 2)/2 = 2; mean 25/12 ≈ 2.08. The others use the midrange, a stack height, or the number of workers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 1 • FORWARD 1 • Q4/5

The dot plot shows the cups of water 12 workers drank before noon. Find the total number of cups drunk.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 1 • FORWARD 1 • Q5/5

36

QUESTION ID
S1-F1-Q5

TYPE
Short answer

CORRECT ANSWER
25

GRADING NOTE $0\cdot1 + 1\cdot2 + 2\cdot5 + 3\cdot3 + 4\cdot1 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 1 • FORWARD 1 • Q5/5

SET 1 • HSS-ID.A.1

FORWARD 2

Choose the display that keeps the information you need • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Which display always shows every individual value of a small data set?

A. a box plot

B. a histogram with bins of width 10

C. a dot plot

D. a box plot with outliers marked

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • FORWARD 2 • Q1/5

38

QUESTION ID

S1-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. a dot plot

GRADING NOTE A dot plot keeps every value; box plots keep five summary values; a histogram keeps only counts per bin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • FORWARD 2 • Q1/5

From a box plot alone, which of these can always be found?

A. the median

B. the number of values

C. the most common value

D. the mean

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • FORWARD 2 • Q2/5

39

QUESTION ID

S1-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. the median

GRADING NOTE The box plot shows the five-number summary only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • FORWARD 2 • Q2/5

True or false?

A histogram with bins 0-10, 10-20, 20-30 shows how many values are exactly 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S1-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • FORWARD 2 • Q3/5

40

QUESTION ID

S1-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It shows only how many values are in the bin 10-20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S1-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • FORWARD 2 • Q3/5

A school records the number of people in each of 400 students' households (whole numbers from 1 to 9). Which display shows how many households have each size and stays readable?

A. a histogram with one bin for each whole number

B. a list of all 400 household sizes

C. a box plot of the household sizes

D. a dot plot with one dot per household

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • FORWARD 2 • Q4/5

41

QUESTION ID

S1-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. a histogram with one bin for each whole number

GRADING NOTE A dot plot of 400 values has stacks too tall to read; a box plot loses the counts; a list is not a display.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • FORWARD 2 • Q4/5

True or false?

Two different data sets can have the same box plot but different dot plots.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • FORWARD 2 • Q5/5

42

QUESTION ID

S1-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 1, 2, 3, 4, 5, 6, 7 and 1, 2, 3.5, 4, 4.5, 6, 7 both have the five-number summary 1, 2, 4, 6, 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 1 • FORWARD 2 • Q5/5

Represent data with histograms

MAIN	Probability & Statistics main question	**HSS-ID.A.1**
BACKWARD 1	Assign values to bins with an endpoint rule	**6.SP.B.4**
BACKWARD 2	Count the values in an interval	**6.SP.B.5.a**
BACKWARD 3	Read a histogram	**6.SP.B.4**
FORWARD 1	Regroup a histogram with wider bins	**HSS-ID.A.1**
FORWARD 2	Locate the median in a histogram	**HSS-ID.A.2**

SET 2 • HSS-ID.A.1

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Commute times (minutes) of 20 employees are 12, 25, 8, 31, 17, 22, 15, 40, 19, 27, 10, 33, 21, 14, 26, 18, 35, 23, 29, 16. A histogram uses the bins 0-10, 10-20, 20-30, 30-40 and 40-50. Each bin includes its left endpoint and excludes its right endpoint. How many values are in the bins 10-20, 30-40 and 40-50?

10-20: _____

30-40: _____

40-50: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • MAIN • Q1/5

45

QUESTION ID

S2-M-Q1

TYPE

Short answer

CORRECT ANSWER

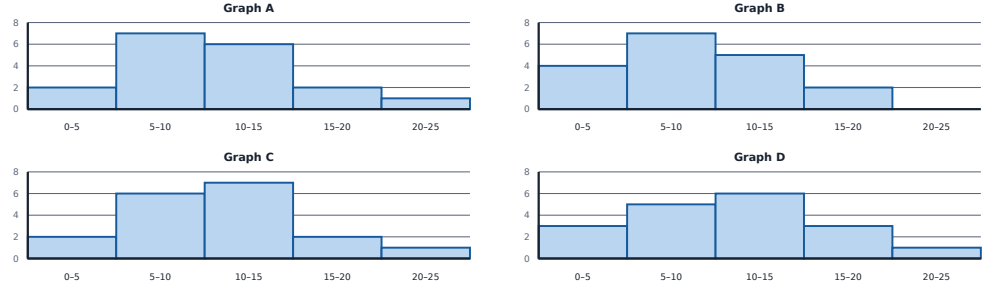
8; 3; 1

GRADING NOTE Form required: all three. Counts 0-10: 1, 10-20: 8 (10 is included), 20-30: 7, 30-40: 3, 40-50: 1 (40 goes in 40-50).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-M-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • MAIN • Q1/5

Waiting times (minutes) of 18 patients are 4, 11, 7, 15, 9, 5, 12, 20, 10, 6, 14, 8, 17, 10, 3, 13, 5, 9. Which histogram shows the data with bins 0-5, 5-10, 10-15, 15-20, 20-25? Each bin includes its left endpoint and excludes its right endpoint. In each histogram, the horizontal axis shows minutes and the vertical axis shows frequency.



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • MAIN • Q2/5

46

QUESTION ID
S2-M-Q2

TYPE
Multiple choice

CORRECT ANSWER
A. Graph A

GRADING NOTE Counts 0-5: 2, 5-10: 7, 10-15: 6, 15-20: 2, 20-25: 1 (5, 10, 15 and 20 go in the bin they start). The others put each endpoint value in the lower bin (4, 7, 5, 2, 0), move one value from 5-10 into 10-15 (2, 6, 7, 2, 1), or miscount three bins (3, 5, 6, 3, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • MAIN • Q2/5

Standing long-jump distances (cm) of 16 students are 152, 168, 145, 175, 160, 158, 171, 149, 163, 180, 155, 166, 150, 172, 161, 157. A histogram uses the bins 140-150, 150-160, 160-170, 170-180 and 180-190. Each bin includes its left endpoint and excludes its right endpoint. How many values are in the bins 150-160 and 170-180?

150-160: _____

170-180: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • MAIN • Q3/5

47

QUESTION ID

S2-M-Q3

TYPE

Short answer

CORRECT ANSWER

5; 3

GRADING NOTE Form required: both. Counts 140-150: 2, 150-160: 5, 160-170: 5, 170-180: 3, 180-190: 1.

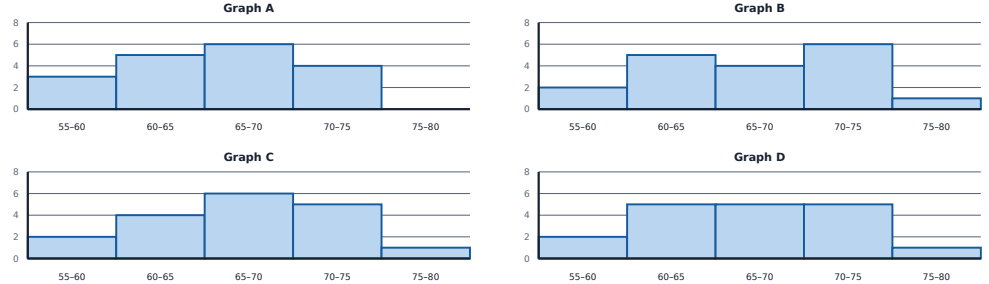
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-M-Q3**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • MAIN • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Daily high temperatures (°F) over 18 days are 61, 58, 70, 66, 72, 59, 64, 75, 68, 63, 71, 60, 67, 65, 74, 69, 62, 70. Which histogram shows the data with bins 55-60, 60-65, 65-70, 70-75, 75-80? Each bin includes its left endpoint and excludes its right endpoint. In each histogram, the horizontal axis shows °F and the vertical axis shows frequency.



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • MAIN • Q4/5

48

QUESTION ID

S2-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE Counts 55-60: 2, 60-65: 5, 65-70: 5, 70-75: 5, 75-80: 1. The others put endpoints such as 60, 65, 70 and 75 in the lower bin (3, 5, 6, 4, 0), move one value from 65-70 into 70-75 (2, 5, 4, 6, 1), or move one value from 60-65 into 65-70 (2, 4, 6, 5, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • MAIN • Q4/5

Page counts of 14 novels are 120, 255, 188, 310, 96, 205, 150, 275, 230, 182, 140, 200, 260, 175. A histogram uses bins of width 50: 50-100, 100-150, ..., 300-350. Each bin includes its left endpoint and excludes its right endpoint. Which bin has the most values (give its left endpoint), and how many values does it have?

Left endpoint of the tallest bin: _____

Values in that bin: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • MAIN • Q5/5

49

QUESTION ID

S2-M-Q5

TYPE

Short answer

CORRECT ANSWER

150; 4

GRADING NOTE Form required: both. Counts 50-100: 1, 100-150: 2, 150-200: 4, 200-250: 3, 250-300: 3, 300-350: 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-M-Q5**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • MAIN • Q5/5

SET 2 • HSS-ID.A.1

BACKWARD 1

Assign values to bins with an endpoint rule • Grade 6 • Statistics & Probability • 6.SP.B.4

Bins 10-20 and 20-30 each include their left endpoint and exclude their right endpoint. Which bin holds 20?

A. 20-30

B. 10-20

C. neither bin

D. both bins

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q1/5

51

QUESTION ID

S2-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. 20-30

GRADING NOTE 20 is the left endpoint of 20-30 and the excluded right end of 10-20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q1/5

True or false?

With bins that include the left endpoint and exclude the right endpoint, the value 30 goes in the bin 20-30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q2/5

52

QUESTION ID

S2-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 30 is excluded from 20-30; it goes in 30-40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q2/5

Bins 0-5, 5-10 and 10-15 include their left endpoint and exclude their right endpoint. Which bin holds 9.9? Give its left endpoint.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q3/5

53

QUESTION ID

S2-B1-Q3

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE $5 \le 9.9 < 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q3/5

How many of the values 4, 5, 9, 10, 12 fall in the bin 5-10, which includes 5 and excludes 10?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q4/5

54

QUESTION ID

S2-B1-Q4

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE 5 and 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q4/5

True or false?

If every bin included both of its endpoints, a value equal to an endpoint shared by two bins would be counted twice.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q5/5

55

QUESTION ID

S2-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE That is why a rule must give each endpoint to exactly one bin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • HSS-ID.A.1

BACKWARD 2

Count the values in an interval • Grade 6 • Statistics & Probability • 6.SP.B.5.a

How many of the values 12, 18, 25, 31, 19, 22 are at least 15 and less than 25?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q1/5

57

QUESTION ID

S2-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE 18, 19, 22.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q1/5

How many of the values 3.5, 4, 6.2, 7, 8 are at least 4 and less than 7?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q2/5

58

QUESTION ID

S2-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE 4 and 6.2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q2/5

True or false?

Exactly two of the values 40, 45, 50, 55 are at least 45 and less than 55.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q3/5

59

QUESTION ID

S2-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 45 and 50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q3/5

How many of the values 2, 9, 11, 14, 20, 21 are at least 10 and less than 20?

A. 3

B. 4

C. 1

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q4/5

60

QUESTION ID

S2-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 2

GRADING NOTE 11 and 14; 20 is not less than 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q4/5

How many of the values 100, 150, 199, 200, 250 are at least 100 and less than 200?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q5/5

61

QUESTION ID

S2-B2-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE 100, 150, 199.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q5

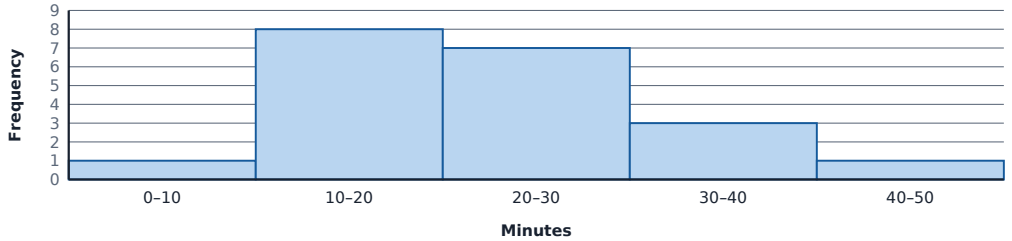
Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q5/5

SET 2 • HSS-ID.A.1

BACKWARD 3

Read a histogram • Grade 6 • Statistics & Probability • 6.SP.B.4

The histogram shows the commute times of some employees. How many employees have commutes in the bin 20-30?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q1/5

63

QUESTION ID

S2-B3-Q1

TYPE

Short answer

CORRECT ANSWER

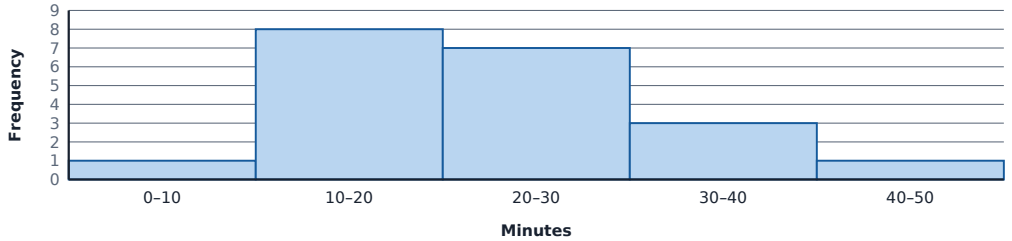
7

GRADING NOTE The 20-30 bar reaches 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q1/5

The histogram shows the commute times of some employees. How many employees are shown in all?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q2/5

64

QUESTION ID

S2-B3-Q2

TYPE

Short answer

CORRECT ANSWER

20

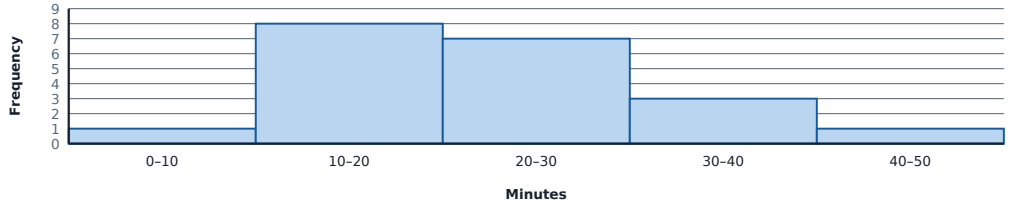
GRADING NOTE 1 + 8 + 7 + 3 + 1 = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q2/5

True or false?

The histogram shows the commute times of some employees. It shows that one employee's commute was exactly 15 minutes.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q3/5

65

QUESTION ID

S2-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

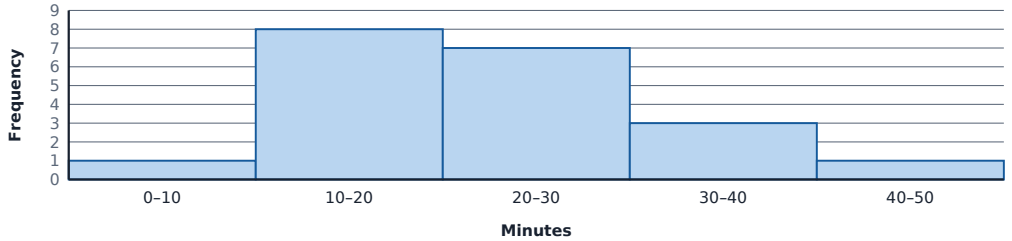
GRADING NOTE A histogram shows counts per bin, not individual values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The histogram shows the commute times of some employees. Which bin has the most values?



A. 10-20 **B.** 0-10 **C.** 20-30 **D.** 30-40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q4/5

66

QUESTION ID

S2-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 10-20

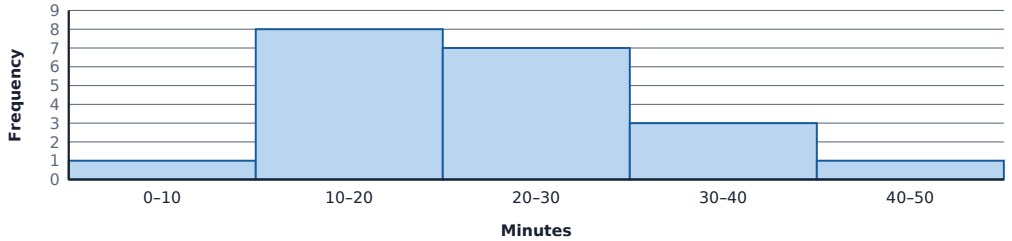
GRADING NOTE Its bar reaches 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

The histogram shows the commute times of some employees. How many employees have commutes of at least 30 minutes?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q5/5

67

QUESTION ID

S2-B3-Q5

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE 3 in 30-40 and 1 in 40-50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q5/5

SET 2 • HSS-ID.A.1

FORWARD 1

Regroup a histogram with wider bins • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

A histogram with bins 0-5, 5-10, 10-15, 15-20 has counts 3, 6, 8, 2. What are the counts for the wider bins 0-10 and 10-20?

0-10: _____

10-20: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • FORWARD 1 • Q1/5

69

QUESTION ID

S2-F1-Q1

TYPE

Short answer

CORRECT ANSWER

9; 10

GRADING NOTE Form required: both. 3 + 6 and 8 + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • FORWARD 1 • Q1/5

A histogram with bins 60-65, 65-70, 70-75, 75-80 has counts 2, 5, 6, 4. What are the counts for 60-70 and 70-80?

60-70: _____

70-80: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • FORWARD 1 • Q2/5

70

QUESTION ID

S2-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7; 10

GRADING NOTE Form required: both. 2 + 5 and 6 + 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • FORWARD 1 • Q2/5

True or false?

Combining each pair of neighboring bins into one bin never changes the total number of values shown.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • FORWARD 1 • Q3/5

71

QUESTION ID

S2-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Every value is still counted exactly once.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • FORWARD 1 • Q3/5

A histogram with bins of width 1 has counts 1, 0, 2, 0, 1, 0, 3, 0, 2, so it looks jagged. Why might bins of width 2 help?

A. Wider bins always make a distribution look symmetric.

B. Wider bins change the data values, so the shape becomes smoother.

C. Wider bins add values to the empty bins, so no bin is 0.

D. Wider bins group nearby values, so the overall shape is easier to see.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • FORWARD 1 • Q4/5

72

QUESTION ID

S2-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Wider bins group nearby values, so the overall shape is easier to see.

GRADING NOTE Regrouping changes only the summary, not the data. The others claim the data change, values are added, or symmetry always appears.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • FORWARD 1 • Q4/5

A histogram with bins of width 10 has counts 4, 9, 5. Can the counts for bins of width 5 be found from it?

A. Yes: halve each count, because each narrower bin is half as wide.

B. Yes: keep the same counts, because the data have not changed.

C. No: each count could be split between the two narrower bins in many ways.

D. No: a histogram with bins of width 10 cannot be redrawn at all.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • FORWARD 1 • Q5/5

73

QUESTION ID

S2-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. No: each count could be split between the two narrower bins in many ways.

GRADING NOTE Combining bins loses information that cannot be recovered. Halving assumes an even split; the others are false.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q5**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 2 • FORWARD 1 • Q5/5

SET 2 • HSS-ID.A.1

FORWARD 2

Locate the median in a histogram • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

A histogram of 21 values has counts 0-10: 4, 10-20: 6, 20-30: 7, 30-40: 4. The median is the 11th value in order. Which bin holds it? Give its left endpoint.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 2 • FORWARD 2 • Q1/5

75

QUESTION ID
S2-F2-Q1

TYPE
Short answer

CORRECT ANSWER
20

GRADING NOTE 4 + 6 = 10 values come before 20, so the 11th is in 20-30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 2 • FORWARD 2 • Q1/5

A histogram of 20 values has counts 50-60: 3, 60-70: 5, 70-80: 8, 80-90: 4. The median is the mean of the 10th and 11th values. Which bin holds both? Give its left endpoint.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 2 • FORWARD 2 • Q2/5

76

QUESTION ID
S2-F2-Q2

TYPE
Short answer

CORRECT ANSWER
70

GRADING NOTE 3 + 5 = 8 values lie below 70; the 9th through 16th are in 70-80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 2 • FORWARD 2 • Q2/5

True or false?

The bin with the tallest bar always holds the median.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 2 • FORWARD 2 • Q3/5

77

QUESTION ID
S2-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Counts 10, 2, 3, 1, 9 (25 values): the tallest bar is the first, but the median (13th value) is in the third bin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 2 • FORWARD 2 • Q3/5

A histogram of 15 values has counts 0-5: 10, 5-10: 2, 10-15: 2, 15-20: 1. Which bin holds the median (the 8th value)?

A. 15-20

B. 10-15

C. 5-10

D. 0-5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 2 • FORWARD 2 • Q4/5

78

QUESTION ID
S2-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. 0-5

GRADING NOTE The first 10 values are in 0-5. The others use the middle bin or later bins.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 2 • FORWARD 2 • Q4/5

A histogram of 15 values has counts 0-10: 2, 10-20: 3, 20-30: 3, 30-40: 7. Which bin holds the median (the 8th value)? Give its left endpoint.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 2 • FORWARD 2 • Q5/5

79

QUESTION ID
S2-F2-Q5

TYPE
Short answer

CORRECT ANSWER
20

GRADING NOTE 2 + 3 = 5 below 20, 8 below 30; the 8th is in 20-30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q5**
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 2 • FORWARD 2 • Q5/5

Represent data with box plots

MAIN	Probability & Statistics main question	**HSS-ID.A.1**
BACKWARD 1	Order data and find the median	**6.SP.B.5.c**
BACKWARD 2	Find quartiles as medians of the halves	**6.SP.B.5.c**
BACKWARD 3	Read a box plot	**6.SP.B.4**
FORWARD 1	Mark outliers on a box plot	**HSS-ID.A.3**
FORWARD 2	Compare two box plots	**HSS-ID.A.2**

SET 3 • HSS-ID.A.1

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Laps swum by 11 swimmers: 12, 15, 9, 18, 14, 11, 16, 20, 13, 10, 17. Give the five-number summary for a box plot. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.

Minimum: _____

Q1: _____

Median: _____

Q3: _____

Maximum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 3 • MAIN • Q1/5

82

QUESTION ID

S3-M-Q1

TYPE

Short answer

CORRECT ANSWER

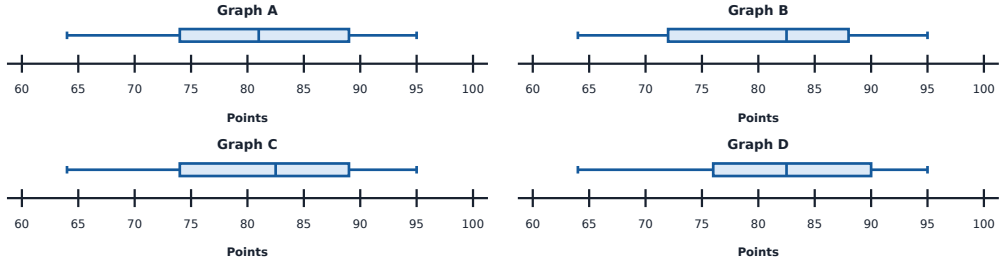
9; 11; 14; 17; 20

GRADING NOTE Form required: all five. Ordered: 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 20; median 14; lower half 9-13 gives Q1 = 11; upper half 15-20 gives Q3 = 17.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-M-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 3 • MAIN • Q1/5

Test scores of 12 students: 72, 85, 90, 64, 78, 88, 95, 70, 81, 76, 84, 92. Which box plot shows the data? Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 3 • MAIN • Q2/5

83

QUESTION ID

S3-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE Ordered: 64, 70, 72, 76, 78, 81, 84, 85, 88, 90, 92, 95; median (81 + 84)/2 = 82.5; Q1 = (72 + 76)/2 = 74; Q3 = (88 + 90)/2 = 89. The others use 81 as the median, the 3rd and 9th values as quartiles, or the wrong middle pair of each half.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 3 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Hours of sleep of 14 teenagers: 7.5, 6, 8, 6.5, 9, 7, 5.5, 8.5, 7, 6.5, 7.5, 8, 6, 7. Give the five-number summary for a box plot. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.

Minimum: _____

Q1: _____

Median: _____

Q3: _____

Maximum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 3 • MAIN • Q3/5

84

QUESTION ID

S3-M-Q3

TYPE

Short answer

CORRECT ANSWER

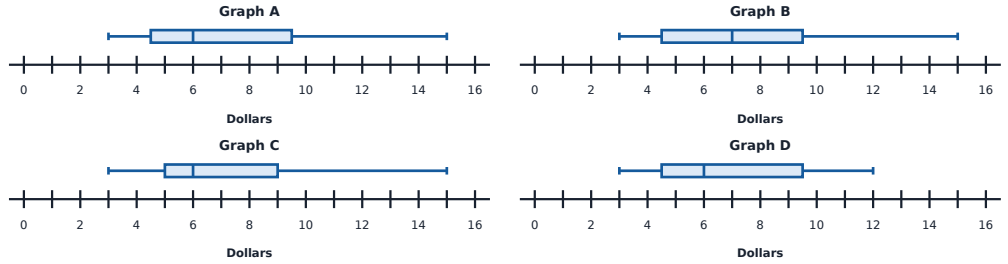
5.5; 6.5; 7; 8; 9

GRADING NOTE Form required: all five. Ordered: 5.5, 6, 6, 6.5, 6.5, 7, 7 | 7, 7.5, 7.5, 8, 8, 8.5, 9; median 7; Q1 = 6.5; Q3 = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 3 • MAIN • Q3/5

Prices (dollars) of 13 menu items: 3, 8, 5, 12, 6, 4, 9, 7, 10, 5, 6, 15, 4. Which box plot shows the data? Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 3 • MAIN • Q4/5

85

QUESTION ID

S3-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE Ordered: 3, 4, 4, 5, 5, 6, 6, 7, 8, 9, 10, 12, 15; median 6 (7th value); Q1 = (4 + 5)/2 = 4.5; Q3 = (9 + 10)/2 = 9.5. The others use 7 as the median, keep the median in both halves (Q1 = 5, Q3 = 9), or end the whisker at 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-M-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 3 • MAIN • Q4/5

Distances (km) 10 students live from school: 2.1, 3.4, 1.8, 5, 2.7, 4.2, 3, 2.4, 3.9, 1.5. Give the five-number summary for a box plot. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.

Minimum: _____

Q1: _____

Median: _____

Q3: _____

Maximum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 3 • MAIN • Q5/5

86

QUESTION ID

S3-M-Q5

TYPE

Short answer

CORRECT ANSWER

1.5; 2.1; 2.85; 3.9; 5

GRADING NOTE Form required: all five. Ordered: 1.5, 1.8, 2.1, 2.4, 2.7 | 3.0, 3.4, 3.9, 4.2, 5.0; median (2.7 + 3.0)/2 = 2.85; Q1 = 2.1; Q3 = 3.9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.1 • Set 3 • MAIN • Q5/5

SET 3 • HSS-ID.A.1

BACKWARD 1

Order data and find the median • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the median of 9, 3, 7, 5, 11.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q1/5

88

QUESTION ID

S3-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Ordered: 3, 5, 7, 9, 11.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q1/5

Find the median of 4, 10, 6, 8.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q2/5

89

QUESTION ID

S3-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Ordered: 4, 6, 8, 10; (6 + 8)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q2/5

True or false?

The median of 2, 9, 4 is 9, the middle value of the list as written.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q3/5

90

QUESTION ID

S3-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Order first: 2, 4, 9; the median is 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q3/5

What is the median of 1, 3, 3, 6, 7, 8?

A. 3

B. 6

C. 4.67

D. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q4/5

91

QUESTION ID

S3-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 4.5

GRADING NOTE (3 + 6)/2 = 4.5. 3 is the mode, 6 a middle value alone, 4.67 the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q4/5

Find the median of 12, 15, 11, 14, 13, 16, 10.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q5/5

92

QUESTION ID

S3-B1-Q5

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Ordered: 10, 11, 12, 13, 14, 15, 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q5/5

SET 3 • HSS-ID.A.1

BACKWARD 2

Find quartiles as medians of the halves • Grade 6 • Statistics & Probability • 6.SP.B.5.c

For 2, 4, 6, 8, 10, 12, 14, 16, find Q1 and Q3 (each the median of a half).

Q1: _____

Q3: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q1/5

94

QUESTION ID

S3-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5; 13

GRADING NOTE Form required: both. Halves 2, 4, 6, 8 and 10, 12, 14, 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q1/5

For 1, 3, 5, 7, 9, 11, 13 (an odd count), leave out the median and find Q1 and Q3.

Q1: _____

Q3: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q2/5

95

QUESTION ID

S3-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3; 11

GRADING NOTE Form required: both. Halves 1, 3, 5 and 9, 11, 13.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q2/5

True or false?

For 1, 2, 3, 4, 5, 6, 7, 8, 9, Q1 is 2.5 when the median 5 is left out of both halves.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q3/5

96

QUESTION ID

S3-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Lower half 1, 2, 3, 4; its median is 2.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q3/5

For 10, 20, 30, 40, 50, 60, what is the interquartile range Q3 − Q1 (Q1 and Q3 are the medians of the halves)?

A. 30

B. 50

C. 20

D. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q4/5

97

QUESTION ID

S3-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 30

GRADING NOTE Q1 = 20, Q3 = 50. 50 is the range; 20 and 35 come from wrong halves.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q4/5

For 5, 6, 8, 9, 12, 15, find Q1 (the median of the lower half).

Q1: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q5/5

98

QUESTION ID

S3-B2-Q5

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Lower half 5, 6, 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

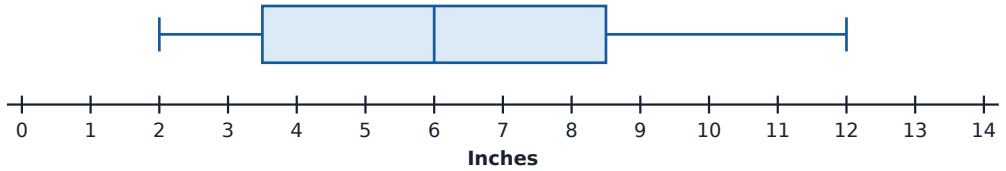
Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q5/5

SET 3 • HSS-ID.A.1

BACKWARD 3

Read a box plot • Grade 6 • Statistics & Probability • 6.SP.B.4

The box plot shows the snowfall in some storms. What is the median snowfall?



Median (inches): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q1/5

100

QUESTION ID

S3-B3-Q1

TYPE

Short answer

CORRECT ANSWER

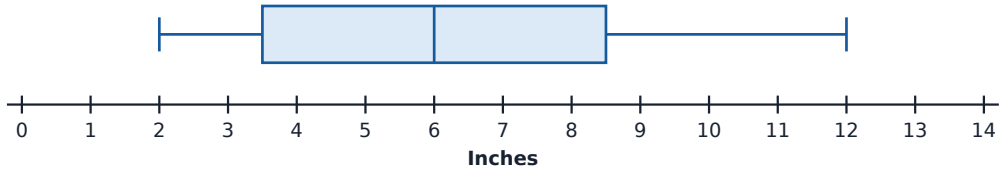
6

GRADING NOTE The line inside the box is at 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q1/5

The box plot shows the snowfall in some storms. What is the interquartile range?



IQR (inches): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q2/5

101

QUESTION ID

S3-B3-Q2

TYPE

Short answer

CORRECT ANSWER

5

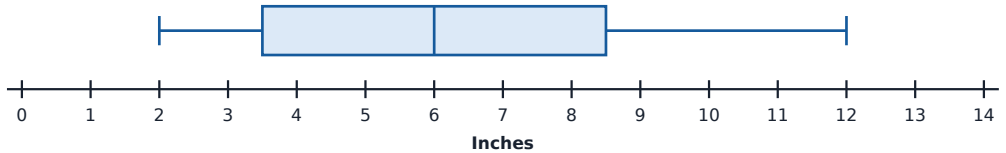
GRADING NOTE Q3 − Q1 = 8.5 − 3.5 = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q2/5

True or false?

The box plot shows the snowfall in some storms. About half of the storms had at least 6 inches.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q3/5

102

QUESTION ID

S3-B3-Q3

TYPE

True or false

CORRECT ANSWER

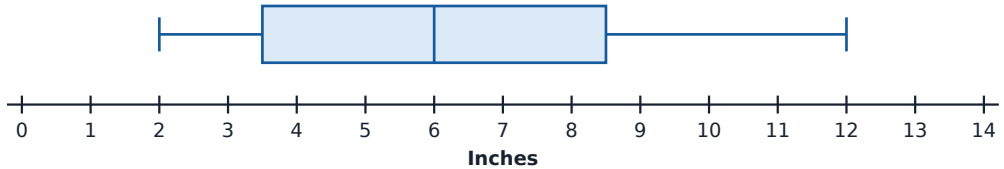
A. True

GRADING NOTE The median is 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q3/5

The box plot shows the snowfall in some storms. What is the range?



A. 6 inches **B.** 10 inches **C.** 12 inches **D.** 5 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q4/5

103

QUESTION ID

S3-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

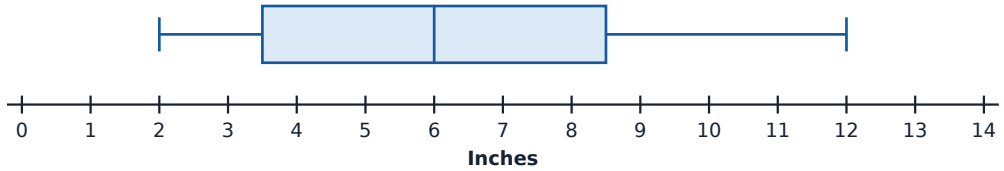
B. 10 inches

GRADING NOTE 12 − 2 = 10. 5 is the IQR, 12 the maximum, 6 the median.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q4/5

The box plot shows the snowfall in some storms. What is Q1?



Q1 (inches): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q5/5

104

QUESTION ID

S3-B3-Q5

TYPE

Short answer

CORRECT ANSWER

3.5

GRADING NOTE The left edge of the box is at 3.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q5/5

SET 3 • HSS-ID.A.1

FORWARD 1

Mark outliers on a box plot • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

A data set has Q1 = 20 and Q3 = 30. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. What are the lower and upper fences?

Lower fence: _____

Upper fence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 3 • FORWARD 1 • Q1/5

106

QUESTION ID
S3-F1-Q1

TYPE
Short answer

CORRECT ANSWER
5; 45

GRADING NOTE Form required: both. IQR 10; 20 − 15 and 30 + 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 3 • FORWARD 1 • Q1/5

Prices of 10 used bikes (dollars): 180, 210, 195, 205, 190, 600, 200, 185, 215, 220. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. How many outliers are there, and where does the upper whisker end?

Number of outliers: _____

Upper whisker ends at: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 3 • FORWARD 1 • Q2/5

107

QUESTION ID

S3-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1; 220

GRADING NOTE Form required: both. Ordered 180, 185, 190, 195, 200, 205, 210, 215, 220, 600; Q1 = 190, Q3 = 215, IQR 25; fences 152.5 and 252.5; only 600 is an outlier; the whisker ends at the largest non-outlier, 220.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 3 • FORWARD 1 • Q2/5

True or false?

In a box plot that marks outliers, a whisker ends at the fence value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S3-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 3 • FORWARD 1 • Q3/5

108

QUESTION ID

S3-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It ends at the most extreme data value inside the fence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S3-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 3 • FORWARD 1 • Q3/5

Race times (seconds) of 9 swimmers: 52, 55, 54, 31, 56, 53, 57, 55, 54. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. Which value is an outlier?

A. 31

B. 57

C. both 31 and 57

D. none

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 3 • FORWARD 1 • Q4/5

109

QUESTION ID

S3-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 31

GRADING NOTE Ordered 31, 52, 53, 54, 54, 55, 55, 56, 57; Q1 = (52 + 53)/2 = 52.5, Q3 = (55 + 56)/2 = 55.5; IQR 3; fences 48 and 60; only 31 is outside.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 3 • FORWARD 1 • Q4/5

A box plot that marks outliers has Q1 = 14 and Q3 = 17, and the largest data values are 18 and 41. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. Where does the upper whisker end?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S3-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 3 • FORWARD 1 • Q5/5

110

QUESTION ID
S3-F1-Q5

TYPE
Short answer

CORRECT ANSWER
18

GRADING NOTE Upper fence 17 + 4.5 = 21.5; 41 is an outlier, 18 is not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q5**

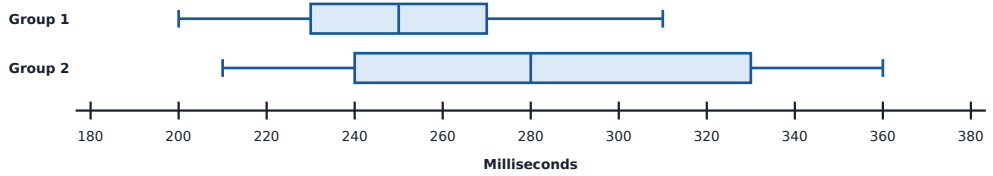
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 3 • FORWARD 1 • Q5/5

SET 3 • HSS-ID.A.1

FORWARD 2

Compare two box plots • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

The box plots show the reaction times of two groups. Which comparison is correct?



A. Group 1 has the greater median and the greater IQR.

B. Group 1 has the greater range and the greater median.

C. The groups have the same median but different IQRs.

D. Group 2 has the greater median and the greater IQR.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 3 • FORWARD 2 • Q1/5

112

QUESTION ID

S3-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. Group 2 has the greater median and the greater IQR.

GRADING NOTE Medians 250 and 280; IQRs 40 and 90. The others reverse or misread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 3 • FORWARD 2 • Q1/5

Box plot 1 has median 40 and IQR 12; box plot 2 has median 46 and IQR 8. By how much do the medians differ, and which has the smaller IQR (enter 1 or 2)?

Difference of medians: _____

Smaller IQR (1 or 2): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 3 • FORWARD 2 • Q2/5

113

QUESTION ID

S3-F2-Q2

TYPE

Short answer

CORRECT ANSWER

6; 2

GRADING NOTE Form required: both. $46 - 40 = 6$; $8 < 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 3 • FORWARD 2 • Q2/5

True or false?

If two box plots have the same median, the two data sets have the same mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 3 • FORWARD 2 • Q3/5

114

QUESTION ID

S3-F2-Q3

TYPE

True or false

CORRECT ANSWER

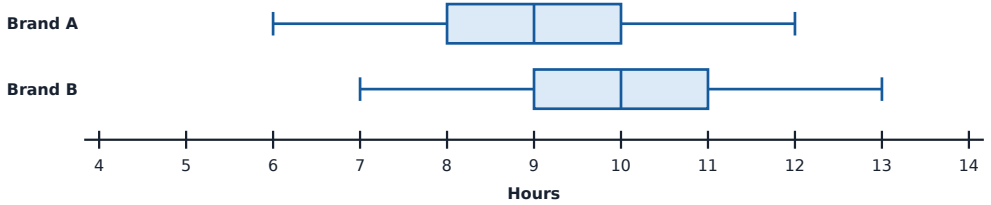
B. False

GRADING NOTE The mean is not shown by a box plot and can differ (for example with different skew).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 3 • FORWARD 2 • Q3/5

The box plots show the battery lives of two brands. About what fraction of brand B batteries last longer than the median life of brand A?



A. about 1/2 **B.** about 3/4 **C.** none **D.** about 1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 3 • FORWARD 2 • Q4/5

115

QUESTION ID
S3-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. about 3/4

GRADING NOTE Brand A's median is 9, which is Brand B's Q1, so about 3/4 of Brand B lies above it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 3 • FORWARD 2 • Q4/5

Two box plots: class A has Q1 = 70 and Q3 = 85; class B has Q1 = 72 and Q3 = 80. Give each IQR.

IQR of A: _____

IQR of B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 3 • FORWARD 2 • Q5/5

116

QUESTION ID

S3-F2-Q5

TYPE

Short answer

CORRECT ANSWER

15; 8

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q5**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 3 • FORWARD 2 • Q5/5

SET 4 • HSS-ID.A.2 • INTERPRETING CATEGORICAL AND QUANTITATIVE DATA

Compare centers with the measure suited to the shape

MAIN	Probability & Statistics main question	**HSS-ID.A.2**
BACKWARD 1	Compute a mean	**6.SP.B.5.c**
BACKWARD 2	Compute a median	**6.SP.B.5.c**
BACKWARD 3	Recognize skew and extreme values	**6.SP.A.2**
FORWARD 1	Find a missing value from a target mean or median	**HSS-ID.A.2**
FORWARD 2	Predict how changes in data move the mean and median	**HSS-ID.A.3**

SET 4 • HSS-ID.A.2

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Sale prices (thousands of dollars) of 8 homes in each of two towns: Town A: 210, 225, 230, 240, 245, 250, 260, 900; Town B: 190, 220, 235, 255, 270, 280, 300, 310. Compare the typical prices with one measure of center suited to the shapes of both data sets. Give that measure for each town.

Town A: _____

Town B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • MAIN • Q1/5

119

QUESTION ID
S4-M-Q1

TYPE
Short answer

CORRECT ANSWER
242.5; 262.5

GRADING NOTE Form required: both medians. Town A has an extreme value (900), so the median is suited; the same measure is used for both towns. Town A: (240 + 245)/2 = 242.5; Town B: (255 + 270)/2 = 262.5. (Town A's mean, 320, is pulled up by 900.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • MAIN • Q1/5

Scores of 7 students in each of two classes: class P: 70, 74, 76, 78, 80, 82, 86; class Q: 66, 70, 72, 74, 76, 78, 82. Which comparison of centers is appropriate and correct?

A. Compare medians: P's median 76 is 2 points above Q's median 74, so P is 2 points higher.

B. Compare the highest scores: P's 86 is 4 points above Q's 82, so P is 4 points higher.

C. Both are roughly symmetric with no outliers, so compare means: P's mean 78 is 4 points above Q's mean 74.

D. Both are roughly symmetric, so compare means: P's mean 76 is 2 points above Q's mean 74.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S4-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • MAIN • Q2/5

120

QUESTION ID
S4-M-Q2

TYPE
Multiple choice

CORRECT ANSWER
C. Both are roughly symmetric with no outliers, so compare means: P's mean 78 is 4 points above Q's mean 74.

GRADING NOTE Means: 546/7 = 78 and 518/7 = 74 (the medians are also 78 and 74). The others miscompute the mean or median of P, or compare maximums, which do not describe the center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S4-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • MAIN • Q2/5

Delivery times (days) of 9 orders from each of two stores: store X: 2, 2, 3, 3, 3, 4, 4, 5, 14; store Y: 2, 3, 3, 4, 4, 5, 5, 6, 6. Compare the typical delivery times with one measure of center suited to the shapes of both data sets. Give that measure for each store.

Store X: _____

Store Y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • MAIN • Q3/5

121

QUESTION ID

S4-M-Q3

TYPE

Short answer

CORRECT ANSWER

3; 4

GRADING NOTE Form required: both medians. Store X has an extreme value (14), so medians are used: 3 and 4 (5th values).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • MAIN • Q3/5

Heights (cm) of 7 seedlings of each of two plants: plant A: 5, 6, 6, 7, 7, 8, 30; plant B: 8, 8, 9, 9, 9, 10, 10. Which comparison of typical heights is appropriate?

A. Compare means: A's mean (about 9.9 cm) is greater, so A's seedlings are typically taller.

B. Compare means: both are about 9 cm, so the seedlings are typically the same height.

C. Compare maximums: A's tallest seedling (30 cm) is taller, so A's seedlings are typically taller.

D. A is skewed by the value 30, so compare medians: B's median 9 cm is greater than A's median 7 cm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • MAIN • Q4/5

122

QUESTION ID

S4-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. A is skewed by the value 30, so compare medians: B's median 9 cm is greater than A's median 7 cm.

GRADING NOTE A's mean 69/7 ≈ 9.86 is pulled up by 30; medians 7 and 9 describe typical seedlings. The others use the mean despite the outlier or use the maximum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Masses (grams) of 6 bags filled by each of two machines: machine 1: 498, 499, 500, 500, 501, 502; machine 2: 503, 504, 505, 505, 506, 507. Compare the typical masses with one measure of center suited to the shapes of both data sets. Give that measure for each machine.

Machine 1: _____

Machine 2: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • MAIN • Q5/5

123

QUESTION ID
S4-M-Q5

TYPE
Short answer

CORRECT ANSWER
500; 505

GRADING NOTE Form required: both. Both sets are symmetric with no outliers, so means suit: 3000/6 = 500 and 3030/6 = 505. (The medians are also 500 and 505, so either measure gives the same values.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • MAIN • Q5/5

SET 4 • HSS-ID.A.2

BACKWARD 1

Compute a mean • Grade 6 • Statistics & Probability • 6.SP.B.5.c

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 4, 7, 10.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q1/5

125

QUESTION ID

S4-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE 21/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 2, 3, 5, 6. Give a decimal.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q2/5

126

QUESTION ID

S4-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE 16/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q2/5

True or false?

The mean of 1, 2, 3, 10 is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q3/5

127

QUESTION ID

S4-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 16/4 = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q3/5

What is the mean of 5, 5, 6, 8?

A. 24

B. 5

C. 6

D. 5.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q4/5

128

QUESTION ID

S4-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 6

GRADING NOTE 24/4. 5 is the mode, 5.5 the median, 24 the sum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q4/5

Find the mean of 1.5, 2.5, 3.5. Give a decimal.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q5/5

129

QUESTION ID

S4-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2.5

GRADING NOTE 7.5/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q5/5

SET 4 • HSS-ID.A.2

BACKWARD 2

Compute a median • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the median of 8, 2, 6.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q1/5

131

QUESTION ID

S4-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Ordered 2, 6, 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q1/5

Find the median of 3, 9, 5, 7.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q2/5

132

QUESTION ID

S4-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Ordered 3, 5, 7, 9; (5 + 7)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q2/5

True or false?

The median of 1, 2, 3, 100 is 2.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q3/5

133

QUESTION ID

S4-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (2 + 3)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q3/5

What is the median of 10, 12, 12, 15, 30?

A. 15.8

B. 12

C. 30

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q4/5

134

QUESTION ID

S4-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 12

GRADING NOTE Third value. 15.8 is the mean; 15 and 30 are other values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q4/5

Find the median of 4.5, 6, 5, 7.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q5/5

135

QUESTION ID

S4-B2-Q5

TYPE

Short answer

CORRECT ANSWER

5.5

GRADING NOTE Ordered 4.5, 5, 6, 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q5/5

SET 4 • HSS-ID.A.2

BACKWARD 3

Recognize skew and extreme values • Grade 6 • Statistics & Probability • 6.SP.A.2

Which data set is skewed to the right?

A. 5, 5, 5, 5, 5

B. 1, 12, 13, 13, 14, 14, 14, 15

C. 1, 2, 2, 3, 3, 3, 4, 15

D. 1, 2, 3, 4, 5, 6, 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q1/5

137

QUESTION ID

S4-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. 1, 2, 2, 3, 3, 3, 4, 15

GRADING NOTE A long tail of high values (15). The others are constant, skewed left (a low tail at 1), or symmetric.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q1/5

True or false?

A data set with a single value far below all the others is skewed to the left.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q2/5

138

QUESTION ID

S4-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The tail points toward the low value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q2/5

Which value is extreme in 20, 22, 21, 23, 22, 60?

A. 60

B. 23

C. none

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q3/5

139

QUESTION ID

S4-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 60

GRADING NOTE 60 is far from the rest.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q3/5

True or false?

In 4, 5, 5, 6, 6, 6, 7, 7, 8 the values are spread evenly on both sides of 6, so the data are roughly symmetric.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q4/5

140

QUESTION ID

S4-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Mirror image around 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q4/5

Incomes in a town include a few very large incomes. The distribution is most likely

A. skewed to the right

B. skewed to the left

C. uniform

D. symmetric

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q5/5

141

QUESTION ID

S4-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. skewed to the right

GRADING NOTE A few large values make a long right tail.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q5/5

SET 4 • HSS-ID.A.2

FORWARD 1

Find a missing value from a target mean or median • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

Four quiz scores are 80, 85, 90, 75. What fifth score makes the mean of all five scores 84?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • FORWARD 1 • Q1/5

143

QUESTION ID
S4-F1-Q1

TYPE
Short answer

CORRECT ANSWER
90

GRADING NOTE $5 \cdot 84 = 420$; $420 - 330 = 90$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • FORWARD 1 • Q1/5

The values 3, 5, 8 and x have mean 7. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • FORWARD 1 • Q2/5

144

QUESTION ID

S4-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE $4 \cdot 7 = 28$; $28 - 16 = 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • FORWARD 1 • Q2/5

True or false?

In a data set of 7 values, replacing the largest value with a larger number can change the median.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • FORWARD 1 • Q3/5

145

QUESTION ID

S4-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The median is the 4th value, which does not change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • FORWARD 1 • Q3/5

The values 4, 6, 9, 10, 12 and one more value x have a median of 8. What is x?

A. 16

B. 7

C. 8

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • FORWARD 1 • Q4/5

146

QUESTION ID

S4-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 7

GRADING NOTE With x = 7: 4, 6, 7, 9, 10, 12 has median (7 + 9)/2 = 8. x = 8 gives 8.5, x = 16 gives 9.5, x = 6 gives 7.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • FORWARD 1 • Q4/5

A team's mean score over 9 games is 70. What score in a 10th game makes the mean of all 10 games 72?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • FORWARD 1 • Q5/5

147

QUESTION ID
S4-F1-Q5

TYPE
Short answer

CORRECT ANSWER
90

GRADING NOTE 720 − 630.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F1-Q5**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • FORWARD 1 • Q5/5

SET 4 • HSS-ID.A.2

FORWARD 2

Predict how changes in data move the mean and median • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

The values are 10, 12, 13, 15, 40. If 40 is replaced by 20, what happens?

A. The mean decreases by 20 and the median stays 13.

B. The mean and the median both decrease by 4.

C. The mean decreases by 4 and the median stays 13.

D. The mean stays the same and the median decreases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 4 • FORWARD 2 • Q1/5

149

QUESTION ID

S4-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. The mean decreases by 4 and the median stays 13.

GRADING NOTE The sum drops by 20, so the mean drops by 20/5 = 4; the middle value 13 is unchanged.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 4 • FORWARD 2 • Q1/5

True or false?

Adding 5 to every value in a data set increases both the mean and the median by 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S4-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 4 • FORWARD 2 • Q2/5

150

QUESTION ID

S4-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Every value, and so every summary of center, moves up by 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 4 • FORWARD 2 • Q2/5

Find the mean and the median of 2, 4, 6, 8, 100.

Mean: _____

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 4 • FORWARD 2 • Q3/5

151

QUESTION ID

S4-F2-Q3

TYPE

Short answer

CORRECT ANSWER

24; 6

GRADING NOTE Form required: both. 120/5 = 24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 4 • FORWARD 2 • Q3/5

Salaries at a company include a few very high salaries. Which statement is most likely true?

A. The median is pulled up more than the mean.

B. The mean salary is greater than the median salary.

C. The median salary is greater than the mean salary.

D. The mean and the median are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 4 • FORWARD 2 • Q4/5

152

QUESTION ID
S4-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. The mean salary is greater than the median salary.

GRADING NOTE High values pull the mean up; the median resists them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 4 • FORWARD 2 • Q4/5

True or false?

Doubling every value in a data set doubles the median but leaves the mean unchanged.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S4-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 4 • FORWARD 2 • Q5/5

153

QUESTION ID
S4-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Both double.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S4-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 4 • FORWARD 2 • Q5/5

Compare spreads with the IQR or the standard deviation

MAIN	Probability & Statistics main question	**HSS-ID.A.2**
BACKWARD 1	Find the range and the interquartile range	**6.SP.B.5.c**
BACKWARD 2	Measure distances from the mean	**6.SP.B.5.c**
BACKWARD 3	Square, average, and take square roots	**8.EE.A.2**
FORWARD 1	Compare population and sample standard deviations	**HSS-ID.A.2**
FORWARD 2	Predict how adding or multiplying changes center and spread	**HSS-ID.A.2**

SET 5 • HSS-ID.A.2

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

Ages of 9 players on each of two teams: team A: 14, 15, 15, 16, 16, 17, 18, 18, 19; team B: 13, 14, 16, 16, 17, 18, 20, 22, 25. Give the interquartile range of each team. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.

IQR of team A: _____

IQR of team B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • MAIN • Q1/5

156

QUESTION ID

S5-M-Q1

TYPE

Short answer

CORRECT ANSWER

3; 6

GRADING NOTE Form required: both. A: Q1 = 15, Q3 = 18, IQR 3. B: Q1 = 15, Q3 = 21, IQR 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • MAIN • Q1/5

Bus delays (minutes) on 5 days: route 1: 4, 6, 8, 10, 12; route 2: 7, 8, 8, 8, 9. Both are symmetric. Give the standard deviation of each route, to the nearest tenth. Use the population standard deviation (divide by n). A calculator or spreadsheet may be used.

Route 1: _____

Route 2: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • MAIN • Q2/5

157

QUESTION ID
S5-M-Q2

TYPE
Short answer

CORRECT ANSWER
2.8; 0.6

GRADING NOTE Form required: both, to the nearest tenth. Route 1: deviations −4, −2, 0, 2, 4; variance 40/5 = 8; √8 ≈ 2.83. Route 2: deviations −1, 0, 0, 0, 1; variance 2/5 = 0.4; √0.4 ≈ 0.63.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • MAIN • Q2/5

Monthly rents (dollars) of 7 apartments in each of two buildings: A: 800, 850, 900, 950, 1000, 1050, 2400; B: 700, 800, 900, 1000, 1100, 1200, 1300. Which comparison of spread is appropriate and correct?

A. Compare standard deviations: A's is greater, so A's typical rents vary more than B's.

B. A has an outlier, so compare IQRs: B's IQR ($400) is twice A's ($200), so B's typical rents vary more.

C. Compare IQRs: A's IQR ($400) is twice B's ($200), so A's typical rents vary more.

D. Compare ranges: A's range ($1,600) is greater than B's ($600), so A's typical rents vary more.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • MAIN • Q3/5

158

QUESTION ID

S5-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. A has an outlier, so compare IQRs: B's IQR ($400) is twice A's ($200), so B's typical rents vary more.

GRADING NOTE Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves. A: Q1 850, Q3 1050, IQR 200; B: Q1 800, Q3 1200, IQR 400. The range and standard deviation of A are inflated by the outlier 2400; the last choice swaps the IQRs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • MAIN • Q3/5

Lengths (cm) of 8 fish from each of two lakes: lake A: 20, 22, 23, 25, 26, 28, 30, 31; lake B: 18, 24, 24, 25, 25, 26, 26, 40. Lake B has outliers. Give the interquartile range of each lake. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.

IQR of lake A: _____

IQR of lake B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • MAIN • Q4/5

159

QUESTION ID
S5-M-Q4

TYPE
Short answer

CORRECT ANSWER
6.5; 2

GRADING NOTE Form required: both. A: Q1 = 22.5, Q3 = 29, IQR 6.5. B: Q1 = 24, Q3 = 26, IQR 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • MAIN • Q4/5

Weekly savings (dollars) of 5 students in each of two groups: group 1: 10, 20, 30, 40, 50; group 2: 25, 28, 30, 32, 35. Both are symmetric. Give the standard deviation of each group, to the nearest tenth. Use the population standard deviation (divide by n). A calculator or spreadsheet may be used.

Group 1: _____

Group 2: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • MAIN • Q5/5

160

QUESTION ID

S5-M-Q5

TYPE

Short answer

CORRECT ANSWER

14.1; 3.4

GRADING NOTE Form required: both, to the nearest tenth. Group 1: variance (400 + 100 + 0 + 100 + 400)/5 = 200; √200 ≈ 14.14. Group 2: deviations −5, −2, 0, 2, 5; variance 58/5 = 11.6; √11.6 ≈ 3.41.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • MAIN • Q5/5

SET 5 • HSS-ID.A.2

BACKWARD 1

Find the range and the interquartile range • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the range of 7, 3, 12, 5, 9.

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q1/5

162

QUESTION ID

S5-B1-Q1

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE 12 − 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q1/5

Q1 = 18 and Q3 = 27. Find the IQR.

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q2/5

163

QUESTION ID

S5-B1-Q2

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE 27 − 18.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q2/5

True or false?

The IQR is the spread of the middle half of the data.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q3/5

164

QUESTION ID

S5-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE From Q1 to Q3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q3/5

For 2, 4, 6, 8, 10, 12, the IQR (with Q1 and Q3 the medians of the halves) is

A. 10

B. 6

C. 4

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q4/5

165

QUESTION ID

S5-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 6

GRADING NOTE Q1 = 4, Q3 = 10. 10 is the range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q4/5

A box plot has minimum 5, Q1 9, median 12, Q3 16, maximum 25. Give the IQR and the range.

IQR: _____

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q5/5

166

QUESTION ID

S5-B1-Q5

TYPE

Short answer

CORRECT ANSWER

7; 20

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q5/5

SET 5 • HSS-ID.A.2

BACKWARD 2

Measure distances from the mean • Grade 6 • Statistics & Probability • 6.SP.B.5.c

The mean of 2, 6, 10 is 6. What are the distances of 2 and of 10 from the mean?

Distance of 2: _____

Distance of 10: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q1/5

168

QUESTION ID

S5-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4; 4

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q1/5

Find the mean absolute deviation of 1, 3, 5, 7.

MAD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q2/5

169

QUESTION ID

S5-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Mean 4; distances 3, 1, 1, 3; mean 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q2/5

True or false?

A data set whose values are all far from the mean has a large spread.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q3/5

170

QUESTION ID

S5-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Spread measures distance from center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q3/5

Which set has the larger spread around its mean?

A. 0, 10, 20

B. cannot tell

C. 9, 10, 11

D. they are the same

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q4/5

171

QUESTION ID

S5-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 0, 10, 20

GRADING NOTE Distances 10, 0, 10 versus 1, 0, 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q4/5

The values 5, 5, 5, 5 have mean 5. What is their mean absolute deviation?

MAD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q5/5

172

QUESTION ID

S5-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0

GRADING NOTE Every distance is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q5/5

SET 5 • HSS-ID.A.2

BACKWARD 3

Square, average, and take square roots • Grade 8 • Expressions & Equations • 8.EE.A.2

Find √16 + √9.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q1/5

174

QUESTION ID

S5-B3-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE 4 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q1/5

The squared distances from a mean are 4, 0, 4. Find their average.

Average: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q2/5

175

QUESTION ID

S5-B3-Q2

TYPE

Short answer

CORRECT ANSWER

8/3

GRADING NOTE 8/3 (about 2.67).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q2/5

True or false?

$\sqrt{(9 + 16)} = \sqrt{9} + \sqrt{16}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q3/5

176

QUESTION ID

S5-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{25} = 5$, but $3 + 4 = 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q3/5

Which is √0.25?

A. 0.5

B. 0.125

C. 0.0625

D. 0.05

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q4/5

177

QUESTION ID

S5-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 0.5

GRADING NOTE $0.5^2 = 0.25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q4/5

Find √8 to the nearest tenth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q5/5

178

QUESTION ID

S5-B3-Q5

TYPE

Short answer

CORRECT ANSWER

2.8

GRADING NOTE 2.828....

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q5/5

SET 5 • HSS-ID.A.2

FORWARD 1

Compare population and sample standard deviations • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

For the data 4, 6, 8, 10, 12, give the sample standard deviation (divide by n − 1), to the nearest hundredth. A calculator or spreadsheet may be used.

s = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 1 • Q1/5

180

QUESTION ID

S5-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3.16

GRADING NOTE Sum of squared deviations 40; 40/4 = 10; √10 ≈ 3.162.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 1 • Q1/5

True or false?

For the same data, the sample standard deviation (divide by n − 1) is never smaller than the population standard deviation (divide by n).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 1 • Q2/5

181

QUESTION ID

S5-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Dividing by the smaller number n − 1 gives a larger or equal value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

For 2, 4, 6, give the population and the sample standard deviations, to the nearest hundredth. A calculator or spreadsheet may be used.

Population: _____

Sample: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 1 • Q3/5

182

QUESTION ID

S5-F1-Q3

TYPE

Short answer

CORRECT ANSWER

1.63; 2.00

GRADING NOTE Form required: both, to the nearest hundredth. Squared deviations 4, 0, 4: $\sqrt{(8/3)} \approx 1.633$ and $\sqrt{(8/2)} = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q3**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 1 • Q3/5

A spreadsheet gives 5.2 for STDEV.S and 4.9 for STDEV.P of the same data. Which statement is correct?

A. STDEV.S divides by n and treats the data as the whole population; STDEV.P divides by n − 1 for a sample.

B. The two functions measure different spreads: STDEV.S the range of the data and STDEV.P their IQR.

C. One of the two values must be an error, because a data set has only one standard deviation.

D. STDEV.S divides by n − 1 and treats the data as a sample; STDEV.P divides by n and treats them as the population.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 1 • Q4/5

183

QUESTION ID
S5-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. STDEV.S divides by n − 1 and treats the data as a sample; STDEV.P divides by n and treats them as the population.

GRADING NOTE The sample formula divides by n − 1, so it is the larger value. The others swap the formulas or deny the difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 1 • Q4/5

Ten values have sum of squared deviations from their mean equal to 90. Give the population and sample variances.

Population variance: _____

Sample variance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 1 • Q5/5

184

QUESTION ID

S5-F1-Q5

TYPE

Short answer

CORRECT ANSWER

9; 10

GRADING NOTE Form required: both. 90/10 and 90/9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q5**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 1 • Q5/5

SET 5 • HSS-ID.A.2

FORWARD 2

Predict how adding or multiplying changes center and spread • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

Every value in a data set is increased by 10. What happens to the standard deviation and the IQR?

A. Both stay the same.

B. The SD stays the same and the IQR increases by 10.

C. Both increase by 10.

D. Both are multiplied by 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 2 • Q1/5

186

QUESTION ID
S5-F2-Q1

TYPE
Multiple choice

CORRECT ANSWER
A. Both stay the same.

GRADING NOTE Shifting every value keeps every distance the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F2-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 2 • Q1/5

A data set has mean 20 and standard deviation 4. Every value is multiplied by 3. Give the new mean and standard deviation.

Mean: _____

Standard deviation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 2 • Q2/5

187

QUESTION ID

S5-F2-Q2

TYPE

Short answer

CORRECT ANSWER

60; 12

GRADING NOTE Form required: both. Both are multiplied by 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F2-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 2 • Q2/5

True or false?

Converting temperatures from degrees to tenths of a degree (multiplying by 10) multiplies the IQR by 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 2 • Q3/5

188

QUESTION ID
S5-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Q1 and Q3 are both multiplied by 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 2 • Q3/5

A data set has median 50 and IQR 8. Each value is changed to 2·value + 5. Give the new median and IQR.

Median: _____

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 2 • Q4/5

189

QUESTION ID

S5-F2-Q4

TYPE

Short answer

CORRECT ANSWER

105; 16

GRADING NOTE Form required: both. 2·50 + 5; the + 5 does not change the IQR.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F2-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 2 • Q4/5

Test scores have SD 6. The teacher adds 3 bonus points to every score. The new SD is

A. 18

B. 9

C. 3

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 2 • Q5/5

190

QUESTION ID

S5-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. 6

GRADING NOTE Adding a constant does not change spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 2 • Q5/5

Interpret differences in shape, center and spread in context

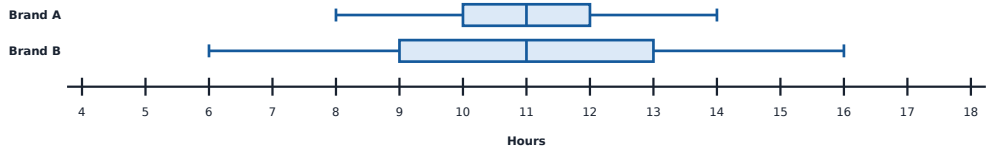
MAIN	Probability & Statistics main question	**HSS-ID.A.3**
BACKWARD 1	Describe the shape of a distribution	**6.SP.A.2**
BACKWARD 2	What center and spread tell about data	**6.SP.A.3**
BACKWARD 3	Compare two distributions informally	**7.SP.B.4**
FORWARD 1	Measure separation in units of spread	**HSS-ID.A.3**
FORWARD 2	Judge whether a difference supports a claim	**HSS-IC.B.6**

SET 6 • HSS-ID.A.3

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

The box plots show the battery lives (hours) of two brands. Which interpretation is correct?



A. Brand A's lives vary more, because its box is narrower than brand B's box is.

B. Typical lives are the same (both medians 11 hours), but brand B's lives vary more (IQR 4 versus 2 hours).

C. Brand B lasts longer, because its longest-lasting battery runs longer than brand A's does.

D. Brand B is better in every way, because its whiskers and its box are both longer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • MAIN • Q1/5

193

QUESTION ID

S6-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Typical lives are the same (both medians 11 hours), but brand B's lives vary more (IQR 4 versus 2 hours).

GRADING NOTE Same median; IQRs 2 and 4. The others use the maximum as the center, read a narrow box as more spread, or call more spread better.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • MAIN • Q1/5

Waiting times at two stores: store 1 is roughly symmetric with mean 6 minutes and standard deviation 1.5 minutes; store 2 is skewed right with median 5 minutes, IQR 6 minutes, and a few waits over 20 minutes. Which interpretation is correct?

A. Store 2 often has a shorter wait, but its waits are much less predictable, with occasional very long waits.

B. Store 2 always has shorter waits than store 1, because its median is 1 minute lower than the mean of store 1.

C. The stores cannot be compared at all, because they are summarized with different measures.

D. Store 1's waits vary more, because its waits are described with a mean and a standard deviation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • MAIN • Q2/5

194

QUESTION ID
S6-M-Q2

TYPE
Multiple choice

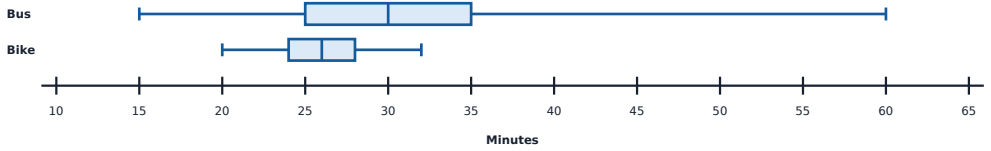
CORRECT ANSWER
A. Store 2 often has a shorter wait, but its waits are much less predictable, with occasional very long waits.

GRADING NOTE Median 5 < mean 6 for a typical wait; IQR 6 and long waits show more variability. The others overstate, invert the spread, or refuse a reasonable comparison.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • MAIN • Q2/5

The box plots show commute times (minutes) by bus and by bike for the same group of workers. Which interpretation is correct?



A. Bus trips are shorter in general, because the bus box plot starts at a lower time than the bike box plot.

B. Bike trips vary more, because the bike box plot is shorter than the bus box plot on the scale.

C. Both ways take the same time, because both box plots contain the times from 25 to 28 minutes.

D. Bike trips are more consistent; bus trips take a little longer in the middle and vary much more, including long trips.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • MAIN • Q3/5

195

QUESTION ID
S6-M-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. Bike trips are more consistent; bus trips take a little longer in the middle and vary much more, including long trips.

GRADING NOTE Medians 30 and 26; IQRs 10 and 4; the bus range is 45. The others use the minimum, overlap, or read a short box as more spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • MAIN • Q3/5

Two classes both have mean score 70 on a test. Class X's scores form two clusters, around 55 and around 85; class Y's form one mound around 70. Which interpretation is correct?

A. Class X did better, because some of its students scored around 85, which is above every score in class Y.

B. Class Y has more spread, because its scores are all close to the middle of the range of scores.

C. The equal means hide a difference: class X seems to contain two groups doing very differently; class Y is more alike.

D. The classes performed the same way, because their means are equal and the mean summarizes the whole class.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S6-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • MAIN • Q4/5

196

QUESTION ID

S6-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. The equal means hide a difference: class X seems to contain two groups doing very differently; class Y is more alike.

GRADING NOTE A bimodal shape suggests two subgroups; the mean alone hides it. The others ignore shape or invert spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S6-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • MAIN • Q4/5

Apples from two farms both have mean weight 150 g; farm A's weights have standard deviation 5 g and farm B's have standard deviation 20 g. Both distributions are roughly symmetric. Which interpretation is correct?

A. Farm B's apples are heavier, because a standard deviation of 20 g is larger than one of 5 g.

B. Farm A's apples are more uniform in size; on average, apples from both farms weigh about the same.

C. Farm A's apples are lighter, because a smaller standard deviation means a smaller mean weight.

D. The farms' apples are the same in every way, because their mean weights are both 150 g.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q5
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • MAIN • Q5/5

197

QUESTION ID
S6-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
B. Farm A's apples are more uniform in size; on average, apples from both farms weigh about the same.

GRADING NOTE Same center, different spread. The others read spread as center or ignore spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q5
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • MAIN • Q5/5

SET 6 • HSS-ID.A.3

BACKWARD 1

Describe the shape of a distribution • Grade 6 • Statistics & Probability • 6.SP.A.2

A histogram has one peak in the middle and falls off evenly on both sides. Its shape is

A. symmetric

B. bimodal

C. skewed left

D. skewed right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q1/5

199

QUESTION ID

S6-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. symmetric

GRADING NOTE Mirror image about the center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q1/5

True or false?

A histogram with two separate peaks is called bimodal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q2/5

200

QUESTION ID

S6-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Two modes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q2/5

A histogram has a long tail toward small values. Its shape is

A. uniform

B. symmetric

C. skewed right

D. skewed left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q3/5

201

QUESTION ID

S6-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. skewed left

GRADING NOTE The tail names the direction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q3/5

True or false?

A distribution whose bars are all about the same height is skewed right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q4/5

202

QUESTION ID

S6-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That is uniform.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q4/5

Waiting times at a busy clinic are mostly short with a few very long waits. The shape is most likely

A. skewed left

B. symmetric

C. bimodal

D. skewed right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q5/5

203

QUESTION ID

S6-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. skewed right

GRADING NOTE Long tail of large values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q5/5

SET 6 • HSS-ID.A.3

BACKWARD 2

What center and spread tell about data • Grade 6 • Statistics & Probability • 6.SP.A.3

True or false?

A measure of center summarizes all the values of a data set with a single number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q1/5

205

GRADING NOTE Definition.

QUESTION ID

S6-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q1/5

True or false?

A measure of spread tells how much the values vary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q2/5

206

QUESTION ID

S6-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q2/5

Two data sets have the same median but different IQRs. Which is true?

A. Their typical values are similar, but one varies more.

B. One of them has a larger typical value than the other.

C. They are identical data sets with the same values.

D. They have the same range and the same maximum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q3/5

207

QUESTION ID

S6-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Their typical values are similar, but one varies more.

GRADING NOTE Center versus spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which describes spread?

A. the median

B. the mean

C. the IQR

D. the mode

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q4/5

208

QUESTION ID

S6-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. the IQR

GRADING NOTE The others are centers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q4/5

True or false?

A data set with a larger mean must have a larger standard deviation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q5/5

209

QUESTION ID

S6-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 5, 5, 5 has mean 5 and SD 0; 1, 2, 3 has mean 2 and SD about 0.82.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q5/5

SET 6 • HSS-ID.A.3

BACKWARD 3

Compare two distributions informally • Grade 7 • Statistics & Probability • 7.SP.B.4

Group 1 has median 12 and IQR 3; group 2 has median 15 and IQR 3. Which is true?

A. Group 2 varies more, since its median is higher.

B. The groups are the same, since the IQRs are equal.

C. Group 2's typical value is 3 higher, with similar spread.

D. Group 1's values are higher, since its IQR is the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q1/5

211

QUESTION ID

S6-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Group 2's typical value is 3 higher, with similar spread.

GRADING NOTE Same IQR, higher median.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q1/5

True or false?

Group 1 has median 20 and IQR 2; group 2 has median 40 and IQR 2. The middle halves (Q1 to Q3) of the two groups cannot overlap.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q2/5

212

QUESTION ID

S6-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each box contains its median and is 2 wide, so box 1 lies within 18 to 22 and box 2 within 38 to 42. The summaries say nothing about the tails, which could still overlap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q2/5

Group A has mean 40 and group B has mean 46. By how much are the means different?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q3/5

213

QUESTION ID

S6-B3-Q3

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE 46 − 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q3/5

Which pair of groups has the smallest gap between medians measured in IQRs (median gap divided by the common IQR)?

A. medians 20 and 30, both IQRs 1

B. medians 20 and 21, both IQRs 10

C. medians 20 and 40, both IQRs 2

D. medians 10 and 50, both IQRs 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q4/5

214

QUESTION ID

S6-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. medians 20 and 21, both IQRs 10

GRADING NOTE 1/10 = 0.1. The others give 20/2 = 10, 10/1 = 10 and 40/5 = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q4/5

True or false?

Two groups with the same center must have the same spread.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q5/5

215

QUESTION ID

S6-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Spread is separate from center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q5/5

SET 6 • HSS-ID.A.3

FORWARD 1

Measure separation in units of spread • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

Two roughly symmetric groups have means 70 and 76; both have standard deviation 4. How many standard deviations apart are the means?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • FORWARD 1 • Q1/5

217

QUESTION ID

S6-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1.5

GRADING NOTE 6/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • FORWARD 1 • Q1/5

Group medians are 30 and 45; both IQRs are 10. How many IQRs apart are the medians?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • FORWARD 1 • Q2/5

218

QUESTION ID
S6-F1-Q2

TYPE
Short answer

CORRECT ANSWER
1.5

GRADING NOTE 15/10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F1-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • FORWARD 1 • Q2/5

Pair 1: means 50 and 55, both SDs 10. Pair 2: means 50 and 53, both SDs 1. Which pair is more clearly separated?

A. pair 1, 5 points apart

B. pair 1, 0.5 SDs apart

C. pair 2, 3 SDs apart

D. they are equally separated

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • FORWARD 1 • Q3/5

219

QUESTION ID
S6-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. pair 2, 3 SDs apart

GRADING NOTE Pair 1 is 0.5 SDs apart, pair 2 is 3 SDs apart. A larger raw gap is not the larger separation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • FORWARD 1 • Q3/5

True or false?

Two approximately normal groups have the same standard deviation, and their means are 0.2 standard deviations apart. Their distributions overlap a great deal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • FORWARD 1 • Q4/5

220

QUESTION ID
S6-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Under the stated normal model with equal spread, a 0.2-SD gap leaves most values of each group in the range of the other. Summaries alone, without a shape, would not decide this.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • FORWARD 1 • Q4/5

Why measure a difference in units of spread?

A. Units of spread show which group has more values in it.

B. Units of spread remove the need to compare the centers of the groups.

C. A gap matters only relative to how much values vary within each group.

D. Units of spread make every difference look larger than it really is.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S6-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • FORWARD 1 • Q5/5

221

QUESTION ID
S6-F1-Q5

TYPE
Multiple choice

CORRECT ANSWER
C. A gap matters only relative to how much values vary within each group.

GRADING NOTE The others are false claims about standardizing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S6-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • FORWARD 1 • Q5/5

SET 6 • HSS-ID.A.3

FORWARD 2

Judge whether a difference supports a claim • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A report says that brand X bulbs "last much longer" than brand Y. Lifetimes of both brands are approximately normal with means 1,010 and 1,000 hours and the same standard deviation, 80 hours. Evaluate.

A. Weak: the claim would be supported only if the standard deviations were different.

B. Strong: both brands have the same standard deviation, so the difference in means must be large.

C. Weak: the means differ by about 0.13 standard deviations, so the lives overlap almost completely.

D. Strong: 1,010 is greater than 1,000, so every brand X bulb lasts longer than any brand Y bulb.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q1/5

223

QUESTION ID

S6-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Weak: the means differ by about 0.13 standard deviations, so the lives overlap almost completely.

GRADING NOTE 10/80 = 0.125 SD; for two normal distributions with equal spread, that gap means near-total overlap. The others overstate or misuse the spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q1/5

True or false?

A difference of 5 points between two group means is 2.5 standard deviations when each group's standard deviation is 2, but only 0.25 standard deviations when it is 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q2/5

224

QUESTION ID
S6-F2-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE 5/2 and 5/20. How strong the evidence is would also depend on sample sizes and design, which are not given.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q2/5

After new training, a team's median race time dropped from 52 to 50 seconds; the IQR was 8 seconds before and after. Which statement correctly describes the size of the drop relative to the spread?

A. It is a quarter of an IQR (2/8), small relative to the athlete-to-athlete spread.

B. It cannot be measured, because the two IQRs are equal to each other.

C. It shows that the athletes slowed down by two seconds on average.

D. It is four IQRs (8/2), large relative to the athlete-to-athlete spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q3/5

225

QUESTION ID
S6-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. It is a quarter of an IQR (2/8), small relative to the athlete-to-athlete spread.

GRADING NOTE 2/8 = 0.25 IQR. Whether 2 seconds matters in a race is a separate judgment that the ratio does not settle. The others invert the ratio, read a lower time as slower, or treat equal IQRs as a problem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q3/5

Two groups have means 120 and 135 and the same standard deviation 10. How many standard deviations apart are they?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q4/5

226

QUESTION ID

S6-F2-Q4

TYPE

Short answer

CORRECT ANSWER

1.5

GRADING NOTE 15/10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q4/5

True or false?

If two groups' means differ by less than one standard deviation, the groups cannot differ in any important way.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q5/5

227

QUESTION ID

S6-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A small standardized difference can still matter in context.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q5/5

Identify outliers and account for their effects

MAIN	Probability & Statistics main question	**HSS-ID.A.3**
BACKWARD 1	Find the interquartile range	**6.SP.B.5.c**
BACKWARD 2	Compute the fences	**6.NS.B.3**
BACKWARD 3	How an extreme value affects the mean and the median	**6.SP.B.5.d**
FORWARD 1	Decide whether an outlier is an error or a real value	**HSS-ID.A.3**
FORWARD 2	Report resistant summaries after checking outliers	**HSS-ID.A.3**

SET 7 • HSS-ID.A.3

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

Points a player scored in 11 games: 12, 15, 14, 18, 16, 13, 17, 15, 41, 14, 16. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. Give the lower fence, the upper fence, and the number of outliers.

Lower fence: _____

Upper fence: _____

Number of outliers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • MAIN • Q1/5

230

QUESTION ID

S7-M-Q1

TYPE

Short answer

CORRECT ANSWER

9.5; 21.5; 1

GRADING NOTE Form required: all three. Ordered 12, 13, 14, 14, 15, 15, 16, 16, 17, 18, 41; Q1 = 14, Q3 = 17, IQR 3; fences 14 − 4.5 = 9.5 and 17 + 4.5 = 21.5; only 41 is an outlier.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-M-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • MAIN • Q1/5

Points a player scored in 11 games: 12, 15, 14, 18, 16, 13, 17, 15, 41, 14, 16. The value 41 is an outlier. Find the mean with and without it, each to the nearest tenth.

Mean with 41: _____

Mean without 41: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • MAIN • Q2/5

231

QUESTION ID
S7-M-Q2

TYPE
Short answer

CORRECT ANSWER
17.4; 15.0

GRADING NOTE Form required: both. 191/11 ≈ 17.36; 150/10 = 15.0. (The median is 15 both ways.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-M-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • MAIN • Q2/5

Hours of screen time 11 students reported for one weekend: 6, 8, 7, 9, 7, 8, 30, 6, 7, 8, 1. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. Give the lower fence, the upper fence, and the number of outliers.

Lower fence: _____

Upper fence: _____

Number of outliers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S7-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • MAIN • Q3/5

232

QUESTION ID

S7-M-Q3

TYPE

Short answer

CORRECT ANSWER

3; 11; 2

GRADING NOTE Form required: all three. Ordered 1, 6, 6, 7, 7, 7, 8, 8, 8, 9, 30; Q1 = 6, Q3 = 8, IQR 2; fences 3 and 11; 1 and 30 are outliers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-M-Q3**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • MAIN • Q3/5

Prices (dollars) of 10 used bikes: 180, 210, 195, 205, 190, 600, 200, 185, 215, 220. The $600 bike is an outlier. Which describes the effect of removing it?

A. The mean and the median both drop by $40.

B. The median drops by $40, while the mean drops by only $2.50.

C. The mean drops by $40, while the median drops by only $2.50.

D. Neither the mean nor the median changes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • MAIN • Q4/5

233

QUESTION ID
S7-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. The mean drops by $40, while the median drops by only $2.50.

GRADING NOTE Mean 2400/10 = 240 becomes 1800/9 = 200; median (200 + 205)/2 = 202.5 becomes 200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • MAIN • Q4/5

Support calls received in each of 10 hours: 3, 5, 4, 6, 5, 4, 22, 5, 3, 4. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. Give the upper fence and the number of outliers.

Upper fence: _____

Number of outliers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • MAIN • Q5/5

234

QUESTION ID

S7-M-Q5

TYPE

Short answer

CORRECT ANSWER

6.5; 1

GRADING NOTE Form required: both. Ordered 3, 3, 4, 4, 4, 5, 5, 5, 6, 22; Q1 = 4, Q3 = 5, IQR 1; fences 2.5 and 6.5; only 22.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • MAIN • Q5/5

SET 7 • HSS-ID.A.3

BACKWARD 1

Find the interquartile range • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Q1 = 12 and Q3 = 20. Find the IQR.

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q1/5

236

QUESTION ID

S7-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE 20 − 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q1/5

For 1, 3, 4, 6, 8, 9, find the IQR (Q1 and Q3 are the medians of the halves).

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q2/5

237

QUESTION ID

S7-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Q1 = 3, Q3 = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q2/5

True or false?

The IQR is the difference between the maximum and the minimum.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q3/5

238

QUESTION ID

S7-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That is the range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q3/5

A box plot has Q1 = 30 and Q3 = 42. The IQR is

A. 36

B. 42

C. 72

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q4/5

239

QUESTION ID

S7-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 12

GRADING NOTE 42 − 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q4/5

For 2, 4, 5, 7, 9, 10, 12 (odd count; leave out the median), find the IQR.

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q5/5

240

QUESTION ID

S7-B1-Q5

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Q1 = 4, Q3 = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q5/5

SET 7 • HSS-ID.A.3

BACKWARD 2

Compute the fences • Grade 6 • The Number System • 6.NS.B.3

Find 1.5 × 6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q1/5

242

QUESTION ID

S7-B2-Q1

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE 6 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find 1.5 × 2.5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q2/5

243

GRADING NOTE 2.5 + 1.25.

QUESTION ID

S7-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q2/5

True or false?

If Q1 = 10 and the IQR is 4, the lower fence is 10 − 6 = 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q3/5

244

QUESTION ID

S7-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 1.5 · 4 = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q3/5

If Q3 = 25 and the IQR is 10, the upper fence is

A. 15

B. 35

C. 40

D. 37.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q4/5

245

QUESTION ID

S7-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 40

GRADING NOTE 25 + 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q4/5

Q1 = 7.5 and the IQR is 3. Find the lower fence.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q5/5

246

QUESTION ID

S7-B2-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE 7.5 − 4.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • HSS-ID.A.3

BACKWARD 3

How an extreme value affects the mean and the median • Grade 6 • Statistics & Probability • 6.SP.B.5.d

In 3, 4, 5, 6, 50, which is pulled more by the value 50?

A. the mean

B. neither

C. the median

D. both equally

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q1/5

248

QUESTION ID

S7-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. the mean

GRADING NOTE Mean 13.6; median 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q1/5

True or false?

The median of 2, 3, 4, 5, 100 is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q2/5

249

QUESTION ID

S7-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Middle value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 2, 3, 4, 5, 100.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q3/5

250

QUESTION ID

S7-B3-Q3

TYPE

Short answer

CORRECT ANSWER

22.8

GRADING NOTE 114/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q3/5

True or false?

Removing an extreme value always changes the median more than the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q4/5

251

QUESTION ID

S7-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Usually the reverse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q4/5

For skewed data with outliers, the better summary of a typical value is usually

A. the maximum

B. the range

C. the median

D. the mean

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q5/5

252

QUESTION ID

S7-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. the median

GRADING NOTE Resistant to extremes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q5/5

SET 7 • HSS-ID.A.3

FORWARD 1

Decide whether an outlier is an error or a real value • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

Heights of 9 adults in a study are listed in centimeters; one value is 17.5. Which is the most reasonable judgment?

A. Outliers are always errors, so every outlier in the data should be removed.

B. It is a real height, so it must be kept exactly as it is in the data.

C. It is probably a recording error (likely 175 cm), so check the record before using it.

D. It is an outlier, so it should be deleted without any further checking.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S7-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 1 • Q1/5

254

QUESTION ID

S7-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. It is probably a recording error (likely 175 cm), so check the record before using it.

GRADING NOTE An adult height of 17.5 cm is impossible; an outlier is checked, not removed automatically.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S7-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 1 • Q1/5

True or false?

A value that the 1.5·IQR rule flags as an outlier must be removed from the data.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S7-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 1 • Q2/5

255

QUESTION ID
S7-F1-Q2

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The rule flags values to examine; a legitimate unusual value stays.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S7-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 1 • Q2/5

Among 30 home prices, one mansion sells for 10 times the median. Which is correct?

A. It must be removed, because the outlier rule requires it to be deleted.

B. It is a legitimate unusual value; report resistant summaries such as the median and the IQR.

C. It is a recording error, because no real home could cost ten times the median price.

D. It should replace the median, because it is the most informative value in the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 1 • Q3/5

256

QUESTION ID

S7-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. It is a legitimate unusual value; report resistant summaries such as the median and the IQR.

GRADING NOTE Real but extreme values are kept; resistant summaries describe the typical home.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 1 • Q3/5

True or false?

An outlier can be a correct measurement of an unusual case.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 1 • Q4/5

257

QUESTION ID

S7-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Outliers are not automatically errors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 1 • Q4/5

A thermometer log shows 18, 19, 21, 20, 210, 19 °C in a classroom. What is the best next step?

A. Check the 210 reading, since a decimal point was likely lost (21.0), before computing summaries.

B. Report the mean including 210, because every reading is data and must be used as recorded.

C. Delete the 19s, because they are the most common values and do not add information.

D. Report the 210 reading as the median, because it is the most unusual value in the list.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 1 • Q5/5

258

QUESTION ID
S7-F1-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. Check the 210 reading, since a decimal point was likely lost (21.0), before computing summaries.

GRADING NOTE 210 °C in a classroom is impossible; check it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 1 • Q5/5

SET 7 • HSS-ID.A.3

FORWARD 2

Report resistant summaries after checking outliers • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

After checking, the $600 bike price is real. Which summaries best describe a typical used bike and the spread?

A. the median and the IQR

B. the mean and the range

C. the maximum and the minimum

D. the mean and the standard deviation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 2 • Q1/5

260

QUESTION ID
S7-F2-Q1

TYPE
Multiple choice

CORRECT ANSWER
A. the median and the IQR

GRADING NOTE Resistant to the outlier.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F2-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 2 • Q1/5

After a recording error is corrected, the data are 12, 15, 14, 18, 16, 13, 17, 15, 14, 14, 16 (the 41 was really 14). Find the new mean to the nearest tenth and the new median.

Mean: _____

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 2 • Q2/5

261

QUESTION ID

S7-F2-Q2

TYPE

Short answer

CORRECT ANSWER

14.9; 15

GRADING NOTE Form required: both. Sum 164/11 ≈ 14.9; ordered middle value 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F2-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 2 • Q2/5

True or false?

In most large data sets, removing one extreme value changes the IQR less than it changes the standard deviation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S7-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 2 • Q3/5

262

QUESTION ID
S7-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Quartiles depend on the middle of the data, so the IQR is usually less sensitive; it can still move when the removal shifts a quartile across a gap in the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S7-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 2 • Q3/5

Which summary is most changed by one extreme value?

A. Q1

B. the standard deviation

C. the median

D. the IQR

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 2 • Q4/5

263

QUESTION ID
S7-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. the standard deviation

GRADING NOTE Squared distances make the SD very sensitive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 2 • Q4/5

Data 5, 6, 6, 7, 8, 30: find the median and the IQR (Q1 and Q3 are the medians of the halves).

Median: _____

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 2 • Q5/5

264

QUESTION ID
S7-F2-Q5

TYPE
Short answer

CORRECT ANSWER
6.5; 2

GRADING NOTE Form required: both. Q1 = 6, Q3 = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F2-Q5**
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 2 • Q5/5

Estimate percentages with the 68-95-99.7 rule

MAIN	Probability & Statistics main question	**HSS-ID.A.4**
BACKWARD 1	Count standard deviations from the mean	**HSS-ID.A.2**
BACKWARD 2	Find a percent of a population	**6.RP.A.3.c**
BACKWARD 3	Use the symmetry of the normal curve	**6.SP.A.2**
FORWARD 1	Find a cutoff from a percent	**HSS-ID.A.4**
FORWARD 2	Compare positions in two populations with standardized values	**HSS-ID.A.4**

SET 8 • HSS-ID.A.4

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Heights of adult women in a population are approximately normal with mean 64 inches and standard deviation 2.5 inches. Use the 68-95-99.7 rule. What percent are between 61.5 and 66.5 inches, and what percent are between 59 and 69 inches?

61.5 to 66.5: _____

59 to 69: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q1/5

267

QUESTION ID

S8-M-Q1

TYPE

Short answer

CORRECT ANSWER

68%; 95%

GRADING NOTE Form required: both, as percents. ±1 SD and ±2 SD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q1/5

Battery lives are approximately normal with mean 40 hours and standard deviation 4 hours. Use the 68-95-99.7 rule. What percent of batteries last more than 48 hours?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q2/5

268

QUESTION ID

S8-M-Q2

TYPE

Short answer

CORRECT ANSWER

2.5%

GRADING NOTE 48 is 2 SD above; (100 − 95)/2 = 2.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q2/5

Commute times are approximately normal with mean 30 minutes and standard deviation 5 minutes. Use the 68-95-99.7 rule. What percent are between 25 and 40 minutes?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q3/5

269

QUESTION ID

S8-M-Q3

TYPE

Short answer

CORRECT ANSWER

81.5%

GRADING NOTE 25 to 30 is 34%; 30 to 40 is 47.5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q3/5

Egg masses are approximately normal with mean 58 g and standard deviation 3 g. Use the 68-95-99.7 rule. What percent are less than 52 g, and what percent are more than 61 g?

Less than 52 g: _____

More than 61 g: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q4/5

270

QUESTION ID

S8-M-Q4

TYPE

Short answer

CORRECT ANSWER

2.5%; 16%

GRADING NOTE Form required: both. 52 is 2 SD below: 2.5%. 61 is 1 SD above: (100 − 68)/2 = 16%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q4/5

Scores on a test are approximately normal with mean 500 and standard deviation 100. Use the 68-95-99.7 rule. What percent are between 300 and 600, and what percent are below 400?

300 to 600: _____

Below 400: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q5/5

271

QUESTION ID

S8-M-Q5

TYPE

Short answer

CORRECT ANSWER

81.5%; 16%

GRADING NOTE Form required: both. 47.5% + 34% = 81.5%; 400 is 1 SD below: 16%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q5/5

SET 8 • HSS-ID.A.4

BACKWARD 1

Count standard deviations from the mean • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

The mean is 50 and the standard deviation is 5. How many standard deviations above the mean is 60?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q1/5

273

QUESTION ID

S8-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE (60 − 50)/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-B1-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q1/5

The mean is 12 and the standard deviation is 3. How many standard deviations below the mean is 9?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q2/5

274

QUESTION ID

S8-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE (12 − 9)/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-B1-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q2/5

True or false?

With mean 100 and standard deviation 15, the value 70 is 2 standard deviations below the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q3/5

275

QUESTION ID

S8-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $100 - 2 \cdot 15 = 70$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q3/5

With mean 64 and standard deviation 2.5, which value is 1 standard deviation above the mean?

A. 62.5

B. 66.5

C. 65

D. 67

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q4/5

276

QUESTION ID

S8-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 66.5

GRADING NOTE 64 + 2.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-B1-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q4/5

The mean is 30 and the standard deviation is 5. What value is 2 standard deviations below the mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q5/5

277

QUESTION ID

S8-B1-Q5

TYPE

Short answer

CORRECT ANSWER

20

GRADING NOTE 30 − 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-B1-Q5**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q5/5

SET 8 • HSS-ID.A.4

BACKWARD 2

Find a percent of a population • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What is 68% of 500?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q1/5

279

QUESTION ID

S8-B2-Q1

TYPE

Short answer

CORRECT ANSWER

340

GRADING NOTE 0.68 · 500.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Half of 5% is what percent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q2/5

280

QUESTION ID

S8-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2.5%

GRADING NOTE 5/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q2/5

True or false?

34% + 13.5% = 47.5%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q3/5

281

QUESTION ID

S8-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Addition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q3/5

If 95% of values are inside an interval, what percent are outside it?

A. 5%

B. 95%

C. 0%

D. 2.5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q4/5

282

QUESTION ID

S8-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 5%

GRADING NOTE 100 − 95; 2.5% is one side only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is 100% minus 68%, divided by 2?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q5/5

283

QUESTION ID

S8-B2-Q5

TYPE

Short answer

CORRECT ANSWER

16%

GRADING NOTE 32/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q5/5

SET 8 • HSS-ID.A.4

BACKWARD 3

Use the symmetry of the normal curve • Grade 6 • Statistics & Probability • 6.SP.A.2

True or false?

In a normal distribution, half of the values are above the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q1/5

285

QUESTION ID

S8-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Symmetric about the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q1/5

True or false?

In a normal distribution, more values are between the mean and 1 SD above it than between the mean and 1 SD below it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q2/5

286

QUESTION ID

S8-B3-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Symmetry makes them equal (34% each).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What percent of a normal distribution lies between the mean and 2 standard deviations above it?

A. 95%

B. 50%

C. 47.5%

D. 34%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q3/5

287

QUESTION ID

S8-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 47.5%

GRADING NOTE Half of 95%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

What percent of a normal distribution lies between 1 SD below the mean and the mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q4/5

288

QUESTION ID

S8-B3-Q4

TYPE

Short answer

CORRECT ANSWER

34%

GRADING NOTE Half of 68%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q4/5

What percent of a normal distribution is more than 1 standard deviation below the mean?

A. 16%

B. 34%

C. 2.5%

D. 32%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q5/5

289

QUESTION ID

S8-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 16%

GRADING NOTE (100 − 68)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q5/5

SET 8 • HSS-ID.A.4

FORWARD 1

Find a cutoff from a percent • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Heights are approximately normal with mean 64 in and standard deviation 2.5 in. Use the 68-95-99.7 rule. Above what height are the tallest 2.5%?

Height (in): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q1/5

291

QUESTION ID

S8-F1-Q1

TYPE

Short answer

CORRECT ANSWER

69

GRADING NOTE Mean + 2 SD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q1/5

Times are approximately normal with mean 30 min and standard deviation 5 min. Use the 68-95-99.7 rule. Below what time are the fastest 16%?

Time (min): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q2/5

292

QUESTION ID

S8-F1-Q2

TYPE

Short answer

CORRECT ANSWER

25

GRADING NOTE Mean − 1 SD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q2/5

True or false?

In a normal distribution with mean 200 and standard deviation 20, the middle 95% is from 160 to 240.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q3/5

293

QUESTION ID

S8-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE ±2 SD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q3/5

Test scores are approximately normal with mean 70 and SD 8. Which score separates the top 16%?

A. 78

B. 74

C. 62

D. 86

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q4/5

294

QUESTION ID

S8-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 78

GRADING NOTE Mean + 1 SD. 86 marks the top 2.5%; 62 the bottom 16%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Lengths are approximately normal with mean 12 cm and standard deviation 0.5 cm. Use the 68-95-99.7 rule. Give the interval that holds the middle 68%.

From: _____

To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q5/5

295

QUESTION ID

S8-F1-Q5

TYPE

Short answer

CORRECT ANSWER

11.5; 12.5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q5**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q5/5

SET 8 • HSS-ID.A.4

FORWARD 2

Compare positions in two populations with standardized values • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Ana scored 85 on a test with mean 75 and SD 5; Ben scored 90 on a test with mean 80 and SD 10. How many standard deviations above the mean is each score?

Ana: _____

Ben: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q1/5

297

QUESTION ID

S8-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2; 1

GRADING NOTE Form required: both. 10/5 and 10/10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q1/5

Using the scores above (Ana 2 SDs above, Ben 1 SD above, both tests normal), which is correct?

A. Ana's score is relatively higher: about 2.5% scored above her, but about 16% scored above Ben.

B. Ben's score is relatively higher, because 90 is a larger number than 85.

C. The scores are equally good, because both are 10 points above their means.

D. They cannot be compared, because the two tests had different means.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q2/5

298

QUESTION ID

S8-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Ana's score is relatively higher: about 2.5% scored above her, but about 16% scored above Ben.

GRADING NOTE Standardized position decides. The others compare raw scores or raw gaps.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q2/5

True or false?

A value 1 SD below the mean in one normal population and a value 1 SD below the mean in another are at the same percentile.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q3/5

299

QUESTION ID
S8-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Both have about 16% below them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q3/5

A 100-m time of 12.0 s comes from a population with mean 13.0 s and SD 0.5 s. How many standard deviations below the mean is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q4/5

300

QUESTION ID

S8-F2-Q4

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE (13.0 − 12.0)/0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q4/5

Two normal populations: in A, mean 50, SD 5; in B, mean 50, SD 10. Which value is more unusual?

A. they are equally unusual

B. 60 in population A

C. 55 in population B

D. 60 in population B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S8-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q5/5

301

QUESTION ID

S8-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 60 in population A

GRADING NOTE 60 is 2 SD above in A, 1 SD in B; 55 is 0.5 SD in B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S8-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q5/5

Estimate normal areas with a table or calculator

MAIN	Probability & Statistics main question	**HSS-ID.A.4**
BACKWARD 1	Compute a standardized value	**7.NS.A.3**
BACKWARD 2	Read areas as proportions	**7.SP.C.5**
BACKWARD 3	Find right tails and areas between from left areas	**HSS-ID.A.4**
FORWARD 1	Find the cutoff for a percentile	**HSS-ID.A.4**
FORWARD 2	Make decisions with normal areas	**HSS-ID.A.4**

SET 9 • HSS-ID.A.4

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

IQ scores are approximately normal with mean 100 and standard deviation 15. What proportion of people score below 118? Use the table values printed below (areas to the left of z). Give four decimal places.

z	1.0	1.2	1.5	2.0
Area to the left	0.8413	0.8849	0.9332	0.9772

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q1/5

304

QUESTION ID

S9-M-Q1

TYPE

Short answer

CORRECT ANSWER

0.8849

GRADING NOTE $z = (118 - 100)/15 = 1.2$; area to the left 0.8849.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q1/5

Lengths of calls are approximately normal with mean 50 seconds and standard deviation 8 seconds. What proportion of calls last more than 60 seconds? Use the table values printed below (areas to the left of z). Give four decimal places.

z	1.0	1.25	1.5	2.0
Area to the left	0.8413	0.8944	0.9332	0.9772

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q2/5

305

QUESTION ID

S9-M-Q2

TYPE

Short answer

CORRECT ANSWER

0.1056

GRADING NOTE $z = 10/8 = 1.25$; $1 - 0.8944 = 0.1056$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q2/5

Daily water use is approximately normal with mean 20 gallons and standard deviation 2.5 gallons. What proportion of days use between 18 and 23 gallons? Use the table values printed below (areas to the left of z). Give four decimal places.

z	−1.0	−0.8	0.8	1.2
Area to the left	0.1587	0.2119	0.7881	0.8849

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q3/5

306

QUESTION ID

S9-M-Q3

TYPE

Short answer

CORRECT ANSWER

0.6730

GRADING NOTE $z = -0.8$ and 1.2; $0.8849 - 0.2119 = 0.6730$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q3/5

Grade-point averages at a school are approximately normal with mean 3.0 and standard deviation 0.4. What proportion are below 2.5? Use the table values printed below (areas to the left of z). Give four decimal places.

z	−1.5	−1.25	−1.0	−0.5
Area to the left	0.0668	0.1056	0.1587	0.3085

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q4/5

307

QUESTION ID

S9-M-Q4

TYPE

Short answer

CORRECT ANSWER

0.1056

GRADING NOTE z = −0.5/0.4 = −1.25; area 0.1056.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q4/5

Heights of adult men are approximately normal with mean 175 cm and standard deviation 7 cm. What proportion are between 168 and 189 cm? Use the table values printed below (areas to the left of z). Give four decimal places.

z	−2.0	−1.0	1.0	2.0
Area to the left	0.0228	0.1587	0.8413	0.9772

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q5/5

308

QUESTION ID

S9-M-Q5

TYPE

Short answer

CORRECT ANSWER

0.8185

GRADING NOTE $z = -1$ and 2; $0.9772 - 0.1587 = 0.8185$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q5/5

SET 9 • HSS-ID.A.4

BACKWARD 1

Compute a standardized value • Grade 7 • The Number System • 7.NS.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Compute (130 − 100)/15.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q1/5

310

QUESTION ID

S9-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE 30/15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q1/5

Compute (42 − 50)/4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q2/5

311

QUESTION ID

S9-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−2

GRADING NOTE −8/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q2/5

True or false?

$(9.5 - 10)/0.5 = -1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q3/5

312

QUESTION ID

S9-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $-0.5/0.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q3/5

Compute (61 − 58)/2.

A. 0.67

B. 3

C. 1.5

D. −1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q4/5

313

QUESTION ID

S9-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 1.5

GRADING NOTE 3/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute (2.2 − 3.0)/0.4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q5/5

314

QUESTION ID

S9-B1-Q5

TYPE

Short answer

CORRECT ANSWER

−2

GRADING NOTE −0.8/0.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • HSS-ID.A.4

BACKWARD 2

Read areas as proportions • Grade 7 • Statistics & Probability • 7.SP.C.5

True or false?

An area of 0.8413 to the left of z = 1 means about 84% of the population is below 1 SD above the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q1/5

316

QUESTION ID

S9-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Area is proportion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q1/5

The table gives area 0.0228 to the left of $z = -2$. What does this mean?

A. Exactly 2 values are below the mean.

B. About 22.8% of values are below the mean.

C. About 2.28% of values are above the mean.

D. About 2.28% of values are below 2 SDs below the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q2/5

317

QUESTION ID

S9-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. About 2.28% of values are below 2 SDs below the mean.

GRADING NOTE Read as a proportion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q2/5

Write 0.1587 as a percent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q3/5

318

QUESTION ID

S9-B2-Q3

TYPE

Short answer

CORRECT ANSWER

15.87%

GRADING NOTE × 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q3/5

True or false?

A proportion of a population can be greater than 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q4/5

319

QUESTION ID

S9-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Proportions are between 0 and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is the largest proportion?

A. 0.5

B. 0.8413

C. 0.9772

D. 0.0228

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q5/5

320

QUESTION ID

S9-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 0.9772

GRADING NOTE Compare decimals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • HSS-ID.A.4

BACKWARD 3

Find right tails and areas between from left areas • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

ANSWER KEY Reduced copy of the question page, with its answer

The area to the left of z = 1.5 is 0.9332. What is the area to the right of z = 1.5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q1/5

322

QUESTION ID
S9-B3-Q1

TYPE
Short answer

CORRECT ANSWER
0.0668

GRADING NOTE 1 − 0.9332.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q1/5

Areas to the left: $z = -1$ gives 0.1587; $z = 1$ gives 0.8413. Find the area between $z = -1$ and $z = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q2/5

323

QUESTION ID

S9-B3-Q2

TYPE

Short answer

CORRECT ANSWER

0.6826

GRADING NOTE 0.8413 − 0.1587.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-B3-Q2**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q2/5

True or false?

The area to the right of z = 0 is 0.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q3/5

QUESTION ID

S9-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Symmetry.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q3/5

The area to the left of z = 0.5 is 0.6915. The area between z = 0 and z = 0.5 is

A. 0.3085

B. 0.6915

C. 0.1915

D. 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q4/5

325

QUESTION ID

S9-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 0.1915

GRADING NOTE 0.6915 − 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-B3-Q4**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q4/5

The area to the left of $z = 2$ is 0.9772. What is the area to the left of $z = -2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q5/5

326

QUESTION ID

S9-B3-Q5

TYPE

Short answer

CORRECT ANSWER

0.0228

GRADING NOTE By symmetry it equals the area to the right of $z = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-B3-Q5**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q5/5

SET 9 • HSS-ID.A.4

FORWARD 1

Find the cutoff for a percentile • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

IQ scores are normal with mean 100 and SD 15. The table gives area 0.9332 to the left of z = 1.5. What score is at the 93.32nd percentile?

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q1/5

328

QUESTION ID

S9-F1-Q1

TYPE

Short answer

CORRECT ANSWER

122.5

GRADING NOTE 100 + 1.5·15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q1/5

Call lengths are normal with mean 50 s and SD 8 s. The table gives area 0.1056 to the left of z = −1.25. What length is at the 10.56th percentile?

Seconds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q2/5

329

QUESTION ID

S9-F1-Q2

TYPE

Short answer

CORRECT ANSWER

40

GRADING NOTE 50 − 1.25·8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q2/5

True or false?

A value with z = 0 is at the 50th percentile of a normal distribution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q3/5

330

QUESTION ID

S9-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The median.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q3/5

A factory wants only the longest 2.28% of rods to be rejected. Rod lengths are normal with mean 10 cm and SD 0.2 cm; the table gives area 0.9772 to the left of $z = 2$. What cutoff should it use?

A. 10.2 cm

B. 12 cm

C. 10.4 cm

D. 9.6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q4/5

331

QUESTION ID

S9-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 10.4 cm

GRADING NOTE 10 + 2·0.2. 9.6 is the lower cutoff; 12 uses z as centimeters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q4/5

Heights are normal with mean 175 cm and SD 7 cm; area 0.8413 is to the left of z = 1. What height separates the tallest 15.87%?

Height (cm): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q5/5

332

QUESTION ID

S9-F1-Q5

TYPE

Short answer

CORRECT ANSWER

182

GRADING NOTE 175 + 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F1-Q5**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q5/5

SET 9 • HSS-ID.A.4

FORWARD 2

Make decisions with normal areas • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

A bolt fits if its diameter is between 9.8 and 10.2 mm. Diameters are normal with mean 10.0 mm and SD 0.1 mm. Table: area 0.9772 left of z = 2 and 0.0228 left of z = −2. What proportion of bolts fit?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q1/5

334

QUESTION ID

S9-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.9544

GRADING NOTE 0.9772 − 0.0228.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q1/5

Fill amounts from both machines are approximately normal. Machine A fills bottles with mean 500 mL, SD 4 mL; machine B with mean 502 mL, SD 1 mL. A bottle is underfilled below 498 mL. Table: area 0.3085 to the left of z = −0.5; the area to the left of z = −4 is less than 0.0001. Which machine underfills fewer bottles?

A. A, because its standard deviation is larger

B. B: under 0.01% versus about 31% for A

C. A, because its mean is closer to 498 mL

D. They underfill the same proportion

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q2/5

335

QUESTION ID

S9-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. B: under 0.01% versus about 31% for A

GRADING NOTE A: z = (498 − 500)/4 = −0.5, area 0.3085. B: z = (498 − 502)/1 = −4, area very close to 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q2/5

True or false?

A process makes items whose lengths are approximately normal. If its standard deviation is reduced while its mean stays the same and the process stays normal, the proportion of items more than 1 mm from the mean goes down.

A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q3/5

336

QUESTION ID
S9-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The same 1 mm cutoff is now more standard deviations away, so each normal tail is smaller. For other shapes a smaller SD alone would not guarantee this.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q3/5

Scores are normal with mean 70 and SD 10. A prize goes to scores of 85 or more. Table: area 0.9332 left of $z = 1.5$. What proportion of students win?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q4/5

337

QUESTION ID

S9-F2-Q4

TYPE

Short answer

CORRECT ANSWER

0.0668

GRADING NOTE $1 - 0.9332$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q4/5

Two normal processes have the same mean. Process 1 has SD 2; process 2 has SD 5. Which has more items within 4 units of the mean?

A. process 2

B. the same

C. cannot tell

D. process 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q5/5

338

QUESTION ID

S9-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. process 1

GRADING NOTE 4 units is 2 SDs for process 1 (95%) and 0.8 SD for process 2 (about 58%).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F2-Q5**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q5/5

Recognize when a normal model is not appropriate

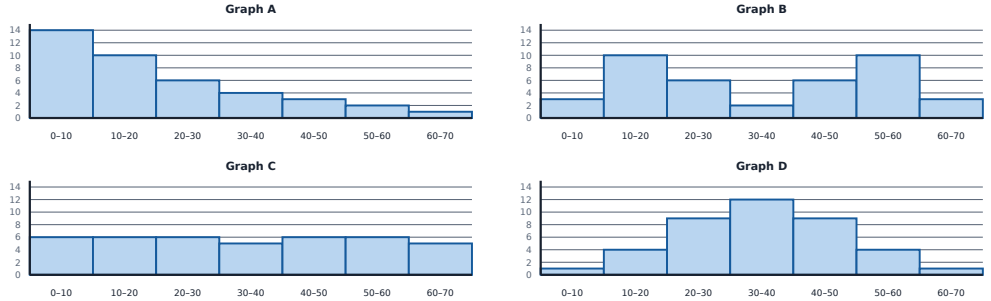
MAIN	Probability & Statistics main question	**HSS-ID.A.4**
BACKWARD 1	Describe the shape of a distribution	**6.SP.A.2**
BACKWARD 2	Relate the mean and median to the shape	**6.SP.B.5.d**
BACKWARD 3	Test a normal model against the data	**HSS-ID.A.4**
FORWARD 1	Compare actual percents with the 68-95-99.7 rule	**HSS-ID.A.4**
FORWARD 2	Choose summaries for data that are not normal	**HSS-ID.A.2**

SET 10 • HSS-ID.A.4

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Each histogram shows 40 measurements on the same scale. For which data set is a normal model most reasonable? In each histogram, the horizontal axis shows value and the vertical axis shows frequency.



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q1/5

341

QUESTION ID

S10-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE The single symmetric mound fits a normal model. The others are skewed right, have two peaks, or are flat.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q1/5

Waiting times at a clinic can never be negative. Their mean is 3 minutes and their standard deviation is 4 minutes. If a normal model with this mean and standard deviation were used, what proportion of waiting times would it place below 0 minutes? Use the table values printed below (areas to the left of z). Give four decimal places.

z	−1.0	−0.75	−0.5	0.5
Area to the left	0.1587	0.2266	0.3085	0.6915

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q2/5

342

QUESTION ID

S10-M-Q2

TYPE

Short answer

CORRECT ANSWER

0.2266

GRADING NOTE $z = (0 - 3)/4 = -0.75$; area 0.2266.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q2/5

Waiting times can never be negative, yet a normal model with their mean (3 minutes) and standard deviation (4 minutes) places about 23% of times below 0. What does this show?

A. The times are normal, but 23% of them were recorded wrong.

B. The standard deviation must have been computed incorrectly.

C. A normal model fits, because the mean is a positive number.

D. A normal model is not appropriate for these waiting times.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q3/5

343

QUESTION ID

S10-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A normal model is not appropriate for these waiting times.

GRADING NOTE A model that predicts impossible values for a large share of the data does not fit. A large SD relative to the mean is typical of right-skewed, bounded data; nothing says it was computed wrongly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q3/5

Which data set is least likely to be approximately normal?

A. the masses of eggs from one large farm

B. repeated careful measurements of one object's length

C. the yearly incomes of all households in a large city

D. the heights of adult women in one country

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q4/5

344

QUESTION ID

S10-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. the yearly incomes of all households in a large city

GRADING NOTE Incomes are strongly skewed right (a few very large values). The others are classic near-normal examples.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q4/5

A box plot of 60 values shows minimum 2, Q1 4, median 5, Q3 9 and maximum 30. Which conclusion is correct?



A. The data are roughly symmetric, so a normal model fits well.

B. The data are skewed left, so a normal model is not a good fit.

C. The data are skewed right, so a normal model is not a good fit.

D. A median of 5 shows that the data must be normal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q5/5

345

QUESTION ID

S10-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. The data are skewed right, so a normal model is not a good fit.

GRADING NOTE The upper whisker (21) and upper box half (4) are far longer than the lower ones (2 and 1): skewed right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q5/5

SET 10 • HSS-ID.A.4

BACKWARD 1

Describe the shape of a distribution • Grade 6 • Statistics & Probability • 6.SP.A.2

True or false?

A histogram with a long tail to the right is called skewed right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q1/5

347

QUESTION ID

S10-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The tail names the skew.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q1/5

What is the shape of the data 1, 1, 2, 2, 2, 3, 3, 4, 6, 9?

A. skewed right

B. skewed left

C. symmetric

D. uniform

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q2/5

348

QUESTION ID

S10-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. skewed right

GRADING NOTE Most values are low, with a tail toward 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q2/5

True or false?

A distribution with two separate peaks is called unimodal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q3/5

349

QUESTION ID

S10-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Two peaks: bimodal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which list of values is roughly symmetric?

A. 1, 10, 18, 19, 19, 20

B. 3, 3, 3, 3, 3, 30

C. 1, 1, 1, 2, 9, 20

D. 2, 4, 5, 5, 6, 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q4/5

350

QUESTION ID

S10-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 2, 4, 5, 5, 6, 8

GRADING NOTE Centered at 5 with equal spread on each side. The others have a long tail.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q4/5

For the data 4, 5, 5, 6, 6, 6, 7, 7, 8, find the mean and the median.

Mean: _____

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q5/5

351

QUESTION ID

S10-B1-Q5

TYPE

Short answer

CORRECT ANSWER

6; 6

GRADING NOTE Form required: both. Symmetric data: mean = median = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • HSS-ID.A.4

BACKWARD 2

Relate the mean and median to the shape • Grade 6 • Statistics & Probability • 6.SP.B.5.d

True or false?

In a distribution skewed right, the mean is usually greater than the median.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q1/5

353

QUESTION ID

S10-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The tail pulls the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q1/5

A data set has mean 52 and median 40. Which shape is most likely?

A. skewed right

B. skewed left

C. uniform

D. symmetric

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q2/5

354

QUESTION ID

S10-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. skewed right

GRADING NOTE Mean well above the median.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

For the data 1, 2, 2, 3, 22, find the mean and the median.

Mean: _____

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q3/5

355

QUESTION ID

S10-B2-Q3

TYPE

Short answer

CORRECT ANSWER

6; 2

GRADING NOTE Form required: both. 30/5 = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q3/5

True or false?

For strongly skewed data, the mean and standard deviation are the best summaries of center and spread.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q4/5

356

QUESTION ID

S10-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The median and IQR resist the tail.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

For data skewed right, which measure best describes a typical value?

A. median

B. mean

C. midrange

D. maximum

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q5/5

357

QUESTION ID

S10-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. median

GRADING NOTE The median is resistant to the tail.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q5/5

SET 10 • HSS-ID.A.4

BACKWARD 3

Test a normal model against the data • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Counts of errors per page (never negative) have mean 8 and standard deviation 4. How many standard deviations below the mean is 0, and what percent would a normal model place below 0 by the 68-95-99.7 rule?

Standard deviations: _____

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q1/5

359

QUESTION ID

S10-B3-Q1

TYPE

Short answer

CORRECT ANSWER

2; 2.5%

GRADING NOTE Form required: both. (8 − 0)/4 = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-B3-Q1**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q1/5

True or false?

A data set with any mean and any standard deviation is approximately normal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q2/5

360

QUESTION ID

S10-B3-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Every data set has a mean and SD; the shape decides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-B3-Q2**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q2/5

Of 20 values with mean 50 and standard deviation 10, 19 lie between 40 and 60. Is this consistent with a normal model?

A. Yes: a normal model predicts about 95% within 1 SD.

B. Yes: every value is within 3 SDs of the mean.

C. No: a normal model predicts about 68% within 1 SD, but 95% are.

D. No: a normal model predicts all values within 1 SD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q3/5

361

QUESTION ID

S10-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. No: a normal model predicts about 68% within 1 SD, but 95% are.

GRADING NOTE 19/20 = 95% within 1 SD, far above 68%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q3/5

A normal model is used for 200 values. About how many values does it predict within 2 standard deviations of the mean?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q4/5

362

QUESTION ID
S10-B3-Q4

TYPE
Short answer

CORRECT ANSWER
190

GRADING NOTE 0.95 · 200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q4/5

True or false?

A normal model with mean 3 and SD 4 places about 16% of its values below −1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q5/5

363

QUESTION ID

S10-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE −1 is 1 SD below the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q5/5

SET 10 • HSS-ID.A.4

FORWARD 1

Compare actual percents with the 68-95-99.7 rule • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Of 40 values, 27 lie within 1 standard deviation of the mean and 38 lie within 2. What percent is each?

Within 1 SD: _____

Within 2 SD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q1/5

365

QUESTION ID

S10-F1-Q1

TYPE

Short answer

CORRECT ANSWER

67.5%; 95%

GRADING NOTE Form required: both. 27/40 and 38/40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q1/5

Of 40 values, 67.5% are within 1 SD of the mean and 95% within 2 SDs, and a histogram is one symmetric mound. Which conclusion is reasonable?

A. A normal model is not reasonable, because 67.5% is not 68%.

B. The data must be skewed right.

C. A normal model is reasonable for these data.

D. No conclusion is possible without the raw values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q2/5

366

QUESTION ID

S10-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. A normal model is reasonable for these data.

GRADING NOTE Percents match the rule closely and the shape fits.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q2/5

True or false?

If 92% of a data set lies within 1 SD of its mean, the data fit a normal model well.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q3/5

367

QUESTION ID

S10-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A normal model gives about 68%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q3/5

Of 50 values, 46 lie within 1 standard deviation of the mean. What percent is that?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q4/5

368

QUESTION ID

S10-F1-Q4

TYPE

Short answer

CORRECT ANSWER

92%

GRADING NOTE 46/50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q4/5

Which result most strongly suggests a data set is not normal?

A. About half of the values lie above the mean.

B. 96% of the values lie within 2 SDs of the mean.

C. 92% of the values lie within 1 SD of the mean.

D. 70% of the values lie within 1 SD of the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q5/5

369

QUESTION ID

S10-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 92% of the values lie within 1 SD of the mean.

GRADING NOTE 92% is far from 68%; the others are close to the rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F1-Q5**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q5/5

SET 10 • HSS-ID.A.4

FORWARD 2

Choose summaries for data that are not normal • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

Home prices in a town are strongly skewed right. Which pair of summaries is most appropriate?

A. mode and maximum

B. mean and range

C. mean and standard deviation

D. median and IQR

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 10 • FORWARD 2 • Q1/5

371

QUESTION ID

S10-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. median and IQR

GRADING NOTE Resistant summaries for skewed data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F2-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 10 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Home prices (thousands of dollars) are 150, 160, 170, 180, 190, 200, 900. Find the median and the IQR. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.

Median: _____

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 10 • FORWARD 2 • Q2/5

372

QUESTION ID

S10-F2-Q2

TYPE

Short answer

CORRECT ANSWER

180; 40

GRADING NOTE Form required: both. Q1 = 160, Q3 = 200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F2-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 10 • FORWARD 2 • Q2/5

True or false?

Using a normal model for strongly skewed data can give misleading percentages.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 10 • FORWARD 2 • Q3/5

373

QUESTION ID
S10-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The model's tails do not match.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 10 • FORWARD 2 • Q3/5

For the prices 150, 160, 170, 180, 190, 200, 900 (thousands), which statement is correct?

A. The value 900 pulls the median above the mean.

B. The mean and the median are equal.

C. The value 900 pulls the mean above the median.

D. The value 900 changes the IQR more than the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 10 • FORWARD 2 • Q4/5

374

QUESTION ID

S10-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. The value 900 pulls the mean above the median.

GRADING NOTE Mean = 1950/7 ≈ 278.6 > 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 10 • FORWARD 2 • Q4/5

True or false?

The standard deviation is resistant to one extreme value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 10 • FORWARD 2 • Q5/5

375

QUESTION ID
S10-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It uses every deviation, squared.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F2-Q5**
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 10 • FORWARD 2 • Q5/5

Joint and marginal frequencies in two-way tables

MAIN	Probability & Statistics main question	**HSS-ID.B.5**
BACKWARD 1	Complete and read a two-way frequency table	**8.SP.A.4**
BACKWARD 2	Write a part of a whole as a fraction, decimal and percent	**7.NS.A.2.d**
FORWARD 1	Build a joint relative-frequency table	**HSS-ID.B.5**
FORWARD 2	Look for an association by comparing rows	**HSS-ID.B.5**

SET 11 • HSS-ID.B.5

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

160 students were asked how they prefer to take a class. What fraction of all students are 10th graders who chose online? Give a fraction in lowest terms.

	Online	In person	Total
9th grade	30	50	80
10th grade	45	35	80
Total	75	85	160

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • MAIN • Q1/5

378

QUESTION ID
S11-M-Q1

TYPE
Short answer

CORRECT ANSWER
9/32

GRADING NOTE 45/160 = 9/32 (joint).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • MAIN • Q1/5

160 students were asked how they prefer to take a class. What is the marginal relative frequency of choosing online? Give a fraction in lowest terms.

	Online	In person	Total
9th grade	30	50	80
10th grade	45	35	80
Total	75	85	160

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • MAIN • Q2/5

379

QUESTION ID
S11-M-Q2

TYPE
Short answer

CORRECT ANSWER
15/32

GRADING NOTE 75/160 = 15/32.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • MAIN • Q2/5

A survey of 120 students: 50 are 7th graders and 70 are 8th graders. 18 of the 7th graders play an instrument, and 20 of the 8th graders do not. How many 7th graders do not play, and how many 8th graders do play?

7th graders who do not play: _____

8th graders who play: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • MAIN • Q3/5

380

QUESTION ID
S11-M-Q3

TYPE
Short answer

CORRECT ANSWER
32; 50

GRADING NOTE Form required: both. 50 − 18 and 70 − 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-M-Q3**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • MAIN • Q3/5

160 students were surveyed (table). Which statement gives a marginal relative frequency?

	Online	In person	Total
9th grade	30	50	80
10th grade	45	35	80
Total	75	85	160

A. 28.125% of all students are 10th graders who chose online.

B. 53.125% of all students chose in person.

C. 56.25% of the 10th graders chose online.

D. 37.5% of the 9th graders chose online.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • MAIN • Q4/5

381

QUESTION ID
S11-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. 53.125% of all students chose in person.

GRADING NOTE 85/160 uses a total over the grand total. The others are a joint frequency and two conditional ones.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • MAIN • Q4/5

A dealer has 100 cars in two lots (table). What percent of all the cars are gray, and what percent of all the cars are red cars in lot B?

	White	Red	Gray	Total
Lot A	18	12	10	40
Lot B	22	8	30	60
Total	40	20	40	100

Gray: _____

Red in lot B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • MAIN • Q5/5

382

QUESTION ID

S11-M-Q5

TYPE

Short answer

CORRECT ANSWER

40%; 8%

GRADING NOTE Form required: both. 40/100 and 8/100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-M-Q5**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • MAIN • Q5/5

SET 11 • HSS-ID.B.5

BACKWARD 1

Complete and read a two-way frequency table • Grade 8 • Statistics & Probability • 8.SP.A.4

120 adults were asked whether they own a dog and whether they walk daily (table). How many walk daily?

	Walks daily	Does not	Total
Has a dog	36	24	60
No dog	15	45	60
Total	51	69	120

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q1/5

384

QUESTION ID

S11-B1-Q1

TYPE

Short answer

CORRECT ANSWER

51

GRADING NOTE 36 + 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q1/5

From the dog-and-walking table, how many adults have no dog and do not walk daily?

	Walks daily	Does not	Total
Has a dog	36	24	60
No dog	15	45	60
Total	51	69	120

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q2/5

385

QUESTION ID

S11-B1-Q2

TYPE

Short answer

CORRECT ANSWER

45

GRADING NOTE Read the cell.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q2/5

Of 50 students, 20 are in band and 12 of the band members are 9th graders. 30 of the 50 students are 9th graders. How many 9th graders are not in band?

A. 30

B. 8

C. 18

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q3/5

386

QUESTION ID

S11-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 18

GRADING NOTE 30 − 12. 8 is band members not in 9th grade.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q3/5

True or false?

In a two-way table, the row totals add up to the grand total.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q4/5

387

QUESTION ID

S11-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Every count is in one row.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q4/5

True or false?

In a two-way table, a cell count can be greater than its row total.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q5/5

388

QUESTION ID

S11-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Cells add to the row total.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • HSS-ID.B.5

BACKWARD 2

Write a part of a whole as a fraction, decimal and percent • Grade 7 • The Number System • 7.NS.A.2.d

ANSWER KEY Reduced copy of the question page, with its answer

Write 45/160 as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q1/5

390

QUESTION ID

S11-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.28125

GRADING NOTE 45 ÷ 160.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 24/120 as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q2/5

391

QUESTION ID

S11-B2-Q2

TYPE

Short answer

CORRECT ANSWER

20%

GRADING NOTE 0.2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q2/5

True or false?

3/8 = 0.375.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q3/5

392

QUESTION ID

S11-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 3 ÷ 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q3/5

Write 15/32 as a percent.

A. 15.32%

B. 47%

C. 2.13%

D. 46.875%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q4/5

393

QUESTION ID

S11-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 46.875%

GRADING NOTE 15 ÷ 32 = 0.46875.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 30/160 in lowest terms.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q5/5

394

QUESTION ID

S11-B2-Q5

TYPE

Short answer

CORRECT ANSWER

3/16

GRADING NOTE Divide by 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q5/5

SET 11 • HSS-ID.B.5

FORWARD 1

Build a joint relative-frequency table • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

Convert the online-class table to relative frequencies of the grand total (160). What are the joint relative frequencies for 9th grade online and 9th grade in person? Give decimals.

	Online	In person	Total
9th grade	30	50	80
10th grade	45	35	80
Total	75	85	160

9th online: _____

9th in person: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 1 • Q1/5

396

QUESTION ID

S11-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.1875; 0.3125

GRADING NOTE Form required: both. 30/160 and 50/160.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F1-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 1 • Q1/5

True or false?

In a table of joint relative frequencies of the grand total, all the cells add to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 1 • Q2/5

397

QUESTION ID

S11-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Every individual is in one cell.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 1 • Q2/5

In a joint relative-frequency table, the 9th-grade cells are 0.1875 and 0.3125. What is the 9th-grade marginal relative frequency?

A. 0.5

B. 0.1875

C. 0.6

D. 0.125

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 1 • Q3/5

398

QUESTION ID
S11-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. 0.5

GRADING NOTE Add the row.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F1-Q3**
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In the car table (100 cars), what are the joint relative frequencies for white cars in lot A and gray cars in lot B? Give decimals.

	White	Red	Gray	Total
Lot A	18	12	10	40
Lot B	22	8	30	60
Total	40	20	40	100

White, lot A: _____

Gray, lot B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 1 • Q4/5

399

QUESTION ID

S11-F1-Q4

TYPE

Short answer

CORRECT ANSWER

0.18; 0.3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F1-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 1 • Q4/5

True or false?

A marginal relative frequency is always smaller than every joint relative frequency in its row.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 1 • Q5/5

400

QUESTION ID

S11-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is the sum of them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 1 • Q5/5

SET 11 • HSS-ID.B.5

FORWARD 2

Look for an association by comparing rows • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

In the online-class table, what percent of 9th graders and what percent of 10th graders chose online?

	Online	In person	Total
9th grade	30	50	80
10th grade	45	35	80
Total	75	85	160

9th: _____

10th: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 2 • Q1/5

402

QUESTION ID

S11-F2-Q1

TYPE

Short answer

CORRECT ANSWER

37.5%; 56.25%

GRADING NOTE Form required: both. 30/80 and 45/80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F2-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 2 • Q1/5

If grade and choice of class were not associated, what would be true?

A. Every cell of the table would contain the same count.

B. The percent choosing online would be about the same in each grade.

C. The number choosing online would be 0 in one grade.

D. The grade totals would have to be unequal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 2 • Q2/5

403

QUESTION ID
S11-F2-Q2

TYPE
Multiple choice

CORRECT ANSWER
B. The percent choosing online would be about the same in each grade.

GRADING NOTE No association: equal conditional rates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 2 • Q2/5

True or false?

Because more 10th graders (45) than 9th graders (30) chose online, the counts alone prove an association, even if the grades had very different sizes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 2 • Q3/5

404

QUESTION ID
S11-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Compare rates, not counts, when totals differ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 2 • Q3/5

In the car table, what percent of lot A cars and what percent of lot B cars are gray?

	White	Red	Gray	Total
Lot A	18	12	10	40
Lot B	22	8	30	60
Total	40	20	40	100

Lot A: _____

Lot B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 2 • Q4/5

405

QUESTION ID

S11-F2-Q4

TYPE

Short answer

CORRECT ANSWER

25%; 50%

GRADING NOTE Form required: both. 10/40 and 30/60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F2-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 2 • Q4/5

Gray makes up 25% of lot A and 50% of lot B. Which is correct?

A. Lot B has exactly twice as many cars as lot A.

B. Gray is the most common color in lot A.

C. Lot and color are associated: gray is more common in lot B.

D. Lot and color are not associated.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 2 • Q5/5

406

QUESTION ID
S11-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
C. Lot and color are associated: gray is more common in lot B.

GRADING NOTE Rates differ. Lot B has 60 cars, A has 40; white (18) is most common in A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 2 • Q5/5

Conditional relative frequencies and association

MAIN	Probability & Statistics main question	**HSS-ID.B.5**
BACKWARD 1	Choose the denominator for a conditional relative frequency	**8.SP.A.4**
BACKWARD 2	Find a percent of a group	**6.RP.A.3.c**
BACKWARD 3	Compare two fractions	**4.NF.A.2**
FORWARD 1	Distinguish the two directions of a conditional rate	**HSS-ID.B.5**
FORWARD 2	Recognize when combining groups reverses a comparison	**HSS-ID.B.5**

SET 12 • HSS-ID.B.5

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

160 students were surveyed (table). What fraction of the 9th graders chose online, and what fraction of the 10th graders chose online? Give fractions in lowest terms.

	Online	In person	Total
9th grade	30	50	80
10th grade	45	35	80
Total	75	85	160

9th graders: _____

10th graders: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • MAIN • Q1/5

409

QUESTION ID

S12-M-Q1

TYPE

Short answer

CORRECT ANSWER

3/8; 9/16

GRADING NOTE Form required: both. 30/80 and 45/80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-M-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • MAIN • Q1/5

In the survey, 37.5% of 9th graders and 56.25% of 10th graders chose online. Which statement is correct?

	Online	In person	Total
9th grade	30	50	80
10th grade	45	35	80
Total	75	85	160

A. 9th graders chose online at a higher rate, so grade and choice are associated.

B. The rates are equal, so grade and choice are not associated.

C. 10th graders chose online at a higher rate, so grade and choice are associated.

D. Each grade has 80 students, so grade and choice cannot be associated.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S12-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • MAIN • Q2/5

410

QUESTION ID

S12-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 10th graders chose online at a higher rate, so grade and choice are associated.

GRADING NOTE Compare the conditional rates. Equal group sizes say nothing about association.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S12-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • MAIN • Q2/5

160 students were surveyed (table). Of the students who chose online, what percent were 10th graders?

	Online	In person	Total
9th grade	30	50	80
10th grade	45	35	80
Total	75	85	160

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • MAIN • Q3/5

411

QUESTION ID
S12-M-Q3

TYPE
Short answer

CORRECT ANSWER
60%

GRADING NOTE 45/75 (the online column).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-M-Q3**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • MAIN • Q3/5

120 adults were surveyed (table). What percent of dog owners walk daily, and what percent of adults without a dog walk daily?

	Walks daily	Does not	Total
Has a dog	36	24	60
No dog	15	45	60
Total	51	69	120

Dog owners: _____

No dog: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • MAIN • Q4/5

412

QUESTION ID
S12-M-Q4

TYPE
Short answer

CORRECT ANSWER
60%; 25%

GRADING NOTE Form required: both. 36/60 and 15/60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-M-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • MAIN • Q4/5

In the survey, 60% of dog owners and 25% of adults without a dog walk daily. Which conclusion is supported?

	Walks daily	Does not	Total
Has a dog	36	24	60
No dog	15	45	60
Total	51	69	120

A. Dog ownership and walking daily are not associated.

B. Owning a dog causes people to walk daily.

C. Owning a dog and walking daily are associated in this sample.

D. Adults without a dog walk daily more often.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • MAIN • Q5/5

413

QUESTION ID
S12-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
C. Owning a dog and walking daily are associated in this sample.

GRADING NOTE Different rates show association; a survey cannot show cause.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • MAIN • Q5/5

SET 12 • HSS-ID.B.5

BACKWARD 1

Choose the denominator for a conditional relative frequency • Grade 8 • Statistics & Probability • 8.SP.A.4

Of the 60 dog owners, 36 walk daily. What fraction of dog owners walk daily? Give lowest terms.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q1/5

415

QUESTION ID

S12-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3/5

GRADING NOTE 36/60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q1/5

True or false?

To find the percent of 9th graders who chose online, divide the 9th-grade online count by the grand total of all students.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q2/5

416

QUESTION ID

S12-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Divide by the 9th-grade total.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q2/5

In the online-class table, which number is the denominator for the percent of online-choosers who are 9th graders?

	Online	In person	Total
9th grade	30	50	80
10th grade	45	35	80
Total	75	85	160

A. 80, the 9th-grade total

B. 75, the online total

C. 30, the cell count

D. 160, the grand total

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q3/5

417

QUESTION ID

S12-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 75, the online total

GRADING NOTE "Of online-choosers" names the column.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 30 out of 80 as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q4/5

418

QUESTION ID

S12-B1-Q4

TYPE

Short answer

CORRECT ANSWER

37.5%

GRADING NOTE 30/80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q4/5

True or false?

A row relative frequency uses that row's total as the denominator.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q5/5

419

QUESTION ID

S12-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • HSS-ID.B.5

BACKWARD 2

Find a percent of a group • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

45 is what percent of 80?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q1/5

421

QUESTION ID

S12-B2-Q1

TYPE

Short answer

CORRECT ANSWER

56.25%

GRADING NOTE 45/80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q1/5

What is 25% of 60?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q2/5

422

QUESTION ID

S12-B2-Q2

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE 0.25 · 60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

18 out of 24 is what percent?

A. 18%

B. 75%

C. 133%

D. 24%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q3/5

423

QUESTION ID

S12-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 75%

GRADING NOTE 18/24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q3/5

True or false?

9/16 = 56.25%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q4/5

424

QUESTION ID

S12-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 9 ÷ 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q4/5

True or false?

45 is 60% of 80.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q5/5

425

QUESTION ID

S12-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 45/80 = 56.25%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q5/5

SET 12 • HSS-ID.B.5

BACKWARD 3

Compare two fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.2

Which is larger, 3/8 or 9/16?

A. 3/8

B. they cannot be compared

C. they are equal

D. 9/16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q1/5

427

QUESTION ID

S12-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. 9/16

GRADING NOTE 3/8 = 6/16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q1/5

True or false?

3/5 > 1/4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q2/5

428

QUESTION ID

S12-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 0.6 > 0.25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q2/5

Write 3/8 as an equivalent fraction with denominator 16. What is the numerator?

Numerator: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q3/5

429

QUESTION ID

S12-B3-Q3

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Multiply by 2/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q3/5

True or false?

2/5 > 3/4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q4/5

430

QUESTION ID

S12-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 0.4 < 0.75.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Order 1/4, 3/5 and 3/8 from least to greatest.

A. 1/4, 3/5, 3/8

B. 3/8, 1/4, 3/5

C. 1/4, 3/8, 3/5

D. 3/5, 3/8, 1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q5/5

431

QUESTION ID

S12-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 1/4, 3/8, 3/5

GRADING NOTE 0.25, 0.375, 0.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q5/5

SET 12 • HSS-ID.B.5

FORWARD 1

Distinguish the two directions of a conditional rate • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In the car table, what percent of the gray cars are in lot B, and what percent of the lot B cars are gray?

	White	Red	Gray	Total
Lot A	18	12	10	40
Lot B	22	8	30	60
Total	40	20	40	100

Gray cars in lot B: _____

Lot B cars that are gray: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 1 • Q1/5

433

QUESTION ID
S12-F1-Q1

TYPE
Short answer

CORRECT ANSWER
75%; 50%

GRADING NOTE Form required: both. 30/40 and 30/60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 1 • Q1/5

True or false?

The percent of lot B cars that are gray equals the percent of gray cars that are in lot B.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 1 • Q2/5

434

QUESTION ID

S12-F1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 50% versus 75%: different denominators.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 1 • Q2/5

In the car table, what percent of the white cars are in lot A?

	White	Red	Gray	Total
Lot A	18	12	10	40
Lot B	22	8	30	60
Total	40	20	40	100

A. 30%

B. 45%

C. 18%

D. 40%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 1 • Q3/5

435

QUESTION ID

S12-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 45%

GRADING NOTE 18/40. 18% is of all cars; 40% is lot A's share; 30% is 12/40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 1 • Q3/5

In the car table, what percent of the lot A cars are red?

	White	Red	Gray	Total
Lot A	18	12	10	40
Lot B	22	8	30	60
Total	40	20	40	100

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 1 • Q4/5

436

QUESTION ID
S12-F1-Q4

TYPE
Short answer

CORRECT ANSWER
30%

GRADING NOTE 12/40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 1 • Q4/5

Which statement uses a column total of the car table as its denominator?

	White	Red	Gray	Total
Lot A	18	12	10	40
Lot B	22	8	30	60
Total	40	20	40	100

A. 12% of all cars are red cars in lot A.

B. Of the lot A cars, 30% are red.

C. Of the red cars, 60% are in lot A.

D. 40% of all cars are in lot A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 1 • Q5/5

437

QUESTION ID
S12-F1-Q5

TYPE
Multiple choice

CORRECT ANSWER
C. Of the red cars, 60% are in lot A.

GRADING NOTE 12/20. The others divide by a row total or the grand total.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 1 • Q5/5

SET 12 • HSS-ID.B.5

FORWARD 2

Recognize when combining groups reverses a comparison • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Use the hospital table. What is the survival rate for mild cases at hospital A and at hospital B?

Hospital	Mild: survived / patients	Severe: survived / patients
A	95 / 100	120 / 400
B	360 / 400	25 / 100

A: _____

B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 2 • Q1/5

439

QUESTION ID

S12-F2-Q1

TYPE

Short answer

CORRECT ANSWER

95%; 90%

GRADING NOTE Form required: both. 95/100 and 360/400.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the hospital table. What is the survival rate for severe cases at hospital A and at hospital B?

Hospital	Mild: survived / patients	Severe: survived / patients
A	95 / 100	120 / 400
B	360 / 400	25 / 100

A: _____

B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 2 • Q2/5

440

QUESTION ID
S12-F2-Q2

TYPE
Short answer

CORRECT ANSWER
30%; 25%

GRADING NOTE Form required: both. 120/400 and 25/100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-F2-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the hospital table. What is the overall survival rate (all 500 patients) at hospital A and at hospital B?

Hospital	Mild: survived / patients	Severe: survived / patients
A	95 / 100	120 / 400
B	360 / 400	25 / 100

A: _____

B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 2 • Q3/5

441

QUESTION ID

S12-F2-Q3

TYPE

Short answer

CORRECT ANSWER

43%; 77%

GRADING NOTE Form required: both. 215/500 and 385/500.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-F2-Q3**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 2 • Q3/5

Hospital A has the higher survival rate for mild cases and for severe cases, but the lower rate overall. Why?

Hospital	Mild: survived / patients	Severe: survived / patients
A	95 / 100	120 / 400
B	360 / 400	25 / 100

A. The two hospitals treated different total numbers of patients.

B. The overall rates were computed with the wrong denominators.

C. A treats many more mild cases, and mild cases survive less often.

D. A treats many more severe cases, and severe cases survive less often.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 2 • Q4/5

442

QUESTION ID

S12-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. A treats many more severe cases, and severe cases survive less often.

GRADING NOTE A: 400 of 500 severe; B: 100 of 500. Both treated 500 patients.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-F2-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 2 • Q4/5

True or false?

If one group has the higher rate within every category, it must have the higher rate overall.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 2 • Q5/5

443

QUESTION ID

S12-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The hospital data are a counterexample (Simpson's paradox).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 2 • Q5/5

Represent two quantitative variables on a scatter plot

MAIN	Probability & Statistics main question	**HSS-ID.B.6**
BACKWARD 1	Plot and read points in the first quadrant	**5.G.A.2**
BACKWARD 2	Describe patterns of association	**8.SP.A.1**
FORWARD 1	Compare the direction, form and strength of associations	**HSS-ID.B.6**
FORWARD 2	Distinguish association from causation	**HSS-ID.C.9**

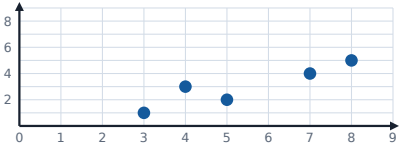
SET 13 • HSS-ID.B.6

MAIN

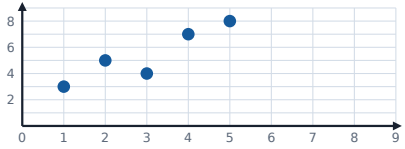
Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6

A seedling was measured weekly: (week, height in cm) = (1, 3), (2, 5), (3, 4), (4, 7), (5, 8). Which scatter plot shows the data?

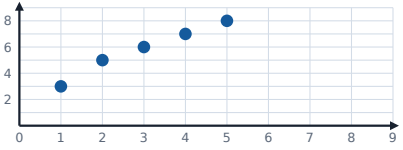
A



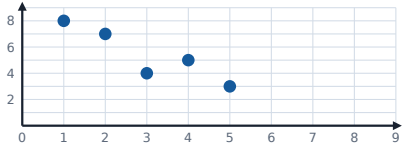
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q1/5

446

QUESTION ID

S13-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE The others swap the coordinates, move the week-3 point to 6, or reverse the heights.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q1/5

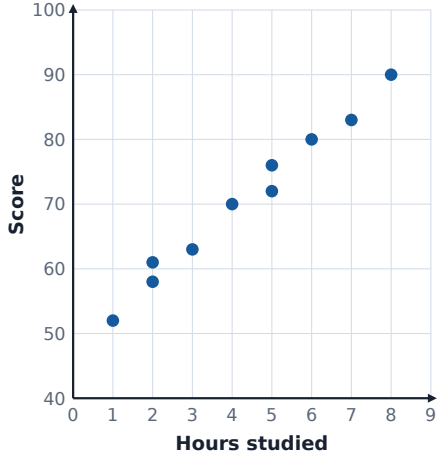
The scatter plot shows hours studied and test scores. Which describes the association?

A. curved, not linear

B. no clear association

C. strong, positive, linear

D. strong, negative, linear



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q2/5

447

QUESTION ID

S13-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. strong, positive, linear

GRADING NOTE Points rise in a narrow band along a line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q2/5

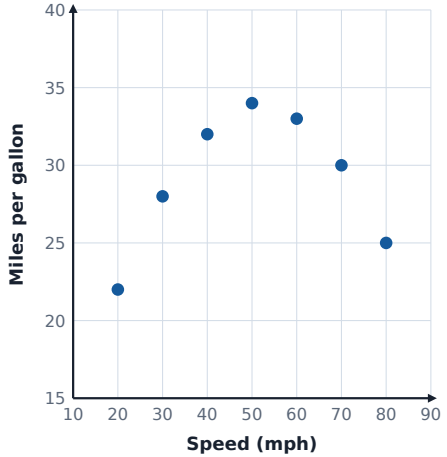
The scatter plot shows a car's speed and fuel efficiency. Which describes the relationship?

A. curved: rises, then falls

B. no association at all

C. strong, negative, linear

D. strong, positive, linear



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q3/5

448

QUESTION ID
S13-M-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. curved: rises, then falls

GRADING NOTE Peaks near 50 mph; a line does not describe it.

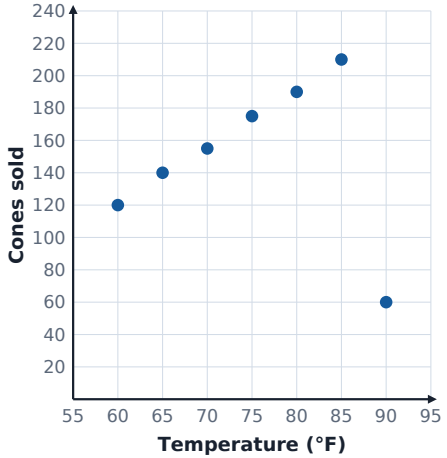
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q3/5

The scatter plot shows daily high temperature and ice-cream cones sold. One day does not follow the pattern. Give its temperature and its cones sold.

Temperature (°F): _____

Cones sold: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q4/5

449

QUESTION ID

S13-M-Q4

TYPE

Short answer

CORRECT ANSWER

90; 60

GRADING NOTE Form required: both. The other six days lie close to a rising line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q4/5

Which pair of variables would most likely show a negative association?

A. the height and arm span of adults

B. hours studied and test score

C. the age of a car and its resale value

D. shoe size and quiz score of adults

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q5/5

450

QUESTION ID

S13-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. the age of a car and its resale value

GRADING NOTE Value falls with age. Height/arm span and study/score are positive; shoe size and quiz score are unrelated.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q5/5

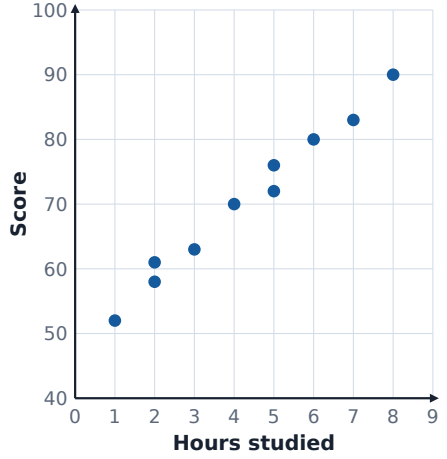
SET 13 • HSS-ID.B.6

BACKWARD 1

Plot and read points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

In the study scatter plot, what score did the student who studied 7 hours get?

Score: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q1/5

452

QUESTION ID

S13-B1-Q1

TYPE

Short answer

CORRECT ANSWER

83

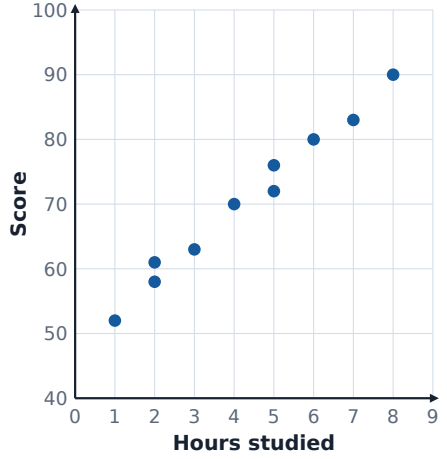
GRADING NOTE Point (7, 83).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q1/5

In the study scatter plot, how many students studied 2 hours?

Number: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q2/5

453

QUESTION ID

S13-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE (2, 58) and (2, 61).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q2/5

In the study data, what does the point (4, 70) mean?

A. a student studied 4 hours and scored 70

B. a student studied 70 hours and scored 4

C. 4 students scored 70

D. the average score was 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q3/5

454

QUESTION ID

S13-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. a student studied 4 hours and scored 70

GRADING NOTE (x, y) = (hours, score).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q3/5

True or false?

In the ordered pair (3, 63), 63 is the x-coordinate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q4/5

455

QUESTION ID

S13-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 63 is the y-coordinate.

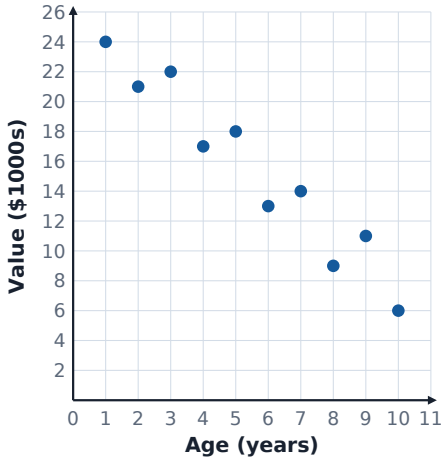
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In the car scatter plot, what is the resale value of the 8-year-old car, in thousands of dollars?

Value: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q5/5

456

QUESTION ID

S13-B1-Q5

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Point (8, 9).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q5/5

SET 13 • HSS-ID.B.6

BACKWARD 2

Describe patterns of association • Grade 8 • Statistics & Probability • 8.SP.A.1

True or false?

When y tends to increase as x increases, the association is positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q1/5

458

QUESTION ID

S13-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q1/5

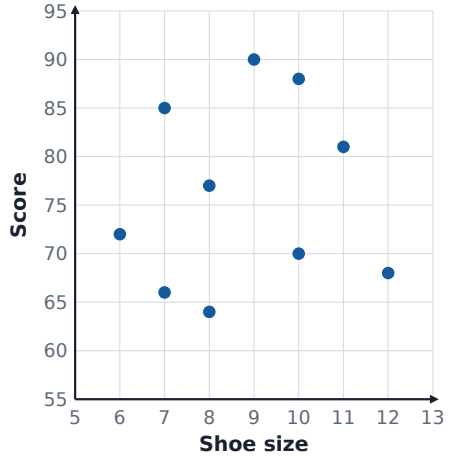
The scatter plot shows shoe size and quiz score. Which describes it?

A. curved

B. strong, positive, linear

C. strong, negative, linear

D. no clear association



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q2/5

459

QUESTION ID
S13-B2-Q2

TYPE
Multiple choice

CORRECT ANSWER
D. no clear association

GRADING NOTE The points form a shapeless cloud.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q2/5

True or false?

A scatter plot whose points lie close to a U-shaped curve shows a linear association.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q3/5

460

QUESTION ID

S13-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That is a nonlinear association.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q3/5

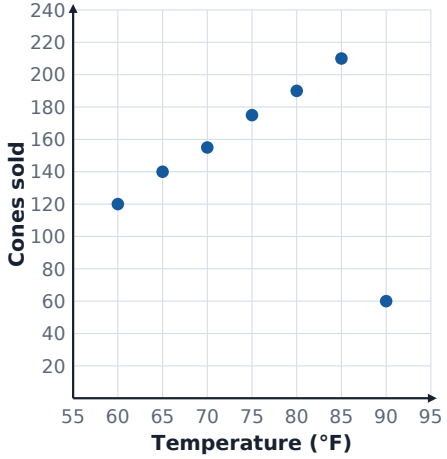
In the ice-cream scatter plot, the point (90, 60) is best described as

A. the trend line

B. an outlier

C. a cluster

D. the y-intercept



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q4/5

461

QUESTION ID
S13-B2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. an outlier

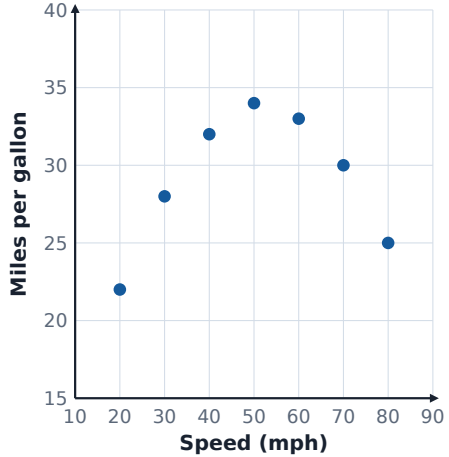
GRADING NOTE It lies far from the pattern of the other points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q4/5

In the fuel-efficiency scatter plot, at what speed is the fuel efficiency greatest?

Speed (mph): _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q5/5

462

QUESTION ID

S13-B2-Q5

TYPE

Short answer

CORRECT ANSWER

50

GRADING NOTE Highest point (50, 34).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q5/5

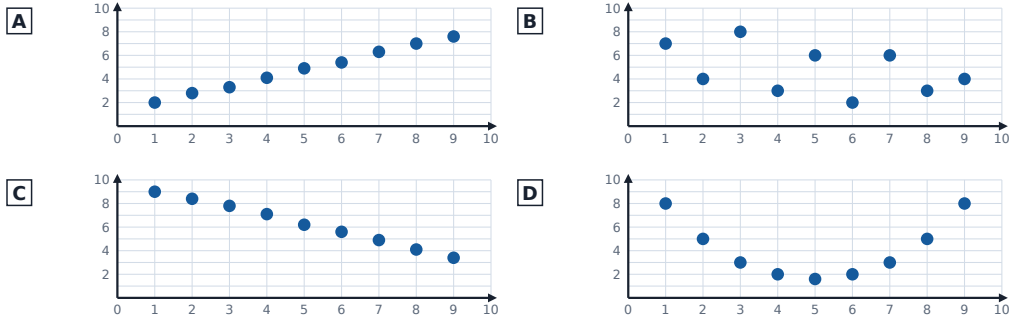
SET 13 • HSS-ID.B.6

FORWARD 1

Compare the direction, form and strength of associations • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Which scatter plot shows a strong, negative, linear association?



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q1/5

464

QUESTION ID

S13-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE The others are positive, weak, or curved.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q1/5

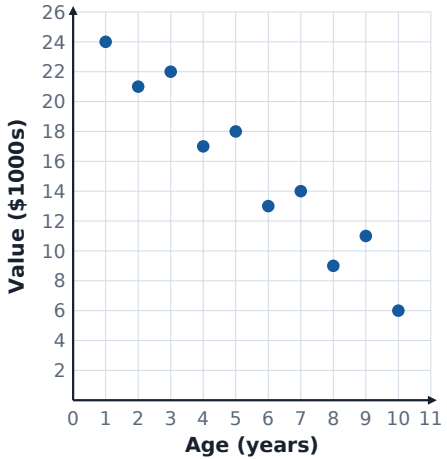
The scatter plot shows the age and resale value of cars. Which describes it?

A. curved, then level

B. strong, negative, linear

C. weak, positive, linear

D. strong, positive, linear



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q2/5

465

QUESTION ID

S13-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. strong, negative, linear

GRADING NOTE Values fall steadily with age.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q2/5

True or false?

A pattern can show a strong association even when it is not linear.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q3/5

466

QUESTION ID

S13-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The fuel data are an example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q3/5

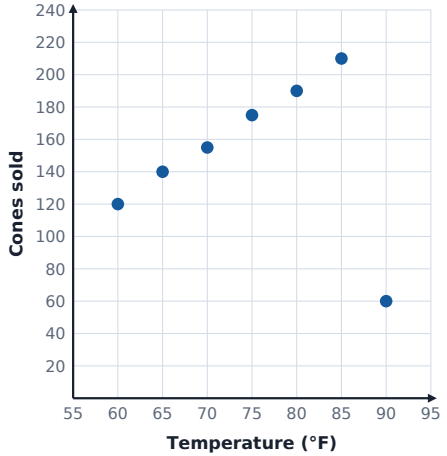
In the ice-cream scatter plot, if the outlier (90, 60) is ignored, the other six days show

A. no association

B. a strong, negative, linear association

C. a U-shaped association

D. a strong, positive, linear association



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q4/5

467

QUESTION ID

S13-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. a strong, positive, linear association

GRADING NOTE The six points rise along a line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q4/5

True or false?

Removing an outlier can never change how strong an association looks.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q5/5

468

QUESTION ID

S13-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The ice-cream data look weak with it and strong without it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q5/5

SET 13 • HSS-ID.B.6

FORWARD 2

Distinguish association from causation • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Cities with more fire stations tend to have more fires. Which is the best explanation?

A. Fires cause cities to build fewer stations.

B. Fire stations cause fires to start.

C. Larger cities have both more stations and more fires.

D. The association must be a mistake in the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 13 • FORWARD 2 • Q1/5

470

QUESTION ID
S13-F2-Q1

TYPE
Multiple choice

CORRECT ANSWER
C. Larger cities have both more stations and more fires.

GRADING NOTE City size is a lurking variable.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F2-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 13 • FORWARD 2 • Q1/5

True or false?

A strong association between two variables proves that one causes the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 13 • FORWARD 2 • Q2/5

471

QUESTION ID
S13-F2-Q2

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Association alone does not show cause.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 13 • FORWARD 2 • Q2/5

To show that a fertilizer causes taller plants, which study is needed?

A. a survey of gardeners about their plants

B. a scatter plot of plant heights found in gardens

C. a randomized experiment with fertilized and unfertilized plants

D. measuring only the plants that received fertilizer

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 13 • FORWARD 2 • Q3/5

472

QUESTION ID
S13-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. a randomized experiment with fertilized and unfertilized plants

GRADING NOTE Random assignment controls other causes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F2-Q3**
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 13 • FORWARD 2 • Q3/5

Ice-cream sales and swimming accidents both rise during the year. What best explains the association?

A. The two variables are unrelated.

B. Swimming accidents cause ice-cream sales.

C. Eating ice cream causes swimming accidents.

D. Hot weather increases both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 13 • FORWARD 2 • Q4/5

473

QUESTION ID

S13-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Hot weather increases both.

GRADING NOTE A lurking variable (temperature).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 13 • FORWARD 2 • Q4/5

True or false?

In an observational study, a lurking variable can create an association between two variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 13 • FORWARD 2 • Q5/5

474

QUESTION ID

S13-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both depend on it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 13 • FORWARD 2 • Q5/5

Use a function fitted to data to solve problems

MAIN	Probability & Statistics main question	**HSS-ID.B.6.a**
BACKWARD 1	Evaluate a function at an input	**HSF-IF.A.2**
BACKWARD 2	Evaluate expressions with exponents	**6.EE.A.2.c**
BACKWARD 3	Solve a linear equation for the input	**HSA-REI.B.3**
FORWARD 1	Judge whether a prediction is reasonable	**HSS-ID.B.6.a**
FORWARD 2	Compare predictions from two fitted models	**HSS-ID.B.6.a**

SET 14 • HSS-ID.B.6.a

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

For a group of students, a line fitted to the data is $y = 2.5x + 12$, where x is hours of practice and y is the predicted number of free throws made out of 50. Predict y for 8 hours of practice.

Predicted y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • MAIN • Q1/5

477

QUESTION ID
S14-M-Q1

TYPE
Short answer

CORRECT ANSWER
32

GRADING NOTE 2.5·8 + 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • MAIN • Q1/5

A bacteria culture is modeled by $P = 200(1.15)^t$, where t is hours. Predict the population after 6 hours, to the nearest whole number. A calculator or spreadsheet may be used.

Population: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q2
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • MAIN • Q2/5

478

QUESTION ID
S14-M-Q2

TYPE
Short answer

CORRECT ANSWER
463

GRADING NOTE $200 \cdot 1.15^6 = 462.61...$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q2
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • MAIN • Q2/5

A ball's height is modeled by $h = -4.9t^2 + 20t + 1.5$ (meters, t in seconds). Predict the height at t = 2 seconds.

Height (m): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • MAIN • Q3/5

479

QUESTION ID
S14-M-Q3

TYPE
Short answer

CORRECT ANSWER
21.9

GRADING NOTE −19.6 + 40 + 1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-M-Q3**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • MAIN • Q3/5

A line fitted to data is y = 0.8x + 5, where x is the temperature (°C) and y is the predicted number of visitors (hundreds). For what temperature does the model predict 29 hundred visitors?

Temperature (°C): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • MAIN • Q4/5

480

QUESTION ID
S14-M-Q4

TYPE
Short answer

CORRECT ANSWER
30

GRADING NOTE 0.8x = 24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • MAIN • Q4/5

A car's value is modeled by $V = 18000(0.85)^t$, where t is years. Predict the value after 4 years, to the nearest dollar. A calculator or spreadsheet may be used.

Value ($): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • MAIN • Q5/5

481

QUESTION ID
S14-M-Q5

TYPE
Short answer

CORRECT ANSWER
9396

GRADING NOTE $0.85^4 = 0.52200625$; $18000 \cdot 0.52200625 = 9396.11$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • MAIN • Q5/5

SET 14 • HSS-ID.B.6.a

BACKWARD 1

Evaluate a function at an input • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

f(x) = 3x − 4. Find f(6).

f(6): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q1/5

483

QUESTION ID

S14-B1-Q1

TYPE

Short answer

CORRECT ANSWER

14

GRADING NOTE 18 − 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q1/5

$g(t) = 100(2)^t$. Find g(3).

g(3): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q2/5

484

QUESTION ID

S14-B1-Q2

TYPE

Short answer

CORRECT ANSWER

800

GRADING NOTE 100 · 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q2/5

True or false?

If $h(t) = -5t^2 + 10t$, then $h(1) = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q3/5

485

QUESTION ID

S14-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $-5 + 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q3/5

f(x) = 0.5x + 7. What is f(10)?

A. 7.5

B. 17

C. 57

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q4/5

486

QUESTION ID

S14-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 12

GRADING NOTE 5 + 7. 57 treats 0.5x as 50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q4/5

True or false?

If $p(t) = 50(1.1)^t$, then $p(0) = 55$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q5/5

487

QUESTION ID

S14-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $p(0) = 50$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • HSS-ID.B.6.a

BACKWARD 2

Evaluate expressions with exponents • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate 1.5^2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q1/5

489

QUESTION ID

S14-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2.25

GRADING NOTE 1.5 · 1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $4 \cdot 3^2 - 6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q2/5

490

QUESTION ID

S14-B2-Q2

TYPE

Short answer

CORRECT ANSWER

30

GRADING NOTE Exponent first: $36 - 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q2/5

True or false?

$2 \cdot 5^2 = 100$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q3/5

491

QUESTION ID

S14-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2 \cdot 25 = 50$; 100 squares 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q3/5

Evaluate $1000(0.9)^2$.

A. 1800

B. 900

C. 81

D. 810

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q4/5

492

QUESTION ID

S14-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 810

GRADING NOTE $0.81 \cdot 1000$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q4/5

True or false?

$(-3)^2 = 9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q5/5

493

QUESTION ID

S14-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(-3)(-3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q5/5

SET 14 • HSS-ID.B.6.a

BACKWARD 3

Solve a linear equation for the input • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve 0.8x + 5 = 21.

x: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q1/5

495

QUESTION ID

S14-B3-Q1

TYPE

Short answer

CORRECT ANSWER

20

GRADING NOTE 0.8x = 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q1/5

Solve 2.5x + 12 = 47.

x: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q2/5

496

QUESTION ID

S14-B3-Q2

TYPE

Short answer

CORRECT ANSWER

14

GRADING NOTE 2.5x = 35.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q2/5

True or false?

The solution of $3x - 4 = 11$ is $x = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q3/5

497

QUESTION ID

S14-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $3x = 15$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q3/5

Solve 1.2x + 6 = 30.

A. 28.8

B. 25

C. 20

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q4/5

498

QUESTION ID

S14-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 20

GRADING NOTE 1.2x = 24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q4/5

True or false?

The solution of $0.5x + 3 = 8$ is $x = 5.5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q5/5

499

QUESTION ID

S14-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $0.5x = 5$, $x = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q5/5

SET 14 • HSS-ID.B.6.a

FORWARD 1

Judge whether a prediction is reasonable • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

The line y = 2.5x + 12 was fitted to students who practiced 0 to 12 hours (y = free throws made out of 50). It predicts y = 87 for 30 hours. Which is correct?

A. Reasonable, because 30 hours is a whole number of hours.

B. Unreasonable, because a fitted line must have a negative slope.

C. Unreasonable: 30 hours is far outside the data, and 87 exceeds 50.

D. Reasonable, because the line fits the data from 0 to 12 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q1/5

501

QUESTION ID
S14-F1-Q1

TYPE
Multiple choice

CORRECT ANSWER
C. Unreasonable: 30 hours is far outside the data, and 87 exceeds 50.

GRADING NOTE Extrapolation; the value is impossible. A positive slope is not the problem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q1/5

A line fitted to data for x from 10 to 40 is y = 3x − 5. Predict y at x = 25, and state whether x = 25 is inside the data range by entering yes or no.

Predicted y: _____

Inside the range (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q2/5

502

QUESTION ID

S14-F1-Q2

TYPE

Short answer

CORRECT ANSWER

70; yes

GRADING NOTE Form required: both. 75 − 5 = 70; 25 is between 10 and 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F1-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q2/5

True or false?

Predicting far outside the range of the data (extrapolation) is usually less reliable than predicting inside it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S14-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q3/5

503

QUESTION ID

S14-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The pattern may not continue.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S14-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q3/5

The model $V = 18000(0.85)^t$ fits a car's value for $t = 0$ to 8 years. What value does it predict at $t = 0$?

V ($): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q4/5

504

QUESTION ID

S14-F1-Q4

TYPE

Short answer

CORRECT ANSWER

18000

GRADING NOTE $0.85^0 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F1-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q4/5

True or false?

The model $h = -4.9t^2 + 20t + 1.5$ gives sensible heights for every value of t, including t = 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q5/5

505

QUESTION ID
S14-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE h(10) = −288.5, below the ground: the ball has landed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q5/5

SET 14 • HSS-ID.B.6.a

FORWARD 2

Compare predictions from two fitted models • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

Two models fit a town's population (thousands), t years after 2020: linear L = 2t + 40 and exponential $E = 40(1.05)^t$. Predict each for t = 2. Round E to two decimal places.

L: _____

E: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q1/5

507

QUESTION ID

S14-F2-Q1

TYPE

Short answer

CORRECT ANSWER

44; 44.10

GRADING NOTE Form required: both. 40 · 1.1025 = 44.1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F2-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q1/5

For the models $L = 2t + 40$ and $E = 40(1.05)^t$, which statement about large t is correct?

A. E eventually predicts less, because 1.05 is close to 1.

B. The two models always predict the same population.

C. L always predicts more, because its slope is 2.

D. E eventually predicts more, because it grows by a constant percent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q2/5

508

QUESTION ID
S14-F2-Q2

TYPE
Multiple choice

CORRECT ANSWER
D. E eventually predicts more, because it grows by a constant percent.

GRADING NOTE Exponential growth eventually exceeds linear growth.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q2/5

Using L = 2t + 40 (thousands), in what year t after 2020 does L predict 60 thousand?

t: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q3/5

509

QUESTION ID
S14-F2-Q3

TYPE
Short answer

CORRECT ANSWER
10

GRADING NOTE 2t = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q3/5

True or false?

If two models fit the data equally well inside the data range, they must give the same prediction far outside it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q4/5

510

QUESTION ID
S14-F2-Q4

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE They can diverge, as L and E do.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q4/5

A residual check shows the exponential model leaves random scatter and the linear model leaves a curved pattern. Which model should be used for predictions?

A. the exponential model

B. neither, because residuals exist

C. the linear model

D. either, equally

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q5/5

511

QUESTION ID
S14-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. the exponential model

GRADING NOTE A curved residual pattern shows the linear form is wrong.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q5/5

Choose and fit a linear, quadratic or exponential function

MAIN	Probability & Statistics main question	**HSS-ID.B.6.a**
BACKWARD 1	Tell equal differences from equal factors	**HSF-LE.A.1.a**
BACKWARD 2	Write a linear function from two points	**8.F.B.4**
BACKWARD 3	Write an exponential function from two points	**HSF-LE.A.2**
FORWARD 1	Fit a model with regression technology	**HSS-ID.B.6.a**
FORWARD 2	Check a fitted model with residuals	**HSS-ID.B.6.b**

SET 15 • HSS-ID.B.6.a

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

The data fit an exponential function $y = a \cdot b^x$ exactly. Find a and b.

x	0	1	2	3	4
y	3	6	12	24	48

a: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S15-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • MAIN • Q1/5

514

QUESTION ID
S15-M-Q1

TYPE
Short answer

CORRECT ANSWER
3; 2

GRADING NOTE Form required: both. y(0) = 3; each value doubles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-M-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • MAIN • Q1/5

The data fit a linear function y = mx + b exactly. Find m and b.

x	0	1	2	3	4
y	5	8	11	14	17

m: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • MAIN • Q2/5

515

QUESTION ID
S15-M-Q2

TYPE
Short answer

CORRECT ANSWER
3; 5

GRADING NOTE Form required: both. Constant difference 3; y(0) = 5.

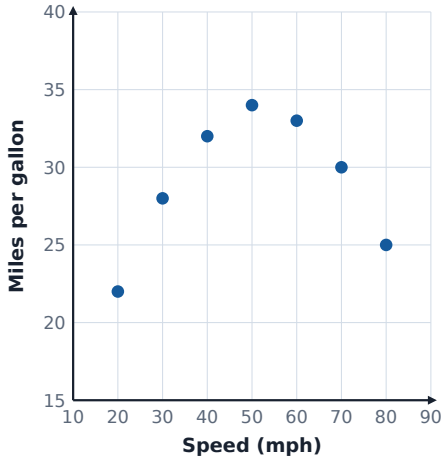
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • MAIN • Q2/5

The scatter plot shows a car's speed and fuel efficiency. Which family of functions best fits the data?

A. quadratic **B.** exponential

C. linear **D.** constant



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • MAIN • Q3/5

516

QUESTION ID

S15-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. quadratic

GRADING NOTE The values rise to a peak and fall: a parabola opening downward.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • MAIN • Q3/5

A town has 12,000 people and its population grows by about 8% each year. Which model fits, with t in years?

A. $P = 12000 + 0.08t$

B. $P = 12000(1.08)^t$

C. $P = 12000 + 1.08t$

D. $P = 12000(0.08)^t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • MAIN • Q4/5

517

QUESTION ID
S15-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. $P = 12000(1.08)^t$

GRADING NOTE Constant percent growth: factor 1.08.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • MAIN • Q4/5

An exponential function $y = a \cdot b^x$ passes through (0, 500) and (1, 400). Find a and b.

a: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • MAIN • Q5/5

518

QUESTION ID

S15-M-Q5

TYPE

Short answer

CORRECT ANSWER

500; 0.8

GRADING NOTE Form required: both. a = 500; b = 400/500.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • MAIN • Q5/5

SET 15 • HSS-ID.B.6.a

BACKWARD 1

Tell equal differences from equal factors • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a

True or false?

In y = 3x + 5, y changes by equal differences over equal intervals of x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q1/5

520

QUESTION ID

S15-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Linear.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q1/5

For x = 0, 1, 2, 3 the values of y are 2, 6, 18, 54. What is the pattern?

A. squared each step

B. add 3 each step

C. add 4 each step

D. multiplied by 3 each step

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q2/5

521

QUESTION ID

S15-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. multiplied by 3 each step

GRADING NOTE 6/2 = 18/6 = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q2/5

True or false?

The values 4, 7, 12, 19 (for x = 0 to 3) have constant first differences.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q3/5

522

QUESTION ID
S15-B1-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Differences 3, 5, 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q3/5

For x = 0 to 4, y = 4, 7, 12, 19, 28. What is the constant second difference?

Second difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q4/5

523

QUESTION ID
S15-B1-Q4

TYPE
Short answer

CORRECT ANSWER
2

GRADING NOTE First differences 3, 5, 7, 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q4/5

Which list of y-values (for x = 0, 1, 2, 3) is linear?

A. 1, 2, 4, 8

B. 0, 1, 4, 9

C. 5, 5.5, 6.5, 8

D. 10, 7, 4, 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q5/5

524

QUESTION ID

S15-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. 10, 7, 4, 1

GRADING NOTE Constant difference −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q5/5

SET 15 • HSS-ID.B.6.a

BACKWARD 2

Write a linear function from two points • Grade 8 • Functions • 8.F.B.4

A line passes through (0, 5) and (2, 11). Find its slope and y-intercept.

Slope: _____

y-intercept: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q1/5

526

QUESTION ID

S15-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3; 5

GRADING NOTE Form required: both. 6/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q1/5

Find the slope of the line through (1, 4) and (5, 12).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q2/5

527

QUESTION ID

S15-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE 8/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q2/5

True or false?

The table x = 0, 1, 2 and y = 7, 4, 1 fits y = −3x + 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q3/5

528

QUESTION ID

S15-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Check each point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q3/5

Which function fits x = 0, 1, 2 and y = 2, 4.5, 7?

A. y = 2x + 2.5

B. y = 2.5x + 2

C. y = 4.5x

D. y = 2.5x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q4/5

529

QUESTION ID

S15-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. y = 2.5x + 2

GRADING NOTE Difference 2.5, start 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q4/5

True or false?

The line through (0, 4) and (1, 10) has slope 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q5/5

530

QUESTION ID

S15-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Slope 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q5/5

SET 15 • HSS-ID.B.6.a

BACKWARD 3

Write an exponential function from two points • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

An exponential function $y = a \cdot b^x$ passes through (0, 200) and (1, 300). Find a and b.

a: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q1/5

532

QUESTION ID

S15-B3-Q1

TYPE

Short answer

CORRECT ANSWER

200; 1.5

GRADING NOTE Form required: both. 300/200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-B3-Q1**

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q1/5

An exponential function $y = a \cdot b^x$ with $b > 0$ passes through (0, 80) and (2, 20). Find a and b.

a: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q2/5

533

QUESTION ID

S15-B3-Q2

TYPE

Short answer

CORRECT ANSWER

80; 0.5

GRADING NOTE Form required: both. $b^2 = 1/4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-B3-Q2**

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q2/5

True or false?

$y = 50(1.2)^x$ passes through (1, 60).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q3/5

534

QUESTION ID

S15-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $50 \cdot 1.2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q3/5

Which exponential function passes through (0, 4) and (1, 12)?

A. $y = 3(4)^x$

B. $y = 4(3)^x$

C. $y = 4 + 8x$

D. $y = 12(4)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q4/5

535

QUESTION ID

S15-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $y = 4(3)^x$

GRADING NOTE a = 4, b = 3. 4 + 8x is linear.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-B3-Q4**

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q4/5

True or false?

$y = 10(0.5)^x$ passes through (2, 5).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q5/5

536

QUESTION ID

S15-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $10 \cdot 0.25 = 2.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q5/5

SET 15 • HSS-ID.B.6.a

FORWARD 1

Fit a model with regression technology • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

Find the least-squares line y = mx + b for the spring data. Round m and b to two decimal places. A calculator or spreadsheet may be used.

Mass (g)	2	4	5	7	8	10
Length (cm)	15	19	24	26	31	35

m: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q1/5

538

QUESTION ID
S15-F1-Q1

TYPE
Short answer

CORRECT ANSWER
2.52; 9.86

GRADING NOTE Form required: both. Linear regression.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F1-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q1/5

The points (0, 2), (1, 5), (2, 10), (3, 17) lie exactly on a quadratic $y = ax^2 + bx + c$. Find a, b and c. A calculator or spreadsheet may be used.

a: _____

b: _____

c: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q2/5

539

QUESTION ID
S15-F1-Q2

TYPE
Short answer

CORRECT ANSWER
1; 2; 2

GRADING NOTE Form required: all three. Second difference 2, so a = 1; c = 2; b = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q2/5

Data on y rise faster and faster as x increases, y is always positive, and each step multiplies y by about 1.3. Which regression should be used?

A. linear

B. exponential

C. none of these

D. quadratic

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q3/5

540

QUESTION ID
S15-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. exponential

GRADING NOTE Nearly constant factor.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q3/5

Using the fitted line y = 2.52x + 9.86, predict y when x = 6.

y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q4/5

541

QUESTION ID

S15-F1-Q4

TYPE

Short answer

CORRECT ANSWER

24.98

GRADING NOTE 15.12 + 9.86.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F1-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q4/5

True or false?

Height-versus-time data for a thrown ball are better fitted by a quadratic than by a line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q5
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q5/5

542

QUESTION ID
S15-F1-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Projectile heights follow a parabola.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q5
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q5/5

SET 15 • HSS-ID.B.6.a

FORWARD 2

Check a fitted model with residuals • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b

For the model y = 2.52x + 9.86, find the residual (observed minus predicted) for the data point (5, 24).

Residual: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q1/5

544

QUESTION ID

S15-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1.54

GRADING NOTE 24 − 22.46.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F2-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q1/5

For the model y = 2.52x + 9.86, find the residual (observed minus predicted) for the data point (4, 19).

Residual: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q2/5

545

QUESTION ID

S15-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−0.94

GRADING NOTE 19 − 19.94.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F2-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q2/5

True or false?

A positive residual means the data point lies above the fitted line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q3/5

546

QUESTION ID
S15-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Observed > predicted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q3/5

A line is fitted to data that actually grow exponentially. What pattern do the residuals usually show?

A. every residual positive

B. a curved (U-shaped) pattern

C. random scatter around 0

D. all residuals equal to 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q4/5

547

QUESTION ID

S15-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. a curved (U-shaped) pattern

GRADING NOTE The line misses the curvature systematically.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q4/5

True or false?

If the residuals of a linear model are small but form a clear curve, a linear model is still the best family.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q5/5

548

QUESTION ID
S15-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A pattern means the family is wrong.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q5/5

Assess the fit of a function with residuals

MAIN	Probability & Statistics main question	**HSS-ID.B.6.b**
BACKWARD 1	Subtract with signed decimals	**7.NS.A.1.c**
BACKWARD 2	Find a predicted value from a model	**HSF-IF.A.2**
BACKWARD 3	Compare an observed value with a prediction	**HSS-ID.B.6.a**
FORWARD 1	Compute least-squares residuals	**HSS-ID.B.6.b**
FORWARD 2	Compare two models with residuals	**HSS-ID.B.6.b**

SET 16 • HSS-ID.B.6.b

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b

The model y = 2x + 1 is used for the data in the table. Find the residuals (observed minus predicted) at x = 1 and at x = 2.

x	1	2	3	4	5
y	3.5	4.6	7.2	8.8	11.1

Residual at x = 1: _____

Residual at x = 2: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • MAIN • Q1/5

551

QUESTION ID

S16-M-Q1

TYPE

Short answer

CORRECT ANSWER

0.5; −0.4

GRADING NOTE Form required: both. 3.5 − 3 and 4.6 − 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • MAIN • Q1/5

A model predicts y = 3x + 10. Find the residuals (observed minus predicted) for the observed points (4, 20) and (6, 30).

Residual for (4, 20): _____

Residual for (6, 30): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • MAIN • Q2/5

552

QUESTION ID

S16-M-Q2

TYPE

Short answer

CORRECT ANSWER

−2; 2

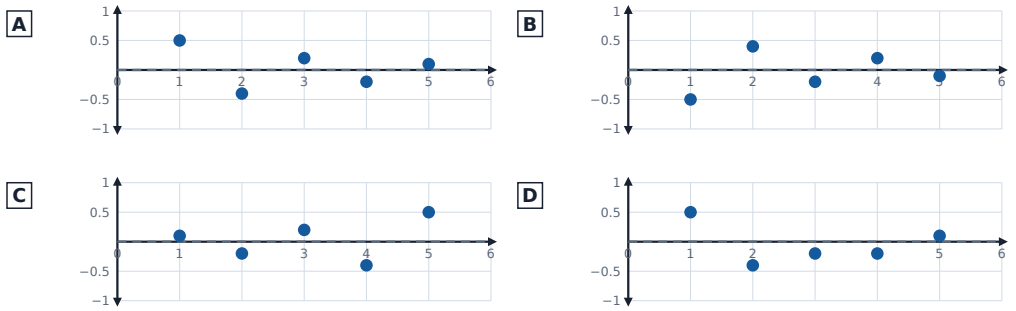
GRADING NOTE Form required: both. 20 − 22 and 30 − 28.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-M-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The model y = 2x + 1 is used for the data in the table (x = 1 to 5; y = 3.5, 4.6, 7.2, 8.8, 11.1). Which plot is the residual plot (observed minus predicted)?



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • MAIN • Q3/5

553

QUESTION ID
S16-M-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. Graph A

GRADING NOTE Residuals 0.5, −0.4, 0.2, −0.2, 0.1. The others use predicted minus observed, reverse the order, or flip the sign at x = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • MAIN • Q3/5

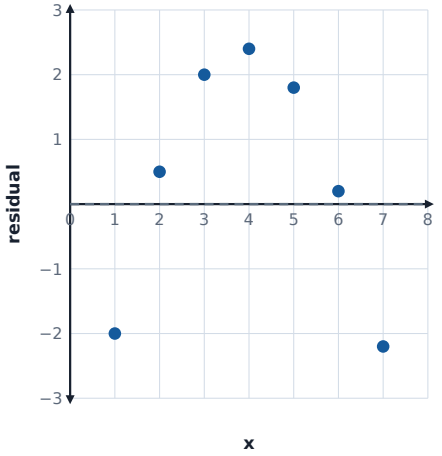
A line was fitted to a data set. The residual plot is shown. Which conclusion is correct?

A. The linear model is appropriate, because some residuals are near 0.

B. The linear model is appropriate, because residuals are both signs.

C. The data must contain an error, because residuals are not all 0.

D. A linear model is not appropriate; the curve suggests a curved model.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • MAIN • Q4/5

554

QUESTION ID
S16-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. A linear model is not appropriate; the curve suggests a curved model.

GRADING NOTE A systematic curve in the residuals shows the line misses the form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • MAIN • Q4/5

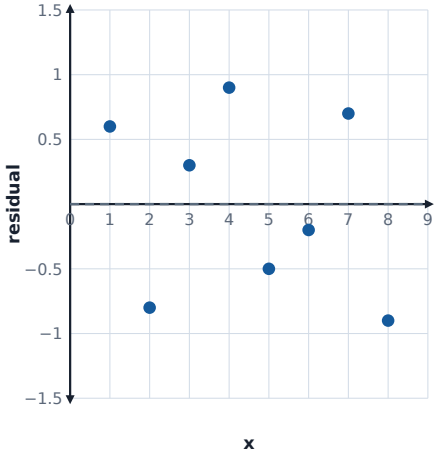
A line was fitted to a different data set, and its residual plot is shown. What does the plot show about the linear model?

A. The linear model is not appropriate, because residuals are not 0.

B. The linear model is appropriate: the residuals show no pattern.

C. A quadratic model is needed, because residuals change sign.

D. The model is perfect, because the residuals add to about 0.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • MAIN • Q5/5

555

QUESTION ID
S16-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
B. The linear model is appropriate: the residuals show no pattern.

GRADING NOTE Random scatter around 0 supports the linear form; residuals are never all 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • MAIN • Q5/5

SET 16 • HSS-ID.B.6.b

BACKWARD 1

Subtract with signed decimals • Grade 7 • The Number System • 7.NS.A.1.c

Compute 3.5 − 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q1/5

557

QUESTION ID

S16-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.5

GRADING NOTE Difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 4.6 − 5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q2/5

558

QUESTION ID

S16-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−0.4

GRADING NOTE Below zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q2/5

True or false?

$20 - 22 = -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q3/5

559

QUESTION ID

S16-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Signed result.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q3/5

Compute 7.2 − 7.5.

A. −0.3

B. −14.7

C. 14.7

D. 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q4/5

560

QUESTION ID

S16-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. −0.3

GRADING NOTE Smaller minus larger is negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q4/5

True or false?

$11 - 11.4 = 0.4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q5/5

561

QUESTION ID

S16-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE -0.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q5/5

SET 16 • HSS-ID.B.6.b

BACKWARD 2

Find a predicted value from a model • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

For y = 2x + 1, find the predicted y at x = 3.

y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q1/5

563

QUESTION ID

S16-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE 6 + 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q1/5

For y = 3x + 10, find the predicted y at x = 4.

y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q2/5

564

QUESTION ID

S16-B2-Q2

TYPE

Short answer

CORRECT ANSWER

22

GRADING NOTE 12 + 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B2-Q2**

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q2/5

True or false?

For y = −1.5x + 20, the predicted y at x = 4 is 14.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q3/5

565

QUESTION ID

S16-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE −6 + 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q3/5

For y = 0.5x + 3, what is the predicted y at x = 8?

A. 8.5

B. 4

C. 7

D. 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q4/5

566

QUESTION ID

S16-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 7

GRADING NOTE 4 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q4/5

True or false?

For y = 4x − 2, the predicted y at x = 0 is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q5/5

567

QUESTION ID

S16-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • HSS-ID.B.6.b

BACKWARD 3

Compare an observed value with a prediction • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

A model predicts y = 2.5x + 12 free throws for x hours of practice. A student practiced 8 hours and made 35. What is the predicted number, and how many more did the student make than predicted?

Predicted: _____

More by: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q1/5

569

QUESTION ID

S16-B3-Q1

TYPE

Short answer

CORRECT ANSWER

32; 3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B3-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q1/5

True or false?

If an observed value is less than the predicted value, the point lies below the fitted line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q2/5

570

QUESTION ID

S16-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Observed < predicted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q2/5

The observed value is 50 and the predicted value is 46. What is observed minus predicted?

A. 4

B. 96

C. −4

D. 46

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q3/5

571

QUESTION ID

S16-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4

GRADING NOTE 50 − 46.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B3-Q3**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q3/5

True or false?

In this collection, a residual is defined as predicted minus observed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q4/5

572

QUESTION ID

S16-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Residual = observed − predicted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q4/5

The observed value is 18 and the predicted value is 20.5. What is observed minus predicted?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q5/5

573

QUESTION ID

S16-B3-Q5

TYPE

Short answer

CORRECT ANSWER

−2.5

GRADING NOTE 18 − 20.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B3-Q5**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q5/5

SET 16 • HSS-ID.B.6.b

FORWARD 1

Compute least-squares residuals • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b

The least-squares line for (1, 2), (2, 5), (3, 4), (4, 7) is y = 1.4x + 1. Find the residuals (observed minus predicted) at x = 1 and x = 2.

At x = 1: _____

At x = 2: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q1/5

575

QUESTION ID

S16-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−0.4; 1.2

GRADING NOTE Form required: both. 2 − 2.4 and 5 − 3.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q1/5

For y = 1.4x + 1 and the points (1, 2), (2, 5), (3, 4), (4, 7), what is the sum of all four residuals?

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q2/5

576

QUESTION ID

S16-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0

GRADING NOTE −0.4 + 1.2 − 1.2 + 0.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q2/5

True or false?

For a least-squares line, the residuals always add to 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q3/5

577

QUESTION ID
S16-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE A property of least squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q3**
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q3/5

For y = 1.4x + 1 and the points (1, 2), (2, 5), (3, 4), (4, 7), which points have the largest residuals in absolute value?

A. (1, 2) only

B. (1, 2) and (4, 7)

C. (4, 7) only

D. (2, 5) and (3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q4/5

578

QUESTION ID

S16-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (2, 5) and (3, 4)

GRADING NOTE |1.2| > |0.4|.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q4/5

True or false?

A residual of −1.2 means the point lies 1.2 units above the line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q5/5

579

QUESTION ID
S16-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Below the line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q5/5

SET 16 • HSS-ID.B.6.b

FORWARD 2

Compare two models with residuals • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b

Two models were fitted to the same data. Linear residuals: 3, −1, −3, −2, 1, 4. Exponential residuals: 0.2, −0.3, 0.1, 0.3, −0.2, −0.1. Which model fits better?

A. the exponential model

B. the linear model

C. neither model can be judged

D. they fit equally

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q1/5

581

QUESTION ID
S16-F2-Q1

TYPE
Multiple choice

CORRECT ANSWER
A. the exponential model

GRADING NOTE The exponential residuals are small and patternless; the linear ones form a U.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the sum of squared residuals for each model: linear 3, −1, −3, −2, 1, 4 and exponential 0.2, −0.3, 0.1, 0.3, −0.2, −0.1.

Linear: _____

Exponential: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q2/5

582

QUESTION ID

S16-F2-Q2

TYPE

Short answer

CORRECT ANSWER

40; 0.28

GRADING NOTE Form required: both. 9+1+9+4+1+16; 0.04+0.09+0.01+0.09+0.04+0.01.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F2-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q2/5

True or false?

Of two models fitted to the same data, the one with the smaller sum of squared residuals fits those data more closely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q3/5

583

QUESTION ID
S16-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Smaller total squared error.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q3/5

A residual plot fans out: the residuals get larger as x increases. What does this show?

A. the slope is negative

B. prediction errors grow as x increases

C. the model is exact

D. the data have no outliers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q4/5

584

QUESTION ID
S16-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. prediction errors grow as x increases

GRADING NOTE Spread of errors depends on x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q4/5

True or false?

A residual plot with a clear U shape shows that the model is a good fit.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q5/5

585

QUESTION ID
S16-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A pattern shows a poor fit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F2-Q5**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q5/5

Fit a linear function to a linear scatter plot

MAIN	Probability & Statistics main question	**HSS-ID.B.6.c**
BACKWARD 1	Fit a line by eye	**8.SP.A.2**
BACKWARD 2	Find slope and intercept from points	**8.F.B.4**
FORWARD 1	Use the properties of the least-squares line	**HSS-ID.B.6.c**
FORWARD 2	Interpret the fitted line's slope and intercept	**HSS-ID.C.7**

SET 17 • HSS-ID.B.6.c

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c

ANSWER KEY Reduced copy of the question page, with its answer

Find the least-squares line y = mx + b for hours studied (x) and score (y). Round m and b to two decimal places. A calculator or spreadsheet may be used.

Hours	1	2	2	3	4	5	5	6	7	8
Score	52	58	61	63	70	72	76	80	83	90

m: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • MAIN • Q1/5

588

QUESTION ID
S17-M-Q1

TYPE
Short answer

CORRECT ANSWER
5.17; 48.28

GRADING NOTE Form required: both. m = 2485/481 ≈ 5.166; b = 23225/481 ≈ 48.285.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • MAIN • Q1/5

Find the least-squares line y = mx + b for car age (x) and value in $1000s (y). Round m and b to two decimal places. A calculator or spreadsheet may be used.

Age (years)	1	2	3	4	5	6	7	8	9	10
Value ($1000s)	24	21	22	17	18	13	14	9	11	6

m: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • MAIN • Q2/5

589

QUESTION ID

S17-M-Q2

TYPE

Short answer

CORRECT ANSWER

−1.88; 25.87

GRADING NOTE Form required: both. m = −311/165 ≈ −1.885; b = 388/15 ≈ 25.867.

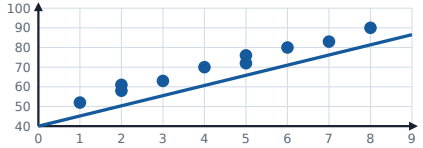
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-M-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • MAIN • Q2/5

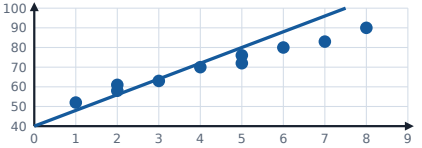
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows the line that best fits the study data (hours, score)?

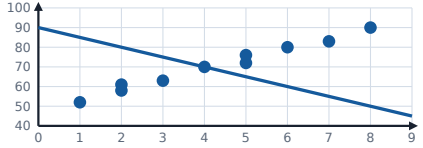
A



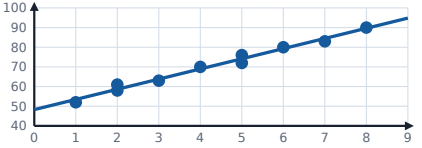
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • MAIN • Q3/5

590

QUESTION ID

S17-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE The least-squares line y = 5.17x + 48.28 runs through the middle. The others are too low, too steep, or falling.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • MAIN • Q3/5

Using the least-squares line y = 5.17x + 48.28 for hours studied and score, predict the score for 3 hours.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • MAIN • Q4/5

591

QUESTION ID
S17-M-Q4

TYPE
Short answer

CORRECT ANSWER
63.79

GRADING NOTE 15.51 + 48.28.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • MAIN • Q4/5

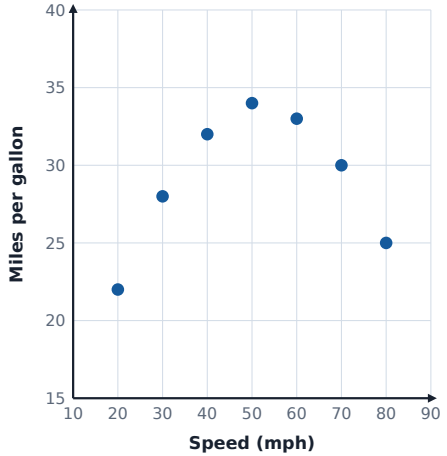
A student wants to fit a least-squares line to the data in this scatter plot. Which statement is correct?

A. A line is appropriate, because r can always be computed.

B. A line is appropriate, because the data have no outliers.

C. A line is not appropriate: the pattern is curved.

D. A line fits well, because the points rise at first.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • MAIN • Q5/5

592

QUESTION ID
S17-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
C. A line is not appropriate: the pattern is curved.

GRADING NOTE The standard applies to scatter plots that suggest a linear association.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • MAIN • Q5/5

SET 17 • HSS-ID.B.6.c

BACKWARD 1

Fit a line by eye • Grade 8 • Statistics & Probability • 8.SP.A.2

True or false?

A least-squares line must have exactly as many data points above it as below it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q1/5

QUESTION ID

S17-B1-Q1

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Least squares (with an intercept) makes the residuals add to 0; the numbers of points above and below can differ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q1/5

Which line passes through (0, 2), (2, 6) and (4, 10)?

A. y = x + 2

B. y = 2x

C. y = 2x + 2

D. y = 4x + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q2/5

595

QUESTION ID

S17-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. y = 2x + 2

GRADING NOTE Check each point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q2/5

A line drawn by eye passes through (0, 50) and (5, 75). Find its slope and y-intercept.

Slope: _____

y-intercept: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q3/5

596

QUESTION ID

S17-B1-Q3

TYPE

Short answer

CORRECT ANSWER

5; 50

GRADING NOTE Form required: both. 25/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q3/5

True or false?

A fitted line must pass through every data point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q4/5

597

QUESTION ID

S17-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It summarizes the trend.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q4/5

Use the line y = 5x + 50 to predict y at x = 4.

y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q5/5

598

QUESTION ID

S17-B1-Q5

TYPE

Short answer

CORRECT ANSWER

70

GRADING NOTE 20 + 50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • HSS-ID.B.6.c

BACKWARD 2

Find slope and intercept from points • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope of the line through (2, 58) and (6, 80).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q1/5

600

QUESTION ID

S17-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5.5

GRADING NOTE 22/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q1/5

A line has slope 5 and passes through (2, 60). Find its y-intercept.

y-intercept: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q2/5

601

QUESTION ID

S17-B2-Q2

TYPE

Short answer

CORRECT ANSWER

50

GRADING NOTE 60 − 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q2/5

True or false?

The line through (0, 26) with slope −2 has y = 16 at x = 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q3/5

602

QUESTION ID

S17-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 26 − 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q3/5

Which equation is the line through (0, 10) and (4, 2)?

A. y = 2x + 10

B. y = −2x + 2

C. y = −0.5x + 10

D. y = −2x + 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q4/5

603

QUESTION ID

S17-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. y = −2x + 10

GRADING NOTE Slope −8/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q4/5

True or false?

The slope of the line through (1, 24) and (10, 6) is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q5/5

604

QUESTION ID

S17-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-18/9 = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q5/5

SET 17 • HSS-ID.B.6.c

FORWARD 1

Use the properties of the least-squares line • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c

Find the mean hours $\bar{x}$ and the mean score $\bar{y}$ for the study data.

Hours	1	2	2	3	4	5	5	6	7	8
Score	52	58	61	63	70	72	76	80	83	90

$\bar{x}$: _____

$\bar{y}$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q1/5

606

QUESTION ID
S17-F1-Q1

TYPE
Short answer

CORRECT ANSWER
4.3; 70.5

GRADING NOTE Form required: both. 43/10 and 705/10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q1/5

True or false?

The least-squares line passes through the point $(\bar{x}, \bar{y})$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q2/5

607

QUESTION ID

S17-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A property of least squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F1-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q2/5

The least-squares line is the line that makes which quantity as small as possible?

A. the number of points off the line

B. the largest single residual

C. the sum of the residuals

D. the sum of the squared residuals

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q3/5

608

QUESTION ID
S17-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. the sum of the squared residuals

GRADING NOTE Definition. The plain sum is 0 for many lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q3/5

A least-squares line has slope 2 and passes through $(\bar{x}, \bar{y}) = (5, 20)$. Find its y-intercept.

y-intercept: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q4/5

609

QUESTION ID
S17-F1-Q4

TYPE
Short answer

CORRECT ANSWER
10

GRADING NOTE 20 − 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F1-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q4/5

True or false?

Adding one point far from the others can never change the least-squares line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q5/5

610

QUESTION ID

S17-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE An influential point can move it a lot.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q5/5

SET 17 • HSS-ID.B.6.c

FORWARD 2

Interpret the fitted line's slope and intercept • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

For hours studied (x) and score (y), the least-squares line is y = 5.17x + 48.28. What does 5.17 mean?

A. A student who studied 0 hours scored exactly 5.17.

B. Each extra hour goes with about 5.17 more points, on average.

C. The score rises 48.28 points for each hour.

D. Each student scored 5.17 points per test.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 17 • FORWARD 2 • Q1/5

612

QUESTION ID
S17-F2-Q1

TYPE
Multiple choice

CORRECT ANSWER
B. Each extra hour goes with about 5.17 more points, on average.

GRADING NOTE Slope: predicted change in y per unit of x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 17 • FORWARD 2 • Q1/5

Using y = 5.17x + 48.28, how much higher is the predicted score for a student who studies 2 more hours?

Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 17 • FORWARD 2 • Q2/5

613

QUESTION ID
S17-F2-Q2

TYPE
Short answer

CORRECT ANSWER
10.34

GRADING NOTE 2 · 5.17.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 17 • FORWARD 2 • Q2/5

For car age (x, years) and value (y, $1000s), $y = -1.88x + 25.87$. What does 25.87 mean?

A. The oldest car is worth about $25,870.

B. A car is predicted to be worth $0 after 25.87 years.

C. Cars lose about $25,870 each year.

D. A new car (age 0) has a predicted value of about $25,870.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 17 • FORWARD 2 • Q3/5

614

QUESTION ID
S17-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. A new car (age 0) has a predicted value of about $25,870.

GRADING NOTE Intercept: predicted y at x = 0 (age 0 is near the data).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q3**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 17 • FORWARD 2 • Q3/5

True or false?

In $y = -1.88x + 25.87$, the slope means every car loses exactly $1,880 each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S17-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 17 • FORWARD 2 • Q4/5

615

QUESTION ID

S17-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is an average rate for the model, not exact for each car.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S17-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 17 • FORWARD 2 • Q4/5

Using $y = -1.88x + 25.87$, how much greater (in \$1000s) is the predicted value of a 3-year-old car than a 7-year-old car?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 17 • FORWARD 2 • Q5/5

616

QUESTION ID
S17-F2-Q5

TYPE
Short answer

CORRECT ANSWER
7.52

GRADING NOTE $4 \cdot 1.88$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q5**
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 17 • FORWARD 2 • Q5/5

Interpret the slope and intercept of a linear model

MAIN	Probability & Statistics main question	**HSS-ID.C.7**
BACKWARD 1	Interpret the rate and the initial value of a linear function	**8.F.B.4**
BACKWARD 2	Use a linear model's equation to answer questions	**8.SP.A.3**
FORWARD 1	Interpret parameters with units and judge meaning	**HSS-ID.C.7**
FORWARD 2	Relate the slope to the correlation coefficient	**HSS-ID.C.8**

SET 18 • HSS-ID.C.7

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

A model for a family's savings is y = 85x + 1200, where x is months and y is dollars. What does 85 mean?

A. The family started with $85 in savings.

B. The family saves $1200 each month.

C. The savings will reach $85 after 1200 months.

D. The savings are predicted to grow by $85 each month.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • MAIN • Q1/5

619

QUESTION ID
S18-M-Q1

TYPE
Multiple choice

CORRECT ANSWER
D. The savings are predicted to grow by $85 each month.

GRADING NOTE Slope: dollars per month.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • MAIN • Q1/5

A model for weekly hours of exercise is $y = 6 - 0.2x$, where x is weekly hours of TV. What does 6 mean?

A. A person who watches 0 hours of TV is predicted to exercise 6 hours.

B. Each hour of TV is predicted to add 6 hours of exercise.

C. Each hour of TV is predicted to cut exercise by 6 hours.

D. A person exercises at most 0.2 hours per week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • MAIN • Q2/5

620

QUESTION ID
S18-M-Q2

TYPE
Multiple choice

CORRECT ANSWER
A. A person who watches 0 hours of TV is predicted to exercise 6 hours.

GRADING NOTE Intercept: predicted y at x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A tree's height is modeled by y = 0.4x + 1.5, with y in meters and x in years since planting. What is the predicted growth per year, and what is the predicted height at planting?

Growth per year (m): _____

Height at planting (m): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • MAIN • Q3/5

621

QUESTION ID
S18-M-Q3

TYPE
Short answer

CORRECT ANSWER
0.4; 1.5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • MAIN • Q3/5

For adults with heights from 150 to 190 cm, arm span is modeled by $y = 1.1x - 15$ (both in cm). Which statement about the intercept -15 is correct?

A. It means a person 0 cm tall has an arm span of -15 cm.

B. It means arm spans are 15 cm less than heights.

C. It is not meaningful: a height of 0 cm is far outside the data.

D. It means the slope of the model is negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • MAIN • Q4/5

622

QUESTION ID
S18-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. It is not meaningful: a height of 0 cm is far outside the data.

GRADING NOTE Extrapolating to x = 0 makes no sense here.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-M-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • MAIN • Q4/5

Weekly gasoline use is modeled by y = 0.35x + 4.2 gallons, where x is miles driven per day. By how many gallons do the predictions differ for two drivers whose daily miles differ by 20?

Gallons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • MAIN • Q5/5

623

QUESTION ID
S18-M-Q5

TYPE
Short answer

CORRECT ANSWER
7

GRADING NOTE 20 · 0.35.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • MAIN • Q5/5

SET 18 • HSS-ID.C.7

BACKWARD 1

Interpret the rate and the initial value of a linear function • Grade 8 • Functions • 8.F.B.4

A plumber charges y = 50x + 80 dollars for x hours. What does 80 mean?

A. the charge per hour

B. a fixed fee charged before any hours

C. the number of hours

D. the total for 1 hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q1/5

625

QUESTION ID

S18-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. a fixed fee charged before any hours

GRADING NOTE Initial value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q1/5

A phone plan costs y = 0.1x + 25 dollars for x minutes. What is the cost per minute, and what is the fixed cost?

Per minute ($): _____

Fixed ($): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q2/5

626

QUESTION ID

S18-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.1; 25

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q2/5

True or false?

In y = 3x + 12, when x increases by 1, y increases by 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q3/5

627

QUESTION ID

S18-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE By 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q3/5

A candle's height is h = 20 − 1.5t centimeters after t hours. What does 1.5 mean?

A. the candle burns 1.5 cm per hour

B. the candle starts 1.5 cm tall

C. the candle burns 20 cm per hour

D. the candle lasts 1.5 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q4/5

628

QUESTION ID

S18-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. the candle burns 1.5 cm per hour

GRADING NOTE Rate of change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q4/5

True or false?

In y = 3x + 12, the value of y when x = 0 is 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q5/5

629

QUESTION ID

S18-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Initial value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • HSS-ID.C.7

BACKWARD 2

Use a linear model's equation to answer questions • Grade 8 • Statistics & Probability • 8.SP.A.3

Plant height is modeled by y = 1.5x + 5 (cm after x weeks). Predict the height at 6 weeks.

Height (cm): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q1/5

631

QUESTION ID

S18-B2-Q1

TYPE

Short answer

CORRECT ANSWER

14

GRADING NOTE 9 + 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q1/5

In the plant model y = 1.5x + 5, one more week of growth goes with

A. 1.5 more weeks

B. 5 cm more height

C. 1.5 cm more height

D. 6.5 cm more height

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q2/5

632

QUESTION ID

S18-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 1.5 cm more height

GRADING NOTE Slope.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q2/5

True or false?

In the plant model y = 1.5x + 5, the slope means the plants were 1.5 cm tall at the start.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q3/5

633

QUESTION ID

S18-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That is the intercept, 5 cm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q3/5

In the plant model y = 1.5x + 5, how much greater is the predicted height at week 10 than at week 2?

Difference (cm): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q4/5

634

QUESTION ID

S18-B2-Q4

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE 8 · 1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q4/5

True or false?

A negative slope means the predicted y decreases as x increases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q5/5

635

QUESTION ID

S18-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Direction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • HSS-ID.C.7

FORWARD 1

Interpret parameters with units and judge meaning • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

Using y = 5.17x + 48.28 (score versus hours studied), what is the predicted score increase for 3 more hours?

Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • FORWARD 1 • Q1/5

637

QUESTION ID
S18-F1-Q1

TYPE
Short answer

CORRECT ANSWER
15.51

GRADING NOTE 3 · 5.17.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-F1-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • FORWARD 1 • Q1/5

A shipping cost is modeled by $y = 0.02x + 3.5$ dollars for x grams. What does 0.02 mean?

A. Each package costs $0.02 no matter what it weighs.

B. Each additional gram adds about $0.02 to the predicted cost.

C. The predicted cost rises by $3.50 for each gram.

D. A package weighing 0 grams is predicted to cost $0.02.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • FORWARD 1 • Q2/5

638

QUESTION ID
S18-F1-Q2

TYPE
Multiple choice

CORRECT ANSWER
B. Each additional gram adds about $0.02 to the predicted cost.

GRADING NOTE Dollars per gram.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • FORWARD 1 • Q2/5

True or false?

The intercept of a linear model always has a meaningful interpretation in context.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • FORWARD 1 • Q3/5

639

QUESTION ID
S18-F1-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Not when x = 0 is outside the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • FORWARD 1 • Q3/5

For ages 20 to 60, maximum heart rate is modeled by $y = 208 - 0.7x$ beats per minute, x in years. Which statement about 208 is correct?

A. It is the predicted heart rate for a 20-year-old person.

B. It is the prediction at age 0, outside the data: not meaningful.

C. It is the decrease in maximum heart rate for each year of age.

D. It is the largest heart rate that any person can have.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • FORWARD 1 • Q4/5

640

QUESTION ID
S18-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. It is the prediction at age 0, outside the data: not meaningful.

GRADING NOTE Age 0 is outside 20 to 60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • FORWARD 1 • Q4/5

Using y = 208 − 0.7x, predict the maximum heart rate at age 40.

Beats per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • FORWARD 1 • Q5/5

641

QUESTION ID

S18-F1-Q5

TYPE

Short answer

CORRECT ANSWER

180

GRADING NOTE 208 − 28.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-F1-Q5**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 18 • FORWARD 1 • Q5/5

SET 18 • HSS-ID.C.7

FORWARD 2

Relate the slope to the correlation coefficient • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

True or false?

The sign of the correlation coefficient matches the sign of the slope of the least-squares line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q1/5

643

QUESTION ID
S18-F2-Q1

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Both measure direction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q1/5

The least-squares line is $y = -1.88x + 25.87$. Which could be the correlation coefficient?

A. 0.97

B. 1.88

C. −1.88

D. −0.97

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q2/5

644

QUESTION ID

S18-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. −0.97

GRADING NOTE Negative, and $|r| \le 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-F2-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q2/5

True or false?

A larger slope always means a stronger correlation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q3/5

645

QUESTION ID
S18-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Slope depends on units; r does not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q3/5

The least-squares slope equals r·(s_y/s_x). If r = 0.8, s_y = 10 and s_x = 4, what is the slope?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q4/5

646

QUESTION ID
S18-F2-Q4

TYPE
Short answer

CORRECT ANSWER
2

GRADING NOTE 0.8 · 2.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-F2-Q4**
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q4/5

The x-values vary and the y-values vary (neither is constant). If the least-squares line has slope 0, what is r?

A. 1

B. −1

C. it cannot be found

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q5/5

647

QUESTION ID
S18-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
D. 0

GRADING NOTE b = r·s_y/s_x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q5/5

Compute the correlation coefficient with technology

MAIN	Probability & Statistics main question	**HSS-ID.C.8**
BACKWARD 1	Find means and deviations	**6.SP.B.5.c**
BACKWARD 2	Read the direction of an association	**HSS-ID.B.6**
FORWARD 1	Compute r from deviations	**HSS-ID.C.8**
FORWARD 2	See how changes to the data affect r	**HSS-ID.C.8**

SET 19 • HSS-ID.C.8

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

Find the correlation coefficient r for hours studied and score. Round to two decimal places. A calculator or spreadsheet may be used.

Hours	1	2	2	3	4	5	5	6	7	8
Score	52	58	61	63	70	72	76	80	83	90

r: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • MAIN • Q1/5

650

QUESTION ID
S19-M-Q1

TYPE
Short answer

CORRECT ANSWER
0.99

GRADING NOTE r = 0.9920.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • MAIN • Q1/5

Find the correlation coefficient r for car age and value. Round to two decimal places. A calculator or spreadsheet may be used.

Age (years)	1	2	3	4	5	6	7	8	9	10
Value ($1000s)	24	21	22	17	18	13	14	9	11	6

r: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • MAIN • Q2/5

651

QUESTION ID
S19-M-Q2

TYPE
Short answer

CORRECT ANSWER
−0.97

GRADING NOTE r = −0.9654.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • MAIN • Q2/5

Find the correlation coefficient r for hours after sunrise (x) and ice thickness in cm (y). Round to two decimal places. A calculator or spreadsheet may be used.

Hours	1	2	3	4	5	6
Thickness (cm)	48	44	41	35	33	28

r: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • MAIN • Q3/5

652

QUESTION ID

S19-M-Q3

TYPE

Short answer

CORRECT ANSWER

−0.99

GRADING NOTE r = −0.9949.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • MAIN • Q3/5

Find r for the temperature and cones-sold data for all 7 days, and again without the 90-degree day. Round each to three decimal places. A calculator or spreadsheet may be used.

Temperature (°F)	60	65	70	75	80	85	90
Cones sold	120	140	155	175	190	210	60

All 7 days: _____

Without the 90-degree day: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • MAIN • Q4/5

653

QUESTION ID
S19-M-Q4

TYPE
Short answer

CORRECT ANSWER
−0.008; 0.999

GRADING NOTE Form required: both. −0.0077 and 0.9992.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-M-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • MAIN • Q4/5

Find the correlation coefficient r for the spring data (mass and length). Round to two decimal places. A calculator or spreadsheet may be used.

Mass (g)	2	4	5	7	8	10
Length (cm)	15	19	24	26	31	35

r: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • MAIN • Q5/5

654

QUESTION ID
S19-M-Q5

TYPE
Short answer

CORRECT ANSWER
0.99

GRADING NOTE r = 0.9881.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • MAIN • Q5/5

SET 19 • HSS-ID.C.8

BACKWARD 1

Find means and deviations • Grade 6 • Statistics & Probability • 6.SP.B.5.c

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 1, 2, 3, 4, 5.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q1/5

656

QUESTION ID

S19-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE 15/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For the values 2, 5, 4, 7 (mean 4.5), find the deviations of 2 and 7 from the mean.

2: _____

7: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q2/5

657

QUESTION ID

S19-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−2.5; 2.5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q2/5

True or false?

The deviations of a data set from its mean always add to 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q3/5

658

QUESTION ID

S19-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Balance property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the mean of 48, 44, 41, 35, 33, 28, rounded to two decimal places?

A. 35

B. 38.17

C. 41

D. 38

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q4/5

659

QUESTION ID

S19-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 38.17

GRADING NOTE 229/6 = 38.1666... ≈ 38.17.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q4/5

True or false?

A value below the mean has a positive deviation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q5/5

660

QUESTION ID

S19-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q5/5

SET 19 • HSS-ID.C.8

BACKWARD 2

Read the direction of an association • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6

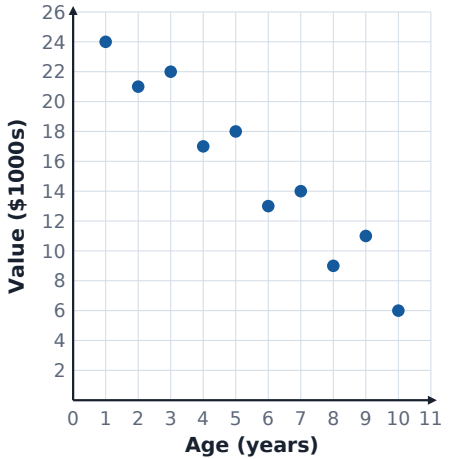
The scatter plot shows car age and value. What is the sign of r?

A. positive

B. cannot be found

C. zero

D. negative



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q1/5

662

QUESTION ID

S19-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. negative

GRADING NOTE Values fall as age rises.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-B2-Q1**

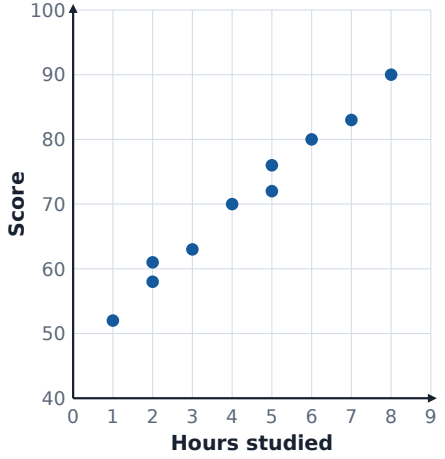
High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q1/5

True or false?

For the hours-studied scatter plot, r is positive.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q2/5

QUESTION ID

S19-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rising pattern.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q2/5

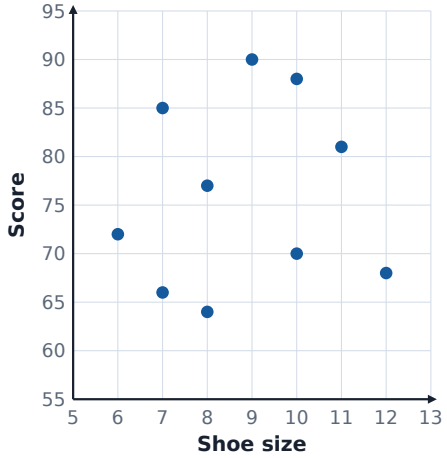
For the shoe-size scatter plot, r is closest to

A. 1

B. 0

C. −1

D. 0.9



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q3/5

664

QUESTION ID

S19-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0

GRADING NOTE No linear pattern.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q3/5

True or false?

For a cloud of points with no pattern, r is close to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q4/5

665

QUESTION ID

S19-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Close to 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q4/5

Which scatter plot would have r close to −1?

A. points along a U-shaped curve

B. a shapeless cloud

C. points falling closely along a line

D. points rising closely along a line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q5/5

666

QUESTION ID

S19-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. points falling closely along a line

GRADING NOTE Strong negative linear.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • HSS-ID.C.8

FORWARD 1

Compute r from deviations • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

ANSWER KEY Reduced copy of the question page, with its answer

For (1, 2), (2, 5), (3, 4), (4, 7): $\Sigma(x - \bar{x})(y - \bar{y}) = 7$, $\Sigma(x - \bar{x})^2 = 5$ and $\Sigma(y - \bar{y})^2 = 13$. Compute $r = 7/\sqrt{(5 \cdot 13)}$, rounded to two decimal places.

r: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 1 • Q1/5

668

QUESTION ID

S19-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.87

GRADING NOTE $7/\sqrt{65} = 0.868$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 1 • Q1/5

True or false?

If all points lie exactly on a line with positive slope, r = 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 1 • Q2/5

669

QUESTION ID
S19-F1-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Perfect positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 1 • Q2/5

For the points (1, 2), (2, 4), (3, 6), what is r?

r: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 1 • Q3/5

670

QUESTION ID
S19-F1-Q3

TYPE
Short answer

CORRECT ANSWER
1

GRADING NOTE Exactly on y = 2x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q3**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 1 • Q3/5

A calculator reports r = 1.4. What should you conclude?

A. The association is very strong and positive.

B. The data have one outlier.

C. A mistake was made: r is always between −1 and 1.

D. The slope of the line is 1.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 1 • Q4/5

671

QUESTION ID

S19-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. A mistake was made: r is always between −1 and 1.

GRADING NOTE $|r| \le 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 1 • Q4/5

True or false?

For the points (1, 6), (2, 4), (3, 2), r = 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 1 • Q5/5

672

QUESTION ID
S19-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE r = −1 (falling line).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 1 • Q5/5

SET 19 • HSS-ID.C.8

FORWARD 2

See how changes to the data affect r • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

True or false?

Changing x from hours to minutes does not change r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 2 • Q1/5

674

QUESTION ID
S19-F2-Q1

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE r has no units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 2 • Q1/5

True or false?

Swapping the roles of x and y changes the value of r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 2 • Q2/5

675

QUESTION ID
S19-F2-Q2

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE r is symmetric.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 2 • Q2/5

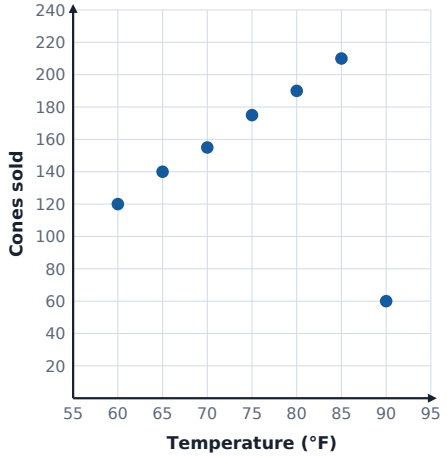
For the cones data, r = 0.999 for six days and −0.008 with the 90-degree day included. What does this show?

A. Temperature causes cone sales to fall.

B. r is not affected by outliers.

C. The six days have no association.

D. One outlier can change r dramatically.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 2 • Q3/5

676

QUESTION ID

S19-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. One outlier can change r dramatically.

GRADING NOTE The single point erases the pattern.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Find r for (1, 3), (2, 5), (3, 7), and for the same points with every y increased by 10. Give both.

Original: _____

After adding 10: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 2 • Q4/5

677

QUESTION ID
S19-F2-Q4

TYPE
Short answer

CORRECT ANSWER
1; 1

GRADING NOTE Form required: both. Adding a constant does not change r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F2-Q4**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 2 • Q4/5

Which change to a data set could change r?

A. adding 5 to every x-value

B. converting y from meters to centimeters

C. adding a new point far from the others

D. doubling every y-value

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 2 • Q5/5

678

QUESTION ID

S19-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. adding a new point far from the others

GRADING NOTE Shifts and positive rescalings leave r unchanged.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 2 • Q5/5

Interpret the correlation coefficient

MAIN	Probability & Statistics main question	**HSS-ID.C.8**
BACKWARD 1	Describe strength and direction from a plot	**HSS-ID.B.6**
BACKWARD 2	Compare numbers on a scale from -1 to 1	**6.NS.C.7.c**
FORWARD 1	Match correlation coefficients to scatter plots	**HSS-ID.C.8**
FORWARD 2	Correlation is not causation	**HSS-ID.C.9**

SET 20 • HSS-ID.C.8

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

For car age and value, r = −0.97. Which interpretation is correct?

A. weak negative linear association

B. strong positive linear association

C. strong negative linear association

D. no association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q1/5

681

QUESTION ID

S20-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. strong negative linear association

GRADING NOTE |r| near 1, negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q1/5

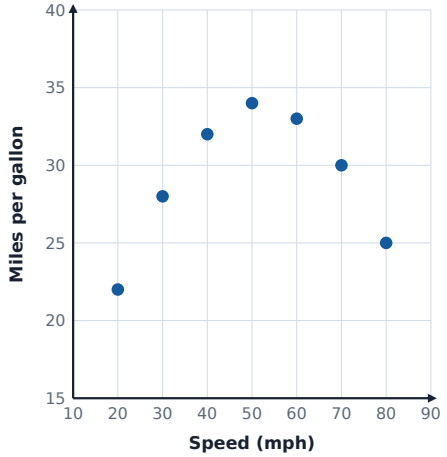
For speed and fuel efficiency, r ≈ 0.245 but the scatter plot shows a clear curve. Which statement is correct?

A. r must have been computed incorrectly.

B. There is a weak linear relationship and nothing more.

C. r measures only linear association, so it misses the curve.

D. There is no relationship between the variables.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q2/5

682

QUESTION ID
S20-M-Q2

TYPE
Multiple choice

CORRECT ANSWER
C. r measures only linear association, so it misses the curve.

GRADING NOTE A strong nonlinear relationship can have r near 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q2/5

Which correlation coefficient shows the strongest linear association?

A. −0.91

B. 0.85

C. −0.40

D. 0.10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q3/5

683

QUESTION ID
S20-M-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. −0.91

GRADING NOTE Largest |r|.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q3/5

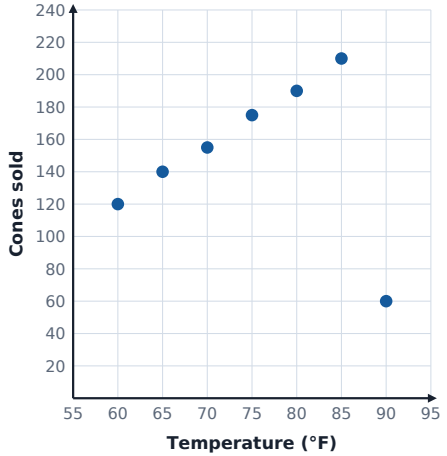
For the cones data, r = −0.008 for all 7 days but 0.999 without the 90-degree day. Which interpretation is best?

A. The six typical days show a strong positive linear association.

B. The 90-degree day is the most typical day.

C. Temperature and cone sales have no relationship on any day.

D. Hotter days have fewer cones sold.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q4/5

684

QUESTION ID

S20-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. The six typical days show a strong positive linear association.

GRADING NOTE The outlier hides the pattern.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q4/5

Which statement about r is correct?

A. r changes when x is converted from hours to minutes.

B. r is negative whenever the data contain an outlier.

C. r can be as large as 5 for very strong data.

D. r has no units and stays the same if the units change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S20-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q5/5

685

QUESTION ID
S20-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
D. r has no units and stays the same if the units change.

GRADING NOTE r is unit-free and between −1 and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-M-Q5**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q5/5

SET 20 • HSS-ID.C.8

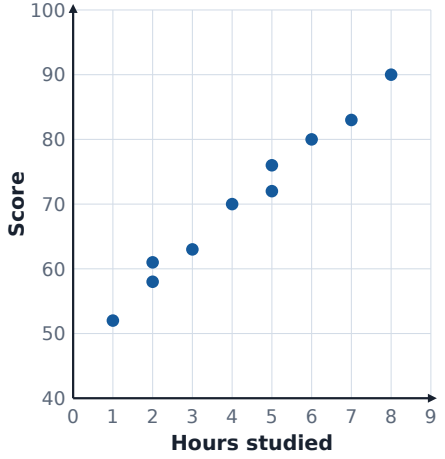
BACKWARD 1

Describe strength and direction from a plot • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6

The scatter plot shows hours studied and score. How strong is the linear association?

A. strong **B.** weak

C. cannot be judged **D.** none



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q1/5

687

QUESTION ID

S20-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. strong

GRADING NOTE Points hug a line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q1/5

True or false?

In a strong association, the points lie close to a line or curve.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q2/5

688

QUESTION ID

S20-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Strength.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q2/5

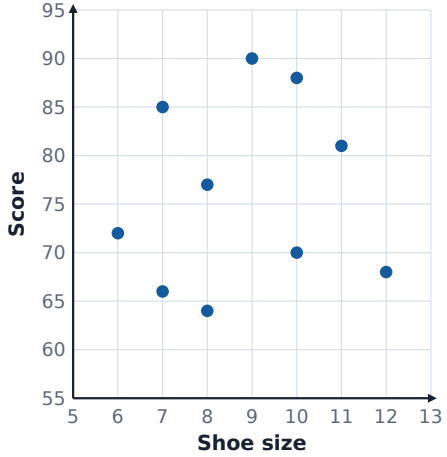
The scatter plot shows shoe size and quiz score. How strong is the association?

A. very weak or none

B. strong negative

C. strong positive

D. perfect



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q3/5

689

QUESTION ID

S20-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. very weak or none

GRADING NOTE No pattern.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q3/5

True or false?

A scatter plot that falls from left to right shows a positive association.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q4/5

690

QUESTION ID

S20-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

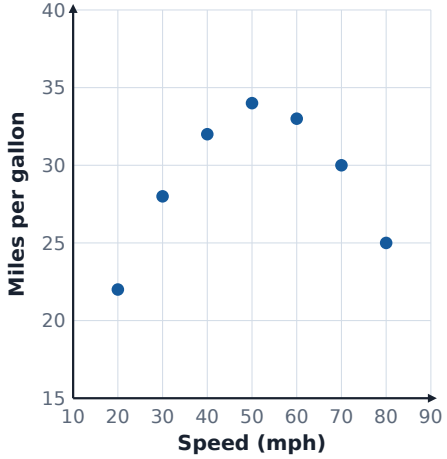
The scatter plot of speed and fuel efficiency has what form?

A. exponential growth

B. linear

C. curved

D. no form



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q5/5

691

QUESTION ID

S20-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. curved

GRADING NOTE Rises then falls.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • HSS-ID.C.8

BACKWARD 2

Compare numbers on a scale from −1 to 1 • Grade 6 • The Number System • 6.NS.C.7.c

ANSWER KEY Reduced copy of the question page, with its answer

What is $|-0.91|$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q1/5

693

QUESTION ID

S20-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.91

GRADING NOTE Distance from 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q1/5

True or false?

$|-0.8| > |0.6|$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q2/5

694

QUESTION ID

S20-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $0.8 > 0.6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q2/5

Which is closest to 0?

A. −0.1

B. −0.6

C. 0.5

D. 0.9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q3/5

695

QUESTION ID

S20-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −0.1

GRADING NOTE Smallest absolute value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q3/5

True or false?

-0.9 is greater than 0.2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q4/5

696

QUESTION ID

S20-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-0.9 < 0.2$, though $|-0.9|$ is larger.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q4/5

Which has the greater absolute value, −0.75 or 0.7?
Enter the number.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q5/5

697

QUESTION ID
S20-B2-Q5

TYPE
Short answer

CORRECT ANSWER
−0.75

GRADING NOTE 0.75 > 0.7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

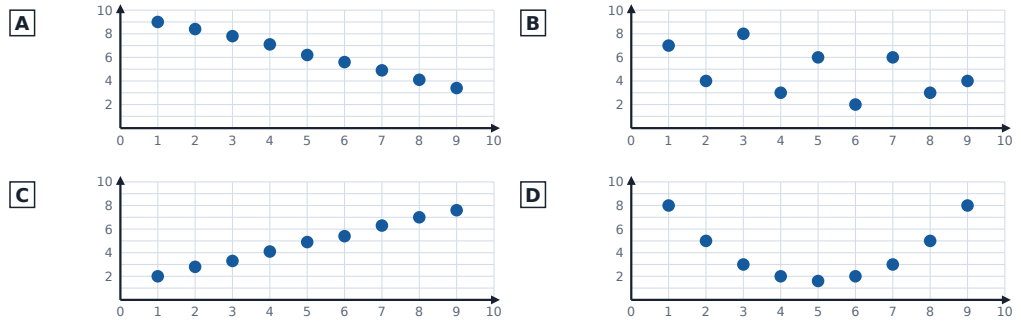
Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q5/5

SET 20 • HSS-ID.C.8

FORWARD 1

Match correlation coefficients to scatter plots • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

Which scatter plot has r ≈ −0.45?



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • FORWARD 1 • Q1/5

699

QUESTION ID

S20-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE Weak negative. The others have r ≈ −1.00, 1.00 and 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F1-Q1**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • FORWARD 1 • Q1/5

A U-shaped scatter plot that is perfectly symmetric has r equal to

A. 0.5

B. 0

C. −1

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • FORWARD 1 • Q2/5

700

QUESTION ID
S20-F1-Q2

TYPE
Multiple choice

CORRECT ANSWER
B. 0

GRADING NOTE The rising and falling halves cancel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F1-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • FORWARD 1 • Q2/5

True or false?

r = −0.45 indicates a stronger linear association than r = 0.30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • FORWARD 1 • Q3/5

701

QUESTION ID
S20-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE 0.45 > 0.30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • FORWARD 1 • Q3/5

Which value of r goes with points falling steadily in a narrow band?

A. 0.99

B. −0.99

C. 0

D. −0.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • FORWARD 1 • Q4/5

QUESTION ID

S20-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. −0.99

GRADING NOTE Strong negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • FORWARD 1 • Q4/5

True or false?

Two data sets with the same r must have scatter plots of the same shape.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • FORWARD 1 • Q5/5

703

QUESTION ID

S20-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Very different plots can share an r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • FORWARD 1 • Q5/5

SET 20 • HSS-ID.C.8

FORWARD 2

Correlation is not causation • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Across towns, the number of libraries and the number of crimes have r = 0.9. Which is the best explanation?

A. Libraries cause crime.

B. r = 0.9 proves a cause in one direction.

C. Crime causes towns to build libraries.

D. Larger towns tend to have more of both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 20 • FORWARD 2 • Q1/5

705

QUESTION ID
S20-F2-Q1

TYPE
Multiple choice

CORRECT ANSWER
D. Larger towns tend to have more of both.

GRADING NOTE Population is a lurking variable.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 20 • FORWARD 2 • Q1/5

True or false?

A correlation of r = 0.95 shows that changes in x cause changes in y.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 20 • FORWARD 2 • Q2/5

706

QUESTION ID

S20-F2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Strong correlation is not evidence of cause by itself.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 20 • FORWARD 2 • Q2/5

Which kind of study can show that x causes y?

A. a scatter plot of past data

B. a study with r close to 1

C. a randomized experiment

D. a survey with many people

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 20 • FORWARD 2 • Q3/5

707

QUESTION ID
S20-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. a randomized experiment

GRADING NOTE Random assignment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 20 • FORWARD 2 • Q3/5

True or false?

A lurking variable can produce a strong correlation between two variables that do not affect each other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 20 • FORWARD 2 • Q4/5

708

QUESTION ID

S20-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both follow it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 20 • FORWARD 2 • Q4/5

Students who sleep more have higher grades (r = 0.6) in an observational study. Which conclusion is justified?

A. Higher grades cause these students to sleep more.

B. Sleep and grades are positively associated in this group.

C. Sleep and grades are unrelated for these students.

D. More sleep causes higher grades for these students.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S20-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 20 • FORWARD 2 • Q5/5

709

QUESTION ID
S20-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
B. Sleep and grades are positively associated in this group.

GRADING NOTE Only association is shown.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S20-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 20 • FORWARD 2 • Q5/5

Distinguish between correlation and causation

MAIN	Probability & Statistics main question	**HSS-ID.C.9**
BACKWARD 1	Recognize a strong correlation	**HSS-ID.C.8**
BACKWARD 2	Describe the association in a scatter plot	**8.SP.A.1**
FORWARD 1	Tell experiments from observational studies	**HSS-IC.B.3**
FORWARD 2	Judge causal claims from experiment results	**HSS-IC.B.5**

SET 21 • HSS-ID.C.9

MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Over several years, a city's sunscreen sales and its number of pool visits are strongly correlated. What best explains this?

A. Swimming causes people to buy sunscreen.

B. Sunny, hot weather raises both.

C. Sunscreen causes people to swim.

D. The data must be wrong.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 21 • MAIN • Q1/5

712

QUESTION ID
S21-M-Q1

TYPE
Multiple choice

CORRECT ANSWER
B. Sunny, hot weather raises both.

GRADING NOTE A lurking variable drives both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 21 • MAIN • Q1/5

Countries with more doctors per person have longer life spans. Which conclusion is justified?

A. Longer life spans cause countries to train more doctors.

B. Adding more doctors in a country will lengthen life spans.

C. Doctors per person and life span are positively associated.

D. Doctors per person and life span are unrelated.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 21 • MAIN • Q2/5

713

QUESTION ID
S21-M-Q2

TYPE
Multiple choice

CORRECT ANSWER
C. Doctors per person and life span are positively associated.

GRADING NOTE Observational data show association only; wealth may drive both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-M-Q2**

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 21 • MAIN • Q2/5

Researchers want to know whether a new study app raises test scores. Which design can show cause?

A. Find the correlation of app use and scores.

B. Survey students who already chose to use the app.

C. Compare the scores of last year's users and nonusers.

D. Randomly assign students to use the app or not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 21 • MAIN • Q3/5

714

QUESTION ID
S21-M-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. Randomly assign students to use the app or not.

GRADING NOTE Random assignment balances other causes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 21 • MAIN • Q3/5

Children with larger shoe sizes have higher reading scores. What is the most likely lurking variable?

A. age

B. reading-book color

C. school lunch menu

D. shoe brand

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 21 • MAIN • Q4/5

715

QUESTION ID
S21-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. age

GRADING NOTE Older children have bigger feet and read better.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 21 • MAIN • Q4/5

Which statement is correct?

A. Association implies causation when r is above 0.9.

B. If two variables are not correlated, neither can cause anything.

C. An observed association alone does not establish a causal relationship.

D. Causation can only be shown with a scatter plot.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 21 • MAIN • Q5/5

716

QUESTION ID
S21-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
C. An observed association alone does not establish a causal relationship.

GRADING NOTE Association is not enough by itself to show cause. The others are false: no r value proves cause, scatter plots cannot show cause, and a lack of correlation does not rule out every causal effect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 21 • MAIN • Q5/5

SET 21 • HSS-ID.C.9

BACKWARD 1

Recognize a strong correlation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

True or false?

r = −0.92 shows a strong linear association.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q1/5

718

QUESTION ID

S21-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE |r| near 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which r shows the weakest linear association?

A. −0.7

B. 0.05

C. 0.6

D. −0.95

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q2/5

719

QUESTION ID

S21-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 0.05

GRADING NOTE Smallest |r|.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-B1-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q2/5

True or false?

A strong correlation means the two variables must be causally related.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q3/5

720

QUESTION ID

S21-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Not necessarily.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q3/5

A data set has r = 0.88. Is the linear association positive or negative? Enter positive or negative.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q4/5

721

QUESTION ID

S21-B1-Q4

TYPE

Short answer

CORRECT ANSWER

positive

GRADING NOTE Sign of r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-B1-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q4/5

r = 0.88 for study time and grades in a survey. What does it describe?

A. a curved association

B. a weak association

C. a strong positive linear association

D. proof that study time causes grades

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q5/5

722

QUESTION ID

S21-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. a strong positive linear association

GRADING NOTE Association only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-B1-Q5**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • HSS-ID.C.9

BACKWARD 2

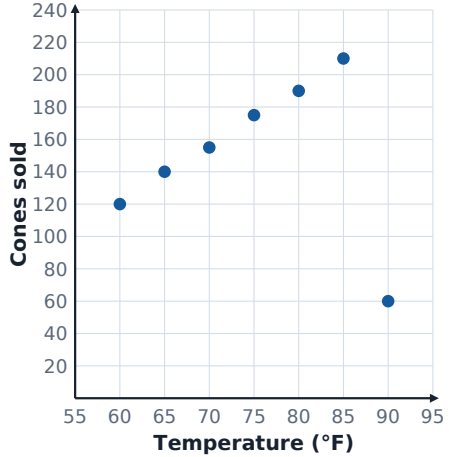
Describe the association in a scatter plot • Grade 8 • Statistics & Probability • 8.SP.A.1

True or false?

In the ice-cream scatter plot, the six typical days show a positive association.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q1/5

724

QUESTION ID

S21-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rising pattern.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q1/5

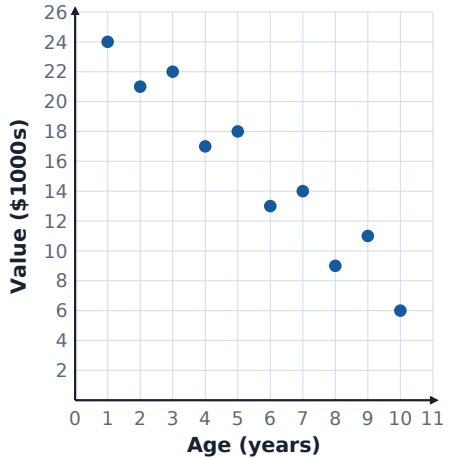
The scatter plot shows car age and value. Which describes it?

A. curved association

B. positive linear association

C. no association

D. negative linear association



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q2/5

725

QUESTION ID

S21-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. negative linear association

GRADING NOTE Falling pattern.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q2/5

True or false?

A scatter plot can show an association between two variables that do not cause each other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q3/5

726

QUESTION ID

S21-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Correlation without cause.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q3/5

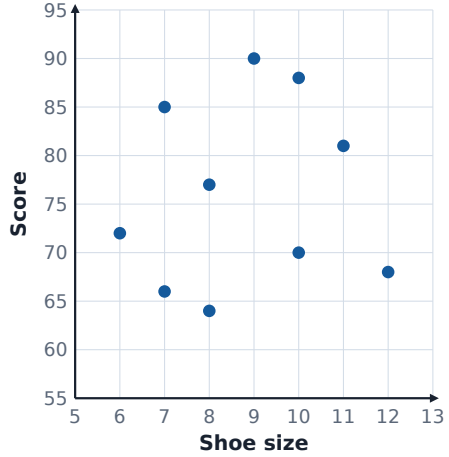
The scatter plot shows shoe size and quiz score of adults. Which describes it?

A. curved

B. no clear association

C. strong negative

D. strong positive



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q4/5

727

QUESTION ID

S21-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. no clear association

GRADING NOTE Cloud.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q4/5

True or false?

An association seen in a scatter plot shows which variable is the cause.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q5/5

728

QUESTION ID

S21-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A plot cannot show cause.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • HSS-ID.C.9

FORWARD 1

Tell experiments from observational studies • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

Researchers record the coffee intake and blood pressure of 500 adults without changing anything. What type of study is this?

A. randomized experiment

B. simulation

C. sample survey of opinions

D. observational study

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q1/5

730

QUESTION ID

S21-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. observational study

GRADING NOTE Nothing is imposed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q1/5

True or false?

In a randomized experiment, the researcher assigns the treatments by chance.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q2/5

731

QUESTION ID

S21-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q2/5

Which study is a randomized experiment?

A. Researchers compare towns that chose different school lunches.

B. Volunteers are randomly assigned to a new diet or the usual diet.

C. A survey asks people which diet they prefer.

D. Doctors record the diets their patients already follow.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q3/5

732

QUESTION ID

S21-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Volunteers are randomly assigned to a new diet or the usual diet.

GRADING NOTE Chance assignment of treatments.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F1-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q3/5

True or false?

An observational study can establish cause as well as a randomized experiment can.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q4/5

733

QUESTION ID

S21-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It cannot control lurking variables.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q4/5

Why does random assignment help support causal conclusions?

A. It removes all variability from the measurements.

B. It guarantees that r will be close to 1.

C. It makes the sample larger than other designs do.

D. It tends to balance other variables between the groups.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q5/5

734

QUESTION ID

S21-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. It tends to balance other variables between the groups.

GRADING NOTE Balancing lurking variables. The others are false: random assignment does not remove variability, fix r, or enlarge the sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F1-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q5/5

SET 21 • HSS-ID.C.9

FORWARD 2

Judge causal claims from experiment results • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

In a randomized experiment, 40 plants got fertilizer and grew a mean of 18 cm; 40 got none and grew a mean of 12 cm. What is the difference in means (fertilizer minus none)?

Difference (cm): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q1/5

736

QUESTION ID

S21-F2-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE 18 − 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q1/5

(Hypothetical simulation results.) In an experiment, 80 plants were randomly assigned, 40 to fertilizer and 40 to none, and the mean growth difference (fertilizer minus none) was 6 cm. Under the no-effect model, re-randomizing the 80 results many times gave differences of 6 cm or more (fertilizer minus none) in only 1% of the re-randomizations. Using a one-sided rule (fewer than 5% counts as significant), what can be concluded?

A. Fertilizer is unrelated to growth.

B. The fertilizer likely caused more growth.

C. The difference is due to chance alone.

D. No conclusion is possible without r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q2/5

737

QUESTION ID

S21-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. The fertilizer likely caused more growth.

GRADING NOTE 1% < 5%: a difference this large is rare under the no-effect model, and random assignment lets it be credited to the fertilizer. The others ignore the evidence or bring in r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q2/5

True or false?

In a randomized experiment, a difference much larger than chance alone would produce supports a causal conclusion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q3/5

738

QUESTION ID
S21-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Random assignment rules out lurking variables.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

An observational study found a large difference between coffee drinkers and others. Why can't it prove cause?

A. Observational studies do not collect any data.

B. Coffee drinkers may differ in other ways that matter.

C. Coffee intake cannot be measured accurately.

D. The difference was too large to be believable.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q4/5

739

QUESTION ID

S21-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Coffee drinkers may differ in other ways that matter.

GRADING NOTE Lurking variables.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q4/5

True or false?

If an experiment's difference could easily happen by chance, the experiment proves the treatment has no effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q5/5

740

QUESTION ID
S21-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It only fails to show an effect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q5/5

Statistics as a process of inference from samples

MAIN	Probability & Statistics main question	**HSS-IC.A.1**
BACKWARD 1	Recognize random and representative samples	**7.SP.A.1**
BACKWARD 2	Use a sample to estimate a population count	**7.SP.A.2**
FORWARD 1	See how sample size affects sampling variability	**HSS-IC.A.1**
FORWARD 2	Attach a margin of error to an estimate	**HSS-IC.B.4**

SET 22 • HSS-IC.A.1

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 200 of a school's 1,500 students found that 62 bike to school. What is the sample proportion who bike? Give a decimal.

Sample proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q1/5

743

QUESTION ID

S22-M-Q1

TYPE

Short answer

CORRECT ANSWER

0.31

GRADING NOTE 62/200. This is the statistic.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q1/5

A random sample of 200 of a school's 1,500 students found that 62 bike to school. What is the parameter of interest?

A. the 200 students who were in the sample

B. the sample proportion 0.31 from the survey

C. the proportion of all 1,500 students who bike

D. the 62 sampled students who said they bike

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q2/5

744

QUESTION ID

S22-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. the proportion of all 1,500 students who bike

GRADING NOTE A parameter describes the population. 0.31 is the statistic; 62 and 200 are counts or people.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q2/5

In a random sample of 200 of a school's 1,500 students, the proportion who bike is 0.31. Use it to estimate how many of the 1,500 students bike.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q3/5

745

QUESTION ID

S22-M-Q3

TYPE

Short answer

CORRECT ANSWER

465

GRADING NOTE 0.31 · 1500.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q3/5

Which sample of a school's students allows an inference about all of the students?

A. the first 200 students to arrive at school one morning

B. a random sample of 200 students chosen from the school roster

C. all 200 students who belong to the school bike club

D. the 200 students who chose to answer an online poll

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q4/5

746

QUESTION ID

S22-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. a random sample of 200 students chosen from the school roster

GRADING NOTE Only random selection makes the sample representative. The others are voluntary or convenience samples.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q4/5

Two random samples of 200 students from the same school give sample proportions 0.31 and 0.27. Why do they differ?

A. One of the two samples must have been chosen incorrectly.

B. Sample proportions can never be used to estimate a parameter.

C. Sample statistics vary from one random sample to another by chance.

D. The population proportion changed between the two samples.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q5/5

747

QUESTION ID

S22-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. Sample statistics vary from one random sample to another by chance.

GRADING NOTE Sampling variability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q5/5

SET 22 • HSS-IC.A.1

BACKWARD 1

Recognize random and representative samples • Grade 7 • Statistics & Probability • 7.SP.A.1

True or false?

A random sample tends to be representative of its population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q1/5

749

QUESTION ID

S22-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each member has the same chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q1/5

A town wants residents' opinion on a new park and has a complete list of all residents. Which method gives a random sample of residents?

A. ask the first 100 people at the park

B. ask the members of the parks committee

C. post an online survey and use the first replies

D. pick 100 names by lottery from the complete resident list

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q2/5

750

QUESTION ID

S22-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. pick 100 names by lottery from the complete resident list

GRADING NOTE Chance selection from the whole population.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q2/5

True or false?

A sample of volunteers who choose to respond is a random sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q3/5

751

QUESTION ID

S22-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Self-selected.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q3/5

A survey of shoppers at a sporting-goods store is used to estimate how much all town residents exercise. What is the main problem?

A. the sample is not representative

B. the sample is a census

C. the sample is too random

D. the sample has no data

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q4/5

752

QUESTION ID

S22-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. the sample is not representative

GRADING NOTE Shoppers there likely exercise more.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q4/5

True or false?

The whole group a study wants to learn about is called the population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q5/5

753

QUESTION ID

S22-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q5/5

SET 22 • HSS-IC.A.1

BACKWARD 2

Use a sample to estimate a population count • Grade 7 • Statistics & Probability • 7.SP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

In a random sample of 50 students, 12 play an instrument. Estimate how many of 600 students play.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q1/5

755

QUESTION ID

S22-B2-Q1

TYPE

Short answer

CORRECT ANSWER

144

GRADING NOTE 12/50 · 600.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q1/5

In a random sample of 40 light bulbs, 2 are defective. Estimate the number defective in 2,000 bulbs.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q2/5

756

QUESTION ID

S22-B2-Q2

TYPE

Short answer

CORRECT ANSWER

100

GRADING NOTE 0.05 · 2000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q2/5

True or false?

If 30% of a random sample prefer pizza, about 30% of the population is estimated to prefer pizza.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q3/5

757

QUESTION ID

S22-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The sample proportion estimates the population proportion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q3/5

In a random sample of 80 voters, 44 support a measure. What is the estimated percent of all voters who support it?

A. 55%

B. 36%

C. 44%

D. 80%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q4/5

758

QUESTION ID

S22-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 55%

GRADING NOTE 44/80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q4/5

True or false?

Estimates from a random sample are always exactly equal to the population value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q5/5

759

QUESTION ID

S22-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They vary by chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • HSS-IC.A.1

FORWARD 1

See how sample size affects sampling variability • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

Many random samples of size 25 and many of size 400 are taken from the same population. Which is true?

A. Larger samples are always biased.

B. Proportions from samples of 25 vary less.

C. Both sizes vary the same amount.

D. Proportions from samples of 400 vary less.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q1/5

761

QUESTION ID

S22-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. Proportions from samples of 400 vary less.

GRADING NOTE Larger n, less variability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q1/5

True or false?

Increasing the size of a random sample reduces bias caused by a poor sampling method.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q2/5

762

QUESTION ID

S22-F1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Size reduces variability, not bias.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q2/5

A random sample of 400 voters has 228 who favor a law. What is the sample proportion? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q3/5

763

QUESTION ID

S22-F1-Q3

TYPE

Short answer

CORRECT ANSWER

0.57

GRADING NOTE 228/400.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F1-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q3/5

Which estimate of a population proportion is likely to be closest to the true value?

A. one from 1,000 volunteers who called in

B. one from a random sample of 50

C. one from 50 friends of the researcher

D. one from a random sample of 1,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q4/5

764

QUESTION ID

S22-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. one from a random sample of 1,000

GRADING NOTE Large and random.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q4/5

True or false?

Sample proportions from repeated random samples center around the population proportion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q5/5

765

QUESTION ID

S22-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Random sampling is unbiased.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F1-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q5/5

SET 22 • HSS-IC.A.1

FORWARD 2

Attach a margin of error to an estimate • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

A random sample gives a sample proportion of 0.57 with a margin of error of 0.05. Give the interval of plausible values for the population proportion.

From: _____

To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q1/5

767

QUESTION ID

S22-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.52; 0.62

GRADING NOTE Form required: both. 0.57 ± 0.05.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q1/5

A poll reports 48% with a margin of error of 3 percentage points. Which value is plausible for the population percent?

A. 52%

B. 55%

C. 44%

D. 50%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q2/5

768

QUESTION ID

S22-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 50%

GRADING NOTE 45% to 51%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q2/5

True or false?

A larger random sample usually gives a smaller margin of error.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q3/5

769

QUESTION ID
S22-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Less variability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q3/5

(Hypothetical simulation results.) Simulated sample proportions have standard deviation 0.025. Using margin of error = 2 × standard deviation, what is the margin of error?

Margin of error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q4/5

770

QUESTION ID

S22-F2-Q4

TYPE

Short answer

CORRECT ANSWER

0.05

GRADING NOTE 2 · 0.025.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q4/5

True or false?

A margin of error accounts for bias from a sample of volunteers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q5/5

771

QUESTION ID

S22-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It covers only chance variation in random samples.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q5/5

Simulate a probability model with random digits

MAIN	Probability & Statistics main question	**HSS-IC.A.2**
BACKWARD 1	Design a simulation for a compound event	**7.SP.C.8.c**
BACKWARD 2	Find a probability as a relative frequency	**7.SP.C.6**
FORWARD 1	Use two-digit groups for probabilities in hundredths	**HSS-IC.A.2**
FORWARD 2	Use simulation results to judge a model	**HSS-IC.A.2**

SET 23 • HSS-IC.A.2

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A player makes 70% of free throws. To simulate, digits 0-6 mean "make" and 7-9 mean "miss". Use the first 10 digits of line 1 of the table as 10 shots. How many shots are made?

Line	Digits	Line	Digits
1	15881 39986 98797 30911	5	17784 86488 76634 40029
2	41707 53645 56815 18844	6	78753 47787 68045 71784
3	72940 55761 69871 68976	7	72671 47513 62701 07981
4	84697 88093 20819 67273	8	93319 08452 05997 28379

Shots made: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q1/5

774

QUESTION ID

S23-M-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Line 1 begins 1588139986; the digits 0-6 among them: 1, 5, 1, 3, 6 → 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A player makes 70% of free throws (digits 0-6 = make, 7-9 = miss). Each line's first 10 digits are one trial of 10 shots, giving 8 trials. In how many trials does the player make 8 or more shots?

Line	Digits	Line	Digits
1	15881 39986 98797 30911	5	17784 86488 76634 40029
2	41707 53645 56815 18844	6	78753 47787 68045 71784
3	72940 55761 69871 68976	7	72671 47513 62701 07981
4	84697 88093 20819 67273	8	93319 08452 05997 28379

Trials: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q2/5

775

QUESTION ID

S23-M-Q2

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Makes per line: 5, 8, 7, 4, 4, 3, 7, 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q2/5

An event has probability 0.4. Which assignment of single random digits simulates it?

A. 4 = success; all other digits = failure

B. 0-3 and 9 = success; 4-8 = failure

C. 0-4 = success; 5-9 = failure

D. 0-3 = success; 4-9 = failure

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q3/5

776

QUESTION ID

S23-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0-3 = success; 4-9 = failure

GRADING NOTE 4 of 10 equally likely digits. The others give 0.5, 0.5 and 0.1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A spinner lands on red with probability 0.25. Read line 5 of the table as ten two-digit numbers; 00-24 means red. How many of the ten spins are red?

Line	Digits	Line	Digits
1	15881 39986 98797 30911	5	17784 86488 76634 40029
2	41707 53645 56815 18844	6	78753 47787 68045 71784
3	72940 55761 69871 68976	7	72671 47513 62701 07981
4	84697 88093 20819 67273	8	93319 08452 05997 28379

Red spins: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q4/5

777

QUESTION ID

S23-M-Q4

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Line 5 pairs: 17, 78, 48, 64, 88, 76, 63, 44, 00, 29.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q4/5

A simulation of 50 families of 4 children (each birth a girl with probability 0.5) gave the results in the table. What proportion of the simulated families had 3 or more girls? Give a decimal.

Girls in 4 births	0	1	2	3	4
Number of simulated trials	3	12	19	13	3

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q5/5

778

QUESTION ID

S23-M-Q5

TYPE

Short answer

CORRECT ANSWER

0.32

GRADING NOTE (13 + 3)/50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q5/5

SET 23 • HSS-IC.A.2

BACKWARD 1

Design a simulation for a compound event • Grade 7 • Statistics & Probability • 7.SP.C.8.c

To simulate a fair coin with random digits, which assignment works?

A. 0-4 = heads; 4-9 = tails

B. 0 = heads; 1-9 = tails

C. even digits = heads; odd digits = tails

D. 0-6 = heads; 7-9 = tails

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q1/5

780

QUESTION ID

S23-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. even digits = heads; odd digits = tails

GRADING NOTE Even and odd digits split 5 and 5. 0-4 and 4-9 overlap at 4; 0 alone gives 0.1; 0-6 gives 0.7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q1/5

In the digits 4 1 7 0 7 5 3 6 4 5, digits 0-4 mean heads. How many heads?

Heads: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q2/5

781

QUESTION ID

S23-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE 4, 1, 0, 3, 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q2/5

True or false?

A trial of a simulation for "number of heads in 10 flips" uses 10 random digits.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q3/5

782

QUESTION ID

S23-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE One digit per flip.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q3/5

To simulate rolling a fair six-sided die with digits, what should be done with 0, 7, 8 and 9?

A. skip them and read the next digit

B. stop the simulation

C. count them as a 6

D. count them as a 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q4/5

783

QUESTION ID

S23-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. skip them and read the next digit

GRADING NOTE Only 1-6 are used, so each face stays equally likely.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q4/5

True or false?

Running more trials of a simulation makes the estimated probability less reliable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q5/5

784

QUESTION ID

S23-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE More trials, more reliable.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q5/5

SET 23 • HSS-IC.A.2

BACKWARD 2

Find a probability as a relative frequency • Grade 7 • Statistics & Probability • 7.SP.C.6

ANSWER KEY Reduced copy of the question page, with its answer

(Hypothetical simulation results.) In 50 simulated trials, an event happened 16 times. What is its relative frequency? Give a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q1/5

786

QUESTION ID

S23-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.32

GRADING NOTE 16/50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q1/5

An event happened in 9 of 200 trials. What is its relative frequency as a percent?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q2/5

787

QUESTION ID

S23-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4.5%

GRADING NOTE 9/200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q2/5

True or false?

A relative frequency of 0.32 means the event happened in 32% of the trials.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q3/5

788

QUESTION ID

S23-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Decimal to percent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q3/5

An event happened in 12 of 60 trials. What is its relative frequency?

A. 0.2

B. 0.6

C. 5

D. 0.12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q4/5

789

QUESTION ID

S23-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 0.2

GRADING NOTE 12/60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q4/5

True or false?

The relative frequency from 10 trials is always exactly equal to the true probability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q5/5

790

QUESTION ID

S23-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It varies, especially in few trials.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • HSS-IC.A.2

FORWARD 1

Use two-digit groups for probabilities in hundredths • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

An event has probability 0.35. Which assignment of two-digit numbers simulates it?

A. 35 = success; others = failure

B. 00-35 = success; 36-99 = failure

C. 00-34 = success; 35-99 = failure

D. 01-34 = success; others = failure

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q1/5

792

QUESTION ID

S23-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. 00-34 = success; 35-99 = failure

GRADING NOTE 00-34 is 35 of the 100 numbers. 00-35 has 36; 01-34 has 34; 35 alone has probability 0.01.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q1/5

Read line 3 as ten two-digit numbers; 00-34 means success. How many successes?

Line	Digits	Line	Digits
1	15881 39986 98797 30911	5	17784 86488 76634 40029
2	41707 53645 56815 18844	6	78753 47787 68045 71784
3	72940 55761 69871 68976	7	72671 47513 62701 07981
4	84697 88093 20819 67273	8	93319 08452 05997 28379

Successes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q2/5

793

QUESTION ID

S23-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Line 3 pairs: 72, 94, 05, 57, 61, 69, 87, 16, 89, 76.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q2/5

True or false?

Two-digit groups 00-99 give 100 equally likely outcomes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q3/5

794

QUESTION ID

S23-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each has probability 0.01.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A fair coin model uses 0-4 = heads. Using the first 10 digits of each of the 8 lines as 8 trials of 10 flips, in how many trials are there 3 or fewer heads?

Line	Digits	Line	Digits
1	15881 39986 98797 30911	5	17784 86488 76634 40029
2	41707 53645 56815 18844	6	78753 47787 68045 71784
3	72940 55761 69871 68976	7	72671 47513 62701 07981
4	84697 88093 20819 67273	8	93319 08452 05997 28379

Trials: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q4/5

795

QUESTION ID

S23-F1-Q4

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Heads per line: 3, 5, 4, 3, 3, 2, 5, 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q4/5

True or false?

To simulate a probability of 1/3 with single digits, use 1-3 = success, 4-9 = failure and skip 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q5/5

796

QUESTION ID

S23-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 3 of 9 used digits.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q5/5

SET 23 • HSS-IC.A.2

FORWARD 2

Use simulation results to judge a model • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

In 100 simulated sets of 10 fair-coin flips (table), in what proportion of the sets were there 9 or more heads? Give a decimal.

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q1/5

798

QUESTION ID

S23-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.02

GRADING NOTE 2/100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A coin claimed to be fair gave 9 heads in 10 flips. Use the table of 100 simulated sets. Decision rule: the model is a fair coin. Count the simulated results that are 9 or more heads (a one-sided check: only an excess of heads counts against the model). If they make up fewer than 5% of the simulated results, the model is not consistent with the data; otherwise it is consistent. Which decision is correct?

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

A. Not consistent: 9 or more heads happened in only 2% of the trials.

B. Not consistent: a fair coin must give exactly 5 heads.

C. Consistent: a fair coin can land heads 9 times.

D. Consistent: 9 or more heads happened in 2% of the trials.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q2/5

799

QUESTION ID

S23-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Not consistent: 9 or more heads happened in only 2% of the trials.

GRADING NOTE 2% < 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q2/5

True or false?

A result that happens in 25% of the simulated trials is consistent with the model under the 5% rule.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q3/5

800

QUESTION ID
S23-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE 25% ≥ 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q3/5

In the table of 100 simulated sets, in what proportion were there 7 or more heads? Give a decimal.

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q4/5

801

QUESTION ID
S23-F2-Q4

TYPE
Short answer

CORRECT ANSWER
0.18

GRADING NOTE (12 + 4 + 2)/100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q4/5

True or false?

If a model is consistent with the data, the simulation proves that the model is true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q5/5

802

QUESTION ID

S23-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It only fails to rule the model out.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q5/5

Decide whether a model is consistent with data

MAIN	Probability & Statistics main question	**HSS-IC.A.2**
BACKWARD 1	Read the results of a simulation	**HSS-IC.A.2**
BACKWARD 2	Compare observed frequencies with a probability model	**7.SP.C.7.b**
FORWARD 1	Decide using results extreme in either direction	**HSS-IC.A.2**
FORWARD 2	Judge a treatment difference by re-randomization	**HSS-IC.B.5**

SET 24 • HSS-IC.A.2

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

The table shows 100 simulated sets of 10 fair-coin flips. In what proportion of the sets were there 8 or more heads? Give a decimal.

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q1/5

805

QUESTION ID
S24-M-Q1

TYPE
Short answer

CORRECT ANSWER
0.06

GRADING NOTE (4 + 2 + 0)/100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q1/5

A coin claimed to be fair gave 8 heads in 10 flips. Use the table of 100 simulated sets. Decision rule: the model is a fair coin. Count the simulated results that are 8 or more heads (a one-sided check: only an excess of heads counts against the model). If they make up fewer than 5% of the simulated results, the model is not consistent with the data; otherwise it is consistent. Which decision is correct?

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

A. Not consistent: a fair coin should give 5 heads every time.

B. Consistent: 8 heads happened in exactly 4% of the sets.

C. Not consistent: 8 or more heads happened in only 6% of the sets.

D. Consistent: 8 or more heads happened in 6% of the simulated sets.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q2/5

806

QUESTION ID

S24-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Consistent: 8 or more heads happened in 6% of the simulated sets.

GRADING NOTE 6% is not fewer than 5%. The others misapply the rule, expect exactly 5 heads, or use "exactly 8" (4%) instead of "8 or more".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

(Hypothetical simulation results.) A die is claimed to be fair. In 60 rolls it showed 18 sixes. In 200 simulated sets of 60 fair rolls, 18 or more sixes happened 4 times. What proportion of the simulated sets is that? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q3/5

807

QUESTION ID
S24-M-Q3

TYPE
Short answer

CORRECT ANSWER
0.02

GRADING NOTE 4/200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q3/5

(Hypothetical simulation results.) A die claimed to be fair showed 18 sixes in 60 rolls. In 200 simulated sets of 60 fair rolls, 18 or more sixes happened 4 times. Decision rule: the model is a fair die. Count the simulated results that are 18 or more sixes (a one-sided check: only an excess of sixes counts against the model). If they make up fewer than 5% of the simulated results, the model is not consistent with the data; otherwise it is consistent. Which decision is correct?

A. Consistent: 18 sixes is possible with a fair die.

B. Not consistent: so many sixes happened in only 2% of simulated sets.

C. Not consistent: a fair die must give exactly 10 sixes.

D. Consistent: 4 simulated sets had 18 or more sixes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q4/5

808

QUESTION ID

S24-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Not consistent: so many sixes happened in only 2% of simulated sets.

GRADING NOTE 2% < 5%. Possible is not the test; the proportion is.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q4/5

(Hypothetical simulation results.) A spinner is claimed to land on red 25% of the time. In 40 spins it landed on red 6 times. In 1,000 simulated sets of 40 spins, 6 or fewer reds happened 120 times. Decision rule: the model is a spinner that lands on red 25% of the time. Count the simulated results that are 6 or fewer reds (a one-sided check: only too few reds counts against the model). If they make up fewer than 5% of the simulated results, the model is not consistent with the data; otherwise it is consistent. Which decision is correct?

A. Consistent: the spinner landed on red exactly 25% of the time.

B. Consistent: 6 or fewer reds happened in 12% of the sets.

C. Not consistent: only 120 of the sets had 6 or fewer.

D. Not consistent: 6 is fewer than the expected 10 reds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q5/5

809

QUESTION ID

S24-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Consistent: 6 or fewer reds happened in 12% of the sets.

GRADING NOTE 120/1000 = 12% ≥ 5%. 6/40 = 15%, not 25%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q5/5

SET 24 • HSS-IC.A.2

BACKWARD 1

Read the results of a simulation • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

ANSWER KEY Reduced copy of the question page, with its answer

In the table of 50 simulated families of 4 children, how many families had exactly 2 girls?

Girls in 4 births	0	1	2	3	4
Number of simulated trials	3	12	19	13	3

Families: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q1/5

811

QUESTION ID

S24-B1-Q1

TYPE

Short answer

CORRECT ANSWER

19

GRADING NOTE Read the table.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-B1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In the table of 50 simulated families, what proportion had 1 or fewer girls? Give a decimal.

Girls in 4 births	0	1	2	3	4
Number of simulated trials	3	12	19	13	3

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q2/5

812

QUESTION ID

S24-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.3

GRADING NOTE (3 + 12)/50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-B1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q2/5

True or false?

The table of 100 simulated sets of 10 fair-coin flips shows 25 sets with exactly 5 heads.

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q3/5

813

QUESTION ID

S24-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Read it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q3/5

In the table of 100 simulated sets of 10 fair-coin flips, how many sets had 3 or fewer heads?

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

A. 12

B. 17

C. 5

D. 29

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q4/5

814

QUESTION ID

S24-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 17

GRADING NOTE 0 + 1 + 4 + 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q4/5

True or false?

In the table of 100 simulated sets of 10 fair-coin flips, more sets had 6 heads than 4 heads.

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q5/5

815

QUESTION ID

S24-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 20 each.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • HSS-IC.A.2

BACKWARD 2

Compare observed frequencies with a probability model • Grade 7 • Statistics & Probability • 7.SP.C.7.b

True or false?

A coin that lands heads 520 times in 1,000 flips is reasonably consistent with a fair coin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q1/5

817

QUESTION ID

S24-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 52% is close to 50%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q1/5

A fair die is rolled 600 times, independently. About how many sixes does the fair model predict?

A. 160

B. 60

C. 100

D. 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q2/5

818

QUESTION ID

S24-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 100

GRADING NOTE 600/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q2/5

How many heads does a fair-coin model predict on average in 10 flips?

Heads: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q3/5

819

QUESTION ID

S24-B2-Q3

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE $10 \cdot 0.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q3/5

True or false?

A fair die rolled 60 times, independently, must show each face exactly 10 times.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q4/5

820

QUESTION ID

S24-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Results vary by chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q4/5

A spinner is claimed to land on red 50% of the time. Which result in 100 spins most suggests the claim is wrong?

A. 82 red

B. 50 red

C. 47 red

D. 53 red

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q5/5

821

QUESTION ID

S24-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 82 red

GRADING NOTE Farthest from 50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q5/5

SET 24 • HSS-IC.A.2

FORWARD 1

Decide using results extreme in either direction • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

In the table of 100 simulated sets of 10 fair flips, what proportion had 9 or more heads or 1 or fewer heads? Give a decimal.

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q1/5

823

QUESTION ID

S24-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.03

GRADING NOTE (2 + 0) + (0 + 1) = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q1/5

A coin claimed to be fair gave 1 head in 10 flips. Use the table of 100 simulated sets of 10 fair-coin flips; results at least as far from 5 heads as 1 head are in either tail. Decision rule: the model is a fair coin. Count the simulated results that are 9 or more heads or 1 or fewer heads (a two-sided check: too many or too few heads both count). If they make up fewer than 5% of the simulated results, the model is not consistent with the data; otherwise it is consistent. Which decision is correct?

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

A. Not consistent: a fair coin cannot give 1 head.

B. Not consistent: such results happened in 3% of the sets.

C. Consistent: 1 head happened in exactly 1 set.

D. Consistent: such results happened in 3% of the sets.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q2/5

QUESTION ID

S24-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Not consistent: such results happened in 3% of the sets.

GRADING NOTE 3% < 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q2/5

True or false?

Under the 5% rule, the smaller the proportion of simulated results as extreme as the data, the stronger the evidence against the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q3/5

825

QUESTION ID

S24-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rarer under the model means stronger evidence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q3/5

A coin claimed to be fair gave 7 heads in 10 flips. In the table, what proportion of sets are at least as far from 5 heads in either direction (7 or more, or 3 or fewer)? Give a decimal.

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q4/5

826

QUESTION ID

S24-F1-Q4

TYPE

Short answer

CORRECT ANSWER

0.35

GRADING NOTE $(12 + 4 + 2) + (12 + 4 + 1) = 35$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q4/5

True or false?

If a model is judged not consistent under the 5% rule, there is no chance that the model is true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q5/5

827

QUESTION ID

S24-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Rare results can still happen.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q5/5

SET 24 • HSS-IC.A.2

FORWARD 2

Judge a treatment difference by re-randomization • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

(Hypothetical simulation results.) In 100 re-randomizations of an experiment's results, a difference in means of 6 or more appeared 3 times. What proportion is that? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q1/5

829

QUESTION ID

S24-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.03

GRADING NOTE 3/100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q1/5

(Hypothetical simulation results.) In a randomized experiment, the treatment group's mean minus the control group's mean was 6. In 100 re-randomizations of the results, differences (treatment minus control) of 6 or more appeared 3 times. Decision rule: the model is the "no effect" model (the treatment does not change the results). Count the simulated results that are differences of 6 or more (a one-sided check). If they make up fewer than 5% of the simulated results, the "no effect" model is not consistent with the data and the difference is significant; otherwise it is consistent, and the difference is not significant. Which decision is correct?

A. The difference is significant: every difference is significant.

B. The difference is not significant: it appeared 3 times.

C. The difference is significant: chance alone gave it 3% of the time.

D. The difference is not significant: re-randomizing changes the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q2/5

830

QUESTION ID

S24-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. The difference is significant: chance alone gave it 3% of the time.

GRADING NOTE 3% < 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q2/5

True or false?

A re-randomization simulation assumes that the treatment has no effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q3/5

831

QUESTION ID

S24-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It reshuffles the responses as if labels did not matter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q3/5

In an experiment, the treatment group's mean was 24 and the control group's mean was 19. What is the difference (treatment minus control)?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q4/5

832

QUESTION ID

S24-F2-Q4

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE 24 − 19.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q4/5

True or false?

If the observed difference appeared in 40% of re-randomizations, the treatment effect is significant under the 5% rule.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q5/5

833

QUESTION ID
S24-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 40% ≥ 5%: not significant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q5/5

SET 25 • HSS-IC.B.3 • MAKING INFERENCES AND JUSTIFYING CONCLUSIONS

Sample surveys, experiments and observational studies

MAIN	Probability & Statistics main question	**HSS-IC.B.3**
BACKWARD 1	Identify the population, sample, parameter and statistic	**HSS-IC.A.1**
BACKWARD 2	Separate association from causation	**HSS-ID.C.9**
FORWARD 1	Identify the parts of an experiment	**HSS-IC.B.3**
FORWARD 2	Choose a study design for a question	**HSS-IC.B.3**

SET 25 • HSS-IC.B.3

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

A researcher randomly assigns 60 volunteers to a new sleep routine or their usual routine and compares their test scores. What kind of study is this?

A. census

B. sample survey

C. experiment

D. observational study

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q1/5

836

QUESTION ID

S25-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. experiment

GRADING NOTE Treatments are imposed and assigned.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q1/5

A polling company asks a random sample of 1,000 adults which candidate they prefer. What kind of study is this?

A. experiment

B. simulation

C. sample survey

D. observational study comparing groups

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q2/5

837

QUESTION ID

S25-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. sample survey

GRADING NOTE It estimates a population characteristic by asking a sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q2/5

Researchers compare lung health of people who already smoke with people who do not. What kind of study is this?

A. experiment

B. census

C. observational study

D. sample survey of opinions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q3/5

838

QUESTION ID

S25-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. observational study

GRADING NOTE Groups form on their own; nothing is imposed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q3/5

Which kind of study is designed mainly to estimate a characteristic of a population?

A. an observational study of two groups

B. a sample survey

C. a simulation of a model

D. an experiment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q4/5

839

QUESTION ID

S25-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. a sample survey

GRADING NOTE Surveys estimate; experiments test treatments.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q4/5

Why can an experiment support a cause-and-effect conclusion when an observational study cannot?

A. The researcher imposes treatments, so other variables can be controlled.

B. Observational studies never collect any numerical data.

C. Experiments do not need any comparison or control group.

D. Experiments always use more subjects than observational studies.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q5/5

840

QUESTION ID

S25-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. The researcher imposes treatments, so other variables can be controlled.

GRADING NOTE Imposed, randomly assigned treatments control lurking variables. The others are false claims about data, sizes or control groups.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q5/5

SET 25 • HSS-IC.B.3

BACKWARD 1

Identify the population, sample, parameter and statistic • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

300 randomly chosen city residents are asked how often they ride the bus. What is the population?

A. the 300 residents

B. all city residents

C. the city's buses

D. bus riders only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q1/5

842

QUESTION ID

S25-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. all city residents

GRADING NOTE The whole group.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

300 randomly chosen city residents are asked how often they ride the bus. What is the sample size?

Size: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q2/5

843

QUESTION ID
S25-B1-Q2

TYPE
Short answer

CORRECT ANSWER
300

GRADING NOTE Count of people asked.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-B1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q2/5

True or false?

A statistic describes a sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q3/5

844

QUESTION ID

S25-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q3/5

True or false?

A census collects data from only part of the population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q4/5

845

QUESTION ID

S25-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A census uses everyone.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q4/5

Which is a parameter?

A. the mean height of 50 sampled students

B. the number 50

C. the proportion in the sample

D. the mean height of all students in a state

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q5/5

846

QUESTION ID

S25-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. the mean height of all students in a state

GRADING NOTE Describes the population.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q5/5

SET 25 • HSS-IC.B.3

BACKWARD 2

Separate association from causation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

True or false?

An association seen without assigning treatments may be caused by a lurking variable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q1/5

848

QUESTION ID

S25-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Observational data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Towns with more ice-cream shops have more drownings. What is a likely lurking variable?

A. hot weather

B. the number of shops

C. the price of ice cream

D. the town's name

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q2/5

849

QUESTION ID

S25-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. hot weather

GRADING NOTE Affects both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q2/5

True or false?

A correlation always implies causation when the study is large.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q3/5

850

QUESTION ID

S25-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Size does not change it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q3/5

Which conclusion requires an experiment?

A. The mean recovery time was 6 days.

B. Recovery time and vitamin use are associated.

C. Half of the patients took the vitamin.

D. A vitamin causes faster recovery.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q4/5

851

QUESTION ID

S25-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. A vitamin causes faster recovery.

GRADING NOTE A causal claim.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q4/5

True or false?

Random assignment helps rule out lurking variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q5/5

852

QUESTION ID

S25-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It balances them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • HSS-IC.B.3

FORWARD 1

Identify the parts of an experiment • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

An experiment compares two fertilizers on 40 plots and measures crop yield. What is the response variable?

A. the 40 plots

B. the two fertilizers

C. the farmer

D. crop yield

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q1/5

854

QUESTION ID

S25-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. crop yield

GRADING NOTE What is measured.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q1/5

An experiment tests 3 doses of a medicine and a placebo. How many treatments are there?

Treatments: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q2/5

855

QUESTION ID

S25-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE 3 + 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q2/5

True or false?

A control group receives the standard treatment or a placebo so the new treatment can be compared.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q3/5

856

QUESTION ID

S25-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Purpose of control.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q3/5

Why do experiments on medicines often give some subjects a placebo?

A. to increase the number of different treatments

B. to make the experiment cheaper to run

C. to account for the effect of believing one is being treated

D. to make sure the results come out positive

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q4/5

857

QUESTION ID

S25-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. to account for the effect of believing one is being treated

GRADING NOTE Placebo effect. The others are false: a placebo does not lower cost, add treatments, or fix the outcome.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q4/5

True or false?

In a double-blind experiment, only the subjects do not know which treatment they receive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q5/5

858

QUESTION ID

S25-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The people who evaluate them do not know either.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q5/5

SET 25 • HSS-IC.B.3

FORWARD 2

Choose a study design for a question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

Question: does a new math app raise test scores? Which design fits best?

A. a randomized experiment

B. a census of all students

C. a sample survey

D. an observational study of app users

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q1/5

860

QUESTION ID

S25-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. a randomized experiment

GRADING NOTE Cause needs an experiment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q1/5

Question: what percent of a town's residents own a bicycle? Which design fits best?

A. a randomized experiment

B. a sample survey

C. a simulation

D. an observational study of cyclists

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q2/5

861

QUESTION ID

S25-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. a sample survey

GRADING NOTE Estimate a population percent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q2/5

Question: do people who live near highways have more asthma? Assigning where people live is not possible. Which design fits best?

A. an observational study

B. a sample survey of opinions

C. a simulation

D. a randomized experiment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q3/5

862

QUESTION ID

S25-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. an observational study

GRADING NOTE Treatment cannot be assigned.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F2-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q3/5

True or false?

It is ethical to randomly assign people to smoke in an experiment on lung disease.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q4/5

863

QUESTION ID

S25-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It would cause harm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q4/5

True or false?

An observational study can be the best choice when assigning a treatment would be unethical.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q5/5

QUESTION ID

S25-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Then observation is what is possible.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q5/5

How randomization relates to each kind of study

MAIN	Probability & Statistics main question	**HSS-IC.B.3**
BACKWARD 1	Classify a study	**HSS-IC.B.3**
BACKWARD 2	Select a simple random sample	**7.SP.A.1**
FORWARD 1	Decide the scope of conclusions	**HSS-IC.B.3**
FORWARD 2	Evaluate a report's conclusion from its design	**HSS-IC.B.6**

SET 26 • HSS-IC.B.3

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

A class has 40 students numbered 01 to 40. Read line 1 of the table as two-digit numbers, skipping 00, numbers above 40 and repeats. Which three students are chosen first?

Line	Digits	Line	Digits
1	15881 39986 98797 30911	5	17784 86488 76634 40029
2	41707 53645 56815 18844	6	78753 47787 68045 71784
3	72940 55761 69871 68976	7	72671 47513 62701 07981
4	84697 88093 20819 67273	8	93319 08452 05997 28379

First: _____

Second: _____

Third: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q1/5

867

QUESTION ID

S26-M-Q1

TYPE

Short answer

CORRECT ANSWER

15; 13; 9

GRADING NOTE Form required: all three, in order. Line 1: 15, 88, 13, 99, 86, 98, 79, 73, 09 → 15, 13, 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q1/5

What is the main purpose of selecting a sample at random in a survey?

A. so the sample is as small as possible

B. so that every answer comes out the same

C. so a treatment can be shown to cause a change

D. so the results can be generalized to the population

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q2/5

868

QUESTION ID

S26-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. so the results can be generalized to the population

GRADING NOTE Random sampling supports generalization. The others describe random assignment (cause), or claim it makes answers identical or the sample small.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q2/5

What is the main purpose of assigning subjects to treatments at random in an experiment?

A. to guarantee that the treatment will work

B. to help balance other variables between the groups

C. to avoid needing any control or comparison group

D. to let the results describe everyone in the country

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q3/5

869

QUESTION ID

S26-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. to help balance other variables between the groups

GRADING NOTE Random assignment tends to balance other variables, so a difference larger than chance variation can be credited to the treatment; it does not guarantee identical groups. The others describe random sampling (generalization) or claim it guarantees success or replaces a control group.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q3/5

A study randomly assigned 80 volunteers to two diets and found a significant difference in weight loss. The volunteers were not a random sample. What can be concluded?

A. The diet causes the same difference for everyone in the country.

B. The diet caused the difference for volunteers like these.

C. Nothing, because the subjects were volunteers.

D. Only that the two diets are associated with weight.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q4/5

870

QUESTION ID

S26-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. The diet caused the difference for volunteers like these.

GRADING NOTE Random assignment gives cause; no random sample, so no broad generalization. The others generalize to the country, ignore the random assignment, or claim only association.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Plots in a field are numbered 01 to 60. Read line 3 of the table as two-digit numbers, skipping 00, numbers above 60 and repeats. Which three plots are assigned fertilizer A first?

Line	Digits	Line	Digits
1	15881 39986 98797 30911	5	17784 86488 76634 40029
2	41707 53645 56815 18844	6	78753 47787 68045 71784
3	72940 55761 69871 68976	7	72671 47513 62701 07981
4	84697 88093 20819 67273	8	93319 08452 05997 28379

First: _____

Second: _____

Third: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q5/5

871

QUESTION ID

S26-M-Q5

TYPE

Short answer

CORRECT ANSWER

5; 57; 16

GRADING NOTE Form required: all three, in order. Line 3: 72, 94, 05, 57, 61, 69, 87, 16 → 5, 57, 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q5/5

SET 26 • HSS-IC.B.3

BACKWARD 1

Classify a study • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

A teacher asks a random sample of 50 students how many hours they sleep. What kind of study is this?

A. observational study comparing treatments

B. sample survey

C. census

D. experiment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q1/5

873

QUESTION ID
S26-B1-Q1

TYPE
Multiple choice

CORRECT ANSWER
B. sample survey

GRADING NOTE Asking a sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q1/5

True or false?

In an experiment, the researcher decides which subjects get which treatment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q2/5

874

QUESTION ID

S26-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Imposed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Doctors compare the heart health of people who already jog with people who do not. What kind of study is this?

A. sample survey

B. simulation

C. observational study

D. experiment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q3/5

875

QUESTION ID

S26-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. observational study

GRADING NOTE Nothing imposed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q3/5

True or false?

A sample survey assigns treatments to people.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q4/5

876

QUESTION ID

S26-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It asks or measures only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q4/5

Which study imposes a treatment?

A. giving half of the volunteers a new drug, chosen by lottery

B. asking a random sample of voters for their opinion

C. counting the cars that pass along a busy road

D. recording the diets that people already choose to eat

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q5/5

877

QUESTION ID

S26-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. giving half of the volunteers a new drug, chosen by lottery

GRADING NOTE Assignment by chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-B1-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q5/5

SET 26 • HSS-IC.B.3

BACKWARD 2

Select a simple random sample • Grade 7 • Statistics & Probability • 7.SP.A.1

True or false?

In a simple random sample, every group of the same size has the same chance of being chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q1/5

879

QUESTION ID

S26-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q1/5

Which method gives a simple random sample of 5 students from a class of 30?

A. draw 5 names from a hat holding all 30 names

B. ask for 5 volunteers to raise their hands

C. choose the 5 tallest students in the class

D. choose the first 5 students to arrive

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q2/5

880

QUESTION ID

S26-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. draw 5 names from a hat holding all 30 names

GRADING NOTE Chance selection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Students are numbered 01 to 30. Reading the two-digit numbers 45 07 29 07 33 18, skipping numbers above 30 and repeats, which three students are chosen?

First: _____

Second: _____

Third: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q3/5

881

QUESTION ID

S26-B2-Q3

TYPE

Short answer

CORRECT ANSWER

7; 29; 18

GRADING NOTE Form required: all three, in order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q3/5

True or false?

Choosing every student whose name starts with A is a random sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q4/5

882

QUESTION ID

S26-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Not chance-based.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q4/5

Why are repeated numbers skipped when selecting a sample from a random-digit table?

A. skipping them saves time in the selection

B. repeated digits are always printing errors

C. repeated numbers are too large to use

D. a person cannot be chosen twice

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q5/5

883

QUESTION ID

S26-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. a person cannot be chosen twice

GRADING NOTE Sampling without replacement. The others are false reasons: repeats are valid digits of the right size.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • HSS-IC.B.3

FORWARD 1

Decide the scope of conclusions • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

A study used a random sample from a population but no random assignment. What can be concluded?

A. Nothing at all can be concluded from such a study.

B. Associations generalize to the population, but not cause.

C. Cause can be concluded, but only for the sample.

D. Cause can be concluded for the whole population.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q1/5

885

QUESTION ID
S26-F1-Q1

TYPE
Multiple choice

CORRECT ANSWER
B. Associations generalize to the population, but not cause.

GRADING NOTE Random sampling → generalize; no random assignment → no cause.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A study used both a random sample from a population and random assignment to treatments. What can be concluded?

A. Cause, but only for the sample.

B. Association only.

C. Cause, generalized to the population.

D. Nothing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q2/5

886

QUESTION ID
S26-F1-Q2

TYPE
Multiple choice

CORRECT ANSWER
C. Cause, generalized to the population.

GRADING NOTE Both kinds of randomization.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q2/5

True or false?

Random assignment allows cause-and-effect conclusions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q3/5

887

QUESTION ID
S26-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Its purpose.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Random sampling allows cause-and-effect conclusions even without random assignment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q4/5

888

QUESTION ID

S26-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It allows generalization only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q4/5

A study used neither random sampling nor random assignment. What can be concluded?

A. Cause for the group studied.

B. Cause for the population.

C. Associations for the population.

D. Associations in the group studied only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q5/5

889

QUESTION ID

S26-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. Associations in the group studied only.

GRADING NOTE No randomization of either kind.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q5/5

SET 26 • HSS-IC.B.3

FORWARD 2

Evaluate a report's conclusion from its design • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A news report says, "People who drink green tea live longer, so drink tea to live longer." The data came from an observational study. What is the main flaw?

A. The sample was too large to be trusted.

B. Green tea intake cannot be measured at all.

C. Observational studies cannot find associations.

D. It claims cause from an observational study.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q1/5

891

QUESTION ID

S26-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. It claims cause from an observational study.

GRADING NOTE Cause needs random assignment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q1/5

True or false?

A report based on a voluntary online poll can be generalized to all adults.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q2/5

892

QUESTION ID

S26-F2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Volunteers are not a random sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q2/5

A randomized experiment on 200 college athletes found a new drink improved sprint times. To whom does the result apply most directly?

A. all adults

B. college athletes like those studied

C. all children

D. nobody

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q3/5

893

QUESTION ID
S26-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. college athletes like those studied

GRADING NOTE The subjects define the scope.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q3/5

True or false?

A report should state how its sample was chosen and whether treatments were randomly assigned.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q4/5

894

QUESTION ID

S26-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Needed to judge the conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q4/5

A survey of 50 gym members found 90% exercise daily, reported as "90% of Americans exercise daily." What is wrong?

A. Surveys can never be used to estimate percents.

B. Gym members are not representative of all Americans.

C. The sample of 50 people is far too large.

D. The percent of 90 was computed incorrectly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q5/5

895

QUESTION ID

S26-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Gym members are not representative of all Americans.

GRADING NOTE Biased sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F2-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q5/5

Estimate a population proportion with a margin of error

MAIN	Probability & Statistics main question	**HSS-IC.B.4**
BACKWARD 1	Compute a sample proportion and name it	**HSS-IC.A.1**
BACKWARD 2	Read a standard deviation as typical variation	**HSS-ID.A.2**
FORWARD 1	See how sample size changes the margin of error	**HSS-IC.B.4**
FORWARD 2	Read margins of error in poll reports	**HSS-IC.B.6**

SET 27 • HSS-IC.B.4

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

ANSWER KEY Reduced copy of the question page, with its answer

In a random sample of 400 adults, 248 favor a new park. What is the sample proportion? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q1/5

898

QUESTION ID
S27-M-Q1

TYPE
Short answer

CORRECT ANSWER
0.62

GRADING NOTE 248/400.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q1/5

(Hypothetical simulation results.) 500 random samples of size 400 were simulated from a population with proportion 0.62. The simulated sample proportions had standard deviation 0.024. Use margin of error = 2 × (standard deviation of the simulated values). What is the margin of error?

Margin of error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q2/5

899

QUESTION ID
S27-M-Q2

TYPE
Short answer

CORRECT ANSWER
0.048

GRADING NOTE 2 · 0.024.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q2/5

A sample proportion is 0.62 and the margin of error is 0.048. Give the interval of plausible values for the population proportion.

From: _____

To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q3/5

900

QUESTION ID

S27-M-Q3

TYPE

Short answer

CORRECT ANSWER

0.572; 0.668

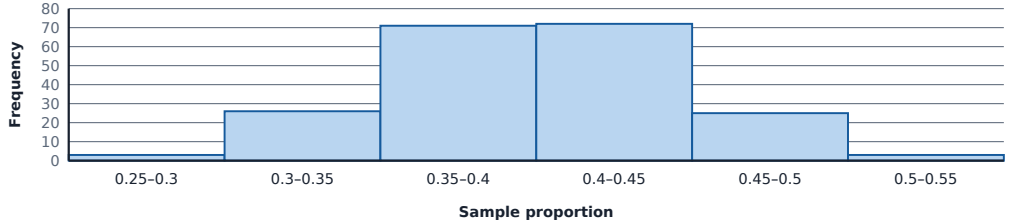
GRADING NOTE Form required: both. 0.62 ± 0.048.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The histogram shows 200 simulated sample proportions. How many of them are between 0.35 and 0.45 (bins 0.35-0.40 and 0.40-0.45)?



Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q4/5

901

QUESTION ID
S27-M-Q4

TYPE
Short answer

CORRECT ANSWER
143

GRADING NOTE 71 + 72.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q4/5

A poll of a random sample reports that 54% of voters support a measure, with a margin of error of 3 percentage points. Which conclusion is best supported?

A. Exactly 54% of all voters support the measure.

B. Support could plausibly be anywhere from 0% to 100%.

C. The poll shows that support is below 50%.

D. A majority plausibly supports it: 51% to 57% are all above 50%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q5/5

902

QUESTION ID
S27-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
D. A majority plausibly supports it: 51% to 57% are all above 50%.

GRADING NOTE 54 ± 3 gives 51% to 57%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q5/5

SET 27 • HSS-IC.B.4

BACKWARD 1

Compute a sample proportion and name it • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

In a random sample of 250 students, 90 walk to school. What is the sample proportion? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q1/5

904

QUESTION ID

S27-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.36

GRADING NOTE 90/250.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-B1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q1/5

True or false?

A sample proportion is a statistic.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q2/5

905

QUESTION ID

S27-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It describes the sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-B1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q2/5

The proportion of all voters in a state who favor a law is a

A. parameter

B. statistic

C. margin of error

D. sample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q3/5

906

QUESTION ID

S27-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. parameter

GRADING NOTE Describes the population.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-B1-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In a random sample of 80 phones, 6 were defective. What is the sample proportion? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q4/5

907

QUESTION ID
S27-B1-Q4

TYPE
Short answer

CORRECT ANSWER
0.075

GRADING NOTE 6/80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-B1-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q4/5

True or false?

Different random samples always give the same sample proportion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q5/5

908

QUESTION ID

S27-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They vary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q5/5

SET 27 • HSS-IC.B.4

BACKWARD 2

Read a standard deviation as typical variation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

True or false?

A standard deviation measures how far values typically are from their mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q1/5

910

QUESTION ID

S27-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Meaning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q1/5

(Hypothetical simulation results.) Simulated proportions from samples of size 100 have SD 0.05; from samples of size 400, SD 0.025. Which vary less?

A. cannot tell

B. the proportions from samples of size 100

C. the proportions from samples of size 400

D. they vary the same

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q2/5

911

QUESTION ID

S27-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. the proportions from samples of size 400

GRADING NOTE Smaller SD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-B2-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 2 × 0.031.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q3/5

912

QUESTION ID

S27-B2-Q3

TYPE

Short answer

CORRECT ANSWER

0.062

GRADING NOTE Doubling.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q3/5

True or false?

A larger standard deviation means the values are more tightly clustered.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q4/5

913

QUESTION ID

S27-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE More spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

About what percent of values in a roughly normal distribution lie within 2 SDs of the mean?

A. 68%

B. 95%

C. 50%

D. 99.7%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q5/5

914

QUESTION ID

S27-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 95%

GRADING NOTE The rule behind the convention.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-B2-Q5**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q5/5

SET 27 • HSS-IC.B.4

FORWARD 1

See how sample size changes the margin of error • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

For a sample proportion near 0.5, the margin of error is about $2 \cdot \sqrt{(0.5 \cdot 0.5/n)}$. Find it for n = 400 and n = 100.

n = 400: _____

n = 100: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q1/5

916

QUESTION ID

S27-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.05; 0.1

GRADING NOTE Form required: both. 2·0.025 and 2·0.05.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q1/5

True or false?

Making the sample 4 times as large cuts the margin of error about in half.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q2/5

917

QUESTION ID

S27-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\sqrt{4} = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q2/5

A pollster wants a smaller margin of error. What should change?

A. take a larger random sample

B. ask only volunteers

C. round the results

D. take a smaller random sample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q3/5

QUESTION ID

S27-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. take a larger random sample

GRADING NOTE Larger n.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F1-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

(Hypothetical simulation results.) Simulated proportions from samples of size 900 have SD 0.016. Use margin of error = 2 × (standard deviation of the simulated values). What is the margin of error?

Margin of error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q4/5

919

QUESTION ID
S27-F1-Q4

TYPE
Short answer

CORRECT ANSWER
0.032

GRADING NOTE 2 · 0.016.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q4/5

True or false?

A larger sample removes the bias of a poorly worded question.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q5/5

920

QUESTION ID
S27-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Size does not fix bias.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q5/5

SET 27 • HSS-IC.B.4

FORWARD 2

Read margins of error in poll reports • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

Candidate A has 48% and candidate B has 45%, each with a margin of error of 4 percentage points. What is the best conclusion?

A. Both candidates have under 50%, so both lose.

B. No clear leader: the two intervals overlap a lot.

C. B is certainly behind A by at least 3 points.

D. A is certainly ahead of B by exactly 3 points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q1/5

922

QUESTION ID

S27-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. No clear leader: the two intervals overlap a lot.

GRADING NOTE 44%−52% and 41%−49% overlap a lot, so a cautious reading is that the poll shows no clear leader. The others claim certainty the poll cannot give, or confuse sample percents with winning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A report gives 62% with a margin of error of 5 percentage points. Give the interval in percents.

From: _____

To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q2/5

923

QUESTION ID

S27-F2-Q2

TYPE

Short answer

CORRECT ANSWER

57%; 67%

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q2/5

True or false?

A margin of error describes only random sampling variation, not bias from how the sample was chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q3/5

924

QUESTION ID

S27-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It assumes a random sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F2-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q3/5

A website poll of 10,000 visitors reports 70% ± 1%. Why is the margin of error misleading?

A. The sample is too small for a margin of error.

B. A margin of error must be at least 5%.

C. Percents cannot have a margin of error.

D. The visitors chose themselves, so the sample is not random.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q4/5

925

QUESTION ID

S27-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. The visitors chose themselves, so the sample is not random.

GRADING NOTE Self-selected sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q4/5

True or false?

If two candidates' intervals overlap, the poll proves that they are tied.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q5/5

926

QUESTION ID

S27-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It shows the difference is not clear, not that they are tied.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q5/5

Estimate a population mean with a margin of error

MAIN	Probability & Statistics main question	**HSS-IC.B.4**
BACKWARD 1	Use a margin of error	**HSS-IC.B.4**
BACKWARD 2	Compute a mean	**6.SP.B.5.c**
FORWARD 1	Compare variability of sample means	**HSS-IC.B.4**
FORWARD 2	Compare two treatments' means	**HSS-IC.B.5**

SET 28 • HSS-IC.B.4

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

A random sample of 8 students reported these hours of sleep: 7, 6.5, 8, 7.5, 6, 9, 7, 8. What is the sample mean?

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q1/5

929

QUESTION ID

S28-M-Q1

TYPE

Short answer

CORRECT ANSWER

7.375

GRADING NOTE 59/8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q1/5

(Hypothetical simulation results.) Simulated random samples of 8 students gave sample means with standard deviation 0.15 hour. Use margin of error = 2 × (standard deviation of the simulated values). What is the margin of error?

Margin of error (hours): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q2/5

930

QUESTION ID
S28-M-Q2

TYPE
Short answer

CORRECT ANSWER
0.3

GRADING NOTE 2 · 0.15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q2/5

A sample mean is 7.4 hours with a margin of error of 0.3 hour. Give the interval of plausible values for the population mean.

From: _____

To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q3/5

931

QUESTION ID

S28-M-Q3

TYPE

Short answer

CORRECT ANSWER

7.1; 7.7

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q3/5

A study reports a mean of 7.4 hours of sleep with a margin of error of 0.3 hour, from a random sample. Which interpretation is correct?

A. Population means from 7.1 to 7.7 hours are plausible.

B. The population mean is exactly 7.4 hours.

C. Half of all students sleep 7.4 hours.

D. Every student sleeps between 7.1 and 7.7 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q4/5

932

QUESTION ID

S28-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Population means from 7.1 to 7.7 hours are plausible.

GRADING NOTE The interval is for the mean, not for individuals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q4/5

(Hypothetical simulation results.) In 1,000 simulated samples, the middle 95% of sample means ran from 41.2 to 44.8. Give the center of this interval and the margin of error (half its width).

Center: _____

Margin of error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q5/5

933

QUESTION ID

S28-M-Q5

TYPE

Short answer

CORRECT ANSWER

43; 1.8

GRADING NOTE Form required: both. (41.2 + 44.8)/2 and (44.8 − 41.2)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q5/5

SET 28 • HSS-IC.B.4

BACKWARD 1

Use a margin of error • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

ANSWER KEY Reduced copy of the question page, with its answer

An estimate is 0.40 with a margin of error of 0.06. Give the interval.

From: _____

To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q1/5

935

QUESTION ID

S28-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.34; 0.46

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-B1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q1/5

True or false?

Margin of error = 2 × 0.05 = 0.1 when the simulated values have SD 0.05.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q2/5

936

QUESTION ID

S28-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Convention.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which estimate has the widest interval?

A. 20 ± 1

B. 20 ± 4

C. 20 ± 2

D. 20 ± 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q3/5

937

QUESTION ID

S28-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 20 ± 4

GRADING NOTE Largest margin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-B1-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q3/5

An interval runs from 12 to 18. What is the margin of error?

Margin of error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q4/5

938

QUESTION ID

S28-B1-Q4

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Half of 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-B1-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q4/5

True or false?

An interval from 0.30 to 0.40 has center 0.30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q5/5

939

QUESTION ID

S28-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Center 0.35.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • HSS-IC.B.4

BACKWARD 2

Compute a mean • Grade 6 • Statistics & Probability • 6.SP.B.5.c

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 4, 6, 8, 10.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q1/5

941

QUESTION ID

S28-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE 28/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 6.5, 7, 8.5, rounded to three decimal places.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q2/5

942

QUESTION ID

S28-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7.333

GRADING NOTE 22/3 = 7.333..., to three decimal places.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q2/5

True or false?

The mean of 2, 4, 9 is 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q3/5

943

QUESTION ID

S28-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 15/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q3/5

What is the mean of 10, 12, 14, 20?

A. 14

B. 56

C. 13

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q4/5

944

QUESTION ID

S28-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 14

GRADING NOTE 56/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q4/5

True or false?

The mean of a data set must be one of its values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q5/5

945

QUESTION ID

S28-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Not necessarily.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q5/5

SET 28 • HSS-IC.B.4

FORWARD 1

Compare variability of sample means • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

Sample means from samples of size 9 have SD 2; from samples of size 36, SD 1. Which statement is correct?

A. Means of larger samples vary more, so the margin is larger.

B. Both sample sizes give the same margin of error.

C. Means of larger samples vary less, so the margin of error is smaller.

D. Sample size has no effect at all on sample means.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q1/5

947

QUESTION ID

S28-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Means of larger samples vary less, so the margin of error is smaller.

GRADING NOTE Quadrupling n halves the SD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q1/5

(Hypothetical simulation results.) Simulated sample means from samples of size 36 have SD 1. Use margin of error = 2 × (standard deviation of the simulated values). What is the margin of error?

Margin of error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q2/5

948

QUESTION ID

S28-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE 2 · 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Individual values vary more than means of samples of those values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q3/5

949

QUESTION ID
S28-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Averaging reduces spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

An estimate of 52 has a margin of error of 2. Is 55 a plausible population mean? Enter yes or no.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q4/5

950

QUESTION ID

S28-F1-Q4

TYPE

Short answer

CORRECT ANSWER

no

GRADING NOTE 50 to 54.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q4/5

True or false?

A margin of error corrects the bias of a convenience sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q5/5

951

QUESTION ID

S28-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A margin of error describes only random sampling variation; it does not fix bias from how the sample was chosen.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F1-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q5/5

SET 28 • HSS-IC.B.4

FORWARD 2

Compare two treatments' means • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

Plant food A: mean growth 13 cm; plant food B: mean growth 10 cm. What is the difference (A minus B)?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q1/5

953

QUESTION ID

S28-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE 13 − 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q1/5

An experiment's difference in means could be due to the treatments or to chance. What tool checks chance?

A. a re-randomization simulation

B. a census

C. a larger margin of error

D. a scatter plot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q2/5

954

QUESTION ID

S28-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. a re-randomization simulation

GRADING NOTE Re-randomization models "no effect".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q2/5

True or false?

A difference in means from a randomized experiment is always caused by the treatment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q3/5

955

QUESTION ID

S28-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It may be chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q3/5

(Hypothetical simulation results.) In 200 re-randomizations, a difference of 3 or more occurred 4 times. What proportion? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q4/5

956

QUESTION ID

S28-F2-Q4

TYPE

Short answer

CORRECT ANSWER

0.02

GRADING NOTE 4/200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q4/5

True or false?

A difference that chance alone produces in only 2% of re-randomizations is significant under a 5% rule.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q5/5

957

QUESTION ID
S28-F2-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE 2% < 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q5/5

Compare two treatments with a randomization simulation

MAIN	Probability & Statistics main question	**HSS-IC.B.5**
BACKWARD 1	Decide whether a model is consistent with data	**HSS-IC.A.2**
BACKWARD 2	Explain random assignment	**HSS-IC.B.3**
BACKWARD 3	Compute means of small groups	**6.SP.B.5.c**
FORWARD 1	Use differences in either direction	**HSS-IC.B.5**
FORWARD 2	Evaluate a report of an experiment	**HSS-IC.B.6**

SET 29 • HSS-IC.B.5

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Ten plants were randomly assigned to plant food A or B. Find the mean growth for each group and the difference (A minus B).

Plant food	Growth of the 5 plants (cm)
A	14, 11, 13, 12, 15
B	10, 12, 9, 11, 8

Mean A: _____

Mean B: _____

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q1/5

960

QUESTION ID

S29-M-Q1

TYPE

Short answer

CORRECT ANSWER

13; 10; 3

GRADING NOTE Form required: all three. 65/5, 50/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q1/5

(Hypothetical simulation results.) In an experiment, 10 plants were randomly assigned to two plant foods (5 each), and the difference in mean growth was 3 cm. The 10 results were re-randomized into two groups of 5 a total of 200 times. A difference of 3 or more occurred 4 times. What proportion of the re-randomizations is that? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q2/5

961

QUESTION ID
S29-M-Q2

TYPE
Short answer

CORRECT ANSWER
0.02

GRADING NOTE 4/200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q2/5

(Hypothetical simulation results.) Ten plants were randomly assigned to plant food A or B (5 each); mean growth A minus B was 3 cm. In 200 re-randomizations, a difference (A minus B) of 3 or more occurred 4 times. Decision rule: the model is the "no difference" model (plant food does not change growth). Count the simulated results that are differences of 3 or more (A minus B; a one-sided check). If they make up fewer than 5% of the simulated results, the model is not consistent with the data and the difference is significant; otherwise it is consistent and the difference is not significant. Which decision is correct?

A. The difference is significant: any difference is significant.

B. The difference is not significant: it happened 4 times by chance.

C. The difference is significant: chance gave it only 2% of the time.

D. The difference is not significant: the groups had only 5 plants.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q3/5

962

QUESTION ID

S29-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. The difference is significant: chance gave it only 2% of the time.

GRADING NOTE 2% < 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q3/5

In an experiment, 10 plants were randomly assigned to plant food A or B (5 each). What does one re-randomization of the 10 growth results do?

A. Repeats the whole experiment with many more plants.

B. Swaps the means of the two original groups of plants.

C. Reshuffles the 10 results into groups of 5, ignoring food.

D. Grows 10 new plants, again using the two plant foods.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q4/5

963

QUESTION ID

S29-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Reshuffles the 10 results into groups of 5, ignoring food.

GRADING NOTE It models the "no difference" assumption.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q4/5

(Hypothetical simulation results.) In another plant-food experiment with random assignment, mean growth A minus B was 1.2 cm. In 200 re-randomizations, differences (A minus B) of 1.2 or more occurred 62 times. Decision rule: the model is the "no difference" model (plant food does not change growth). Count the simulated results that are differences of 1.2 or more (A minus B; a one-sided check). If they make up fewer than 5% of the simulated results, the model is not consistent with the data and the difference is significant; otherwise it is consistent and the difference is not significant. Which decision is correct?

A. Not significant: re-randomizations can never be significant.

B. Significant: 62 is a large number of re-randomizations.

C. Not significant: chance gave such a difference 31% of the time.

D. Significant: the difference 1.2 is greater than 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q5/5

964

QUESTION ID

S29-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. Not significant: chance gave such a difference 31% of the time.

GRADING NOTE 62/200 = 31% ≥ 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q5/5

SET 29 • HSS-IC.B.5

BACKWARD 1

Decide whether a model is consistent with data • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

True or false?

A result that would happen rarely under a model is evidence against the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q1/5

966

QUESTION ID

S29-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Logic of the test.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q1/5

(Hypothetical simulation results.) A result at least as extreme happened in 3 of 150 simulated trials. What proportion? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q2/5

967

QUESTION ID

S29-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.02

GRADING NOTE 3/150.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-B1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q2/5

Under the 5% rule, a result as extreme as the data occurred in 12% of trials. The decision is

A. the model is proven

B. consistent with the model

C. not consistent with the model

D. no decision

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q3/5

968

QUESTION ID

S29-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. consistent with the model

GRADING NOTE 12% ≥ 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q3/5

True or false?

Under the 5% rule, 4% of trials means the model is consistent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q4/5

969

QUESTION ID

S29-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 4% < 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which proportion of simulated results gives the strongest evidence against a model?

A. 0.001

B. 0.2

C. 0.04

D. 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q5/5

970

QUESTION ID

S29-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 0.001

GRADING NOTE Smallest.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q5/5

SET 29 • HSS-IC.B.5

BACKWARD 2

Explain random assignment • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

True or false?

Random assignment tends to make the treatment groups similar before treatment, although chance differences remain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q1/5

QUESTION ID

S29-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It tends to balance other variables between the groups; it does not make them identical.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-B2-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q1/5

In an experiment, plants were assigned to two plant foods at random. Why?

A. so that more of the plants grow well

B. so that no plant needs to be measured

C. so growth differences can be credited to food

D. so the two group means come out equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q2/5

973

QUESTION ID

S29-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. so growth differences can be credited to food

GRADING NOTE Purpose.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-B2-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q2/5

True or false?

In an experiment, subjects choose which treatment they receive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q3/5

974

QUESTION ID

S29-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The researcher assigns.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q3/5

Ten plants are numbered 1-10. Which assignment is random?

A. put the 5 tallest plants on food A

B. let the gardener choose

C. draw 5 numbers from a hat for food A

D. put plants 1-5 on food A

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q4/5

975

QUESTION ID

S29-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. draw 5 numbers from a hat for food A

GRADING NOTE Chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-B2-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q4/5

True or false?

Random assignment is what allows an experiment to support a conclusion about cause.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q5/5

976

QUESTION ID
S29-B2-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Key idea.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q5/5

SET 29 • HSS-IC.B.5

BACKWARD 3

Compute means of small groups • Grade 6 • Statistics & Probability • 6.SP.B.5.c

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 14, 11, 13, 12, 15.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q1/5

978

QUESTION ID

S29-B3-Q1

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE 65/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 10, 12, 9, 11, 8.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q2/5

979

QUESTION ID

S29-B3-Q2

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE 50/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q2/5

True or false?

The difference of the means 13 and 10 is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q3/5

980

QUESTION ID

S29-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 13 − 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q3/5

Find the mean of 2.5, 3, 3.5, 5.

A. 3

B. 3.5

C. 3.25

D. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q4/5

981

QUESTION ID

S29-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 3.5

GRADING NOTE 14/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q4/5

True or false?

The mean of 8, 9, 10, 11, 12 is 11.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q5/5

982

QUESTION ID

S29-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q5/5

SET 29 • HSS-IC.B.5

FORWARD 1

Use differences in either direction • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

(Hypothetical simulation results.) In 200 re-randomizations, differences of 3 or more occurred 4 times and differences of −3 or less occurred 5 times. What proportion is at least 3 away from 0 in either direction? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q1/5

984

QUESTION ID

S29-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.045

GRADING NOTE 9/200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q1/5

True or false?

Under a 5% rule, a proportion of 0.045 means the difference is significant.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q2/5

985

QUESTION ID

S29-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $0.045 < 0.05$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q2/5

With more plants per group, re-randomized differences tend to be

A. always positive

B. closer to 0

C. farther from 0

D. exactly 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q3/5

986

QUESTION ID

S29-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. closer to 0

GRADING NOTE Less variability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F1-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q3/5

Treatment mean 31, control mean 28. What is the difference (treatment minus control)?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q4/5

987

QUESTION ID

S29-F1-Q4

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE 31 − 28.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q4/5

True or false?

A significant difference proves that the treatment helps every individual.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q5/5

988

QUESTION ID

S29-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is about the groups on average.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q5/5

SET 29 • HSS-IC.B.5

FORWARD 2

Evaluate a report of an experiment • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A company reports: "Our drink improved scores in a study." The study gave the drink to 10 volunteers and compared them with 10 who chose not to drink it. What is the main flaw?

A. Drinks cannot be studied in an experiment.

B. Subjects chose their own group: no random assignment.

C. Twenty subjects is too many for a study.

D. Scores of two groups cannot be compared.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q1/5

990

QUESTION ID

S29-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Subjects chose their own group: no random assignment.

GRADING NOTE Self-selection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q1/5

True or false?

A report should say whether a difference could plausibly be due to chance.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q2/5

991

QUESTION ID
S29-F2-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Significance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q2/5

(Hypothetical simulation results.) A report says a new method raised mean scores from 70 to 71, "a significant improvement." Re-randomization shows such differences 40% of the time. What is the best response?

A. The new method actually lowered scores.

B. Proven true: the study was an experiment.

C. Supported: 71 is higher than 70.

D. Not supported: chance often gives such differences.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q3/5

992

QUESTION ID
S29-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. Not supported: chance often gives such differences.

GRADING NOTE 40% ≥ 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F2-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q3/5

True or false?

A randomized experiment on adults automatically applies to children.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q4/5

993

QUESTION ID

S29-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Scope is the subjects.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which detail most strengthens a report's causal claim?

A. subjects were randomly assigned to treatments

B. the study had a long and technical name

C. the researchers felt confident in the result

D. the report used colorful, eye-catching graphs

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q5/5

994

QUESTION ID
S29-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. subjects were randomly assigned to treatments

GRADING NOTE Design.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q5/5

Evaluate the conclusions of reports based on data

MAIN	Probability & Statistics main question	**HSS-IC.B.6**
BACKWARD 1	Know what each design can support	**HSS-IC.B.3**
BACKWARD 2	Recognize lurking variables	**HSS-ID.C.9**
FORWARD 1	Identify the type of bias in a report	**HSS-IC.B.6**
FORWARD 2	Weigh several flaws and decide what a report shows	**HSS-IC.B.6**

SET 30 • HSS-IC.B.6

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A report says: "Teens who eat breakfast get better grades, so eating breakfast raises grades." The data came from a survey of 2,000 randomly chosen teens. What is the main problem?

A. A survey shows association, not that breakfast causes better grades.

B. A random sample of 2,000 teens is far too small for any survey.

C. Grades cannot be compared between different groups of teens.

D. Random samples cannot be used to describe a population of teens.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q1/5

997

QUESTION ID

S30-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. A survey shows association, not that breakfast causes better grades.

GRADING NOTE Cause needs an experiment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A magazine asked readers to mail in whether they liked their jobs; 80% of the 3,000 replies said no. The magazine reports that "80% of workers dislike their jobs." What is the main problem?

A. Readers chose to reply, so the sample is likely biased.

B. Three thousand replies is too few to report.

C. The percent 80% was computed incorrectly.

D. Job satisfaction cannot be surveyed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q2/5

998

QUESTION ID

S30-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Readers chose to reply, so the sample is likely biased.

GRADING NOTE Voluntary response.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A survey asked, "Don't you agree that the unfair new tax should be repealed?" What is the problem?

A. The question is too short to be answered.

B. The wording pushes people toward one answer.

C. The sample must have been too large.

D. Tax questions cannot appear in surveys.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q3/5

999

QUESTION ID
S30-M-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. The wording pushes people toward one answer.

GRADING NOTE Leading question: response bias.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A randomized experiment found that a reading program raised scores of 3rd graders in one district. Which conclusion is best supported?

A. The program will raise scores for students of every age everywhere.

B. The program caused higher scores for students like those in the study.

C. The program is associated with scores but did not cause any change.

D. Nothing can be concluded, because only one district was studied.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q4/5

1000

QUESTION ID
S30-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. The program caused higher scores for students like those in the study.

GRADING NOTE Cause for the subjects; limited generalization.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A report on 12 patients says a new treatment "works for everyone," because 9 of the 12 improved. Which criticism is most important?

A. Twelve patients is too many to study at once.

B. The fraction 9/12 should have been written 12/9.

C. No comparison group, and 3 of the 12 did not improve.

D. Improvement cannot be measured in patients.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q5/5

1001

QUESTION ID

S30-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. No comparison group, and 3 of the 12 did not improve.

GRADING NOTE No control; and "everyone" contradicts the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q5/5

SET 30 • HSS-IC.B.6

BACKWARD 1

Know what each design can support • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

True or false?

Of a sample survey, an observational study and a randomized experiment, the randomized experiment is the design built to support a cause-and-effect conclusion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q1/5

1003

QUESTION ID

S30-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Random assignment makes the groups comparable; the other two designs do not assign treatments.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which design lets results be generalized to a population?

A. a group of volunteers

B. a convenience sample

C. an experiment on friends

D. a random sample from that population

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q2/5

1004

QUESTION ID

S30-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. a random sample from that population

GRADING NOTE Random sampling.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

An observational study can prove cause if it is large enough.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q3/5

1005

QUESTION ID

S30-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Size does not help.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A study used volunteers and random assignment. Its conclusions can include

A. nothing at all

B. cause, for the whole population

C. cause, for subjects like the volunteers

D. association only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q4/5

1006

QUESTION ID

S30-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. cause, for subjects like the volunteers

GRADING NOTE Assignment yes, sampling no.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-B1-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Random sampling and random assignment serve different purposes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q5/5

1007

QUESTION ID

S30-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Generalize versus cause.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • HSS-IC.B.6

BACKWARD 2

Recognize lurking variables • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Cities with more police officers have more crimes. What is a likely lurking variable?

A. the number of officers

B. crime reports

C. city population

D. police uniforms

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q1/5

1009

QUESTION ID

S30-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. city population

GRADING NOTE Size drives both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-B2-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q1/5

True or false?

A lurking variable affects both variables in an association.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q2/5

1010

QUESTION ID
S30-B2-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Students who take music lessons score higher in math. Which is a possible lurking variable?

A. family income

B. the music teacher's name

C. the number of notes played

D. math test length

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q3/5

1011

QUESTION ID

S30-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. family income

GRADING NOTE May affect both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If an association is strong, there can be no lurking variable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q4/5

1012

QUESTION ID

S30-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Strength does not exclude one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q4/5

True or false?

Random assignment protects an experiment from lurking variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q5/5

1013

QUESTION ID

S30-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It balances them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • HSS-IC.B.6

FORWARD 1

Identify the type of bias in a report • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A phone survey calls only landlines during work hours. Which bias is most likely?

A. no bias, because calls are made at random

B. response bias from the wording of questions

C. voluntary response bias from self-selection

D. undercoverage of people away or with only cell phones

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q1/5

1015

QUESTION ID

S30-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. undercoverage of people away or with only cell phones

GRADING NOTE Some groups cannot be reached.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q1/5

Many people chosen for a survey refuse to answer. Which bias is this?

A. nonresponse bias

B. measurement error

C. wording bias

D. voluntary response bias

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q2/5

1016

QUESTION ID

S30-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. nonresponse bias

GRADING NOTE Chosen but missing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A larger sample removes bias caused by a poor sampling method.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q3/5

1017

QUESTION ID

S30-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It only reduces variability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q3/5

A survey of mall shoppers is used to describe all residents' shopping habits. Which bias is this?

A. random sampling error only, with no bias

B. a convenience sample that is not representative

C. wording bias from a leading question

D. nonresponse bias from refusals to answer

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q4/5

1018

QUESTION ID

S30-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. a convenience sample that is not representative

GRADING NOTE Easy to reach, not representative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q4/5

True or false?

People may give socially acceptable answers instead of true ones when asked sensitive questions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q5/5

1019

QUESTION ID

S30-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Response bias.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F1-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q5/5

SET 30 • HSS-IC.B.6

FORWARD 2

Weigh several flaws and decide what a report shows • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

An app maker reports: "Users who used our app for 30 days lost 8 pounds on average." The users were volunteers and there was no comparison group. Which conclusion is justified?

A. Only that these volunteers lost weight; the app's role is unclear.

B. The app causes weight loss in everyone who uses it.

C. Every single user of the app lost exactly 8 pounds.

D. The study has shown that the app has no effect on weight.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q1/5

1021

QUESTION ID

S30-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Only that these volunteers lost weight; the app's role is unclear.

GRADING NOTE No control, no random sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q1/5

True or false?

A report that gives no information about how its sample was chosen should be treated with caution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q2/5

1022

QUESTION ID

S30-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Cannot judge bias.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which change would most improve the app study?

A. ask the users whether they liked the app

B. randomly assign volunteers to use the app or not

C. extend the study from 30 days to 31 days

D. report the weight loss as a percent instead

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q3/5

1023

QUESTION ID

S30-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. randomly assign volunteers to use the app or not

GRADING NOTE Comparison and random assignment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F2-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A study funded by a company is always wrong about that company's product.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q4/5

1024

QUESTION ID

S30-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Funding is a reason to check the design, not proof of error.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Two reports disagree. Report 1 is a randomized experiment with 400 subjects; report 2 is a voluntary online poll of 5,000. Which deserves more weight for a causal claim?

A. equal weight

B. report 1

C. report 2

D. neither

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q5/5

1025

QUESTION ID

S30-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. report 1

GRADING NOTE Design beats size.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q5/5

SET 31 • HSS-IC.B.6 • MAKING INFERENCES AND JUSTIFYING CONCLUSIONS

Evaluate the statistics and displays in reports

MAIN	Probability & Statistics main question	**HSS-IC.B.6**
BACKWARD 1	Use margins of error to compare results	**HSS-IC.B.4**
BACKWARD 2	Compute percent change	**7.RP.A.3**
FORWARD 1	Spot misleading features of displays	**HSS-IC.B.6**
FORWARD 2	Reason about conditional rates in reports	**HSS-CP.A.5**

SET 31 • HSS-IC.B.6

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A bar graph's vertical axis starts at 50. One bar shows 52 and another shows 55. How tall is the second bar compared with the first (ratio of drawn heights), and what is the ratio of the actual values? Give both ratios to two decimal places.

Ratio of drawn heights: _____

Ratio of values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q1/5

1028

QUESTION ID

S31-M-Q1

TYPE

Short answer

CORRECT ANSWER

2.50; 1.06

GRADING NOTE Form required: both, to two decimal places. (55 − 50)/(52 − 50) = 2.5; 55/52 = 1.0577.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q1/5

An ad says a risk "increased 80%," from 1 in 10,000 to 1.8 in 10,000. Which statement is correct?

A. The absolute risk rose by only 0.8 in 10,000.

B. Most people will now be affected by it.

C. The risk is now 80 out of every 100 people.

D. The risk went from 1% of people to 80%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q2/5

1029

QUESTION ID

S31-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. The absolute risk rose by only 0.8 in 10,000.

GRADING NOTE Relative change sounds large; absolute change is tiny.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q2/5

A poll shows candidate A at 51% and B at 49%, with a margin of error of 3 percentage points. A headline says "Poll shows A has a reliable lead among all voters!" What is the best evaluation?

A. Misleading: the gap is well within the margin of error.

B. Correct: A has a majority of all the voters.

C. Misleading: B is certainly ahead of A.

D. Correct: 51% is more than 49%, so A leads.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q3/5

1030

QUESTION ID

S31-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Misleading: the gap is well within the margin of error.

GRADING NOTE 48%−54% and 46%−52% overlap a lot, so the poll does not show a reliable lead. The others are false: 51% vs 49% is a sample result, not the population; a sample does not show a majority of all voters; and B is not certainly ahead.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q3/5

A report gives the mean income in a town as $95,000, but most households earn under $50,000 and a few earn millions. What is the best evaluation?

A. The report must have added incomes incorrectly.

B. The mean is the best summary for skewed data.

C. The range is the best measure of center.

D. The median would describe a typical household better.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q4/5

1031

QUESTION ID

S31-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. The median would describe a typical household better.

GRADING NOTE Right-skewed: the mean is pulled up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q4/5

A pictograph shows sales doubling by drawing a square icon twice as tall and twice as wide. How many times as large is the new icon's area?

Times as large: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q5/5

1032

QUESTION ID

S31-M-Q5

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE 2 · 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q5/5

SET 31 • HSS-IC.B.6

BACKWARD 1

Use margins of error to compare results • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

ANSWER KEY Reduced copy of the question page, with its answer

An estimate is 51% with a margin of error of 3 points. Give the interval.

From: _____

To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q1/5

1034

QUESTION ID

S31-B1-Q1

TYPE

Short answer

CORRECT ANSWER

48%; 54%

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-B1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q1/5

True or false?

If two intervals overlap a lot, a cautious reading is not to claim from them which value is larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q2/5

1035

QUESTION ID

S31-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Overlap is only a cautious reading; a formal comparison would use the margin of error of the difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which pair of estimates (each ± 2 points) clearly differs?

A. 48% and 50%

B. 40% and 50%

C. 49% and 51%

D. 45% and 47%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q3/5

1036

QUESTION ID

S31-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 40% and 50%

GRADING NOTE 38-42 and 48-52 do not overlap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-B1-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q3/5

True or false?

A margin of error of 3 points means the estimate is off by exactly 3 points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q4/5

1037

QUESTION ID

S31-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It bounds plausible values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q4/5

An interval is 46% to 52%. What is the estimate at its center?

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q5/5

1038

QUESTION ID

S31-B1-Q5

TYPE

Short answer

CORRECT ANSWER

49%

GRADING NOTE Midpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-B1-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q5/5

SET 31 • HSS-IC.B.6

BACKWARD 2

Compute percent change • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A value rises from 1 to 1.8. What is the percent increase?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q1/5

1040

QUESTION ID

S31-B2-Q1

TYPE

Short answer

CORRECT ANSWER

80%

GRADING NOTE 0.8/1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A price rises from $40 to $50. What is the percent increase?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q2/5

1041

QUESTION ID

S31-B2-Q2

TYPE

Short answer

CORRECT ANSWER

25%

GRADING NOTE 10/40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A rise from 2 to 3 is a 50% increase.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q3/5

1042

QUESTION ID

S31-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A count falls from 200 to 150. What is the percent decrease?

A. 33%

B. 75%

C. 50%

D. 25%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q4/5

1043

QUESTION ID

S31-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 25%

GRADING NOTE 50/200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A rise from 50 to 55 is a 5% increase.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q5/5

1044

QUESTION ID

S31-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 5/50 = 10%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q5/5

SET 31 • HSS-IC.B.6

FORWARD 1

Spot misleading features of displays • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A line graph of monthly sales has a vertical axis from 980 to 1,000. What effect does this have?

A. Large changes look very small.

B. It hides the months.

C. Small changes look very large.

D. It has no effect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q1/5

1046

QUESTION ID

S31-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Small changes look very large.

GRADING NOTE Truncated axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q1/5

True or false?

A pie chart whose percents add to 130% is misleading or in error.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q2/5

1047

QUESTION ID

S31-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Parts of one whole add to 100%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A graph's vertical axis starts at 90. Bars show 92 and 98. What is the ratio of the drawn heights?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q3/5

1048

QUESTION ID

S31-F1-Q3

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE 8/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F1-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q3/5

True or false?

Using a 3-D effect on a pie chart never changes how large slices appear.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q4/5

1049

QUESTION ID

S31-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Perspective distorts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which display choice is most honest for comparing two counts of 52 and 55?

A. a bar graph with the axis starting at 51

B. a 3-D pie chart

C. a bar graph with the axis starting at 0

D. icons of different widths

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q5/5

1050

QUESTION ID

S31-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. a bar graph with the axis starting at 0

GRADING NOTE Zero baseline.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q5/5

SET 31 • HSS-IC.B.6

FORWARD 2

Reason about conditional rates in reports • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5

ANSWER KEY Reduced copy of the question page, with its answer

A report says "most car accidents happen within 5 miles of home, so driving near home is dangerous." What is the main flaw?

A. Car accidents cannot be counted accurately by anyone.

B. A distance of five miles is too short to be studied.

C. Cars are always safer when driven far from home.

D. It uses counts without the amount of driving near and far from home.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q1/5

1052

QUESTION ID

S31-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. It uses counts without the amount of driving near and far from home.

GRADING NOTE A fair comparison needs exposure, such as accidents per mile driven near and far from home; the report gives counts only. The others are unsupported claims.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q1/5

True or false?

Knowing that 90% of people in a hospital are sick does not tell you the chance that a sick person is in the hospital.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q2/5

1053

QUESTION ID

S31-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE P(sick given in the hospital) differs from P(in the hospital given sick).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q2/5

A test catches 95% of people who have a rare disease. A report concludes that 95% of people who test positive have it. What is the error?

A. Rare diseases cannot be tested for at all.

B. It confuses P(positive given disease) with P(disease given positive).

C. A test must catch 100% of cases before it is reported.

D. The 95% should have been written as a fraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q3/5

1054

QUESTION ID
S31-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. It confuses P(positive given disease) with P(disease given positive).

GRADING NOTE Conditional probabilities are not symmetric.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q3/5

True or false?

A higher number of accidents involving red cars proves that red cars are more dangerous.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q4/5

1055

QUESTION ID

S31-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It depends on how many red cars there are.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q4/5

To compare the safety of two car colors fairly, which statistic is needed?

A. the number of different car colors sold

B. the average price of the cars of each color

C. accidents per car (or per mile) for each color

D. the total number of accidents for each color

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q5/5

1056

QUESTION ID

S31-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. accidents per car (or per mile) for each color

GRADING NOTE A rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q5/5

Describe sample spaces and events

MAIN	Probability & Statistics main question	**HSS-CP.A.1**
BACKWARD 1	List a sample space with a list, table or tree	**7.SP.C.8.b**
BACKWARD 2	Find probabilities in a uniform model	**7.SP.C.7.a**
FORWARD 1	Describe events in multi-stage sample spaces	**HSS-CP.A.1**
FORWARD 2	Combine events with "and", "or" and "not"	**HSS-CP.A.1**

SET 32 • HSS-CP.A.1

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A fair coin is flipped and a fair six-sided die is rolled, independently. How many outcomes are in the sample space?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q1/5

1059

QUESTION ID

S32-M-Q1

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE 2 · 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two fair coins are flipped independently; the sample space is HH, HT, TH, TT. How many outcomes are in the event "at least one head"?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q2/5

1060

QUESTION ID

S32-M-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE HH, HT, TH.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q2/5

A fair coin is flipped twice, independently, and the sequence of results is recorded. Which is the sample space?

A. {H, T}

B. {HH, HT, TT}

C. {HH, TT}

D. {HH, HT, TH, TT}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q3/5

1061

QUESTION ID

S32-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. {HH, HT, TH, TT}

GRADING NOTE Order matters: HT and TH differ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q3/5

Two fair dice are rolled independently. How many of the 36 equally likely outcomes are in the event "the sum is 7"?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q4/5

1062

QUESTION ID

S32-M-Q4

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE (1,6), (2,5), (3,4), (4,3), (5,2), (6,1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q4/5

A spinner has 8 equal sections numbered 1 to 8. Event A is "an even number greater than 3". How many outcomes are in A, and what is P(A)? Give a fraction in lowest terms.

Outcomes in A: _____

P(A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q5/5

1063

QUESTION ID

S32-M-Q5

TYPE

Short answer

CORRECT ANSWER

3; 3/8

GRADING NOTE Form required: both. {4, 6, 8}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q5/5

SET 32 • HSS-CP.A.1

BACKWARD 1

List a sample space with a list, table or tree • Grade 7 • Statistics & Probability • 7.SP.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

A bag has a red and a blue marble; a fair coin is flipped. How many outcomes (color, coin) are there?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q1/5

1065

QUESTION ID

S32-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE 2 · 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q1/5

True or false?

A tree diagram for flipping a coin 3 times (heads or tails each time) has 8 final branches.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q2/5

1066

QUESTION ID

S32-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 2 · 2 · 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A sandwich shop offers 3 breads and 4 fillings. How many bread-filling choices are there?

A. 12

B. 81

C. 7

D. 34

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q3/5

1067

QUESTION ID

S32-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 12

GRADING NOTE $3 \cdot 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q3/5

True or false?

The sample space for rolling two six-sided dice (as ordered pairs) has 12 outcomes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q4/5

1068

QUESTION ID

S32-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 36.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A spinner with 3 equal colors is spun twice. How many outcomes are in the sample space?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q5/5

1069

QUESTION ID

S32-B1-Q5

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE 3 · 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q5/5

SET 32 • HSS-CP.A.1

BACKWARD 2

Find probabilities in a uniform model • Grade 7 • Statistics & Probability • 7.SP.C.7.a

ANSWER KEY Reduced copy of the question page, with its answer

A fair die is rolled. What is P(a number greater than 4)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q1/5

1071

QUESTION ID

S32-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE {5, 6}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q1/5

True or false?

In a uniform model, every outcome has the same probability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q2/5

1072

QUESTION ID

S32-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A letter is chosen at random from the word MATH. What is P(vowel)?

A. 1/2

B. 1/3

C. 1/4

D. 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q3/5

1073

QUESTION ID

S32-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1/4

GRADING NOTE A only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A number is chosen at random from 1 to 20. What is P(multiple of 5)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q4/5

1074

QUESTION ID

S32-B2-Q4

TYPE

Short answer

CORRECT ANSWER

1/5

GRADING NOTE 5, 10, 15, 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A spinner with sections of different sizes is a uniform model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q5/5

1075

QUESTION ID

S32-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Outcomes are not equally likely.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q5/5

SET 32 • HSS-CP.A.1

FORWARD 1

Describe events in multi-stage sample spaces • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Three fair coins are flipped independently. How many outcomes are in the sample space, and how many have exactly 2 heads?

Outcomes: _____

Exactly 2 heads: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q1/5

1077

QUESTION ID

S32-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8; 3

GRADING NOTE Form required: both. HHT, HTH, THH.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Three fair coins are flipped independently. What is P(exactly 2 heads)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q2/5

1078

QUESTION ID
S32-F1-Q2

TYPE
Short answer

CORRECT ANSWER
3/8

GRADING NOTE 3 of 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q2/5

True or false?

For two fair dice rolled independently, the event "doubles" contains 6 outcomes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q3/5

1079

QUESTION ID

S32-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (1,1) ... (6,6).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For two fair dice rolled independently, which event contains exactly 4 outcomes?

A. the sum is 7

B. the sum is 12

C. doubles

D. the sum is 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q4/5

1080

QUESTION ID

S32-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. the sum is 5

GRADING NOTE (1,4), (2,3), (3,2), (4,1). Sum 7 has 6; doubles 6; sum 12 has 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Two fair dice are rolled independently. How many outcomes have a sum of 10 or more?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q5/5

1081

QUESTION ID

S32-F1-Q5

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Sum 10: 3; sum 11: 2; sum 12: 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F1-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q5/5

SET 32 • HSS-CP.A.1

FORWARD 2

Combine events with "and", "or" and "not" • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A fair six-sided die is rolled. A = "even", B = "greater than 3". How many outcomes are in "A and B"?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q1/5

1083

QUESTION ID

S32-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE {4, 6}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A fair six-sided die is rolled. A = "even", B = "greater than 3". How many outcomes are in "A or B"?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q2/5

1084

QUESTION ID

S32-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE {2, 4, 5, 6}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q2/5

True or false?

The event "not A" contains every outcome of the sample space that is not in A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q3/5

1085

QUESTION ID

S32-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Complement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A fair six-sided die is rolled. A = "even". Which set is "not A"?

A. {2, 4, 6}

B. {1, 2, 3}

C. { }

D. {1, 3, 5}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q4/5

1086

QUESTION ID

S32-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. {1, 3, 5}

GRADING NOTE Odd outcomes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q4/5

True or false?

The event "A and B" always contains at least as many outcomes as "A or B".

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q5/5

1087

QUESTION ID

S32-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Never more.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q5/5

Unions, intersections and complements of events

MAIN	Probability & Statistics main question	**HSS-CP.A.1**
BACKWARD 1	Count the outcomes in an event	**HSS-CP.A.1**
BACKWARD 2	Find a probability by counting	**7.SP.C.8.a**
FORWARD 1	Use a Venn diagram of counts	**HSS-CP.A.1**
FORWARD 2	Apply the Addition Rule	**HSS-CP.B.7**

SET 33 • HSS-CP.A.1

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A fair die is rolled. A = "even" and B = "greater than 3". How many outcomes are in A and B, and what is P(A and B)? Give a fraction in lowest terms.

Outcomes: _____

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q1/5

1090

QUESTION ID

S33-M-Q1

TYPE

Short answer

CORRECT ANSWER

2; 1/3

GRADING NOTE Form required: both. {4, 6}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A fair die is rolled. A = "even" and B = "greater than 3". How many outcomes are in A or B, and what is P(A or B)? Give a fraction in lowest terms.

Outcomes: _____

P(A or B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q2/5

1091

QUESTION ID

S33-M-Q2

TYPE

Short answer

CORRECT ANSWER

4; 2/3

GRADING NOTE Form required: both. {2, 4, 5, 6}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q2/5

A fair die is rolled. A = "even". What is P(not A)? Give a fraction in lowest terms.

P(not A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q3/5

1092

QUESTION ID

S33-M-Q3

TYPE

Short answer

CORRECT ANSWER

1/2

GRADING NOTE {1, 3, 5}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A fair die is rolled. A = "even" and B = "greater than 3". Which describes the event {1, 3}?

A. A or B

B. A and B

C. not (A or B)

D. not A

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q4/5

1093

QUESTION ID

S33-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. not (A or B)

GRADING NOTE Neither even nor greater than 3. "not A" is {1, 3, 5}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A card is drawn at random from a standard 52-card deck. H = "heart", F = "face card (jack, queen or king)". Find P(H and F) and P(H or F). Give a fraction in lowest terms.

P(H and F): _____

P(H or F): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q5/5

1094

QUESTION ID

S33-M-Q5

TYPE

Short answer

CORRECT ANSWER

3/52; 11/26

GRADING NOTE Form required: both. 3 heart face cards; 13 + 12 − 3 = 22, and 22/52 = 11/26.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q5/5

SET 33 • HSS-CP.A.1

BACKWARD 1

Count the outcomes in an event • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A number from 1 to 10 is chosen. How many outcomes are in "multiple of 3"?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q1/5

1096

QUESTION ID

S33-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE 3, 6, 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-B1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q1/5

True or false?

For one roll of a fair six-sided die, the event "less than 3" is {1, 2}.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q2/5

1097

QUESTION ID

S33-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE List.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A letter is chosen from the word PROBABILITY. How many outcomes (letter positions) are in "the letter B"?

A. 11

B. 1

C. 3

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q3/5

1098

QUESTION ID

S33-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2

GRADING NOTE B appears twice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

How many cards in a standard 52-card deck are face cards?

Cards: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q4/5

1099

QUESTION ID
S33-B1-Q4

TYPE
Short answer

CORRECT ANSWER
12

GRADING NOTE 3 per suit · 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-B1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q4/5

True or false?

A standard deck has 4 aces and 13 hearts, so it has 17 cards that are aces or hearts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q5/5

1100

QUESTION ID

S33-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The ace of hearts is counted twice: 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • HSS-CP.A.1

BACKWARD 2

Find a probability by counting • Grade 7 • Statistics & Probability • 7.SP.C.8.a

ANSWER KEY Reduced copy of the question page, with its answer

Two fair coins are flipped independently. What is P(two heads)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q1/5

1102

QUESTION ID

S33-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1/4

GRADING NOTE 1 of 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A fair six-sided die is rolled. What is P(1 or 2)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q2/5

1103

QUESTION ID

S33-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE 2 of 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q2/5

True or false?

The probability of an event plus the probability of its complement is 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q3/5

1104

QUESTION ID

S33-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Complements.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q3/5

A card is drawn at random from a standard 52-card deck. What is P(king)?

A. 1/4

B. 1/52

C. 4/13

D. 1/13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q4/5

1105

QUESTION ID

S33-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1/13

GRADING NOTE 4/52.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If P(A) = 0.3, then P(not A) = 0.3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q5/5

1106

QUESTION ID

S33-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 0.7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q5/5

SET 33 • HSS-CP.A.1

FORWARD 1

Use a Venn diagram of counts • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

Of 30 students, 18 play soccer, 12 play basketball and 5 play both. How many play soccer only, and how many play neither?

Soccer only: _____

Neither: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q1/5

1108

QUESTION ID

S33-F1-Q1

TYPE

Short answer

CORRECT ANSWER

13; 5

GRADING NOTE Form required: both. 18 − 5; 30 − (18 + 12 − 5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Of 30 students, 18 play soccer, 12 play basketball and 5 play both. A student is chosen at random. What is P(soccer or basketball)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q2/5

1109

QUESTION ID

S33-F1-Q2

TYPE

Short answer

CORRECT ANSWER

5/6

GRADING NOTE 25/30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q2/5

True or false?

In a Venn diagram, the overlap of two circles shows "A and B".

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q3/5

1110

QUESTION ID

S33-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Intersection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Of 30 students, 18 play soccer, 12 play basketball and 5 play both. What is P(not soccer)?

A. 3/5

B. 2/5

C. 7/30

D. 1/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q4/5

1111

QUESTION ID

S33-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 2/5

GRADING NOTE 12/30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q4/5

True or false?

Of 30 students, 18 play soccer, 12 play basketball and 5 play both. Adding 18 and 12 gives the number who play soccer or basketball.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q5/5

1112

QUESTION ID

S33-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The 5 are counted twice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q5/5

SET 33 • HSS-CP.A.1

FORWARD 2

Apply the Addition Rule • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.5, P(B) = 0.4 and P(A and B) = 0.2. Find P(A or B).

P(A or B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q1/5

1114

QUESTION ID

S33-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.7

GRADING NOTE 0.5 + 0.4 − 0.2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A card is drawn at random from a standard 52-card deck. Find P(king or heart). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q2/5

1115

QUESTION ID

S33-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4/13

GRADING NOTE 4/52 + 13/52 − 1/52 = 16/52.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q2/5

True or false?

If A and B cannot happen together, P(A or B) = P(A) + P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q3/5

1116

QUESTION ID

S33-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE P(A and B) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.6, P(B) = 0.3, P(A or B) = 0.7. What is P(A and B)?

A. 0.9

B. 0.2

C. 0.1

D. 0.18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q4/5

1117

QUESTION ID

S33-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0.2

GRADING NOTE 0.6 + 0.3 − 0.7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q4/5

True or false?

P(A or B) can be greater than 1 if A and B are both likely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q5/5

1118

QUESTION ID

S33-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Never above 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q5/5

Decide independence with the product rule

MAIN	Probability & Statistics main question	**HSS-CP.A.2**
BACKWARD 1	Multiply probabilities	**5.NF.B.4**
BACKWARD 2	Find P(A and B) by counting	**HSS-CP.A.1**
FORWARD 1	Use independence for several events	**HSS-CP.A.2**
FORWARD 2	Interpret independence with conditional probability	**HSS-CP.A.3**

SET 34 • HSS-CP.A.2

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.4, P(B) = 0.5 and P(A and B) = 0.2. Find P(A)·P(B), and decide whether A and B are independent (enter yes or no).

P(A)·P(B): _____

Independent (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q1/5

1121

QUESTION ID

S34-M-Q1

TYPE

Short answer

CORRECT ANSWER

0.2; yes

GRADING NOTE Form required: both. 0.2 = 0.2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.3, P(B) = 0.6 and P(A and B) = 0.25. Find P(A)·P(B), and decide whether A and B are independent (enter yes or no).

P(A)·P(B): _____

Independent (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q2/5

1122

QUESTION ID

S34-M-Q2

TYPE

Short answer

CORRECT ANSWER

0.18; no

GRADING NOTE Form required: both. 0.18 ≠ 0.25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A and B are independent with P(A) = 0.7 and P(B) = 0.2. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q3/5

1123

QUESTION ID
S34-M-Q3

TYPE
Short answer

CORRECT ANSWER
0.14

GRADING NOTE 0.7 · 0.2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q3/5

A fair six-sided die is rolled. A = "even" and B = "1 or 2". Are A and B independent?

A. No: A and B share the outcome 2.

B. No: P(A and B) = 1/6, not 1/2.

C. Yes: P(A and B) = 1/6 = (1/2)(1/3).

D. Yes: A and B have no outcomes in common.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q4/5

1124

QUESTION ID

S34-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Yes: P(A and B) = 1/6 = (1/2)(1/3).

GRADING NOTE A and B = {2}. Sharing an outcome does not decide independence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In the survey table, a person is chosen at random. M = "man" and Y = "said yes". Find P(M and Y), and decide whether M and Y are independent (enter yes or no).

	Yes	No	Total
Men	24	16	40
Women	36	24	60
Total	60	40	100

P(M and Y): _____

Independent (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q5/5

1125

QUESTION ID

S34-M-Q5

TYPE

Short answer

CORRECT ANSWER

0.24; yes

GRADING NOTE Form required: both. 24/100; P(M)·P(Y) = 0.4 · 0.6 = 0.24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q5/5

SET 34 • HSS-CP.A.2

BACKWARD 1

Multiply probabilities • Grade 5 • Number & Operations–Fractions • 5.NF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Compute (1/2)(1/3). Give a fraction in lowest terms.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q1/5

1127

QUESTION ID

S34-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1/6

GRADING NOTE 1/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 0.4 · 0.5.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q2/5

1128

QUESTION ID

S34-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.2

GRADING NOTE Decimal product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

(2/3)(3/4) = 1/2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q3/5

1129

QUESTION ID

S34-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 6/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 0.3 · 0.6.

A. 1.8

B. 0.36

C. 0.18

D. 0.9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q4/5

1130

QUESTION ID

S34-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 0.18

GRADING NOTE 3 · 6 = 18, two decimal places.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q4/5

True or false?

$0.7 \cdot 0.2 = 1.4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q5/5

1131

QUESTION ID

S34-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 0.14.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • HSS-CP.A.2

BACKWARD 2

Find P(A and B) by counting • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A fair six-sided die is rolled. What is P(even and greater than 3)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q1/5

1133

QUESTION ID

S34-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE {4, 6}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-B2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A fair coin is flipped and a fair die is rolled, independently. What is P(heads and 6)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q2/5

1134

QUESTION ID

S34-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1/12

GRADING NOTE 1 of 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-B2-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q2/5

True or false?

For one roll of a fair six-sided die, P(odd and even) = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q3/5

1135

QUESTION ID

S34-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE No outcome is both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two fair coins are flipped independently. What is P(first is heads and second is tails)?

A. 3/4

B. 1/4

C. 1

D. 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q4/5

1136

QUESTION ID

S34-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1/4

GRADING NOTE HT only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-B2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q4/5

True or false?

For one roll of a fair six-sided die, P(less than 3 and odd) = 1/3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q5/5

1137

QUESTION ID

S34-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE {1}: 1/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q5/5

SET 34 • HSS-CP.A.2

FORWARD 1

Use independence for several events • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A fair coin is flipped 3 times, independently. What is P(all three heads)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q1/5

1139

QUESTION ID

S34-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1/8

GRADING NOTE $(1/2)^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Each of 3 independent lights works with probability 0.9. What is P(all 3 work), and P(at least one fails)?

All work: _____

At least one fails: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q2/5

1140

QUESTION ID

S34-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.729; 0.271

GRADING NOTE Form required: both. 0.9^3; $1 - 0.729$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q2/5

True or false?

For independent events, P(A and B and C) = P(A)·P(B)·P(C).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q3/5

1141

QUESTION ID

S34-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Extends the product rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q3/5

A fair die is rolled twice, independently. What is P(two sixes)?

A. 1/6

B. 1/36

C. 2/6

D. 1/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q4/5

1142

QUESTION ID

S34-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1/36

GRADING NOTE (1/6)(1/6).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q4/5

True or false?

If P(A and B) = P(A)·P(B), then A and B cannot happen together.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q5/5

1143

QUESTION ID

S34-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Independent events can happen together; that describes mutually exclusive events.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q5/5

SET 34 • HSS-CP.A.2

FORWARD 2

Interpret independence with conditional probability • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

P(A and B) = 0.2 and P(B) = 0.5. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q1/5

1145

QUESTION ID

S34-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.4

GRADING NOTE 0.2/0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q1/5

True or false?

If A and B are independent, P(A given B) = P(A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q2/5

1146

QUESTION ID

S34-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Knowing B does not change A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q2/5

P(A) = 0.4 and P(A given B) = 0.4. What can be concluded?

A. A and B are independent.

B. P(B) = 0.4.

C. B always happens.

D. A and B are mutually exclusive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q3/5

1147

QUESTION ID

S34-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A and B are independent.

GRADING NOTE Equal conditional and unconditional probability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F2-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.3, P(B) = 0.6 and P(A and B) = 0.18. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q4/5

1148

QUESTION ID

S34-F2-Q4

TYPE

Short answer

CORRECT ANSWER

0.3

GRADING NOTE 0.18/0.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q4/5

True or false?

If P(A given B) is greater than P(A), then A and B are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q5/5

1149

QUESTION ID

S34-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE B changes A's chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q5/5

Compute conditional probability as P(A and B)/P(B)

MAIN	Probability & Statistics main question	**HSS-CP.A.3**
BACKWARD 1	Divide decimals and fractions	**6.NS.B.3**
BACKWARD 2	Count outcomes in a restricted sample space	**HSS-CP.A.1**
FORWARD 1	Compute conditional probability in both directions	**HSS-CP.A.3**
FORWARD 2	Test independence with conditional probability	**HSS-CP.A.3**

SET 35 • HSS-CP.A.3

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

P(A and B) = 0.12 and P(B) = 0.4. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q1/5

1152

QUESTION ID

S35-M-Q1

TYPE

Short answer

CORRECT ANSWER

0.3

GRADING NOTE 0.12/0.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q1/5

A fair die is rolled. What is P(6 given the roll is even)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q2/5

1153

QUESTION ID

S35-M-Q2

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE P(6)/P(even) = (1/6)/(1/2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Two fair dice are rolled independently. What is P(doubles given the sum is 8)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q3/5

1154

QUESTION ID
S35-M-Q3

TYPE
Short answer

CORRECT ANSWER
1/5

GRADING NOTE Sum 8: (2,6), (3,5), (4,4), (5,3), (6,2); one is doubles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q3/5

On a given day, P(rain) = 0.3 and P(rain and wind) = 0.12. What is P(wind given rain)?

A. 0.4

B. 2.5

C. 0.12

D. 0.036

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q4/5

1155

QUESTION ID

S35-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 0.4

GRADING NOTE 0.12/0.3. 0.036 multiplies; 2.5 inverts the division.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

P(B) = 0.25 and P(A given B) = 0.6. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q5/5

1156

QUESTION ID

S35-M-Q5

TYPE

Short answer

CORRECT ANSWER

0.15

GRADING NOTE 0.6 · 0.25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q5/5

SET 35 • HSS-CP.A.3

BACKWARD 1

Divide decimals and fractions • Grade 6 • The Number System • 6.NS.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Compute 0.12 ÷ 0.4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q1/5

1158

QUESTION ID

S35-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.3

GRADING NOTE 1.2 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 0.18 ÷ 0.6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q2/5

1159

QUESTION ID

S35-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.3

GRADING NOTE 1.8 ÷ 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q2/5

True or false?

$0.2 \div 0.5 = 0.4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q3/5

1160

QUESTION ID

S35-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 2 ÷ 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 0.05 ÷ 0.2.

A. 0.25

B. 0.01

C. 2.5

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q4/5

1161

QUESTION ID

S35-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 0.25

GRADING NOTE 0.5 ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q4/5

True or false?

$0.3 \div 0.6 = 2.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q5/5

1162

QUESTION ID

S35-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • HSS-CP.A.3

BACKWARD 2

Count outcomes in a restricted sample space • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

A fair six-sided die is rolled and shows an even number. How many outcomes remain possible?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q1/5

1164

QUESTION ID

S35-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE {2, 4, 6}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-B2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two fair dice are rolled independently and the sum is 8. How many outcomes give a sum of 8?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q2/5

1165

QUESTION ID

S35-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE (2,6) to (6,2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-B2-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q2/5

True or false?

A card is drawn from a standard 52-card deck. Knowing that it is a heart leaves 13 possible cards.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q3/5

1166

QUESTION ID
S35-B2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The hearts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q3/5

A fair six-sided die is rolled and shows a number greater than 2. How many of the remaining outcomes are odd?

A. 2

B. 3

C. 4

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q4/5

1167

QUESTION ID

S35-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2

GRADING NOTE {3, 4, 5, 6}: 3 and 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-B2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q4/5

True or false?

A card is drawn from a standard 52-card deck. Knowing that it is a face card leaves 13 possible cards.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q5/5

1168

QUESTION ID
S35-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q5/5

SET 35 • HSS-CP.A.3

FORWARD 1

Compute conditional probability in both directions • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A card is drawn at random from a standard 52-card deck. Find P(king given face card) and P(face card given king). Give a fraction in lowest terms.

P(king given face card): _____

P(face card given king): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q1/5

1170

QUESTION ID

S35-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1/3; 1

GRADING NOTE Form required: both. 4/12 and 4/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q1/5

True or false?

P(A given B) and P(B given A) are always equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q2/5

1171

QUESTION ID

S35-F1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The king example shows otherwise.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.5, P(B) = 0.2 and P(A and B) = 0.1. Find P(A given B) and P(B given A).

P(A given B): _____

P(B given A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q3/5

1172

QUESTION ID

S35-F1-Q3

TYPE

Short answer

CORRECT ANSWER

0.5; 0.2

GRADING NOTE Form required: both. 0.1/0.2 and 0.1/0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-F1-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q3/5

Which formula gives P(B given A)?

A. P(A)/P(B)

B. P(A)·P(B)

C. P(A and B)/P(A)

D. P(A and B)/P(B)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q4/5

QUESTION ID

S35-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. P(A and B)/P(A)

GRADING NOTE Divide by the condition's probability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q4/5

True or false?

P(A given B) = P(B given A) exactly when P(A) = P(B), provided P(A and B) > 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q5/5

1174

QUESTION ID

S35-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both equal P(A and B) divided by equal numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q5/5

SET 35 • HSS-CP.A.3

FORWARD 2

Test independence with conditional probability • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.6, P(B) = 0.5, P(A and B) = 0.3. Find P(A given B), and decide whether A and B are independent (enter yes or no).

P(A given B): _____

Independent (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q1/5

1176

QUESTION ID

S35-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.6; yes

GRADING NOTE Form required: both. 0.3/0.5 = P(A).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q1/5

True or false?

If P(A given B) = P(A), then also P(B given A) = P(B), provided P(A) > 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q2/5

1177

QUESTION ID

S35-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Symmetric.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Two cards are drawn at random, without replacement, from a standard 52-card deck. Are "first card is an ace" and "second card is an ace" independent?

A. No: two aces can never be drawn one after another.

B. Yes: the deck always contains exactly 4 aces.

C. Yes: each card in the deck is equally likely to be drawn.

D. No: the first draw changes the chance for the second.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q3/5

1178

QUESTION ID
S35-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. No: the first draw changes the chance for the second.

GRADING NOTE P(second ace given first ace) = 3/51 ≠ 4/52.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.4, P(B) = 0.5, P(A and B) = 0.1. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q4/5

1179

QUESTION ID
S35-F2-Q4

TYPE
Short answer

CORRECT ANSWER
0.2

GRADING NOTE 0.1/0.5 ≠ 0.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q4/5

True or false?

If P(A given B) = 0.2 and P(A) = 0.4, then A and B are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q5/5

1180

QUESTION ID

S35-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Not equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q5/5

Interpret independence through conditional probability

MAIN	Probability & Statistics main question	**HSS-CP.A.3**
BACKWARD 1	Compute a conditional probability	**HSS-CP.A.3**
BACKWARD 2	Test independence with the product rule	**HSS-CP.A.2**
FORWARD 1	Check both directions of independence	**HSS-CP.A.3**
FORWARD 2	Approximate conditional probabilities from data	**HSS-CP.A.4**

SET 36 • HSS-CP.A.3

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.5, P(B) = 0.4 and P(A and B) = 0.2. Find P(A given B), and decide whether A and B are independent (enter yes or no).

P(A given B): _____

Independent (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q1/5

1183

QUESTION ID

S36-M-Q1

TYPE

Short answer

CORRECT ANSWER

0.5; yes

GRADING NOTE Form required: both. 0.2/0.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.3, P(B) = 0.5 and P(A and B) = 0.1. Find P(A given B), and decide whether A and B are independent (enter yes or no).

P(A given B): _____

Independent (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q2/5

1184

QUESTION ID

S36-M-Q2

TYPE

Short answer

CORRECT ANSWER

0.2; no

GRADING NOTE Form required: both. 0.1/0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q2/5

A and B are independent. Which statement is correct?

A. Whenever B happens, then A must also happen.

B. P(A) and P(B) must be equal to each other.

C. A and B can never happen at the same time as each other.

D. Knowing that B happened does not change the probability of A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q3/5

1185

QUESTION ID

S36-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Knowing that B happened does not change the probability of A.

GRADING NOTE Meaning of independence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two cards are drawn at random from a standard 52-card deck. Find P(second is an ace given the first is an ace) when the first card is replaced, and when it is not replaced. Give a fraction in lowest terms.

With replacement: _____

Without replacement: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q4/5

1186

QUESTION ID

S36-M-Q4

TYPE

Short answer

CORRECT ANSWER

1/13; 1/17

GRADING NOTE Form required: both. 4/52 and 3/51.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q4/5

Which pair of events is independent?

A. a student studies; the student passes the test

B. first card drawn is an ace; second card (not replaced) is an ace

C. it rains today; the ground is wet today

D. a die shows 6; a separate coin shows heads

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q5/5

1187

QUESTION ID

S36-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. a die shows 6; a separate coin shows heads

GRADING NOTE Separate devices do not affect each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q5/5

SET 36 • HSS-CP.A.3

BACKWARD 1

Compute a conditional probability • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

P(A and B) = 0.06 and P(B) = 0.3. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q1/5

1189

QUESTION ID

S36-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.2

GRADING NOTE 0.06/0.3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-B1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A and B) = 0.25 and P(A) = 0.5. Find P(B given A).

P(B given A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q2/5

1190

QUESTION ID

S36-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.5

GRADING NOTE 0.25/0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-B1-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q2/5

True or false?

P(A given B) = P(A and B)/P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q3/5

1191

QUESTION ID

S36-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A and B) = 0.1, P(B) = 0.4. Find P(A given B).

A. 0.4

B. 0.25

C. 4

D. 0.04

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q4/5

1192

QUESTION ID

S36-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0.25

GRADING NOTE 0.1/0.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-B1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q4/5

True or false?

P(A given B) = P(A and B)·P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q5/5

1193

QUESTION ID

S36-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Divide, not multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q5/5

SET 36 • HSS-CP.A.3

BACKWARD 2

Test independence with the product rule • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2

True or false?

If P(A) = 0.5, P(B) = 0.4 and P(A and B) = 0.2, A and B are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q1/5

1195

QUESTION ID

S36-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $0.5 \cdot 0.4 = 0.2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.2 and P(B) = 0.5. If A and B are independent, what is P(A and B)?

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q2/5

1196

QUESTION ID

S36-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.1

GRADING NOTE Product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-B2-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q2/5

P(A) = 0.3, P(B) = 0.4, P(A and B) = 0.2. Are A and B independent?

A. no, because 0.2 > 0

B. yes, because 0.3 < 0.4

C. yes, because 0.2 < 1

D. no, because 0.3 · 0.4 = 0.12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q3/5

1197

QUESTION ID

S36-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. no, because 0.3 · 0.4 = 0.12

GRADING NOTE 0.12 ≠ 0.2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-B2-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q3/5

True or false?

Mutually exclusive events with positive probabilities are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q4/5

1198

QUESTION ID

S36-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $P(A \text{ and } B) = 0 \neq P(A)\cdot P(B)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q4/5

P(A) = 0.6 and P(B) = 0.5 for independent events. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q5/5

1199

QUESTION ID

S36-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0.3

GRADING NOTE 0.6 · 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q5/5

SET 36 • HSS-CP.A.3

FORWARD 1

Check both directions of independence • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.6, P(B) = 0.25 and P(A and B) = 0.15. Find P(A given B) and P(B given A).

P(A given B): _____

P(B given A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q1/5

1201

QUESTION ID

S36-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.6; 0.25

GRADING NOTE Form required: both. Both equal the unconditional values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q1/5

True or false?

If P(A given B) = P(A), then A and B are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q2/5

1202

QUESTION ID

S36-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition (when P(B) > 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.5, P(A given B) = 0.5, and P(B) = 0.3. What is P(B given A)?

A. 0.15

B. 0.6

C. 0.5

D. 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q3/5

1203

QUESTION ID

S36-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0.3

GRADING NOTE Independence works both ways.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F1-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A and B are independent, P(A) = 0.7 and P(B) = 0.4. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q4/5

1204

QUESTION ID

S36-F1-Q4

TYPE

Short answer

CORRECT ANSWER

0.7

GRADING NOTE Independence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q4/5

True or false?

If P(B given A) = 0.25 and P(B) = 0.5, then A and B are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q5/5

1205

QUESTION ID

S36-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Not equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q5/5

SET 36 • HSS-CP.A.3

FORWARD 2

Approximate conditional probabilities from data • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

In the survey table, a person is chosen at random. Find P(yes given man) and P(yes).

	Yes	No	Total
Men	24	16	40
Women	36	24	60
Total	60	40	100

P(yes given man): _____

P(yes): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q1/5

1207

QUESTION ID

S36-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.6; 0.6

GRADING NOTE Form required: both. 24/40 and 60/100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q1/5

True or false?

In the survey table, gender and answer are independent.

	Yes	No	Total
Men	24	16	40
Women	36	24	60
Total	60	40	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q2/5

1208

QUESTION ID

S36-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE P(yes given man) = P(yes).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In the driving table, find P(drives given junior) and P(drives).

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

P(drives given junior): _____

P(drives): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q3/5

1209

QUESTION ID

S36-F2-Q3

TYPE

Short answer

CORRECT ANSWER

0.5; 0.45

GRADING NOTE Form required: both. 60/120 and 90/200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F2-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In the driving table, grade and driving are independent.

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q4/5

1210

QUESTION ID

S36-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $0.5 \neq 0.45$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In the driving table, which is P(junior given drives)?

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

A. 1/2 **B.** 3/10 **C.** 2/3 **D.** 9/20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q5/5

1211

QUESTION ID
S36-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
C. 2/3

GRADING NOTE 60/90. 1/2 reverses the condition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q5/5

Construct two-way frequency tables

MAIN	Probability & Statistics main question	**HSS-CP.A.4**
BACKWARD 1	Fill in a two-way frequency table	**8.SP.A.4**
BACKWARD 2	Read joint and marginal frequencies	**HSS-ID.B.5**
FORWARD 1	Check statements against a built table	**HSS-CP.A.4**
FORWARD 2	Use the table as a sample space	**HSS-CP.A.4**

SET 37 • HSS-CP.A.4

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Of 200 students, 120 are juniors and 80 are seniors. 90 students drive to school, including 60 juniors. Complete the two-way table: juniors who do not drive, seniors who drive, seniors who do not drive.

Juniors, do not drive: _____

Seniors, drive: _____

Seniors, do not drive: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q1/5

1214

QUESTION ID

S37-M-Q1

TYPE

Short answer

CORRECT ANSWER

60; 30; 50

GRADING NOTE Form required: all three. 120 − 60; 90 − 60; 80 − 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Of 150 adults, 90 own a car, 40 own a bike, and 25 own both. In a two-way table (car yes/no by bike yes/no), how many own a car but no bike, and how many own neither?

Car, no bike: _____

Neither: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q2/5

1215

QUESTION ID
S37-M-Q2

TYPE
Short answer

CORRECT ANSWER
65; 45

GRADING NOTE Form required: both. 90 − 25; 150 − (90 + 40 − 25).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q2/5

Of 200 students, 120 are juniors, and 60 of the juniors drive. What percent of juniors drive?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q3/5

1216

QUESTION ID

S37-M-Q3

TYPE

Short answer

CORRECT ANSWER

50%

GRADING NOTE 60/120.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q3/5

80 patients took part in a study. Which description matches the table?

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. 50 patients got the placebo and the other 30 got the drug.

B. 50 patients improved, and 35 of those improved patients got the drug.

C. 50 got the drug (35 improved); 12 of the placebo patients improved.

D. 35 patients got the drug and 12 patients got a placebo instead.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q4/5

1217

QUESTION ID

S37-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 50 got the drug (35 improved); 12 of the placebo patients improved.

GRADING NOTE Drug row 35 + 15 = 50; placebo row 12 + 18 = 30. 47 improved, not 50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q4/5

80 patients: 50 got a drug and 30 a placebo; 35 drug patients and 12 placebo patients improved. How many drug patients and how many placebo patients did not improve?

Drug, did not: _____

Placebo, did not: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q5/5

1218

QUESTION ID

S37-M-Q5

TYPE

Short answer

CORRECT ANSWER

15; 18

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q5/5

SET 37 • HSS-CP.A.4

BACKWARD 1

Fill in a two-way frequency table • Grade 8 • Statistics & Probability • 8.SP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A table has row totals 40 and 60. One cell in the first row is 25. What is the other cell in that row?

Cell: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q1/5

1220

QUESTION ID

S37-B1-Q1

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE 40 − 25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q1/5

True or false?

In a two-way table, the column totals add to the grand total.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q2/5

1221

QUESTION ID

S37-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Every object counted once.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Of 50 people, 30 are adults. 12 of the adults wear glasses. How many adults do not?

A. 18

B. 12

C. 38

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q3/5

1222

GRADING NOTE 30 − 12.

QUESTION ID

S37-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A two-way table with two columns classifies 100 objects. One column total is 45. What is the other column total?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q4/5

1223

QUESTION ID

S37-B1-Q4

TYPE

Short answer

CORRECT ANSWER

55

GRADING NOTE 100 − 45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A cell can be found from its row total and the other cells in that row.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q5/5

1224

QUESTION ID

S37-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Subtraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q5/5

SET 37 • HSS-CP.A.4

BACKWARD 2

Read joint and marginal frequencies • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In the drug table, how many patients improved in total?

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

Improved: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q1/5

1226

QUESTION ID

S37-B2-Q1

TYPE

Short answer

CORRECT ANSWER

47

GRADING NOTE 35 + 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-B2-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In the drug table, what fraction of all patients got the drug and improved? Give lowest terms.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q2/5

1227

QUESTION ID

S37-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7/16

GRADING NOTE 35/80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-B2-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q2/5

True or false?

In the drug table, 30 patients got the placebo.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q3/5

1228

QUESTION ID

S37-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 12 + 18.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In the drug table, which number is a marginal total?

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. 12 **B.** 50 **C.** 35 **D.** 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q4/5

1229

QUESTION ID

S37-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 50

GRADING NOTE The drug row total.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-B2-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q4/5

True or false?

In the drug table, 35/47 is the fraction of drug patients who improved.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q5/5

1230

QUESTION ID

S37-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 35/50; 35/47 is the fraction of improvers who got the drug.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q5/5

SET 37 • HSS-CP.A.4

FORWARD 1

Check statements against a built table • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

True or false?

In the drug table, a greater percent of drug patients than placebo patients improved.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q1/5

1232

QUESTION ID

S37-F1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 70% versus 40%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In the drug table, what percent of drug patients improved, and what percent of placebo patients improved?

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

Drug: _____

Placebo: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q2/5

1233

QUESTION ID

S37-F1-Q2

TYPE

Short answer

CORRECT ANSWER

70%; 40%

GRADING NOTE Form required: both. 35/50 and 12/30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q2/5

A report says "most patients who improved got the drug." Using the drug table, is this correct?

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. No: only 35 patients got the drug.

B. Yes: 70% of all patients improved.

C. Yes: 35 of the 47 who improved got the drug.

D. No: 50 patients got the drug.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q3/5

1234

QUESTION ID

S37-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Yes: 35 of the 47 who improved got the drug.

GRADING NOTE 35/47 ≈ 74%. Only 47 of 80 improved (58.75%).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Of 120 voters, 70 are under 40. 45 of all voters support a plan, including 30 of those under 40. How many voters 40 or older do not support it?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q4/5

1235

QUESTION ID

S37-F1-Q4

TYPE

Short answer

CORRECT ANSWER

35

GRADING NOTE 50 older; 15 support; 35 do not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q4/5

True or false?

A two-way table can be built when each object is classified by only one category.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q5/5

1236

QUESTION ID

S37-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It needs two categorical variables.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F1-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q5/5

SET 37 • HSS-CP.A.4

FORWARD 2

Use the table as a sample space • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A patient is chosen at random from the drug table. What is P(improved)? Give a fraction in lowest terms.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q1/5

1238

QUESTION ID

S37-F2-Q1

TYPE

Short answer

CORRECT ANSWER

47/80

GRADING NOTE 47/80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A patient is chosen at random from the drug table. What is P(improved given drug)? Give a fraction in lowest terms.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q2/5

1239

QUESTION ID

S37-F2-Q2

TYPE

Short answer

CORRECT ANSWER

7/10

GRADING NOTE 35/50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q2/5

True or false?

In the drug table, improving and treatment are independent.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q3/5

1240

QUESTION ID

S37-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $7/10 \neq 47/80$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A patient is chosen at random from the drug table. What is P(drug given improved)?

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. 47/80 **B.** 35/47 **C.** 7/16 **D.** 35/50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q4/5

1241

QUESTION ID
S37-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. 35/47

GRADING NOTE Condition on the improved column.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A patient is chosen at random from the drug table. What is P(placebo and did not improve)? Give a fraction in lowest terms.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q5/5

1242

QUESTION ID

S37-F2-Q5

TYPE

Short answer

CORRECT ANSWER

9/40

GRADING NOTE 18/80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F2-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q5/5

Use two-way tables to find conditional probabilities and judge independence

MAIN	Probability & Statistics main question	**HSS-CP.A.4**
BACKWARD 1	Compute conditional relative frequencies	**HSS-ID.B.5**
BACKWARD 2	Compare P(A given B) with P(A)	**HSS-CP.A.3**
FORWARD 1	Use larger two-way tables	**HSS-CP.A.4**
FORWARD 2	Explain table results in everyday language	**HSS-CP.A.5**

SET 38 • HSS-CP.A.4

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

A student is chosen at random from the driving table. Find P(drives given junior) and P(drives given senior). Give decimals.

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

Given junior: _____

Given senior: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q1/5

1245

QUESTION ID

S38-M-Q1

TYPE

Short answer

CORRECT ANSWER

0.5; 0.375

GRADING NOTE Form required: both. 60/120 and 30/80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In the driving table, are "drives" and "junior" independent?

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

A. No: P(drives given junior) ≠ P(drives given senior).

B. Yes: every student is a junior or a senior.

C. Yes: 60 juniors drive and 60 do not.

D. No: more juniors than seniors were surveyed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q2/5

1246

QUESTION ID

S38-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. No: P(drives given junior) ≠ P(drives given senior).

GRADING NOTE 0.5 versus 0.375. Group sizes alone do not decide.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A student is chosen at random from the sports table. Find P(plays a sport) and P(plays a sport given 9th grade). Give decimals.

	Plays a sport	Does not	Total
9th grade	30	20	50
10th grade	42	28	70
Total	72	48	120

P(sport): _____

P(sport given 9th): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q3/5

1247

QUESTION ID

S38-M-Q3

TYPE

Short answer

CORRECT ANSWER

0.6; 0.6

GRADING NOTE Form required: both. 72/120 and 30/50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In the sports table, are "plays a sport" and "9th grade" independent?

	Plays a sport	Does not	Total
9th grade	30	20	50
10th grade	42	28	70
Total	72	48	120

A. Yes: P(sport given 9th) = P(sport) = 0.6.

B. No: 30 is not equal to 42.

C. No: the grades have different sizes.

D. Yes: more students play than do not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q4/5

1248

QUESTION ID

S38-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Yes: P(sport given 9th) = P(sport) = 0.6.

GRADING NOTE Compare rates, not counts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A student who drives is chosen at random from the driving table. What is P(junior given drives)? Give a fraction in lowest terms.

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q5/5

1249

QUESTION ID

S38-M-Q5

TYPE

Short answer

CORRECT ANSWER

2/3

GRADING NOTE 60/90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q5/5

SET 38 • HSS-CP.A.4

BACKWARD 1

Compute conditional relative frequencies • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In the driving table, what fraction of seniors drive? Give lowest terms.

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q1/5

1251

QUESTION ID

S38-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3/8

GRADING NOTE 30/80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-B1-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In the sports table, 42/70 of 10th graders play a sport.

	Plays a sport	Does not	Total
9th grade	30	20	50
10th grade	42	28	70
Total	72	48	120

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q2/5

1252

QUESTION ID

S38-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Row 10th.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In the sports table, what fraction of students who play a sport are in 10th grade?

	Plays a sport	Does not	Total
9th grade	30	20	50
10th grade	42	28	70
Total	72	48	120

A. 5/12 **B.** 7/12 **C.** 7/20 **D.** 3/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q3/5

1253

QUESTION ID

S38-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 7/12

GRADING NOTE 42/72.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In the driving table, what percent of all students drive?

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q4/5

1254

QUESTION ID

S38-B1-Q4

TYPE

Short answer

CORRECT ANSWER

45%

GRADING NOTE 90/200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-B1-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A row conditional frequency divides by the grand total.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q5/5

1255

QUESTION ID
S38-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE By the row total.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q5/5

SET 38 • HSS-CP.A.4

BACKWARD 2

Compare P(A given B) with P(A) • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

True or false?

If P(A given B) = P(A), A and B are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q1/5

1257

QUESTION ID

S38-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.45 and P(A given B) = 0.5. Are A and B independent?

A. only if P(B) = 0.5

B. yes

C. no

D. cannot tell

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q2/5

1258

QUESTION ID

S38-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. no

GRADING NOTE Not equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-B2-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A and B) = 0.3 and P(B) = 0.5. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q3/5

1259

QUESTION ID

S38-B2-Q3

TYPE

Short answer

CORRECT ANSWER

0.6

GRADING NOTE 0.3/0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-B2-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

P(A given B) is always less than P(A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q4/5

1260

QUESTION ID

S38-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It can be greater, equal or less.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A and B) = 0.18 and P(B) = 0.3. If P(A) = 0.6, are A and B independent?

A. yes

B. only if P(A) = 0.3

C. cannot tell

D. no

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q5/5

1261

QUESTION ID

S38-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. yes

GRADING NOTE 0.18/0.3 = 0.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-B2-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q5/5

SET 38 • HSS-CP.A.4

FORWARD 1

Use larger two-way tables • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A student is chosen at random from the music table. Find P(rock) and P(rock given high school). Give decimals.

	Pop	Rock	Other	Total
Middle school	16	24	10	50
High school	24	36	15	75
Total	40	60	25	125

P(rock): _____

P(rock given high school): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q1/5

1263

QUESTION ID

S38-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.48; 0.48

GRADING NOTE Form required: both. 60/125 and 36/75.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q1/5

True or false?

In the music table, school level and favorite type of music are independent.

	Pop	Rock	Other	Total
Middle school	16	24	10	50
High school	24	36	15	75
Total	40	60	25	125

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q2/5

1264

QUESTION ID

S38-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each row has the same proportions 0.32, 0.48, 0.2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In the music table, what is P(middle school given pop)? Give a fraction in lowest terms.

	Pop	Rock	Other	Total
Middle school	16	24	10	50
High school	24	36	15	75
Total	40	60	25	125

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q3/5

1265

QUESTION ID

S38-F1-Q3

TYPE

Short answer

CORRECT ANSWER

2/5

GRADING NOTE 16/40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-F1-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In the music table, what is P(other given middle school)?

	Pop	Rock	Other	Total
Middle school	16	24	10	50
High school	24	36	15	75
Total	40	60	25	125

A. 1/5 **B.** 2/25 **C.** 2/5 **D.** 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q4/5

1266

QUESTION ID

S38-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 1/5

GRADING NOTE 10/50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q4/5

True or false?

In the music table, P(pop given middle school) is greater than P(pop given high school).

	Pop	Rock	Other	Total
Middle school	16	24	10	50
High school	24	36	15	75
Total	40	60	25	125

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q5/5

1267

QUESTION ID

S38-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Both 0.32.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q5/5

SET 38 • HSS-CP.A.4

FORWARD 2

Explain table results in everyday language • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5

In the driving table, P(drives given junior) = 0.5 and P(drives given senior) = 0.375. Which sentence says this in everyday language?

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

A. Most of the students who drive to school are seniors.

B. Half of the juniors drive, but fewer than 4 in 10 seniors drive.

C. Juniors and seniors are equally likely to drive.

D. Half of all the students who drive are juniors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q1/5

1269

QUESTION ID

S38-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Half of the juniors drive, but fewer than 4 in 10 seniors drive.

GRADING NOTE Condition on grade.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q1/5

True or false?

"Among students who drive, two-thirds are juniors" describes P(junior given drives).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q2/5

1270

QUESTION ID

S38-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The condition is "drives".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which statement describes independence in the sports table?

	Plays a sport	Does not	Total
9th grade	30	20	50
10th grade	42	28	70
Total	72	48	120

A. A student's grade does not change the chance of playing a sport.

B. Every student in the table plays exactly one sport.

C. The two grades have the same number of students.

D. Students in 10th grade play more sports than 9th graders.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q3/5

1271

QUESTION ID

S38-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A student's grade does not change the chance of playing a sport.

GRADING NOTE Same rate in each grade.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q3/5

True or false?

"Most juniors drive" and "most drivers are juniors" mean the same thing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q4/5

1272

QUESTION ID

S38-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Different conditions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q4/5

"Among seniors, 3 in 8 drive." Which probability is this?

A. P(senior given drives)

B. P(drives given senior)

C. P(senior and drives)

D. P(senior)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q5/5

1273

QUESTION ID

S38-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. P(drives given senior)

GRADING NOTE Condition: senior.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q5/5

Conditional probability and independence in everyday language

MAIN	Probability & Statistics main question	**HSS-CP.A.5**
BACKWARD 1	Compute and read conditional probabilities	**HSS-CP.A.3**
BACKWARD 2	Read conditional rates in a table	**HSS-CP.A.4**
FORWARD 1	Translate statements and compute	**HSS-CP.A.5**
FORWARD 2	Find a conditional probability as a fraction of outcomes	**HSS-CP.B.6**

SET 39 • HSS-CP.A.5

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5

Which statement describes a conditional probability?

A. One in four cards in a deck is a heart.

B. Among smokers, the chance of lung disease is higher.

C. A fair coin lands heads half of the time.

D. The chance of rain tomorrow is 30 percent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q1/5

1276

QUESTION ID

S39-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Among smokers, the chance of lung disease is higher.

GRADING NOTE "Among smokers" names the condition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q1/5

"Most people with lung disease were smokers." Which probability does this sentence say is high?

A. P(lung disease)

B. P(smoker given lung disease)

C. P(smoker and lung disease)

D. P(lung disease given smoker)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q2/5

1277

QUESTION ID

S39-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. P(smoker given lung disease)

GRADING NOTE The condition is "people with lung disease".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q2/5

Which pair of events is most likely to be independent?

A. heavy rain today; a baseball game being canceled

B. a student's shirt color; the school bus being late

C. studying hard for a test; passing that test

D. a person's height; the person's shoe size

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q3/5

1278

QUESTION ID

S39-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. a student's shirt color; the school bus being late

GRADING NOTE No plausible influence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A fair coin has landed heads 5 times in a row. What is the probability of heads on the next flip?

A. 1/32

B. 1/2

C. less than 1/2

D. more than 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q4/5

1279

QUESTION ID

S39-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1/2

GRADING NOTE Flips are independent; 1/32 is P(5 heads in a row) before flipping.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q4/5

In a town, 40% of adults own a dog, and among dog owners 70% walk daily. What percent of all adults are dog owners who walk daily?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q5/5

1280

QUESTION ID

S39-M-Q5

TYPE

Short answer

CORRECT ANSWER

28%

GRADING NOTE 0.4 · 0.7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q5/5

SET 39 • HSS-CP.A.5

BACKWARD 1

Compute and read conditional probabilities • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

P(A and B) = 0.28 and P(A) = 0.4. Find P(B given A).

P(B given A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q1/5

1282

QUESTION ID

S39-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.7

GRADING NOTE 0.28/0.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-B1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q1/5

True or false?

"The probability of B given A" means the chance of B when A is known to happen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q2/5

1283

QUESTION ID

S39-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Meaning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q2/5

P(passing given studying) is the probability that

A. a student passes

B. a student passes, among students who study

C. a student studies, among students who pass

D. a student both studies and passes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q3/5

1284

QUESTION ID

S39-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. a student passes, among students who study

GRADING NOTE Condition: studying.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

P(B) = 0.5 and P(A given B) = 0.3. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q4/5

1285

QUESTION ID

S39-B1-Q4

TYPE

Short answer

CORRECT ANSWER

0.15

GRADING NOTE 0.5 · 0.3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-B1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q4/5

True or false?

P(rain given clouds) is the same as P(clouds given rain).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q5/5

1286

QUESTION ID

S39-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Different conditions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • HSS-CP.A.5

BACKWARD 2

Read conditional rates in a table • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

In the driving table, what fraction of juniors drive? Give lowest terms.

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q1/5

1288

QUESTION ID

S39-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1/2

GRADING NOTE 60/120.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-B2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q1/5

True or false?

In the driving table, "among drivers, 2 in 3 are juniors" is correct.

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q2/5

1289

QUESTION ID
S39-B2-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE 60/90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In the driving table, which sentence is correct?

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

A. Among seniors, 3 in 8 drive.

B. Among seniors, 1 in 2 drive.

C. Among juniors, 3 in 8 drive.

D. Among drivers, 3 in 8 are seniors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q3/5

1290

QUESTION ID

S39-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Among seniors, 3 in 8 drive.

GRADING NOTE 30/80. Drivers who are seniors: 30/90 = 1/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-B2-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In the driving table, what fraction of drivers are seniors? Give lowest terms.

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q4/5

1291

QUESTION ID

S39-B2-Q4

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE 30/90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-B2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q4/5

True or false?

In the driving table, "among seniors, most drive" is correct.

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q5/5

1292

QUESTION ID

S39-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3/8 < 1/2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-B2-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q5/5

SET 39 • HSS-CP.A.5

FORWARD 1

Translate statements and compute • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5

ANSWER KEY Reduced copy of the question page, with its answer

"If it is a school day, there is an 80% chance that Sam takes the bus." Which notation matches?

A. P(bus given school day) = 0.8

B. P(bus) = 0.8

C. P(school day given bus) = 0.8

D. P(bus and school day) = 0.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q1/5

1294

QUESTION ID

S39-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. P(bus given school day) = 0.8

GRADING NOTE The condition follows "if".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

5 of 7 days are school days, and on a school day Sam takes the bus with probability 0.8. What is P(school day and bus)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q2/5

1295

QUESTION ID

S39-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4/7

GRADING NOTE (5/7)(4/5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q2/5

True or false?

"Knowing it is a school day changes the chance that Sam takes the bus" means the events are not independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q3/5

1296

QUESTION ID
S39-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Dependence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A test is positive for 95% of people with a disease.
Which sentence states this correctly?

A. 95% of all people have the disease.

B. 95% of all people test positive.

C. Among people with the disease, 95% test positive.

D. Among people who test positive, 95% have the disease.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q4/5

1297

QUESTION ID
S39-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. Among people with the disease, 95% test positive.

GRADING NOTE Condition: has the disease.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q4/5

True or false?

"Among people who test positive, 95% have the disease" follows from "95% of people with the disease test positive".

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q5/5

1298

QUESTION ID

S39-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Reversing the condition is not allowed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q5/5

SET 39 • HSS-CP.A.5

FORWARD 2

Find a conditional probability as a fraction of outcomes • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A fair six-sided die is rolled. Given the roll is odd, what is P(the roll is 5)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q1/5

1300

GRADING NOTE {1, 3, 5}.

QUESTION ID

S39-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q1/5

1300

ANSWER KEY Reduced copy of the question page, with its answer

A card drawn at random from a standard 52-card deck is red. What is P(it is a king)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q2/5

1301

QUESTION ID

S39-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1/13

GRADING NOTE 2 of 26.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q2/5

True or false?

P(A given B) can be found as (outcomes in both A and B)/(outcomes in B) in a uniform model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q3/5

1302

QUESTION ID

S39-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Fraction of B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q3/5

A number from 1 to 10 is even. What is P(it is greater than 6)?

A. 1/2

B. 2/5

C. 2/3

D. 1/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q4/5

1303

QUESTION ID

S39-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 2/5

GRADING NOTE {2, 4, 6, 8, 10}: 8 and 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q4/5

True or false?

Given the roll of a fair six-sided die is even, P(roll is 3) = 1/6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q5/5

1304

QUESTION ID

S39-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q5/5

Find conditional probability as a fraction of outcomes

MAIN	Probability & Statistics main question	**HSS-CP.B.6**
BACKWARD 1	Count outcomes in events	**HSS-CP.A.1**
BACKWARD 2	Use the definition of conditional probability	**HSS-CP.A.3**
FORWARD 1	Condition on "at least one"	**HSS-CP.B.6**
FORWARD 2	Multiply along a chain of conditions	**HSS-CP.B.8**

SET 40 • HSS-CP.B.6

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6

Two fair dice are rolled independently. The first die shows 5. What is P(the sum is 10 or more)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q1/5

1307

QUESTION ID

S40-M-Q1

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE Second die 5 or 6: 2 of 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A card drawn at random from a standard 52-card deck is red. What is P(it is an ace)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q2/5

1308

QUESTION ID
S40-M-Q2

TYPE
Short answer

CORRECT ANSWER
1/13

GRADING NOTE 2 of 26.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q2/5

A patient from the drug table is chosen at random and got the placebo. What is P(improved)? Give a fraction in lowest terms.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q3/5

1309

QUESTION ID

S40-M-Q3

TYPE

Short answer

CORRECT ANSWER

2/5

GRADING NOTE 12 of 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A number from 1 to 10 is chosen at random and is odd. What is P(it is prime)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q4/5

1310

QUESTION ID

S40-M-Q4

TYPE

Short answer

CORRECT ANSWER

3/5

GRADING NOTE {1, 3, 5, 7, 9}: 3, 5, 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q4/5

From the drug table, P(improved given placebo) = 2/5. What does this mean in the model?

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. Of the patients who improved, 2 in 5 got the placebo.

B. 2 in 5 of all patients got the placebo and improved.

C. Of the patients given the placebo, 2 in 5 improved.

D. 2 in 5 of all patients got the placebo.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q5/5

1311

QUESTION ID

S40-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. Of the patients given the placebo, 2 in 5 improved.

GRADING NOTE The placebo group is the sample space.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q5/5

SET 40 • HSS-CP.B.6

BACKWARD 1

Count outcomes in events • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

How many red cards are in a standard 52-card deck?

Cards: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q1/5

1313

QUESTION ID

S40-B1-Q1

TYPE

Short answer

CORRECT ANSWER

26

GRADING NOTE Hearts and diamonds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-B1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q1/5

Two fair dice rolled independently: how many outcomes have the first die equal to 5?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q2/5

1314

QUESTION ID

S40-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE (5,1) to (5,6).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-B1-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q2/5

True or false?

A standard 52-card deck has 2 red aces.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q3/5

1315

QUESTION ID

S40-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Hearts and diamonds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

How many numbers from 1 to 10 are odd and prime?

A. 2

B. 5

C. 3

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q4/5

1316

QUESTION ID

S40-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 3

GRADING NOTE 3, 5, 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-B1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q4/5

True or false?

A standard 52-card deck has 4 red kings.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q5/5

1317

QUESTION ID

S40-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • HSS-CP.B.6

BACKWARD 2

Use the definition of conditional probability • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

P(A and B) = 1/12 and P(B) = 1/4. Find P(A given B).
Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q1/5

1319

QUESTION ID
S40-B2-Q1

TYPE
Short answer

CORRECT ANSWER
1/3

GRADING NOTE (1/12)/(1/4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-B2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q1/5

True or false?

In a uniform model, P(A and B)/P(B) equals (count of A and B)/(count of B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q2/5

1320

QUESTION ID

S40-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The total cancels.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A and B) = 0.1 and P(B) = 0.5. Find P(A given B).

A. 5

B. 0.5

C. 0.05

D. 0.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q3/5

1321

QUESTION ID

S40-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0.2

GRADING NOTE 0.1/0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-B2-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A and B) = 2/36 and P(B) = 6/36. Find P(A given B). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q4/5

1322

QUESTION ID

S40-B2-Q4

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE 2/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-B2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q4/5

True or false?

P(A given B) = P(A and B)/P(A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q5/5

1323

QUESTION ID

S40-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Divide by P(B).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q5/5

SET 40 • HSS-CP.B.6

FORWARD 1

Condition on "at least one" • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Two fair dice are rolled independently and at least one shows 6. What is P(the sum is 7)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q1/5

1325

QUESTION ID

S40-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2/11

GRADING NOTE 11 outcomes contain a 6; (1,6) and (6,1) sum to 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A family has two children; each is equally likely to be a girl or a boy. Given at least one is a girl, what is P(both are girls)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q2/5

1326

QUESTION ID

S40-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE GG, GB, BG.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q2/5

True or false?

Two fair coins are flipped independently. Given at least one shows heads, P(both heads) = 1/2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q3/5

1327

QUESTION ID

S40-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q3/5

Two fair dice are rolled independently and the sum is at least 11. What is P(doubles)?

A. 1/36

B. 1/6

C. 1/2

D. 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q4/5

1328

QUESTION ID

S40-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1/3

GRADING NOTE (5,6), (6,5), (6,6).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q4/5

True or false?

Conditioning on an event removes the outcomes outside it from the sample space.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q5/5

1329

QUESTION ID

S40-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Restriction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q5/5

SET 40 • HSS-CP.B.6

FORWARD 2

Multiply along a chain of conditions • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.8

ANSWER KEY Reduced copy of the question page, with its answer

A bag has 4 red and 2 blue marbles. Two are drawn at random without replacement. What is P(both red)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q1/5

1331

QUESTION ID

S40-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2/5

GRADING NOTE (4/6)(3/5) = 12/30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q1/5

True or false?

P(A and B) = P(A)·P(B given A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q2/5

1332

QUESTION ID

S40-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE General Multiplication Rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.8 and P(B given A) = 0.25. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q3/5

1333

QUESTION ID

S40-F2-Q3

TYPE

Short answer

CORRECT ANSWER

0.2

GRADING NOTE 0.8 · 0.25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q3**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q3/5

Two cards are drawn at random, without replacement, from a standard 52-card deck. What is P(second is a heart given the first is a heart)?

A. 1/4

B. 13/52

C. 12/51

D. 13/51

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q4/5

1334

QUESTION ID
S40-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. 12/51

GRADING NOTE 12 hearts left of 51 cards.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q4/5

True or false?

For draws without replacement, P(second red given first red) equals P(first red).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q5/5

1335

QUESTION ID
S40-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE One red is gone.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q5/5

Apply the Addition Rule

MAIN	Probability & Statistics main question	**HSS-CP.B.7**
BACKWARD 1	Count a union without double counting	**HSS-CP.A.1**
BACKWARD 2	Add and subtract fractions	**5.NF.A.1**
FORWARD 1	Use the rule in reverse and with exclusive events	**HSS-CP.B.7**
FORWARD 2	Combine the Addition Rule with complements	**HSS-CP.B.7**

SET 41 • HSS-CP.B.7

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.45, P(B) = 0.35 and P(A and B) = 0.15. Find P(A or B).

P(A or B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q1/5

1338

QUESTION ID

S41-M-Q1

TYPE

Short answer

CORRECT ANSWER

0.65

GRADING NOTE 0.45 + 0.35 − 0.15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q1/5

A card is drawn at random from a standard 52-card deck. Find P(red or king). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q2/5

1339

QUESTION ID

S41-M-Q2

TYPE

Short answer

CORRECT ANSWER

7/13

GRADING NOTE 26 + 4 − 2 = 28; 28/52.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Two fair dice are rolled independently. Find P(doubles or sum of 8). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q3/5

1340

QUESTION ID
S41-M-Q3

TYPE
Short answer

CORRECT ANSWER
5/18

GRADING NOTE 6 + 5 − 1 = 10 outcomes; 10/36.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Of 200 students, 120 take Spanish, 70 take art, and 40 take both. A student is chosen at random. What is P(Spanish or art)?

A. 3/4

B. 7/10

C. 19/20

D. 1/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q4/5

1341

QUESTION ID

S41-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 3/4

GRADING NOTE 150/200. 19/20 forgets to subtract the 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q4/5

For a randomly chosen shopper, P(buys bread or milk) = 0.65. What does this mean?

A. About 65% of shoppers buy bread, milk, or both.

B. About 65% of shoppers buy both bread and milk.

C. About 65% of shoppers buy neither one.

D. About 65% of shoppers buy exactly one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q5/5

1342

QUESTION ID

S41-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. About 65% of shoppers buy bread, milk, or both.

GRADING NOTE "Or" includes both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q5/5

SET 41 • HSS-CP.B.7

BACKWARD 1

Count a union without double counting • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A fair six-sided die is rolled. How many outcomes are "even or less than 3"?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q1/5

1344

QUESTION ID

S41-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE {1, 2, 4, 6}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q1/5

True or false?

In a standard 52-card deck, the number of cards that are hearts or face cards is 13 + 12 − 3 = 22.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q2/5

1345

QUESTION ID

S41-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Remove the 3 heart face cards counted twice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q2/5

In a standard 52-card deck, how many cards are aces or spades?

A. 4

B. 13

C. 16

D. 17

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q3/5

1346

QUESTION ID

S41-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 16

GRADING NOTE $4 + 13 - 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

How many numbers from 1 to 20 are multiples of 2 or multiples of 5?

Numbers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q4/5

1347

QUESTION ID

S41-B1-Q4

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE 10 + 4 − 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-B1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q4/5

True or false?

The number of outcomes in "A or B" is always the count of A plus the count of B.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q5/5

1348

QUESTION ID

S41-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Subtract the overlap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • HSS-CP.B.7

BACKWARD 2

Add and subtract fractions • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Compute 1/2 + 1/13 − 1/26. Give a fraction in lowest terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q1/5

1350

QUESTION ID

S41-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7/13

GRADING NOTE 13/26 + 2/26 − 1/26.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 1/6 + 5/36 − 1/36. Give a fraction in lowest terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q2/5

1351

QUESTION ID
S41-B2-Q2

TYPE
Short answer

CORRECT ANSWER
5/18

GRADING NOTE 6/36 + 5/36 − 1/36.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q2/5

True or false?

1/4 + 1/3 = 7/12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q3/5

1352

QUESTION ID

S41-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 3/12 + 4/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 3/5 + 7/20 − 1/5.

A. 17/20

B. 1

C. 3/4

D. 9/20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q4/5

1353

GRADING NOTE 12/20 + 7/20 − 4/20.

QUESTION ID

S41-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q4/5

True or false?

1/2 + 1/3 = 2/5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q5/5

1354

QUESTION ID

S41-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 5/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q5/5

SET 41 • HSS-CP.B.7

FORWARD 1

Use the rule in reverse and with exclusive events • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.5, P(B) = 0.4 and P(A or B) = 0.7. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q1/5

1356

QUESTION ID

S41-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.2

GRADING NOTE 0.5 + 0.4 − 0.7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q1/5

True or false?

If A and B are mutually exclusive, P(A or B) = P(A) + P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q2/5

1357

QUESTION ID

S41-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE P(A and B) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A fair six-sided die is rolled. Find P(1 or 6). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q3/5

1358

QUESTION ID
S41-F1-Q3

TYPE
Short answer

CORRECT ANSWER
1/3

GRADING NOTE Exclusive: 1/6 + 1/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F1-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q3/5

P(A) = 0.3, P(B) = 0.5, and A and B are independent. What is P(A or B)?

A. 0.15

B. 0.5

C. 0.65

D. 0.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q4/5

1359

QUESTION ID

S41-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 0.65

GRADING NOTE 0.3 + 0.5 − 0.15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q4/5

True or false?

If P(A) = 0.6 and P(B) = 0.7, then A and B could be mutually exclusive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q5/5

1360

QUESTION ID

S41-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE P(A) + P(B) > 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q5/5

SET 41 • HSS-CP.B.7

FORWARD 2

Combine the Addition Rule with complements • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7

ANSWER KEY Reduced copy of the question page, with its answer

P(A or B) = 0.65. What is P(neither A nor B)?

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q1/5

1362

QUESTION ID

S41-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.35

GRADING NOTE 1 − 0.65.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Of 200 students, 120 take Spanish, 70 take art and 40 take both. What is P(neither)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q2/5

1363

QUESTION ID

S41-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1/4

GRADING NOTE 1 − 3/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q2/5

True or false?

P(not (A or B)) = 1 − P(A or B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q3/5

1364

QUESTION ID

S41-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Complement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q3/5

A card is drawn at random from a standard 52-card deck. What is P(neither red nor a king)?

A. 6/13

B. 1/2

C. 7/13

D. 11/26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q4/5

1365

QUESTION ID

S41-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 6/13

GRADING NOTE 1 − 7/13.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q4/5

True or false?

P(neither A nor B) = 1 − P(A) − P(B) in every case.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q5/5

1366

QUESTION ID
S41-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Only if A and B cannot overlap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q5/5

Apply the general Multiplication Rule (+)

MAIN	Probability & Statistics main question	**HSS-CP.B.8**
BACKWARD 1	Rearrange the conditional probability formula	**HSS-CP.A.3**
BACKWARD 2	Multiply fractions	**5.NF.B.4**
BACKWARD 3	Multiply for independent events	**HSS-CP.A.2**
FORWARD 1	Combine orders and longer chains	**HSS-CP.B.8**
FORWARD 2	Count arrangements to find probabilities	**HSS-CP.B.9**

SET 42 • HSS-CP.B.8

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.8

A bag has 5 red and 3 blue marbles. Two are drawn at random without replacement. What is P(both red)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q1/5

1369

QUESTION ID

S42-M-Q1

TYPE

Short answer

CORRECT ANSWER

5/14

GRADING NOTE (5/8)(4/7) = 20/56.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-M-Q1**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A bag has 5 red and 3 blue marbles. Two are drawn at random without replacement. What is P(first red and second blue)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q2/5

1370

QUESTION ID
S42-M-Q2

TYPE
Short answer

CORRECT ANSWER
15/56

GRADING NOTE (5/8)(3/7).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-M-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q2/5

Two cards are drawn at random, without replacement, from a standard 52-card deck. What is P(both aces)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q3/5

1371

QUESTION ID
S42-M-Q3

TYPE
Short answer

CORRECT ANSWER
1/221

GRADING NOTE (4/52)(3/51).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-M-Q3**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q3/5

P(A) = 0.6 and P(B given A) = 0.5. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q4/5

1372

QUESTION ID
S42-M-Q4

TYPE
Short answer

CORRECT ANSWER
0.3

GRADING NOTE 0.6 · 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-M-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q4/5

A committee has 3 boys and 2 girls. Two members are chosen at random, one after the other. What is P(both girls)?

A. 2/5

B. 1/10

C. 4/25

D. 1/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q5/5

1373

QUESTION ID
S42-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
B. 1/10

GRADING NOTE (2/5)(1/4). 4/25 assumes replacement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q5/5

SET 42 • HSS-CP.B.8

BACKWARD 1

Rearrange the conditional probability formula • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

True or false?

From P(B given A) = P(A and B)/P(A), it follows that P(A and B) = P(A)·P(B given A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q1/5

1375

QUESTION ID

S42-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Multiply both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.4 and P(B given A) = 0.25. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q2/5

1376

QUESTION ID

S42-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.1

GRADING NOTE Product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-B1-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

P(B) = 0.5 and P(A given B) = 0.6. What is P(A and B)?

A. 0.1

B. 1.1

C. 0.83

D. 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q3/5

1377

QUESTION ID

S42-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0.3

GRADING NOTE 0.5 · 0.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-B1-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A and B) = 0.12 and P(A) = 0.3. Find P(B given A).

P(B given A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q4/5

1378

QUESTION ID

S42-B1-Q4

TYPE

Short answer

CORRECT ANSWER

0.4

GRADING NOTE 0.12/0.3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q4/5

True or false?

P(A and B) = P(A) + P(B given A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q5/5

1379

QUESTION ID

S42-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • HSS-CP.B.8

BACKWARD 2

Multiply fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Compute (5/8)(4/7). Give a fraction in lowest terms.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q1/5

1381

QUESTION ID

S42-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5/14

GRADING NOTE 20/56.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute (4/52)(3/51). Give a fraction in lowest terms.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q2/5

1382

QUESTION ID

S42-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1/221

GRADING NOTE 12/2652.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

(2/5)(1/4) = 1/10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q3/5

1383

QUESTION ID

S42-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 2/20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q3/5

Compute (3/8)(2/7).

A. 3/28

B. 6/15

C. 3/56

D. 5/15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q4/5

1384

QUESTION ID

S42-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 3/28

GRADING NOTE 6/56.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

(1/2)(1/3) = 2/5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q5/5

1385

QUESTION ID

S42-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q5/5

SET 42 • HSS-CP.B.8

BACKWARD 3

Multiply for independent events • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A fair die is rolled twice, independently. What is P(both 6)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q1/5

1387

QUESTION ID

S42-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1/36

GRADING NOTE (1/6)(1/6).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-B3-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q1/5

True or false?

With replacement, P(red then red) from 5 red and 3 blue is (5/8)(5/8).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q2/5

1388

QUESTION ID

S42-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Independent draws.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A fair coin is flipped and a fair die is rolled, independently. P(tails and even)?

A. 2/3

B. 1/4

C. 1/12

D. 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q3/5

1389

QUESTION ID

S42-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1/4

GRADING NOTE (1/2)(1/2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-B3-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Independent events have P(A) = 0.5 and P(B) = 0.3.
Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q4/5

1390

QUESTION ID
S42-B3-Q4

TYPE
Short answer

CORRECT ANSWER
0.15

GRADING NOTE 0.5 · 0.3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-B3-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q4/5

True or false?

Without replacement, draws are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q5/5

1391

QUESTION ID

S42-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Without replacement changes the second draw.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q5/5

SET 42 • HSS-CP.B.8

FORWARD 1

Combine orders and longer chains • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.8

ANSWER KEY Reduced copy of the question page, with its answer

A bag has 5 red and 3 blue marbles. Two are drawn at random without replacement. What is P(one of each color)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q1/5

1393

QUESTION ID

S42-F1-Q1

TYPE

Short answer

CORRECT ANSWER

15/28

GRADING NOTE (5/8)(3/7) + (3/8)(5/7) = 30/56.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Three cards are drawn at random, without replacement, from a standard 52-card deck. What is P(all three are hearts)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q2/5

1394

QUESTION ID

S42-F1-Q2

TYPE

Short answer

CORRECT ANSWER

11/850

GRADING NOTE (13/52)(12/51)(11/50) = 1716/132600.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F1-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q2/5

True or false?

P(A and B) = P(B)·P(A given B) gives the same answer as P(A)·P(B given A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q3/5

1395

QUESTION ID
S42-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Both equal P(A and B).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q3/5

A bag has 4 green and 6 yellow marbles. Two are drawn at random without replacement. What is P(both yellow)?

A. 9/25

B. 2/5

C. 3/5

D. 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q4/5

1396

QUESTION ID

S42-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1/3

GRADING NOTE (6/10)(5/9) = 30/90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F1-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q4/5

True or false?

P(first red and second blue) equals P(first blue and second red) for 5 red and 3 blue drawn without replacement.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q5/5

1397

QUESTION ID
S42-F1-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Both 15/56.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q5/5

SET 42 • HSS-CP.B.8

FORWARD 2

Count arrangements to find probabilities • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

ANSWER KEY Reduced copy of the question page, with its answer

How many ways can 2 marbles be chosen from 8 (order does not matter)?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q1/5

1399

QUESTION ID

S42-F2-Q1

TYPE

Short answer

CORRECT ANSWER

28

GRADING NOTE $8 \cdot 7/2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F2-Q1**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How many ways can 2 red marbles be chosen from 5 red?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q2/5

1400

QUESTION ID
S42-F2-Q2

TYPE
Short answer

CORRECT ANSWER
10

GRADING NOTE 5·4/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F2-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q2/5

True or false?

P(both red) = 10/28 agrees with (5/8)(4/7).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q3/5

1401

QUESTION ID

S42-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both 5/14.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q3/5

How many ordered ways can 2 of 8 marbles be drawn one after the other?

A. 16

B. 28

C. 64

D. 56

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q4/5

1402

QUESTION ID

S42-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 56

GRADING NOTE $8 \cdot 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q4/5

True or false?

Counting with combinations always gives a different probability than the Multiplication Rule.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q5/5

1403

QUESTION ID

S42-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Both methods agree.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q5/5

Count ordered arrangements with permutations (+)

MAIN	Probability & Statistics main question	**HSS-CP.B.9**
BACKWARD 1	Count outcomes with a tree or the counting principle	**7.SP.C.8.b**
BACKWARD 2	Multiply several whole numbers	**5.NBT.B.5**
FORWARD 1	Count arrangements with conditions	**HSS-CP.B.9**
FORWARD 2	Count selections where order does not matter	**HSS-CP.B.9**

SET 43 • HSS-CP.B.9

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

Five runners race. In how many ways can first, second and third place be awarded?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q1/5

1406

QUESTION ID

S43-M-Q1

TYPE

Short answer

CORRECT ANSWER

60

GRADING NOTE 5 · 4 · 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-M-Q1**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q1/5

In how many ways can 6 different books be arranged on a shelf?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q2/5

1407

QUESTION ID

S43-M-Q2

TYPE

Short answer

CORRECT ANSWER

720

GRADING NOTE 6!.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-M-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q2/5

A 4-digit code uses the digits 0-9 with no digit repeated. How many codes are possible?

Codes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q3/5

1408

QUESTION ID
S43-M-Q3

TYPE
Short answer

CORRECT ANSWER
5040

GRADING NOTE $10 \cdot 9 \cdot 8 \cdot 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q3/5

What is the value of 8P2 (the number of ordered arrangements of 2 objects from 8)?

A. 56

B. 28

C. 16

D. 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q4/5

1409

QUESTION ID
S43-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. 56

GRADING NOTE 8 · 7. 28 is 8C2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-M-Q4**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q4/5

A club of 10 members chooses a president, a vice president and a treasurer. How many ways are possible?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q5/5

1410

QUESTION ID
S43-M-Q5

TYPE
Short answer

CORRECT ANSWER
720

GRADING NOTE $10 \cdot 9 \cdot 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-M-Q5**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q5/5

SET 43 • HSS-CP.B.9

BACKWARD 1

Count outcomes with a tree or the counting principle • Grade 7 • Statistics & Probability • 7.SP.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

An outfit is a shirt (4 choices) and pants (3 choices).
How many outfits?

Outfits: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q1/5

1412

QUESTION ID
S43-B1-Q1

TYPE
Short answer

CORRECT ANSWER
12

GRADING NOTE 4 · 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q1/5

True or false?

A tree for 3 stages with 2 choices each has 8 paths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q2/5

1413

QUESTION ID

S43-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 2^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A meal is a main (5 choices), a side (3) and a drink (2). How many meals?

A. 30

B. 60

C. 15

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q3/5

1414

QUESTION ID

S43-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 30

GRADING NOTE 5 · 3 · 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A license code is a letter (26 choices) followed by a digit (10 choices). How many codes?

Codes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q4/5

1415

GRADING NOTE 26 · 10.

QUESTION ID

S43-B1-Q4

TYPE

Short answer

CORRECT ANSWER

260

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q4/5

True or false?

With 3 shirts and 2 pants there are 5 outfits.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q5/5

1416

QUESTION ID

S43-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q5/5

SET 43 • HSS-CP.B.9

BACKWARD 2

Multiply several whole numbers • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 10 · 9 · 8.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q1/5

1418

QUESTION ID

S43-B2-Q1

TYPE

Short answer

CORRECT ANSWER

720

GRADING NOTE 90 · 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute $6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q2/5

1419

QUESTION ID

S43-B2-Q2

TYPE

Short answer

CORRECT ANSWER

720

GRADING NOTE 6!.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$5 \cdot 4 \cdot 3 = 60$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q3/5

1420

QUESTION ID

S43-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $20 \cdot 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 7 · 6 · 5.

A. 343

B. 180

C. 42

D. 210

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q4/5

1421

QUESTION ID

S43-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 210

GRADING NOTE 42 · 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q4/5

True or false?

$4 \cdot 3 \cdot 2 \cdot 1 = 10$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q5/5

1422

QUESTION ID

S43-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q5/5

SET 43 • HSS-CP.B.9

FORWARD 1

Count arrangements with conditions • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

ANSWER KEY Reduced copy of the question page, with its answer

In how many ways can the letters of MATH be arranged if M must be first?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q1/5

1424

QUESTION ID
S43-F1-Q1

TYPE
Short answer

CORRECT ANSWER
6

GRADING NOTE 3! for the rest.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q1**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In how many ways can 5 people stand in a line if Ana and Ben must be at the two ends?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q2/5

1425

QUESTION ID
S43-F1-Q2

TYPE
Short answer

CORRECT ANSWER
12

GRADING NOTE 2 ways for the ends · 3! = 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q2/5

True or false?

nPr = n!/(n − r)!.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q3/5

1426

QUESTION ID

S43-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q3/5

How many 3-letter arrangements (no repeats) can be made from the letters A, B, C, D, E if the first letter is a vowel?

A. 60

B. 12

C. 48

D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q4/5

1427

QUESTION ID

S43-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 24

GRADING NOTE 2 vowels · 4 · 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q4/5

True or false?

The number of ways to arrange 4 people in a line is 4 + 3 + 2 + 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S43-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q5/5

1428

QUESTION ID
S43-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $4 \cdot 3 \cdot 2 \cdot 1 = 24$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S43-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q5/5

SET 43 • HSS-CP.B.9

FORWARD 2

Count selections where order does not matter • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

How many ways can 3 of 5 students be chosen for a team (order does not matter)?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q1/5

1430

QUESTION ID
S43-F2-Q1

TYPE
Short answer

CORRECT ANSWER
10

GRADING NOTE 60 ordered / 3! = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F2-Q1**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q1/5

True or false?

Each unordered group of 3 people can be ordered in 6 ways.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q2/5

1431

QUESTION ID

S43-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 3!.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q2/5

Which situation is counted with combinations rather than permutations?

A. choosing 3 pizza toppings

B. awarding gold, silver and bronze

C. making a 4-digit code

D. arranging books on a shelf

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q3/5

1432

QUESTION ID
S43-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. choosing 3 pizza toppings

GRADING NOTE Order of toppings does not matter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A permutation count is 120 for choosing 3 of 6 in order. How many unordered groups is that?

Groups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q4/5

1433

QUESTION ID
S43-F2-Q4

TYPE
Short answer

CORRECT ANSWER
20

GRADING NOTE 120/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q4/5

True or false?

There are more combinations than permutations of 3 objects from 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q5/5

1434

QUESTION ID

S43-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Fewer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q5/5

Count unordered selections with combinations (+)

MAIN	Probability & Statistics main question	**HSS-CP.B.9**
BACKWARD 1	Count ordered arrangements	**HSS-CP.B.9**
BACKWARD 2	Represent and count outcomes of compound events	**7.SP.C.8.b**
FORWARD 1	Count selections from separate groups	**HSS-CP.B.9**
FORWARD 2	Turn counts into probabilities	**HSS-CP.B.9**

SET 44 • HSS-CP.B.9

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

How many ways can 3 students be chosen from 5 (order does not matter)?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S44-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q1/5

1437

QUESTION ID

S44-M-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE 5C3 = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-M-Q1**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q1/5

A pizza shop has 8 toppings. How many 3-topping pizzas are possible (no repeated toppings)?

Pizzas: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q2/5

1438

QUESTION ID
S44-M-Q2

TYPE
Short answer

CORRECT ANSWER
56

GRADING NOTE 8C3 = 56.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-M-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q2/5

How many 4-person committees can be chosen from 10 people?

Committees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q3/5

1439

QUESTION ID

S44-M-Q3

TYPE

Short answer

CORRECT ANSWER

210

GRADING NOTE 10C4 = 210.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-M-Q3**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the value of 6C2?

A. 15

B. 12

C. 36

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q4/5

1440

QUESTION ID

S44-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 15

GRADING NOTE 6·5/2. 30 is 6P2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Seven people each shake hands once with every other person. How many handshakes are there?

Handshakes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q5/5

1441

QUESTION ID
S44-M-Q5

TYPE
Short answer

CORRECT ANSWER
21

GRADING NOTE 7C2 = 21.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q5/5

SET 44 • HSS-CP.B.9

BACKWARD 1

Count ordered arrangements • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

ANSWER KEY Reduced copy of the question page, with its answer

How many ordered ways can 3 of 5 students be lined up?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q1/5

1443

QUESTION ID
S44-B1-Q1

TYPE
Short answer

CORRECT ANSWER
60

GRADING NOTE $5 \cdot 4 \cdot 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-B1-Q1**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q1/5

True or false?

5P3 = 60.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q2/5

1444

QUESTION ID

S44-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 5 · 4 · 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q2/5

How many ordered pairs of different people can be formed from 7 people?

A. 42

B. 21

C. 49

D. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q3/5

1445

QUESTION ID

S44-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 42

GRADING NOTE $7 \cdot 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

How many ways can 3 people be arranged in order?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q4/5

1446

QUESTION ID

S44-B1-Q4

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE 3!.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q4/5

True or false?

10P4 = 210.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q5/5

1447

QUESTION ID

S44-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 5040; 210 is 10C4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • HSS-CP.B.9

BACKWARD 2

Represent and count outcomes of compound events • Grade 7 • Statistics & Probability • 7.SP.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

How many 2-letter groups (order does not matter) can be chosen from the letters A, B, C?

Groups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q1/5

1449

GRADING NOTE AB, AC, BC.

QUESTION ID

S44-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q1/5

True or false?

From A, B, C, D, the groups AB and BA are the same group.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q2/5

1450

QUESTION ID

S44-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Unordered.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q2/5

How many 2-letter groups (order does not matter) can be chosen from A, B, C, D?

A. 8

B. 6

C. 4

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q3/5

1451

QUESTION ID

S44-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6

GRADING NOTE AB, AC, AD, BC, BD, CD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q3/5

How many ordered pairs of different letters can be made from A, B, C, D?

Pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q4/5

1452

QUESTION ID

S44-B2-Q4

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE 4 · 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q4/5

True or false?

An organized list of groups should include both AB and BA when order does not matter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q5/5

1453

QUESTION ID

S44-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE List each group once.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q5/5

SET 44 • HSS-CP.B.9

FORWARD 1

Count selections from separate groups • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

A committee needs 3 of 5 boys and 2 of 4 girls. How many committees are possible?

Committees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q1/5

1455

QUESTION ID

S44-F1-Q1

TYPE

Short answer

CORRECT ANSWER

60

GRADING NOTE $10 \cdot 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How many ways can 2 hearts be chosen from the 13 hearts in a standard 52-card deck?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q2/5

1456

QUESTION ID

S44-F1-Q2

TYPE

Short answer

CORRECT ANSWER

78

GRADING NOTE 13C2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q2/5

True or false?

nCr = n!/(r!(n − r)!).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q3/5

1457

QUESTION ID

S44-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

How many 5-card hands can be chosen from a standard 52-card deck (order does not matter)?

A. 2,598,960

B. 380,204,032

C. 260

D. 311,875,200

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q4/5

1458

QUESTION ID
S44-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. 2,598,960

GRADING NOTE 52C5. 311,875,200 is 52P5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q4/5

True or false?

Choosing 2 from 4 girls and 3 from 5 boys gives 6 + 10 = 16 committees.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q5/5

1459

QUESTION ID

S44-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Multiply: 60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q5/5

SET 44 • HSS-CP.B.9

FORWARD 2

Turn counts into probabilities • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

ANSWER KEY Reduced copy of the question page, with its answer

3 people are chosen at random from 5 boys and 4 girls.
How many groups are possible in all?

Groups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q1/5

1461

QUESTION ID
S44-F2-Q1

TYPE
Short answer

CORRECT ANSWER
84

GRADING NOTE 9C3 = 84.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

3 people are chosen at random, without replacement (every group of 3 equally likely), from 5 boys and 4 girls. What is P(all three are girls)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q2/5

1462

QUESTION ID
S44-F2-Q2

TYPE
Short answer

CORRECT ANSWER
1/21

GRADING NOTE 4C3/9C3 = 4/84.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q2/5

True or false?

In a uniform model, P(event) = (number of outcomes in the event)/(number of outcomes in all).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q3/5

1463

QUESTION ID
S44-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Counting probability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q3/5

2 cards are drawn at random, without replacement, from a standard 52-card deck. What is P(both hearts)?

A. 1/17

B. 1/16

C. 1/4

D. 13/52

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q4/5

1464

QUESTION ID
S44-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. 1/17

GRADING NOTE 78/1326.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q4**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q4/5

True or false?

P(all three are boys) from 5 boys and 4 girls is 3/9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q5/5

1465

QUESTION ID

S44-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 10/84 = 5/42.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q5/5

Compute probabilities with permutations and combinations (+)

MAIN	Probability & Statistics main question	**HSS-CP.B.9**
BACKWARD 1	Count with permutations and combinations	**HSS-CP.B.9**
BACKWARD 2	Multiply conditional probabilities along draws	**HSS-CP.B.8**
FORWARD 1	Use complements with counting	**HSS-CP.B.9**
FORWARD 2	Define a random variable from a counting model	**HSS-MD.A.1**

SET 45 • HSS-CP.B.9

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

A lottery draws 3 different numbers from 1 to 10 (order does not matter). What is the probability that one ticket with 3 numbers wins? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q1/5

1468

QUESTION ID
S45-M-Q1

TYPE
Short answer

CORRECT ANSWER
1/120

GRADING NOTE 1/10C3 = 1/120.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-M-Q1**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A 3-person group is chosen at random from 5 boys and 4 girls. What is P(all girls)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q2/5

1469

QUESTION ID
S45-M-Q2

TYPE
Short answer

CORRECT ANSWER
1/21

GRADING NOTE 4/84.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Five people line up in random order. What is P(Ann is first and Bo is second)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q3/5

1470

QUESTION ID
S45-M-Q3

TYPE
Short answer

CORRECT ANSWER
1/20

GRADING NOTE 3!/5! = 6/120.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q3/5

Two cards are dealt at random, without replacement, from a standard 52-card deck. What is P(both are hearts)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q4/5

1471

QUESTION ID

S45-M-Q4

TYPE

Short answer

CORRECT ANSWER

1/17

GRADING NOTE 13C2/52C2 = 78/1326.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q4/5

A 3-person group is chosen at random from 5 boys and 4 girls. What is P(exactly one girl)?

A. 1/3

B. 5/21

C. 10/21

D. 4/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q5/5

1472

QUESTION ID

S45-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 10/21

GRADING NOTE 4C1 · 5C2 / 84 = 40/84.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-M-Q5**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q5/5

SET 45 • HSS-CP.B.9

BACKWARD 1

Count with permutations and combinations • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

ANSWER KEY Reduced copy of the question page, with its answer

Compute 10C3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q1/5

1474

QUESTION ID

S45-B1-Q1

TYPE

Short answer

CORRECT ANSWER

120

GRADING NOTE 720/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 5!.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q2/5

1475

QUESTION ID
S45-B1-Q2

TYPE
Short answer

CORRECT ANSWER
120

GRADING NOTE 5 · 4 · 3 · 2 · 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-B1-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q2/5

True or false?

9C3 = 84.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q3/5

1476

QUESTION ID

S45-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 504/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q3/5

Compute 52C2.

A. 676

B. 1,326

C. 2,652

D. 104

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q4/5

1477

QUESTION ID

S45-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1,326

GRADING NOTE 52 · 51/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q4/5

True or false?

5C2 = 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q5/5

1478

QUESTION ID

S45-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q5/5

SET 45 • HSS-CP.B.9

BACKWARD 2

Multiply conditional probabilities along draws • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.8

Using the Multiplication Rule, find P(both hearts) for two cards dealt at random, without replacement, from a standard 52-card deck. Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q1/5

1480

QUESTION ID
S45-B2-Q1

TYPE
Short answer

CORRECT ANSWER
1/17

GRADING NOTE (13/52)(12/51).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-B2-Q1**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q1/5

True or false?

(4/9)(3/8)(2/7) = 1/21.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q2/5

1481

QUESTION ID

S45-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 24/504.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q2/5

From 5 boys and 4 girls, two are chosen in order without replacement. P(both boys)?

A. 25/81

B. 1/4

C. 5/9

D. 5/18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q3/5

1482

QUESTION ID

S45-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5/18

GRADING NOTE (5/9)(4/8).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-B2-Q3**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Five people line up at random. P(Ann first)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q4/5

1483

QUESTION ID
S45-B2-Q4

TYPE
Short answer

CORRECT ANSWER
1/5

GRADING NOTE 1 of 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-B2-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q4/5

True or false?

A group has 5 boys and 4 girls. Two people are chosen one after the other, at random and without replacement. Given that the first is a girl, P(second is a girl) = 4/9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q5/5

1484

QUESTION ID
S45-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 3/8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q5/5

SET 45 • HSS-CP.B.9

FORWARD 1

Use complements with counting • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

ANSWER KEY Reduced copy of the question page, with its answer

A 3-person group is chosen at random from 5 boys and 4 girls. What is P(no girls)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q1/5

1486

QUESTION ID

S45-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5/42

GRADING NOTE 10/84.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A 3-person group is chosen at random from 5 boys and 4 girls. What is P(at least one girl)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q2/5

1487

QUESTION ID
S45-F1-Q2

TYPE
Short answer

CORRECT ANSWER
37/42

GRADING NOTE 1 − 5/42.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F1-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q2/5

True or false?

P(at least one) = 1 − P(none).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q3/5

1488

QUESTION ID

S45-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Complement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q3/5

Four cards are dealt at random, without replacement, from a standard 52-card deck. What is P(no aces)? (48C4 = 194,580 and 52C4 = 270,725.)

A. 194,580/52

B. 4/52

C. 1/13

D. 38,916/54,145

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q4/5

1489

QUESTION ID

S45-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 38,916/54,145

GRADING NOTE 194,580/270,725 in lowest terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F1-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

P(at least one girl) = 3 · (4/9).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q5/5

1490

QUESTION ID

S45-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Not a valid method.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q5/5

SET 45 • HSS-CP.B.9

FORWARD 2

Define a random variable from a counting model • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

X = the number of girls in a group of 3 chosen at random (without replacement, every group equally likely) from 5 boys and 4 girls. What values can X take? Give the smallest and the largest.

Smallest: _____

Largest: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q1/5

1492

QUESTION ID

S45-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0; 3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F2-Q1**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

X = the number of girls in a group of 3 chosen at random (without replacement, every group equally likely) from 5 boys and 4 girls. Find P(X = 0) and P(X = 3). Give a fraction in lowest terms.

P(X = 0): _____

P(X = 3): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q2/5

1493

QUESTION ID

S45-F2-Q2

TYPE

Short answer

CORRECT ANSWER

5/42; 1/21

GRADING NOTE Form required: both. 10/84 and 4/84.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F2-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q2/5

True or false?

X = the number of girls in a group of 3 chosen at random (without replacement, every group equally likely) from 5 boys and 4 girls. The probabilities P(X = 0), P(X = 1), P(X = 2), P(X = 3) add to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S45-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q3/5

1494

QUESTION ID

S45-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 10 + 40 + 30 + 4 = 84.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S45-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q3/5

X = the number of girls in a group of 3 chosen at random (without replacement, every group equally likely) from 5 boys and 4 girls. What is P(X = 2)?

A. 5/14

B. 2/9

C. 1/21

D. 10/21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q4/5

1495

QUESTION ID
S45-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. 5/14

GRADING NOTE 4C2 · 5C1/84 = 30/84.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F2-Q4**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q4/5

True or false?

A random variable assigns a probability, not a number, to each outcome.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q5/5

1496

QUESTION ID
S45-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It assigns a number; the distribution gives probabilities.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q5/5

Define a random variable and graph its distribution (+)

MAIN	Probability & Statistics main question	**HSS-MD.A.1**
BACKWARD 1	Describe outcomes in a sample space	**HSS-CP.A.1**
BACKWARD 2	Read and draw a distribution display	**HSS-ID.A.1**
FORWARD 1	Find probabilities of ranges of values	**HSS-MD.A.1**
FORWARD 2	Compute the expected value of a random variable	**HSS-MD.A.2**

SET 46 • HSS-MD.A.1

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.1

Three fair coins are flipped independently. X = number of heads. Find P(X = 0), P(X = 1), P(X = 2) and P(X = 3). Give a fraction in lowest terms.

P(X = 0): _____

P(X = 1): _____

P(X = 2): _____

P(X = 3): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q1/5

1499

QUESTION ID

S46-M-Q1

TYPE

Short answer

CORRECT ANSWER

1/8; 3/8; 3/8; 1/8

GRADING NOTE Form required: all four. Outcomes with 0, 1, 2, 3 heads: 1, 3, 3, 1 of 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two fair dice are rolled independently and X = the sum. How many different values can X take, and what is P(X = 7)? Give a fraction in lowest terms.

Number of values: _____

P(X = 7): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q2/5

1500

QUESTION ID

S46-M-Q2

TYPE

Short answer

CORRECT ANSWER

11; 1/6

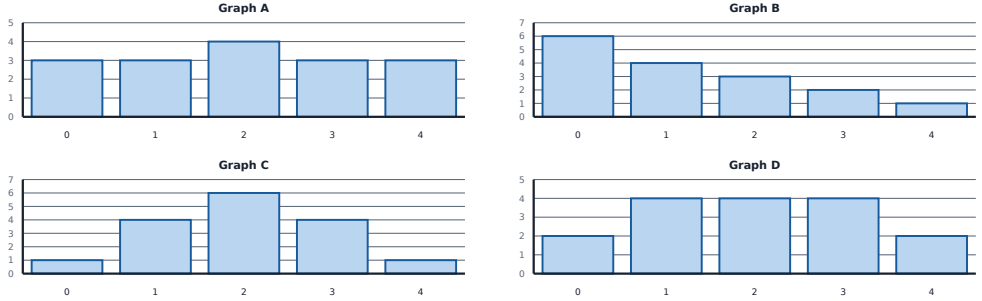
GRADING NOTE Form required: both. 2 to 12; 6 of 36.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-M-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Four fair coins are flipped independently and X = number of heads. Each bar graph shows, for X = 0 to 4, how many of the 16 equally likely outcomes give that value. Which graph shows the distribution of X? In each histogram, the horizontal axis shows number of heads and the vertical axis shows frequency.



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q3/5

1501

QUESTION ID
S46-M-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. Graph C

GRADING NOTE Counts 1, 4, 6, 4, 1 (4C0 to 4C4). The others are flat, skewed, or have the wrong middle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which list could be a probability distribution for X = 1, 2, 3?

A. 0.3, 0.4, 0.4

B. 0.2, 0.2, 0.2

C. 0.5, −0.1, 0.6

D. 0.2, 0.5, 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q4/5

1502

QUESTION ID

S46-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 0.2, 0.5, 0.3

GRADING NOTE Each between 0 and 1 and summing to 1. The others sum to 1.1, include a negative, or sum to 0.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

X takes the values 0, 1, 2, 3 with P(X = 0) = 0.1, P(X = 1) = 0.35 and P(X = 3) = 0.15. Find P(X = 2).

P(X = 2): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q5/5

1503

QUESTION ID
S46-M-Q5

TYPE
Short answer

CORRECT ANSWER
0.4

GRADING NOTE 1 − 0.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q5/5

SET 46 • HSS-MD.A.1

BACKWARD 1

Describe outcomes in a sample space • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Three fair coins are flipped independently. How many outcomes have exactly 1 head?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q1/5

1505

QUESTION ID
S46-B1-Q1

TYPE
Short answer

CORRECT ANSWER
3

GRADING NOTE HTT, THT, TTH.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-B1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q1/5

True or false?

Three fair coins are flipped independently. The 8 outcomes of the sample space are equally likely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q2/5

1506

QUESTION ID

S46-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 2^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q2/5

Two fair dice are rolled independently. How many outcomes have a sum of 4?

A. 3

B. 2

C. 6

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q3/5

1507

QUESTION ID

S46-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3

GRADING NOTE (1,3), (2,2), (3,1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-B1-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Four fair coins are flipped independently. How many outcomes are there?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q4/5

1508

QUESTION ID

S46-B1-Q4

TYPE

Short answer

CORRECT ANSWER

16

GRADING NOTE 2^4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-B1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q4/5

True or false?

For two fair dice rolled independently, the sums 2 and 12 each come from 2 outcomes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q5/5

1509

QUESTION ID

S46-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE One each.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q5/5

SET 46 • HSS-MD.A.1

BACKWARD 2

Read and draw a distribution display • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In a bar graph of a distribution, taller bars show more likely values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q1/5

1511

QUESTION ID

S46-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Height is frequency.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q1/5

A bar graph shows counts 1, 4, 6, 4, 1 for X = 0 to 4. What is its shape?

A. skewed right

B. uniform

C. skewed left

D. symmetric

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q2/5

1512

QUESTION ID

S46-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. symmetric

GRADING NOTE Mirror image.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A bar graph has counts 1, 4, 6, 4, 1 for X = 0 to 4. What is the total count?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q3/5

1513

QUESTION ID

S46-B2-Q3

TYPE

Short answer

CORRECT ANSWER

16

GRADING NOTE Sum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-B2-Q3**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q3/5

True or false?

A bar graph with counts 6, 4, 3, 2, 1 is symmetric.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q4/5

1514

QUESTION ID

S46-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Skewed right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Counts are 1, 4, 6, 4, 1 for X = 0 to 4. Which value of X is most likely?

A. 1

B. 2

C. 0

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q5/5

1515

QUESTION ID

S46-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 2

GRADING NOTE Tallest bar.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-B2-Q5**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • HSS-MD.A.1

FORWARD 1

Find probabilities of ranges of values • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.1

For X = number of heads in 3 flips, find P(X ≥ 2). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q1/5

1517

QUESTION ID
S46-F1-Q1

TYPE
Short answer

CORRECT ANSWER
1/2

GRADING NOTE 3/8 + 1/8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For X = sum of two fair dice rolled independently, find P(X ≤ 4). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q2/5

1518

QUESTION ID
S46-F1-Q2

TYPE
Short answer

CORRECT ANSWER
1/6

GRADING NOTE 1 + 2 + 3 = 6 outcomes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q2/5

True or false?

The probabilities of all values of a random variable add to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q3/5

1519

QUESTION ID
S46-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Distribution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q3/5

X takes the values 0 to 3 with probabilities 0.1, 0.35, 0.4, 0.15. What is P(X > 1)?

A. 0.9

B. 0.45

C. 0.4

D. 0.55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q4/5

1520

QUESTION ID
S46-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. 0.55

GRADING NOTE 0.4 + 0.15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F1-Q4**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$P(X \geq 1) = 1 - P(X = 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q5/5

1521

QUESTION ID

S46-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $1 - P(X = 0)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q5/5

SET 46 • HSS-MD.A.1

FORWARD 2

Compute the expected value of a random variable • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

X = number of heads when 2 fair coins are flipped independently. Find the expected value of X.

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q1/5

1523

QUESTION ID
S46-F2-Q1

TYPE
Short answer

CORRECT ANSWER
1

GRADING NOTE 0(1/4) + 1(1/2) + 2(1/4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q1**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q1/5

True or false?

The expected value is the probability-weighted average of the values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q2/5

1524

QUESTION ID
S46-F2-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

X takes 1, 2, 3 with probabilities 0.2, 0.5, 0.3. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q3/5

1525

QUESTION ID
S46-F2-Q3

TYPE
Short answer

CORRECT ANSWER
2.1

GRADING NOTE 0.2 + 1 + 0.9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q3/5

X = result of spinning a spinner with 4 equal sections numbered 1 to 4. What is E(X)?

A. 2

B. 4

C. 10

D. 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q4/5

1526

QUESTION ID
S46-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. 2.5

GRADING NOTE 10/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q4**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q4/5

True or false?

The expected value must be one of the values X can take.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q5/5

1527

QUESTION ID
S46-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE E(X) = 2.5 for the spinner.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q5/5

Calculate and interpret expected value (+)

MAIN	Probability & Statistics main question	**HSS-MD.A.2**
BACKWARD 1	Compute a mean	**6.SP.B.5.c**
BACKWARD 2	Read a probability distribution	**HSS-MD.A.1**
BACKWARD 3	Multiply and add decimals	**6.NS.B.3**
FORWARD 1	Expected value with gains and losses	**HSS-MD.A.2**
FORWARD 2	Find expected values of theoretical distributions	**HSS-MD.A.3**

SET 47 • HSS-MD.A.2

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

X = number of heads when 3 fair coins are flipped independently (table). Find E(X).

x	0	1	2	3
P(X = x)	1/8	3/8	3/8	1/8

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q1/5

1530

QUESTION ID
S47-M-Q1

TYPE
Short answer

CORRECT ANSWER
1.5

GRADING NOTE 0(1/8) + 1(3/8) + 2(3/8) + 3(1/8) = 12/8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-M-Q1**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find E(X) for the distribution in the table.

x	0	1	2	3
P(X = x)	0.1	0.35	0.4	0.15

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q2/5

1531

QUESTION ID
S47-M-Q2

TYPE
Short answer

CORRECT ANSWER
1.6

GRADING NOTE 0.35 + 0.8 + 0.45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q2/5

X = the number rolled on a fair six-sided die. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q3/5

1532

QUESTION ID
S47-M-Q3

TYPE
Short answer

CORRECT ANSWER
3.5

GRADING NOTE (1 + 2 + ... + 6)/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-M-Q3**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For the distribution in the table, E(X) = 1.6. What does this mean?

x	0	1	2	3
P(X = x)	0.1	0.35	0.4	0.15

A. The single most likely value of X is 1.6.

B. X will equal exactly 1.6 on most of the trials.

C. Over many trials, the average value of X is about 1.6.

D. X is 1.6 or less with a probability of 1.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q4/5

1533

QUESTION ID

S47-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Over many trials, the average value of X is about 1.6.

GRADING NOTE Long-run mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q4/5

A game pays $10 with probability 0.2, $2 with probability 0.3, and $0 otherwise. Find the expected payout.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q5/5

1534

QUESTION ID
S47-M-Q5

TYPE
Short answer

CORRECT ANSWER
2.6

GRADING NOTE 2 + 0.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q5/5

SET 47 • HSS-MD.A.2

BACKWARD 1

Compute a mean • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean of 1, 2, 3, 4, 5, 6.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q1/5

1536

QUESTION ID

S47-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3.5

GRADING NOTE 21/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 0, 1, 1, 1, 2, 2, 2, 3.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q2/5

1537

QUESTION ID

S47-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1.5

GRADING NOTE 12/8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q2/5

True or false?

A value listed 3 times counts 3 times in the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q3/5

1538

QUESTION ID

S47-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Weights.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 2, 2, 2, 8.

A. 14

B. 5

C. 2

D. 3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q4/5

1539

QUESTION ID

S47-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 3.5

GRADING NOTE 14/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q4/5

True or false?

The mean of 0, 0, 0, 4 is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q5/5

1540

QUESTION ID

S47-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 4/4 = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q5/5

SET 47 • HSS-MD.A.2

BACKWARD 2

Read a probability distribution • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

In the distribution table with values 0 to 3, what is P(X = 2)?

x	0	1	2	3
P(X = x)	0.1	0.35	0.4	0.15

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q1/5

1542

QUESTION ID

S47-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.4

GRADING NOTE Read the table.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q1/5

True or false?

A probability distribution lists every value of X with its probability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q2/5

1543

QUESTION ID
S47-B2-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q2/5

In the table (X = number of heads when 3 fair coins are flipped independently), which values of X are equally likely?

x	0	1	2	3
P(X = x)	1/8	3/8	3/8	1/8

A. 2 and 3 **B.** 0 and 1 **C.** all four **D.** 1 and 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q3/5

1544

QUESTION ID
S47-B2-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. 1 and 2

GRADING NOTE P(X = 1) = P(X = 2) = 3/8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q3/5

A distribution has P(X = 5) = 0.3 and P(X = 10) = 0.7.
What is P(X = 5) + P(X = 10)?

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q4/5

1545

QUESTION ID
S47-B2-Q4

TYPE
Short answer

CORRECT ANSWER
1

GRADING NOTE Total probability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q4/5

True or false?

A distribution may have probabilities adding to 1.2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S47-B2-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q5/5

1546

QUESTION ID
S47-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Must add to 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-B2-Q5**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • HSS-MD.A.2

BACKWARD 3

Multiply and add decimals • Grade 6 • The Number System • 6.NS.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Compute 2(0.4) + 3(0.15).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q1/5

1548

QUESTION ID

S47-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1.25

GRADING NOTE 0.8 + 0.45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 10(0.2) + 2(0.3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q2/5

1549

GRADING NOTE 2 + 0.6.

QUESTION ID

S47-B3-Q2

TYPE

Short answer

CORRECT ANSWER

2.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q2/5

True or false?

1(0.35) + 2(0.4) = 1.15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q3/5

1550

QUESTION ID

S47-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 0.35 + 0.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 5(0.3) + 10(0.7).

A. 8.5

B. 7

C. 1.5

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q4/5

1551

QUESTION ID

S47-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 8.5

GRADING NOTE 1.5 + 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q4/5

True or false?

$4(0.25) + 2(0.5) = 1.5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q5/5

1552

GRADING NOTE $1 + 1 = 2$.

QUESTION ID

S47-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q5/5

SET 47 • HSS-MD.A.2

FORWARD 1

Expected value with gains and losses • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A game: win $5 with probability 0.2, lose $2 with probability 0.8. Find the expected gain.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q1/5

1554

QUESTION ID
S47-F1-Q1

TYPE
Short answer

CORRECT ANSWER
−0.6

GRADING NOTE 1 − 1.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F1-Q1**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q1/5

True or false?

A negative expected gain means a player loses money on average over many plays.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q2/5

1555

QUESTION ID
S47-F1-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Long run.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

X has values −3, 0, 6 with probabilities 0.5, 0.2, 0.3. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q3/5

1556

QUESTION ID
S47-F1-Q3

TYPE
Short answer

CORRECT ANSWER
0.3

GRADING NOTE −1.5 + 1.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F1-Q3**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q3/5

A game has expected gain −$0.60. About how much does a player expect to lose in 100 plays?

A. $600

B. $6

C. $0.60

D. $60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q4/5

1557

QUESTION ID
S47-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. $60

GRADING NOTE 100 · 0.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F1-Q4**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q4/5

True or false?

If E(X) = 0, the player breaks even on every single play.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q5/5

1558

QUESTION ID
S47-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Only on average.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q5/5

SET 47 • HSS-MD.A.2

FORWARD 2

Find expected values of theoretical distributions • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A student guesses at random on 6 true-false questions (each answer is True or False, and each guess is correct with probability 1/2, independently of the others). X = number correct. Find P(X = 3). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q1/5

1560

QUESTION ID
S47-F2-Q1

TYPE
Short answer

CORRECT ANSWER
5/16

GRADING NOTE 6C3/64 = 20/64.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A student guesses at random on 6 true-false questions (each answer is True or False, and each guess is correct with probability 1/2, independently of the others). Find the expected number correct.

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q2/5

1561

QUESTION ID
S47-F2-Q2

TYPE
Short answer

CORRECT ANSWER
3

GRADING NOTE 6 · 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q2/5

True or false?

For n independent trials with success probability p, the expected number of successes is np.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q3/5

1562

QUESTION ID
S47-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Binomial mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q3/5

A student guesses at random on 10 four-choice questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others). What is the expected number correct?

A. 2.5

B. 0.25

C. 10

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q4/5

1563

QUESTION ID
S47-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. 2.5

GRADING NOTE 10 · 1/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q4/5

True or false?

A student guesses at random on 4 true-false questions (each answer is True or False, and each guess is correct with probability 1/2, independently of the others). The probability of getting all 4 right is 1/4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q5/5

1564

QUESTION ID
S47-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 1/16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F2-Q5**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q5/5

Develop a theoretical probability distribution (+)

MAIN	Probability & Statistics main question	**HSS-MD.A.3**
BACKWARD 1	Count arrangements of successes	**HSS-CP.B.9**
BACKWARD 2	Multiply probabilities of independent trials	**HSS-CP.A.2**
FORWARD 1	Build the whole theoretical distribution	**HSS-MD.A.3**
FORWARD 2	Compare a theoretical distribution with data	**HSS-MD.A.4**

SET 48 • HSS-MD.A.3

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A student guesses at random on 4 true-false questions (each answer is True or False, and each guess is correct with probability 1/2, independently of the others). X = number correct. Find P(X = 2). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q1/5

1567

QUESTION ID
S48-M-Q1

TYPE
Short answer

CORRECT ANSWER
3/8

GRADING NOTE 4C2 · $(1/2)^4$ = 6/16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A student guesses at random on 4 true-false questions (each answer is True or False, and each guess is correct with probability 1/2, independently of the others).
X = number correct. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q2/5

1568

QUESTION ID
S48-M-Q2

TYPE
Short answer

CORRECT ANSWER
2

GRADING NOTE 4 · 1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A student guesses at random on 5 questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others). Find P(all 5 correct) and the expected number correct. Give each answer as a fraction in lowest terms.

P(all correct): _____

Expected number correct: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q3/5

1569

QUESTION ID
S48-M-Q3

TYPE
Short answer

CORRECT ANSWER
1/1024; 5/4

GRADING NOTE Form required: both, as fractions. $(1/4)^5$; $5 \cdot 1/4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A student guesses at random on 3 four-choice questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others). What is P(exactly 1 correct)?

A. 27/64

B. 9/64

C. 1/4

D. 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q4/5

1570

QUESTION ID

S48-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 27/64

GRADING NOTE $3 \cdot (1/4)(3/4)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A student guesses at random on 3 four-choice questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others). What is P(at least 1 correct)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q5/5

1571

QUESTION ID
S48-M-Q5

TYPE
Short answer

CORRECT ANSWER
37/64

GRADING NOTE $1 - (3/4)^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q5/5

SET 48 • HSS-MD.A.3

BACKWARD 1

Count arrangements of successes • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

ANSWER KEY Reduced copy of the question page, with its answer

In how many ways can exactly 2 of 4 answers be correct?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q1/5

1573

QUESTION ID

S48-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE 4C2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-B1-Q1**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q1/5

In how many ways can exactly 1 of 3 answers be correct?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q2/5

1574

QUESTION ID
S48-B1-Q2

TYPE
Short answer

CORRECT ANSWER
3

GRADING NOTE 3C1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-B1-Q2**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q2/5

True or false?

5C5 = 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q3/5

1575

QUESTION ID

S48-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE One way to get all correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q3/5

In how many ways can exactly 3 of 5 answers be correct?

A. 10

B. 60

C. 15

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q4/5

1576

QUESTION ID

S48-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 10

GRADING NOTE 5C3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q4/5

True or false?

4C2 = 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q5/5

1577

QUESTION ID

S48-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • HSS-MD.A.3

BACKWARD 2

Multiply probabilities of independent trials • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Compute $(1/2)^4$. Give a fraction in lowest terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q1/5

1579

QUESTION ID

S48-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1/16

GRADING NOTE 1/16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-B2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute (1/4)(3/4)(3/4). Give a fraction in lowest terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q2/5

1580

QUESTION ID
S48-B2-Q2

TYPE
Short answer

CORRECT ANSWER
9/64

GRADING NOTE 9/64.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-B2-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q2/5

True or false?

When each guess is made at random and independently, being right on one question does not change the chance of being right on another.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q3/5

1581

QUESTION ID
S48-B2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE That is what independence means.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

P(correct) = 1/4 on each of 2 independent guesses. P(both correct)?

A. 1/8

B. 1/2

C. 1/16

D. 1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q4/5

1582

QUESTION ID

S48-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 1/16

GRADING NOTE $(1/4)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q4/5

True or false?

$(3/4)^3 = 9/12$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q5/5

1583

QUESTION ID

S48-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 27/64.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q5/5

SET 48 • HSS-MD.A.3

FORWARD 1

Build the whole theoretical distribution • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A student guesses at random on 3 four-choice questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others); X = number correct. Find P(X = 0), P(X = 1), P(X = 2), P(X = 3). Give a fraction in lowest terms.

P(X = 0): _____

P(X = 1): _____

P(X = 2): _____

P(X = 3): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q1/5

1585

QUESTION ID

S48-F1-Q1

TYPE

Short answer

CORRECT ANSWER

27/64; 27/64; 9/64; 1/64

GRADING NOTE Form required: all four. $3Ck\ (1/4)^k\ (3/4)^{3-k}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A student guesses at random on 3 four-choice questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others); X = number correct. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q2/5

1586

QUESTION ID
S48-F1-Q2

TYPE
Short answer

CORRECT ANSWER
0.75

GRADING NOTE 3 · 1/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q2/5

True or false?

The four probabilities 27/64, 27/64, 9/64, 1/64 add to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q3/5

1587

QUESTION ID
S48-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE 64/64.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q3/5

A student guesses at random on 3 four-choice questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others); X = number correct. Which value of X is least likely?

A. 3

B. 2

C. 0

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q4/5

1588

QUESTION ID
S48-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. 3

GRADING NOTE 1/64.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q4/5

True or false?

A student guesses at random on 3 four-choice questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others); X = number correct. X = 1 is more likely than X = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q5/5

1589

QUESTION ID
S48-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Equal: 27/64 each.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q5/5

SET 48 • HSS-MD.A.3

FORWARD 2

Compare a theoretical distribution with data • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A fair six-sided die is rolled 60 times, independently. How many sixes does the theoretical distribution predict?

Sixes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q1/5

1591

QUESTION ID
S48-F2-Q1

TYPE
Short answer

CORRECT ANSWER
10

GRADING NOTE 60/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In 60 rolls, a die showed 6 sixes. What is the empirical probability of a six? Give a decimal.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q2/5

1592

QUESTION ID
S48-F2-Q2

TYPE
Short answer

CORRECT ANSWER
0.1

GRADING NOTE 6/60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q2**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q2/5

True or false?

An empirical probability is found from observed data.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q3/5

1593

QUESTION ID
S48-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q3/5

In 100 households, the numbers of TVs gave an average of 1.9. This average is

A. a theoretical probability from a model

B. always equal to 2 for any survey

C. the expected value of the empirical distribution

D. a margin of error for the survey

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q4/5

1594

QUESTION ID
S48-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. the expected value of the empirical distribution

GRADING NOTE Mean from data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q4**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q4/5

True or false?

Empirical and theoretical probabilities for the same event are always equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q5/5

1595

QUESTION ID
S48-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Data vary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q5/5

Develop an empirical probability distribution (+)

MAIN	Probability & Statistics main question	**HSS-MD.A.4**
BACKWARD 1	Find relative frequencies	**7.SP.C.6**
BACKWARD 2	Compute an expected value	**HSS-MD.A.2**
FORWARD 1	Compare empirical and theoretical results	**HSS-MD.A.4**
FORWARD 2	Find the expected payoff of a game of chance	**HSS-MD.B.5.a**

SET 49 • HSS-MD.A.4

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.4

ANSWER KEY Reduced copy of the question page, with its answer

100 households were surveyed (table). X = number of TVs in a randomly chosen household. Find P(X = 2) and P(X ≥ 3). Give decimals.

TVs in household	0	1	2	3	4
Households	5	30	40	20	5

P(X = 2): _____

P(X ≥ 3): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • MAIN • Q1/5

1598

QUESTION ID
S49-M-Q1

TYPE
Short answer

CORRECT ANSWER
0.4; 0.25

GRADING NOTE Form required: both. 40/100; (20 + 5)/100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • MAIN • Q1/5

Using the TV survey, find the expected number of TVs in a household.

TVs in household	0	1	2	3	4
Households	5	30	40	20	5

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • MAIN • Q2/5

1599

QUESTION ID
S49-M-Q2

TYPE
Short answer

CORRECT ANSWER
1.9

GRADING NOTE (30 + 80 + 60 + 20)/100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-M-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The 100 surveyed households were a random sample of a town's households. Using the sample mean, 1.9 TVs per household, as the estimate, about how many TVs are expected in 500 households of the town?

TVs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • MAIN • Q3/5

1600

QUESTION ID
S49-M-Q3

TYPE
Short answer

CORRECT ANSWER
950

GRADING NOTE 500 · 1.9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The TV survey gives E(X) = 1.9. Which interpretation is correct?

TVs in household	0	1	2	3	4
Households	5	30	40	20	5

A. The surveyed households averaged about 1.9 TVs.

B. Most of the households have exactly 1.9 TVs.

C. Every household surveyed has 1 or 2 TVs.

D. There is a 1.9% chance of owning a TV.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • MAIN • Q4/5

1601

QUESTION ID

S49-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. The surveyed households averaged about 1.9 TVs.

GRADING NOTE Mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A store recorded complaints for 50 days (table). Find the expected number of complaints in a day.

Complaints in a day	0	1	2	3
Days	20	15	10	5

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • MAIN • Q5/5

1602

QUESTION ID
S49-M-Q5

TYPE
Short answer

CORRECT ANSWER
1

GRADING NOTE (15 + 20 + 15)/50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • MAIN • Q5/5

SET 49 • HSS-MD.A.4

BACKWARD 1

Find relative frequencies • Grade 7 • Statistics & Probability • 7.SP.C.6

ANSWER KEY Reduced copy of the question page, with its answer

In 100 households, 40 have 2 TVs. What is the relative frequency? Give a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 49 • BACKWARD 1 • Q1/5

1604

QUESTION ID

S49-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.4

GRADING NOTE 40/100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 49 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

On 20 of 50 days there were no complaints. What is the relative frequency? Give a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 49 • BACKWARD 1 • Q2/5

1605

GRADING NOTE 20/50.

QUESTION ID

S49-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 49 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Relative frequencies of all categories add to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 49 • BACKWARD 1 • Q3/5

1606

QUESTION ID

S49-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE All observations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 49 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

On 5 of 50 days there were 3 complaints. Relative frequency?

A. 5

B. 0.1

C. 0.05

D. 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 49 • BACKWARD 1 • Q4/5

1607

QUESTION ID

S49-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0.1

GRADING NOTE 5/50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 49 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A relative frequency of 0.25 means 25 observations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 49 • BACKWARD 1 • Q5/5

1608

QUESTION ID

S49-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 25% of them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 49 • BACKWARD 1 • Q5/5

SET 49 • HSS-MD.A.4

BACKWARD 2

Compute an expected value • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

ANSWER KEY Reduced copy of the question page, with its answer

X takes 0, 1, 2 with probabilities 0.5, 0.3, 0.2. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 49 • BACKWARD 2 • Q1/5

1610

QUESTION ID
S49-B2-Q1

TYPE
Short answer

CORRECT ANSWER
0.7

GRADING NOTE 0.3 + 0.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-B2-Q1**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 49 • BACKWARD 2 • Q1/5

True or false?

$E(X) = \Sigma\, x \cdot P(X = x)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 49 • BACKWARD 2 • Q2/5

1611

QUESTION ID
S49-B2-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 49 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

X takes 1 and 3 with probabilities 0.5 each. E(X)?

A. 4

B. 2

C. 1.5

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 49 • BACKWARD 2 • Q3/5

1612

QUESTION ID
S49-B2-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. 2

GRADING NOTE 0.5 + 1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 49 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

X takes 0, 2, 4 with probabilities 0.25, 0.5, 0.25. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 49 • BACKWARD 2 • Q4/5

1613

QUESTION ID
S49-B2-Q4

TYPE
Short answer

CORRECT ANSWER
2

GRADING NOTE Symmetric.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-B2-Q4**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 49 • BACKWARD 2 • Q4/5

True or false?

E(X) is found by adding the probabilities.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S49-B2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 49 • BACKWARD 2 • Q5/5

1614

QUESTION ID

S49-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Multiply each value by its probability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S49-B2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • HSS-MD.A.4

FORWARD 1

Compare empirical and theoretical results • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A die was rolled 60 times: the faces 1-6 appeared 7, 13, 9, 11, 14, 6 times. Find the empirical P(6) and the theoretical P(6) for a fair die. Give decimals rounded to two places.

Empirical: _____

Theoretical: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • FORWARD 1 • Q1/5

1616

QUESTION ID

S49-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.10; 0.17

GRADING NOTE Form required: both, to two decimal places. 6/60; 1/6 = 0.1666...

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • FORWARD 1 • Q1/5

Using the 60 rolls (faces 1-6 appeared 7, 13, 9, 11, 14, 6 times), find the empirical expected value of a roll.

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • FORWARD 1 • Q2/5

1617

QUESTION ID
S49-F1-Q2

TYPE
Short answer

CORRECT ANSWER
3.5

GRADING NOTE (7 + 26 + 27 + 44 + 70 + 36)/60 = 210/60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F1-Q2**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • FORWARD 1 • Q2/5

True or false?

For repeated independent rolls of a fair die, the empirical probabilities tend to get closer to 1/6 as the number of rolls grows.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • FORWARD 1 • Q3/5

1618

QUESTION ID
S49-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Law of large numbers. For a biased die they approach its own probabilities, not 1/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which result would most suggest the die is unfair?

A. In 600 rolls, a 6 appeared 104 times.

B. In 6,000 rolls, a 6 appeared 1,800 times.

C. In 60 rolls, a 6 appeared 6 times.

D. In 6 rolls, no 6 appeared.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • FORWARD 1 • Q4/5

1619

QUESTION ID

S49-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. In 6,000 rolls, a 6 appeared 1,800 times.

GRADING NOTE 30% in many rolls is far from 1/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • FORWARD 1 • Q4/5

True or false?

An empirical distribution from 10 observations is as reliable as one from 10,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • FORWARD 1 • Q5/5

1620

QUESTION ID
S49-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE More data, more reliable.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • FORWARD 1 • Q5/5

SET 49 • HSS-MD.A.4

FORWARD 2

Find the expected payoff of a game of chance • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.5.a

ANSWER KEY Reduced copy of the question page, with its answer

A $1 ticket wins $200 with probability 1/500 (and nothing otherwise). Find the expected net gain per ticket.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 49 • FORWARD 2 • Q1/5

1622

QUESTION ID

S49-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−0.6

GRADING NOTE 0.4 − 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q1**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 49 • FORWARD 2 • Q1/5

True or false?

A game with negative expected net gain loses money for the player in the long run.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 49 • FORWARD 2 • Q2/5

1623

QUESTION ID
S49-F2-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Long run.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 49 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A spinner: win $6 with probability 1/3, lose $3 otherwise. Find the expected gain.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 49 • FORWARD 2 • Q3/5

1624

QUESTION ID
S49-F2-Q3

TYPE
Short answer

CORRECT ANSWER
0

GRADING NOTE 2 − 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q3**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 49 • FORWARD 2 • Q3/5

A game's expected net gain is −$0.60. Which interpretation is correct?

A. The player wins $0.60 on about half the plays.

B. Every player loses exactly $0.60 on each play.

C. The game is fair, since some plays win.

D. Over many plays, a player loses about $0.60 per play.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 49 • FORWARD 2 • Q4/5

1625

QUESTION ID
S49-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. Over many plays, a player loses about $0.60 per play.

GRADING NOTE Average.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 49 • FORWARD 2 • Q4/5

True or false?

A game with expected net gain $0 is called fair.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 49 • FORWARD 2 • Q5/5

1626

QUESTION ID
S49-F2-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 49 • FORWARD 2 • Q5/5

Weigh the payoffs of a game of chance (+)

MAIN	Probability & Statistics main question	**HSS-MD.B.5.a**
BACKWARD 1	Compute an expected value	**HSS-MD.A.2**
BACKWARD 2	Add and subtract signed numbers	**7.NS.A.3**
FORWARD 1	Set a price that makes a game fair	**HSS-MD.B.5.a**
FORWARD 2	Compare decisions by expected value	**HSS-MD.B.5.b**

SET 50 • HSS-MD.B.5.a

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.5.a

A lottery ticket costs $2 and wins $500 with probability 1/1000; otherwise it wins nothing. Find the expected net gain per ticket.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • MAIN • Q1/5

1629

QUESTION ID
S50-M-Q1

TYPE
Short answer

CORRECT ANSWER
−1.5

GRADING NOTE 500/1000 − 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-M-Q1**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • MAIN • Q1/5

A game: roll a fair die. If it shows 6, you win $5; otherwise you lose $1. Find the expected gain per play.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • MAIN • Q2/5

1630

QUESTION ID
S50-M-Q2

TYPE
Short answer

CORRECT ANSWER
0

GRADING NOTE $5(1/6) - 1(5/6) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A raffle sells 500 tickets at $5 each for one $1,000 prize. Find the expected net gain per ticket.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • MAIN • Q3/5

1631

QUESTION ID
S50-M-Q3

TYPE
Short answer

CORRECT ANSWER
−3

GRADING NOTE 1000/500 − 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • MAIN • Q3/5

A lottery ticket has expected net gain −$1.50. Which interpretation is correct?

A. The lottery is fair, because some tickets win a prize.

B. Every ticket loses exactly $1.50 for the person who buys it.

C. Over many tickets, a buyer loses about $1.50 per ticket on average.

D. Buyers win $1.50 on about half of the tickets they buy.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • MAIN • Q4/5

1632

QUESTION ID
S50-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. Over many tickets, a buyer loses about $1.50 per ticket on average.

GRADING NOTE Long-run average.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A spinner wins $8 with probability 1/4 and loses $2 otherwise. Find the expected gain per spin.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • MAIN • Q5/5

1633

QUESTION ID
S50-M-Q5

TYPE
Short answer

CORRECT ANSWER
0.5

GRADING NOTE 2 − 1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • MAIN • Q5/5

SET 50 • HSS-MD.B.5.a

BACKWARD 1

Compute an expected value • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

X = 10 with probability 0.1 and X = 0 otherwise. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 50 • BACKWARD 1 • Q1/5

1635

QUESTION ID
S50-B1-Q1

TYPE
Short answer

CORRECT ANSWER
1

GRADING NOTE 10 · 0.1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-B1-Q1**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 50 • BACKWARD 1 • Q1/5

True or false?

If X = 10 with probability 0.1 and 0 otherwise, then E(X) = 10(0.1) + 0(0.9).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 50 • BACKWARD 1 • Q2/5

1636

QUESTION ID
S50-B1-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 50 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

X = 4 with probability 0.25 and X = −1 otherwise. E(X)?

A. 1

B. 0.25

C. −0.75

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 50 • BACKWARD 1 • Q3/5

1637

QUESTION ID

S50-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0.25

GRADING NOTE 1 − 0.75.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 50 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

X = 100 with probability 0.01 and 0 otherwise. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 50 • BACKWARD 1 • Q4/5

1638

QUESTION ID
S50-B1-Q4

TYPE
Short answer

CORRECT ANSWER
1

GRADING NOTE 100 · 0.01.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 50 • BACKWARD 1 • Q4/5

True or false?

E(X) = 5 when X = 10 with probability 0.2 and X = 0 otherwise.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 50 • BACKWARD 1 • Q5/5

1639

QUESTION ID

S50-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 50 • BACKWARD 1 • Q5/5

SET 50 • HSS-MD.B.5.a

BACKWARD 2

Add and subtract signed numbers • Grade 7 • The Number System • 7.NS.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Compute 0.5 − 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 50 • BACKWARD 2 • Q1/5

1641

QUESTION ID

S50-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−1.5

GRADING NOTE Negative result.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 50 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 5(1/6) − 1(5/6).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 50 • BACKWARD 2 • Q2/5

1642

QUESTION ID

S50-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0

GRADING NOTE 5/6 − 5/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 50 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$2 - 1.5 = 0.5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 50 • BACKWARD 2 • Q3/5

1643

QUESTION ID

S50-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 50 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute $2 - 5$.

A. 3

B. -7

C. 7

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 50 • BACKWARD 2 • Q4/5

1644

QUESTION ID

S50-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. −3

GRADING NOTE Negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 50 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$-1.5 + 2 = -3.5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 50 • BACKWARD 2 • Q5/5

1645

QUESTION ID

S50-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 50 • BACKWARD 2 • Q5/5

SET 50 • HSS-MD.B.5.a

FORWARD 1

Set a price that makes a game fair • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.5.a

A game pays $12 if a fair die shows 6 and nothing otherwise. What price makes the game fair?

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • FORWARD 1 • Q1/5

1647

QUESTION ID
S50-F1-Q1

TYPE
Short answer

CORRECT ANSWER
2

GRADING NOTE Expected payout 12/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • FORWARD 1 • Q1/5

A raffle has one $1,000 prize and 400 tickets. What ticket price makes the raffle fair?

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • FORWARD 1 • Q2/5

1648

QUESTION ID
S50-F1-Q2

TYPE
Short answer

CORRECT ANSWER
2.5

GRADING NOTE 1000/400.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F1-Q2**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • FORWARD 1 • Q2/5

True or false?

A game is fair when the price equals the expected payout.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • FORWARD 1 • Q3/5

1649

QUESTION ID
S50-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Expected net 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • FORWARD 1 • Q3/5

A game costs $3 and pays $10 with probability 0.25 (and nothing otherwise). What is the expected net gain over 100 plays?

A. −$0.50

B. $50

C. $250

D. −$50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • FORWARD 1 • Q4/5

1650

QUESTION ID
S50-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. −$50

GRADING NOTE 100(2.5 − 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • FORWARD 1 • Q4/5

True or false?

A raffle with 400 tickets and a $1,000 prize is fair at $5 per ticket.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • FORWARD 1 • Q5/5

1651

QUESTION ID

S50-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Fair at $2.50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • FORWARD 1 • Q5/5

SET 50 • HSS-MD.B.5.a

FORWARD 2

Compare decisions by expected value • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.5.b

ANSWER KEY Reduced copy of the question page, with its answer

Option X gains $1,000 with probability 0.6 and loses $500 with probability 0.4. Option Y gains $300 for sure. Find the expected gain of each.

X: _____

Y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 50 • FORWARD 2 • Q1/5

1653

QUESTION ID

S50-F2-Q1

TYPE

Short answer

CORRECT ANSWER

400; 300

GRADING NOTE Form required: both. 600 − 200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 50 • FORWARD 2 • Q1/5

Option X: expected gain $400 but a 40% chance of losing $500. Option Y: $300 for sure. Which statement is correct?

A. X is certain to gain $400 for whoever chooses it.

B. X has the higher expected value, but Y has no risk of loss.

C. X and Y have exactly the same expected value.

D. Y has the higher expected value of the two.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 50 • FORWARD 2 • Q2/5

1654

QUESTION ID
S50-F2-Q2

TYPE
Multiple choice

CORRECT ANSWER
B. X has the higher expected value, but Y has no risk of loss.

GRADING NOTE Expected value and risk.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 50 • FORWARD 2 • Q2/5

True or false?

Expected value describes the long-run average, not what happens in one decision.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 50 • FORWARD 2 • Q3/5

1655

QUESTION ID
S50-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Meaning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 50 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Insurance costs $200. Without it, a $1,500 repair is needed with probability 0.1 (otherwise no repair is needed). What is the expected repair cost without insurance?

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 50 • FORWARD 2 • Q4/5

1656

QUESTION ID
S50-F2-Q4

TYPE
Short answer

CORRECT ANSWER
150

GRADING NOTE 1500 · 0.1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F2-Q4**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 50 • FORWARD 2 • Q4/5

True or false?

Buying insurance always has a greater expected value than not buying it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 50 • FORWARD 2 • Q5/5

1657

QUESTION ID
S50-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Insurers set prices above expected costs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 50 • FORWARD 2 • Q5/5

Evaluate and compare strategies by expected value (+)

MAIN	Probability & Statistics main question	**HSS-MD.B.5.b**
BACKWARD 1	Find the expected payoff of one choice	**HSS-MD.B.5.a**
BACKWARD 2	Read a distribution and its mean	**HSS-MD.A.2**
FORWARD 1	Find break-even probabilities	**HSS-MD.B.5.b**
FORWARD 2	Make fair decisions with probability	**HSS-MD.B.6**

SET 51 • HSS-MD.B.5.b

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.5.b

Plan A: premium $1,200, deductible $500. Plan B: premium $900, deductible $1,000. In a year, one claim larger than $1,000 occurs with probability 0.2, and otherwise there is no claim. When the claim occurs, the owner pays the plan's deductible. Find the expected yearly cost of each plan (premium plus expected deductible).

Plan A: _____

Plan B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • MAIN • Q1/5

1660

QUESTION ID

S51-M-Q1

TYPE

Short answer

CORRECT ANSWER

1300; 1100

GRADING NOTE Form required: both. 1200 + 100; 900 + 200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • MAIN • Q1/5

Plan A has expected cost $1,300 and plan B $1,100 per year. Which plan is better on the basis of expected cost?

A. Plan B

B. It cannot be decided

C. Plan A

D. They are equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • MAIN • Q2/5

1661

QUESTION ID
S51-M-Q2

TYPE
Multiple choice

CORRECT ANSWER
A. Plan B

GRADING NOTE Lower expected cost.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-M-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Option X gains $2,000 with probability 0.3 and loses $400 with probability 0.7. Option Y gains $250 for sure. Find the expected gain of each.

X: _____

Y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • MAIN • Q3/5

1662

QUESTION ID

S51-M-Q3

TYPE

Short answer

CORRECT ANSWER

320; 250

GRADING NOTE Form required: both. 600 − 280.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

X has expected gain $320 with a 70% chance of losing $400; Y gains $250 for sure. Which statement is correct?

A. Y is better on expected value, so everyone should prefer Y.

B. X is certain to gain $320 for the person who chooses it.

C. X and Y have the same expected value, so they are equal.

D. X is better on average, but someone who cannot risk a loss may prefer Y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • MAIN • Q4/5

1663

QUESTION ID
S51-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. X is better on average, but someone who cannot risk a loss may prefer Y.

GRADING NOTE Expected value plus risk.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A phone warranty costs $150. Without it, a repair costing $800 is needed with probability 0.1 (otherwise no repair is needed). What is the expected repair cost without the warranty, and how much more is the warranty than that?

Expected cost: _____

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • MAIN • Q5/5

1664

QUESTION ID
S51-M-Q5

TYPE
Short answer

CORRECT ANSWER
80; 70

GRADING NOTE Form required: both. 800 · 0.1; 150 − 80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • MAIN • Q5/5

SET 51 • HSS-MD.B.5.b

BACKWARD 1

Find the expected payoff of one choice • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.5.a

ANSWER KEY Reduced copy of the question page, with its answer

A choice gains $200 with probability 0.5 and $0 otherwise. Find the expected gain.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 51 • BACKWARD 1 • Q1/5

1666

QUESTION ID
S51-B1-Q1

TYPE
Short answer

CORRECT ANSWER
100

GRADING NOTE 200 · 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 51 • BACKWARD 1 • Q1/5

A choice loses $1,000 with probability 0.2 and $0 otherwise. Find the expected loss.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 51 • BACKWARD 1 • Q2/5

1667

QUESTION ID
S51-B1-Q2

TYPE
Short answer

CORRECT ANSWER
200

GRADING NOTE 1000 · 0.2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-B1-Q2**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 51 • BACKWARD 1 • Q2/5

True or false?

A sure gain of $300 has expected value $300.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 51 • BACKWARD 1 • Q3/5

1668

QUESTION ID
S51-B1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Probability 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 51 • BACKWARD 1 • Q3/5

A bet wins $50 with probability 0.3 and loses $20 otherwise. Expected gain?

A. $30

B. −$14

C. $15

D. $1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 51 • BACKWARD 1 • Q4/5

1669

QUESTION ID
S51-B1-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. $1

GRADING NOTE 15 − 14.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-B1-Q4**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 51 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A choice with expected gain $0 can never make money.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 51 • BACKWARD 1 • Q5/5

1670

QUESTION ID
S51-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It averages 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 51 • BACKWARD 1 • Q5/5

SET 51 • HSS-MD.B.5.b

BACKWARD 2

Read a distribution and its mean • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

True or false?

The option with the larger expected value is better on average.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S51-B2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 51 • BACKWARD 2 • Q1/5

1672

QUESTION ID

S51-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Long run.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S51-B2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 51 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

X = 1000 with probability 0.6 and −500 with probability 0.4. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 51 • BACKWARD 2 • Q2/5

1673

QUESTION ID
S51-B2-Q2

TYPE
Short answer

CORRECT ANSWER
400

GRADING NOTE 600 − 200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-B2-Q2**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 51 • BACKWARD 2 • Q2/5

Which has the larger expected value: $50 for sure, or $120 with probability 0.5 (and $0 otherwise)?

A. cannot tell

B. they are equal

C. $120 with probability 0.5

D. $50 for sure

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 51 • BACKWARD 2 • Q3/5

1674

QUESTION ID
S51-B2-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. $120 with probability 0.5

GRADING NOTE 60 > 50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 51 • BACKWARD 2 • Q3/5

Cost = 500 with probability 0.2 and 0 otherwise. Find the expected cost.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 51 • BACKWARD 2 • Q4/5

1675

QUESTION ID
S51-B2-Q4

TYPE
Short answer

CORRECT ANSWER
100

GRADING NOTE 500 · 0.2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 51 • BACKWARD 2 • Q4/5

True or false?

Expected cost = cost × (1 − probability).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S51-B2-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 51 • BACKWARD 2 • Q5/5

1676

QUESTION ID
S51-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Cost × probability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S51-B2-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 51 • BACKWARD 2 • Q5/5

SET 51 • HSS-MD.B.5.b

FORWARD 1

Find break-even probabilities • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.5.b

ANSWER KEY Reduced copy of the question page, with its answer

A $150 warranty covers an $800 repair. For what repair probability is the expected repair cost equal to $150? Give a decimal.

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • FORWARD 1 • Q1/5

1678

QUESTION ID
S51-F1-Q1

TYPE
Short answer

CORRECT ANSWER
0.1875

GRADING NOTE 150/800.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • FORWARD 1 • Q1/5

True or false?

If the repair probability is below the break-even value, skipping the warranty has the lower expected cost.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • FORWARD 1 • Q2/5

1679

QUESTION ID
S51-F1-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Expected repair cost < 150.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Plan A costs $1,200 plus a $500 deductible; plan B costs $900 plus a $1,000 deductible. For what claim probability p are the expected costs equal?

p: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • FORWARD 1 • Q3/5

1680

QUESTION ID

S51-F1-Q3

TYPE

Short answer

CORRECT ANSWER

0.6

GRADING NOTE 1200 + 500p = 900 + 1000p.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • FORWARD 1 • Q3/5

With claim probability 0.8, which plan has the lower expected cost (A: 1200 + 500p; B: 900 + 1000p)?

A. Plan A

B. Plan B

C. They are equal

D. Neither

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • FORWARD 1 • Q4/5

1681

QUESTION ID
S51-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. Plan A

GRADING NOTE A 1600, B 1700.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • FORWARD 1 • Q4/5

True or false?

The better plan never depends on the claim probability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • FORWARD 1 • Q5/5

1682

QUESTION ID
S51-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It switches at p = 0.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • FORWARD 1 • Q5/5

SET 51 • HSS-MD.B.5.b

FORWARD 2

Make fair decisions with probability • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6

Four friends want to decide fairly with one fair die who gets a ticket. Which method is fair?

A. Assign 1-4 to the friends; on a 5 or 6, roll again.

B. Assign 1-4 and give 5 and 6 to the first friend.

C. Give the ticket to whoever guesses the roll.

D. Give 1-3 to one friend and 4-6 to the other three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 51 • FORWARD 2 • Q1/5

1684

QUESTION ID
S51-F2-Q1

TYPE
Multiple choice

CORRECT ANSWER
A. Assign 1-4 to the friends; on a 5 or 6, roll again.

GRADING NOTE Rolling again until a 1-4 comes up keeps 1/4 each; the others favor someone or do not use chance fairly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 51 • FORWARD 2 • Q1/5

True or false?

A method is fair if every person has the same probability of being chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 51 • FORWARD 2 • Q2/5

1685

QUESTION ID

S51-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 51 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Ana has 4 tickets, Ben 6 and Cy 10 in a drawing of 20 tickets; one ticket is drawn, each equally likely. What is P(Ana wins)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 51 • FORWARD 2 • Q3/5

1686

QUESTION ID
S51-F2-Q3

TYPE
Short answer

CORRECT ANSWER
1/5

GRADING NOTE 4/20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F2-Q3**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 51 • FORWARD 2 • Q3/5

In a game with two fair dice rolled independently, A wins if the sum is 7 or less; otherwise B wins. Is it fair?

A. No: A wins with probability 7/12.

B. No: A wins with probability 6/11.

C. Yes: A has 6 sums and B has 5.

D. Yes: each player has 1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 51 • FORWARD 2 • Q4/5

1687

QUESTION ID
S51-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. No: A wins with probability 7/12.

GRADING NOTE 1 + 2 + 3 + 4 + 5 + 6 = 21 of 36 outcomes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 51 • FORWARD 2 • Q4/5

True or false?

Letting the oldest person choose is a fair random method.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 51 • FORWARD 2 • Q5/5

1688

QUESTION ID
S51-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Not random.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 51 • FORWARD 2 • Q5/5

Use probabilities to make fair decisions (+)

MAIN	Probability & Statistics main question	**HSS-MD.B.6**
BACKWARD 1	Simulate with random digits	**HSS-IC.A.2**
BACKWARD 2	Find probabilities in a uniform model	**7.SP.C.7.a**
FORWARD 1	Design fair methods for awkward numbers	**HSS-MD.B.6**
FORWARD 2	Analyze a decision with a medical test	**HSS-MD.B.7**

SET 52 • HSS-MD.B.6

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Three friends want to decide fairly who gets the last ticket. Which method is fair?

A. Roll a die: 1 Ana, 2 Ben, 3-6 Cy.

B. Roll a die: 1-2 Ana, 3-4 Ben, 5-6 Cy.

C. Flip a coin: heads Ana, tails Ben; Cy only on an edge.

D. Ana chooses, since she asked first.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • MAIN • Q1/5

1691

QUESTION ID
S52-M-Q1

TYPE
Multiple choice

CORRECT ANSWER
B. Roll a die: 1-2 Ana, 3-4 Ben, 5-6 Cy.

GRADING NOTE Each person gets 2 of 6 outcomes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

One of four people, A, B, C and D, is chosen with random digits: 0-1 = A, 2-3 = B, 4-5 = C, 6-7 = D, and 8-9 are skipped. Reading line 2 of the table from the left, who is chosen? Enter A, B, C or D.

Line	Digits	Line	Digits
1	15881 39986 98797 30911	5	17784 86488 76634 40029
2	41707 53645 56815 18844	6	78753 47787 68045 71784
3	72940 55761 69871 68976	7	72671 47513 62701 07981
4	84697 88093 20819 67273	8	93319 08452 05997 28379

Person: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • MAIN • Q2/5

1692

QUESTION ID
S52-M-Q2

TYPE
Short answer

CORRECT ANSWER
C

GRADING NOTE The first digit of line 2 is 4 → C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Two players each flip a fair coin, independently. Ana wins if the coins match; Ben wins if they differ. What is P(Ana wins)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • MAIN • Q3/5

1693

QUESTION ID
S52-M-Q3

TYPE
Short answer

CORRECT ANSWER
1/2

GRADING NOTE HH or TT: 2 of 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-M-Q3**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • MAIN • Q3/5

Two fair dice are rolled independently. Player A wins if the sum is 6, 7 or 8; otherwise B wins. Is the game fair?

A. Yes: the sums are split fairly.

B. Yes: each player wins with probability 1/2.

C. No: A wins with probability 3/11.

D. No: A wins with probability 4/9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S52-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • MAIN • Q4/5

1694

QUESTION ID

S52-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. No: A wins with probability 4/9.

GRADING NOTE 16 of 36 outcomes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S52-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In a drawing, Ana has 3 tickets, Ben 5 and Cy 2, out of 10; one ticket is drawn, each equally likely. What is P(Ben wins)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • MAIN • Q5/5

1695

QUESTION ID
S52-M-Q5

TYPE
Short answer

CORRECT ANSWER
1/2

GRADING NOTE 5/10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-M-Q5**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • MAIN • Q5/5

SET 52 • HSS-MD.B.6

BACKWARD 1

Simulate with random digits • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

True or false?

Each digit 0-9 in a random-digit table is equally likely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 52 • BACKWARD 1 • Q1/5

1697

QUESTION ID

S52-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Uniform.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 52 • BACKWARD 1 • Q1/5

To choose among 4 people with single digits, which assignment is fair?

A. 0-2, 3-5, 6-8 and 9

B. 0-1, 2-3, 4-5, 6-9

C. 0, 1, 2, 3 and stop

D. 0-1, 2-3, 4-5, 6-7 and skip 8-9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 52 • BACKWARD 1 • Q2/5

1698

QUESTION ID

S52-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 0-1, 2-3, 4-5, 6-7 and skip 8-9

GRADING NOTE Two digits each, extras skipped.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-B1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 52 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Using 0-1 = A, 2-3 = B, 4-5 = C, 6-7 = D, skip 8-9, what does the digit 9 mean?

Answer (enter skip or a letter): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 52 • BACKWARD 1 • Q3/5

1699

QUESTION ID

S52-B1-Q3

TYPE

Short answer

CORRECT ANSWER

skip

GRADING NOTE Skipped.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-B1-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 52 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Skipping digits 8 and 9 makes the assignment unfair.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 52 • BACKWARD 1 • Q4/5

1700

QUESTION ID

S52-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The others stay equally likely.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 52 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

One of four people is chosen with random digits: 0-1 = A, 2-3 = B, 4-5 = C, 6-7 = D, and 8-9 are skipped (read the next digit). The digits read 8, 9, 3. Who is chosen?

A. B

B. A

C. no one

D. D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 52 • BACKWARD 1 • Q5/5

1701

QUESTION ID

S52-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. B

GRADING NOTE 8 and 9 skipped; 3 → B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-B1-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 52 • BACKWARD 1 • Q5/5

SET 52 • HSS-MD.B.6

BACKWARD 2

Find probabilities in a uniform model • Grade 7 • Statistics & Probability • 7.SP.C.7.a

ANSWER KEY Reduced copy of the question page, with its answer

A fair six-sided die is rolled. What is P(an odd number)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 52 • BACKWARD 2 • Q1/5

1703

QUESTION ID

S52-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1/2

GRADING NOTE 3/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 52 • BACKWARD 2 • Q1/5

True or false?

With a fair coin, heads and tails each have probability 1/2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 52 • BACKWARD 2 • Q2/5

1704

QUESTION ID

S52-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Uniform.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 52 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Names of 5 people are in a hat. P(a particular person is drawn)?

A. 1/2

B. 1/4

C. 5

D. 1/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 52 • BACKWARD 2 • Q3/5

1705

QUESTION ID

S52-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 1/5

GRADING NOTE Uniform.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 52 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two fair coins are flipped independently. What is P(they match)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 52 • BACKWARD 2 • Q4/5

1706

QUESTION ID

S52-B2-Q4

TYPE

Short answer

CORRECT ANSWER

1/2

GRADING NOTE HH, TT.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 52 • BACKWARD 2 • Q4/5

True or false?

A spinner with one half red and two quarters blue and green is fair among the three colors.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 52 • BACKWARD 2 • Q5/5

1707

QUESTION ID

S52-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1/2, 1/4, 1/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 52 • BACKWARD 2 • Q5/5

SET 52 • HSS-MD.B.6

FORWARD 1

Design fair methods for awkward numbers • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6

Five people must be chosen from fairly with one fair die. Which method works?

A. Assign 1-5 to the people and give 6 to the first person.

B. Assign 1-5 to the people; on a 6, roll again.

C. Assign 1-6 and let the sixth face mean "everyone".

D. Roll and divide the result by 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • FORWARD 1 • Q1/5

1709

QUESTION ID
S52-F1-Q1

TYPE
Multiple choice

CORRECT ANSWER
B. Assign 1-5 to the people; on a 6, roll again.

GRADING NOTE Rolling again until a 1-5 comes up keeps 1/5 each; the others favor one person or do not pick exactly one person.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Five people are assigned the faces 1-5 of a fair die; on a 6 the die is rolled again, until a 1-5 comes up. What is each person's probability of being chosen? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • FORWARD 1 • Q2/5

1710

QUESTION ID
S52-F1-Q2

TYPE
Short answer

CORRECT ANSWER
1/5

GRADING NOTE Uniform over 5 faces.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F1-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A method that uses one roll of a fair die and gives one person 2 of the 6 faces and each other person 1 face is unfair.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • FORWARD 1 • Q3/5

1711

QUESTION ID
S52-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE 2/6 vs 1/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

One of 3 people is chosen with random two-digit numbers 00-99. Each person gets the same number of values, as many of the 100 numbers as possible are used, and a skipped number means reading the next number. How many numbers does each person get?

Numbers each: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • FORWARD 1 • Q4/5

1712

QUESTION ID
S52-F1-Q4

TYPE
Short answer

CORRECT ANSWER
33

GRADING NOTE $3 \cdot 33 = 99$ numbers are used (for example 00-98) and one (99) is skipped. Fewer per person would also be fair but would use fewer of the numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • FORWARD 1 • Q4/5

True or false?

Drawing names one at a time from a hat without replacement is unfair to the last name.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S52-F1-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • FORWARD 1 • Q5/5

1713

QUESTION ID
S52-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Each order position is equally likely.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S52-F1-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • FORWARD 1 • Q5/5

SET 52 • HSS-MD.B.6

FORWARD 2

Analyze a decision with a medical test • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Of 2,000 people, 1% have a disease. How many have it?

People: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 52 • FORWARD 2 • Q1/5

1715

QUESTION ID
S52-F2-Q1

TYPE
Short answer

CORRECT ANSWER
20

GRADING NOTE 0.01 · 2000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 52 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A test is positive for 95% of the 20 people with a disease. How many of them test positive?

People: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 52 • FORWARD 2 • Q2/5

1716

QUESTION ID
S52-F2-Q2

TYPE
Short answer

CORRECT ANSWER
19

GRADING NOTE $0.95 \cdot 20$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F2-Q2**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 52 • FORWARD 2 • Q2/5

True or false?

A test can give positive results for people who do not have the disease.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 52 • FORWARD 2 • Q3/5

1717

QUESTION ID
S52-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE False positives.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 52 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Of 1,980 healthy people, 10% test positive. How many false positives are there?

A. 10

B. 198

C. 19

D. 1,782

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 52 • FORWARD 2 • Q4/5

1718

QUESTION ID
S52-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. 198

GRADING NOTE 0.1 · 1980. 1,782 is the number who correctly test negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 52 • FORWARD 2 • Q4/5

True or false?

If a test is positive for 95% of sick people, then 95% of positives are sick.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 52 • FORWARD 2 • Q5/5

1719

QUESTION ID
S52-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The condition is reversed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 52 • FORWARD 2 • Q5/5

Analyze decisions about medical testing (+)

MAIN	Probability & Statistics main question	**HSS-MD.B.7**
BACKWARD 1	Read a two-way table as a sample space	**HSS-CP.A.4**
BACKWARD 2	Find a conditional probability as a fraction of outcomes	**HSS-CP.B.6**
FORWARD 1	See how prevalence changes the answer	**HSS-MD.B.7**
FORWARD 2	Analyze strategies in games and quality control	**HSS-MD.B.7**

SET 53 • HSS-MD.B.7

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

ANSWER KEY Reduced copy of the question page, with its answer

1,000 people are tested. 2% have a disease; the test is positive for 90% of people with the disease. How many have the disease, and how many of them test positive?

Have the disease: _____

Test positive: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • MAIN • Q1/5

1722

QUESTION ID

S53-M-Q1

TYPE

Short answer

CORRECT ANSWER

20; 18

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Of the 980 people without the disease, 5% test positive. How many false positives are there?

People: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • MAIN • Q2/5

1723

QUESTION ID
S53-M-Q2

TYPE
Short answer

CORRECT ANSWER
49

GRADING NOTE 0.05 · 980.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the table for 1,000 people. A person tests positive. What is P(has disease given positive)? Give a fraction in lowest terms.

	Tests positive	Tests negative	Total
Has disease	18	2	20
Does not	49	931	980
Total	67	933	1000

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • MAIN • Q3/5

1724

QUESTION ID
S53-M-Q3

TYPE
Short answer

CORRECT ANSWER
18/67

GRADING NOTE 18/(18 + 49).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In the table, P(disease given positive) = 18/67, about 27%. What does this show?

	Tests positive	Tests negative	Total
Has disease	18	2	20
Does not	49	931	980
Total	67	933	1000

A. Most people who test positive do not have the disease.

B. Most people who have the disease test negative.

C. The test is wrong 73% of the time for everyone.

D. A positive test means a 90% chance of disease.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • MAIN • Q4/5

1725

QUESTION ID
S53-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. Most people who test positive do not have the disease.

GRADING NOTE Low prevalence makes false positives outnumber true ones.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the table. A person tests negative. What is P(does not have the disease given negative)? Give a fraction in lowest terms.

	Tests positive	Tests negative	Total
Has disease	18	2	20
Does not	49	931	980
Total	67	933	1000

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • MAIN • Q5/5

1726

QUESTION ID
S53-M-Q5

TYPE
Short answer

CORRECT ANSWER
931/933

GRADING NOTE 931/(931 + 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • MAIN • Q5/5

SET 53 • HSS-MD.B.7

BACKWARD 1

Read a two-way table as a sample space • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

In the test table, how many people test positive in all?

	Tests positive	Tests negative	Total
Has disease	18	2	20
Does not	49	931	980
Total	67	933	1000

People: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 53 • BACKWARD 1 • Q1/5

1728

QUESTION ID

S53-B1-Q1

TYPE

Short answer

CORRECT ANSWER

67

GRADING NOTE 18 + 49.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-B1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 53 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In the test table, how many people test negative in all?

	Tests positive	Tests negative	Total
Has disease	18	2	20
Does not	49	931	980
Total	67	933	1000

People: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 53 • BACKWARD 1 • Q2/5

1729

QUESTION ID

S53-B1-Q2

TYPE

Short answer

CORRECT ANSWER

933

GRADING NOTE 2 + 931.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-B1-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 53 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In the test table, 20 people have the disease.

	Tests positive	Tests negative	Total
Has disease	18	2	20
Does not	49	931	980
Total	67	933	1000

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 53 • BACKWARD 1 • Q3/5

1730

QUESTION ID

S53-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Row total.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-B1-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 53 • BACKWARD 1 • Q3/5

In the test table, which cell counts false positives?

	Tests positive	Tests negative	Total
Has disease	18	2	20
Does not	49	931	980
Total	67	933	1000

A. does not have the disease, tests positive

B. does not have the disease, tests negative

C. has the disease, tests negative

D. has the disease, tests positive

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 53 • BACKWARD 1 • Q4/5

1731

QUESTION ID

S53-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. does not have the disease, tests positive

GRADING NOTE 49.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 53 • BACKWARD 1 • Q4/5

True or false?

In the test table, false negatives number 18.

	Tests positive	Tests negative	Total
Has disease	18	2	20
Does not	49	931	980
Total	67	933	1000

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 53 • BACKWARD 1 • Q5/5

1732

QUESTION ID

S53-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-B1-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 53 • BACKWARD 1 • Q5/5

SET 53 • HSS-MD.B.7

BACKWARD 2

Find a conditional probability as a fraction of outcomes • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Of 67 people who test positive, 18 have the disease.
What fraction is that?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 53 • BACKWARD 2 • Q1/5

1734

QUESTION ID
S53-B2-Q1

TYPE
Short answer

CORRECT ANSWER
18/67

GRADING NOTE 18/67.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-B2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 53 • BACKWARD 2 • Q1/5

True or false?

P(disease given positive) uses the positive column as the sample space.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 53 • BACKWARD 2 • Q2/5

1735

QUESTION ID

S53-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Condition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 53 • BACKWARD 2 • Q2/5

Of 20 people with the disease, 18 test positive.
P(positive given disease)?

A. 18/67

B. 9/10

C. 2/20

D. 1/50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 53 • BACKWARD 2 • Q3/5

1736

QUESTION ID

S53-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 9/10

GRADING NOTE 18/20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-B2-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 53 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Of 980 healthy people, 49 test positive. What is P(positive given healthy)? Give a decimal.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 53 • BACKWARD 2 • Q4/5

1737

QUESTION ID

S53-B2-Q4

TYPE

Short answer

CORRECT ANSWER

0.05

GRADING NOTE 49/980.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-B2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 53 • BACKWARD 2 • Q4/5

True or false?

P(positive given disease) = P(disease given positive).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 53 • BACKWARD 2 • Q5/5

1738

QUESTION ID

S53-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 9/10 vs 18/67.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • HSS-MD.B.7

FORWARD 1

See how prevalence changes the answer • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

ANSWER KEY Reduced copy of the question page, with its answer

In a population of 1,000 people, 10% have a disease. A test is positive for 90% of the people with the disease and for 5% of the people without it. How many true positives and how many false positives are there?

True positives: _____

False positives: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 1 • Q1/5

1740

QUESTION ID

S53-F1-Q1

TYPE

Short answer

CORRECT ANSWER

90; 45

GRADING NOTE Form required: both. 0.9 · 100; 0.05 · 900.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

With 90 true positives and 45 false positives, what is P(disease given positive)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 1 • Q2/5

1741

QUESTION ID
S53-F1-Q2

TYPE
Short answer

CORRECT ANSWER
2/3

GRADING NOTE 90/135.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 1 • Q2/5

True or false?

The same test gives a higher P(disease given positive) when the disease is more common.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 1 • Q3/5

1742

QUESTION ID

S53-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 18/67 versus 2/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 1 • Q3/5

Why do doctors often order a second test after a positive result for a rare disease?

A. Negative results are never correct.

B. The first test is always wrong.

C. A second test lowers the prevalence.

D. Many first positives are false positives.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 1 • Q4/5

1743

QUESTION ID

S53-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Many first positives are false positives.

GRADING NOTE Low prevalence makes many positives false. The others are false: tests are not always wrong, negatives can be correct, and testing does not change prevalence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 1 • Q4/5

True or false?

For a rare disease, a positive result makes having the disease certain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 1 • Q5/5

1744

QUESTION ID
S53-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE About 27% here.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 1 • Q5/5

SET 53 • HSS-MD.B.7

FORWARD 2

Analyze strategies in games and quality control • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A box of 10 parts has 2 defective. An inspector checks 2 parts chosen at random, without replacement (every pair equally likely). What is P(no defective part is found)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 2 • Q1/5

1746

QUESTION ID
S53-F2-Q1

TYPE
Short answer

CORRECT ANSWER
28/45

GRADING NOTE 8C2/10C2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F2-Q1**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 2 • Q1/5

True or false?

Checking more parts raises the chance of finding a defective part.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 2 • Q2/5

1747

QUESTION ID
S53-F2-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Fewer escape.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A test question has 4 choices and one correct answer; a correct answer earns 1 point, a wrong answer loses 1/3 point, and a blank earns 0. A guess picks at random among the choices still in play. What is the expected score of a guess with no choices eliminated, and of a guess after one wrong choice is eliminated? Give a fraction in lowest terms.

Random guess: _____

After eliminating one: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 2 • Q3/5

1748

QUESTION ID

S53-F2-Q3

TYPE

Short answer

CORRECT ANSWER

0; 1/9

GRADING NOTE Form required: both. 1/4 − (3/4)(1/3); 1/3 − (2/3)(1/3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A test question has 4 choices and one correct answer: +1 point if correct, −1/3 point if wrong, 0 if left blank. A guess picks at random among the choices still in play. When does a guess have a positive expected score?

A. always, even with no information

B. after eliminating at least one wrong choice

C. only on the last question of the test

D. never, under any circumstances

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 2 • Q4/5

1749

QUESTION ID
S53-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. after eliminating at least one wrong choice

GRADING NOTE No elimination: 1/4 − (3/4)(1/3) = 0, the same as a blank. One elimination: 1/3 − (2/3)(1/3) = 1/9 > 0. The others ignore the elimination or the zero expected value of a blind guess.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A strategy with a higher expected value always wins in a single try.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 2 • Q5/5

1750

QUESTION ID

S53-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only on average.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 2 • Q5/5

Analyze strategies and decisions with probability (+)

MAIN	Probability & Statistics main question	**HSS-MD.B.7**
BACKWARD 1	Compare expected values of choices	**HSS-MD.A.2**
BACKWARD 2	Count selections to find probabilities	**HSS-CP.B.9**
FORWARD 1	Weigh probabilities against consequences	**HSS-MD.B.7**
FORWARD 2	Combine several probability tools in one decision	**HSS-MD.B.7**

SET 54 • HSS-MD.B.7

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A hockey team trails by 1 goal with 2 minutes left, and its only goal is to avoid losing, which requires tying. Keeping the goalie gives a 0.08 chance of tying; pulling the goalie for an extra skater gives a 0.20 chance (allowing another goal still means a loss, as before). By how many percentage points does pulling the goalie raise the chance of tying, and how many times as large is that chance?

Percentage points: _____

Times as large: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • MAIN • Q1/5

1753

QUESTION ID
S54-M-Q1

TYPE
Short answer

CORRECT ANSWER
12; 2.5

GRADING NOTE Form required: both. 20 − 8 = 12 points; 0.20/0.08 = 2.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The team loses unless it ties. Keeping the goalie gives a 0.08 chance of tying; pulling gives 0.20. Which strategy is better?

A. Pull the goalie: it gives the better chance of avoiding a loss.

B. Keep the goalie: allowing another goal would be worse.

C. Keep the goalie: 0.08 is the safer probability.

D. Either one: both chances are less than one-half.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • MAIN • Q2/5

1754

QUESTION ID
S54-M-Q2

TYPE
Multiple choice

CORRECT ANSWER
A. Pull the goalie: it gives the better chance of avoiding a loss.

GRADING NOTE A second goal against changes nothing: a loss is a loss.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A box of 12 parts has 3 defective. An inspector checks 2 parts chosen at random, without replacement (every pair equally likely). What is P(no defective part is found)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • MAIN • Q3/5

1755

QUESTION ID
S54-M-Q3

TYPE
Short answer

CORRECT ANSWER
6/11

GRADING NOTE 9C2/12C2 = 36/66.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-M-Q3**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A test question has 5 choices and one correct answer; a correct answer earns 1 point, a wrong answer loses 1/4 point, and a blank earns 0. A guess picks at random among the choices still in play. Find the expected score of a guess with no choices eliminated, and of a guess after eliminating one wrong choice. Give a fraction in lowest terms.

Random guess: _____

After eliminating one: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • MAIN • Q4/5

1756

QUESTION ID
S54-M-Q4

TYPE
Short answer

CORRECT ANSWER
0; 1/16

GRADING NOTE Form required: both. 1/5 − (4/5)(1/4); 1/4 − (3/4)(1/4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A company can test 2 or 5 parts from each box of 10 (2 defective). With 2 tested, P(missing both defectives) = 28/45; with 5 tested, it is 2/9. Which is correct?

A. Testing 5 parts lowers the chance of missing the defects.

B. Testing 5 parts always finds both defects.

C. Testing 2 parts finds the defects more often.

D. The number tested does not change anything.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • MAIN • Q5/5

1757

QUESTION ID
S54-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. Testing 5 parts lowers the chance of missing the defects.

GRADING NOTE With 5 tested, P(no defective found) = 8C5/10C5 = 56/252 = 2/9, less than 28/45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • MAIN • Q5/5

SET 54 • HSS-MD.B.7

BACKWARD 1

Compare expected values of choices • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

Strategy A scores 3 with probability 0.5 and 0 otherwise. Find its expected score.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 54 • BACKWARD 1 • Q1/5

1759

QUESTION ID
S54-B1-Q1

TYPE
Short answer

CORRECT ANSWER
1.5

GRADING NOTE $3 \cdot 0.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 54 • BACKWARD 1 • Q1/5

True or false?

Between two strategies, the one with the larger expected value is better on average.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S54-B1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 54 • BACKWARD 1 • Q2/5

1760

QUESTION ID

S54-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Long run.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S54-B1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 54 • BACKWARD 1 • Q2/5

Strategy B scores 2 for sure; strategy A scores 3 with probability 0.5 and 0 otherwise. Which is better on average?

A. cannot tell

B. A

C. B

D. equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 54 • BACKWARD 1 • Q3/5

1761

QUESTION ID
S54-B1-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. B

GRADING NOTE $2 > 1.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-B1-Q3**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 54 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A guess earns 1 with probability 1/3 and loses 1/3 otherwise. Expected score? Give a fraction in lowest terms.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 54 • BACKWARD 1 • Q4/5

1762

QUESTION ID
S54-B1-Q4

TYPE
Short answer

CORRECT ANSWER
1/9

GRADING NOTE 1/3 − 2/9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 54 • BACKWARD 1 • Q4/5

True or false?

An expected score of 0 means the strategy always scores 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 54 • BACKWARD 1 • Q5/5

1763

QUESTION ID
S54-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Average.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 54 • BACKWARD 1 • Q5/5

SET 54 • HSS-MD.B.7

BACKWARD 2

Count selections to find probabilities • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

ANSWER KEY Reduced copy of the question page, with its answer

Compute 10C2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 54 • BACKWARD 2 • Q1/5

1765

QUESTION ID

S54-B2-Q1

TYPE

Short answer

CORRECT ANSWER

45

GRADING NOTE 10 · 9/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 54 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 8C2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 54 • BACKWARD 2 • Q2/5

1766

QUESTION ID

S54-B2-Q2

TYPE

Short answer

CORRECT ANSWER

28

GRADING NOTE 8 · 7/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 54 • BACKWARD 2 • Q2/5

True or false?

8C5 = 56.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 54 • BACKWARD 2 • Q3/5

1767

QUESTION ID

S54-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 8C3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 54 • BACKWARD 2 • Q3/5

Compute 10C5.

A. 30240

B. 120

C. 252

D. 50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 54 • BACKWARD 2 • Q4/5

1768

QUESTION ID
S54-B2-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. 252

GRADING NOTE Combinations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-B2-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 54 • BACKWARD 2 • Q4/5

True or false?

P(no defectives in 2 of 10 with 2 defective) = 8/10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 54 • BACKWARD 2 • Q5/5

1769

QUESTION ID

S54-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 28/45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 54 • BACKWARD 2 • Q5/5

SET 54 • HSS-MD.B.7

FORWARD 1

Weigh probabilities against consequences • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A product test costs $50 per box and finds the defects with probability 0.8. An undetected defective box costs $400. For a defective box, what is the expected cost of shipping without testing, and with testing (test cost plus expected cost of a missed defect)?

Without testing: _____

With testing: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 1 • Q1/5

1771

QUESTION ID
S54-F1-Q1

TYPE
Short answer

CORRECT ANSWER
400; 130

GRADING NOTE Form required: both. 50 + 0.2 · 400.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 1 • Q1/5

True or false?

A strategy can have a better expected value even if it sometimes produces the worst outcome.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 1 • Q2/5

1772

QUESTION ID
S54-F1-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Expected value averages outcomes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F1-Q2**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 1 • Q2/5

On a test, each question has 4 choices and one correct answer: +1 point if correct, −1/3 point if wrong, 0 if blank. A guess picks at random among the choices still in play. Which statement about strategy is correct?

A. A blind guess averages −1/12, less than a blank, so never guess without eliminating.

B. A blind guess and a blank both average 0; a guess after one elimination averages 1/9.

C. Eliminating one wrong choice leaves the average score of a guess at 0.

D. A blind guess averages 1/4, more than a blank, so guess on every question.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 1 • Q3/5

1773

QUESTION ID
S54-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. A blind guess and a blank both average 0; a guess after one elimination averages 1/9.

GRADING NOTE Blind: 1/4 − (3/4)(1/3) = 0, the same as a blank (so blind guessing neither helps nor hurts on average). After one elimination: 1/3 − (2/3)(1/3) = 1/9. The others give wrong expected values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A question has 4 choices and one correct answer: +1 point if correct, −1/3 point if wrong, 0 if blank. Two wrong choices are eliminated, and a guess picks at random between the 2 left. What is the expected score of the guess? Give a fraction in lowest terms.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 1 • Q4/5

1774

QUESTION ID
S54-F1-Q4

TYPE
Short answer

CORRECT ANSWER
1/3

GRADING NOTE 1/2 − (1/2)(1/3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 1 • Q4/5

True or false?

Probability analysis can tell exactly what will happen in one game.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 1 • Q5/5

1775

QUESTION ID
S54-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It describes chances.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 1 • Q5/5

SET 54 • HSS-MD.B.7

FORWARD 2

Combine several probability tools in one decision • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A quiz has 5 questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others), and a student guesses on all of them. What is the expected number correct, and P(all wrong)? Give each answer as a fraction in lowest terms.

Expected correct: _____

P(all wrong): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 2 • Q1/5

1777

QUESTION ID

S54-F2-Q1

TYPE

Short answer

CORRECT ANSWER

5/4; 243/1024

GRADING NOTE Form required: both, as fractions. $5 \cdot 1/4$; $(3/4)^5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

P(at least one correct) = 1 − P(all wrong).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 2 • Q2/5

1778

QUESTION ID

S54-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Complement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A quiz has 5 questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others), and a student guesses on all of them. What is P(at least one correct)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 2 • Q3/5

1779

QUESTION ID
S54-F2-Q3

TYPE
Short answer

CORRECT ANSWER
781/1024

GRADING NOTE 1 − 243/1024.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 2 • Q3/5

A team must choose between a play that gains 2 points with probability 0.5 (and 0 otherwise) and one that gains 3 points with probability 0.3 (and 0 otherwise). Which has the higher expected gain?

A. equal

B. cannot tell

C. the 2-point play

D. the 3-point play

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 2 • Q4/5

1780

QUESTION ID

S54-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. the 2-point play

GRADING NOTE 1.0 versus 0.9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 2 • Q4/5

True or false?

On one final play, option P scores 2 points with probability 0.5 (otherwise 0) and option Q scores 3 points with probability 0.3 (otherwise 0). The team needs at least 3 points on this play to win. Option P is the better choice.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 2 • Q5/5

1781

QUESTION ID

S54-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Option P can never reach 3 points, so its chance of winning is 0; option Q wins with probability 0.3, even though P has the higher expected gain (1.0 versus 0.9).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 2 • Q5/5